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Matter and its classification

1. When two or more elements combine chemically with one another ...... is formed ?

(A) Element (B) Mixture (C) Fluid (D) Compound

2. Which one of the following statements is correct ?

(A) Two or more than two atoms of the element combine and form compound.

(B) The atoms retain their own property when form a compound.

(C) Each substance of a mixture loses its original property.

(D) Each substance of a mixture can be separated by physical or chemical methods.

3. Which one of the following is heterogeneous mixture ?

(A) Air (B) Brass (C) NaCl + Fe (D) Salt solution

4. Which one of the following pairs have both are present a compound and mixture

(A) NH_3 and salt solution (B) Lemon juice and Liquidgum

(C) Ice cream and NaCl (D) Gun powder and plaster of paris.

5. Which one of the following is not an example of homogeneous mixture ?

(A) Sugar solution (B) $O_2 + N_2$ gases (C) Salt + Sand (D) Zn + Cu alloy

6. Co stands for _____ while CO stands for ______ .

(A) The atoms of the element cobalt; the atoms of the compound carbon monoxide

(B) The atoms of the element carbon monoxide

(C) The atom of the element cobalt; the molecules of the compound carbon monoxide

(D) The molecules and atoms of element carbon

7. Classify each of the following as an element, a compound, or a mixture.

(a) water (b) iron (c) ice-cream (d) sugar

(e) toothpaste (f) silicon dioxide (g) sulfur (h) cement

(i) air (j) magnesium oxide

Formula of ionic compounds

8. Which of the following is the formula of the compound nickel bisulphate?

(A) Ni HSO_4 (B) Ni_2HSO_4 (C) Ni_2SO_4 (D) $Ni(HSO_4)_2$

9. What is the chemical name of the substance whose formulae is $Na(NH_4)HPO_4$?

(A) Sodium hydrogen phosphate (B) Ammonium hydrogen phosphate

(C) Sodium ammonium hydrogen phosphate (D) None of these

10. Which of the following is the formula of the compound stannic phosphate?

(A) $Sn_3(PO_4)_4$ (B) $Sn_2(PO_3)_2$ (C) $Sn_3(PO_3)_2$ (D) $Sn_2(PO_3)_4$

11. Which of the following is the formula of the compound magnesium phosphite?

(A) Mg_2PO_3 (B) $Mg_2(PO_3)_4$ (C) $Mg_3(PO_3)_2$ (D) None of these

12. Which of the following is the chemical name of $Ba(ClO_3)_2$?

(A) Barium chloride (B) Barium chlorate (C) Barium chlorite (D) Barium hypochlorite

13. Which of the following is the formula of barium peroxide ?

(A) Ba_2O (B) Ba_2O_2 (C) BaO_2 (D) BaO_3

Matrix-Match Type

14. Match the column

(A) Barium nitrate	(P) Cation is bivalent
(B) Silver chromate	(Q) Anion is bivalent
(C) Sodium Hydrogen phosphate	(R) no. of cations > no. of anions in one formula unit
(D) Magnesium phosphate	(S) no. of cations = no. of anions in one formula unit
	(T) Total no. of ions per formula unit is 3

Paragraph Type

We know that ionic compounds formed by combination of cation and anion.

By using NH_4^+, CrO_4^{-2}, HCO_3^-, Ca^{2+}, SO_4^{-2}

Answer the following:

15. Name the compound having least number of ions per formula unit.

(A) Ammonium chromate (B) Calcium bicarbonate

(C) Calcium chromate (D) Ammonium sulphate

16. Name the compound having least number of ions and minimum positive charge per formula unit.

(A) Ammonium chromate (B) Calcium chromate

(C) Ammonium bicarbonate (D) Calcium sulphate

17. The chloride of a metal has the formula MCl_3. The formula of its phosphate will be-

(A) M_2PO_4 (B) MPO_4 (C) M_3PO_4 (D) $M(PO_4)_2$

ATOMIC MASS UNIT AND AVEGRAGE ATOMIC AND MOLECULAR MASS

18. Atomic weight of Ne is 20.2. Ne is mixture of Ne^{20} and Ne^{22}, Relative abundance of heavier isotope is

(A) 90 (B) 20 (C) 40 (D) 10

19. Mass of one atom of the element A is 3.9854×10^{-23} g. How many atoms are contained in 1 g of the element A?

(A) 2.509×10^{23} (B) 6.022×10^{23} (C) 12.044×10^{23} (D) None

20. The average atomic mass of a mixture containing 79 mole % of ^{24}Mg and remaining 21 mole % of ^{25}Mg and ^{26}Mg , is 24.31. then **% mole** of ^{26}Mg is

(A) 5 (B) 20 (C) 10 (D) 15

21. The actual weight of a molecule of water is

(A) 18 g (B) 2.99×10^{-23} g

(C) both (A) & (B) are correct (D) none of these

22. Number of oxygen molecules having weight equal to weight of 20 molecules of SO_3 is equal to

(A) 100 (B) 50 (C) 15 (D) 8

Protons , Neutrons and Electrons calculations

1. Number of protons, neutrons & electrons in the element $_{89}X^{231}$ is

(A) 89, 231, 89 (B) 89, 89, 242 (C) 89, 142, 89 (D) 89, 71, 89

2. The charge on the atom containing 17 protons, 18 neutrons and 18 electrons is

(A) +1 (B) –2 (C) –1 (D) Zero

3. In an atom $_{13}Al^{27}$. number of protons is(a) electron is (b) and neutron is (c). Hence ratio will be [in order c : b : a]

(A) 13 : 14 : 13 (B) 13 : 13 : 14 (C) 14 : 13 : 13 (D) 14 : 13 : 14

4. A and B are two elements which have same atomic weight and are having atomic number 27 and 30 respectively. If the atomic weight of A is 57 then number of neutron in B is

(A) 27 (B) 33 (C) 30 (D) 40

5. The atomic mass 25 had 13 neutron's in its nucleus. What its ion can be

(A) Mn^{+2} (B) Cr^{+3} (C) Al^{+3} (D) Mg^{+2}

6. The sum of number of neutrons and protons in all of the isotopes of hydrogen is

(A) 3 (B) 4 (C) 5 (D) 6

7. Choose the false statement about deuterium

(A) It is an isotope of hydrogen (B) It contains $[(1\ e^-) + (1\ P^+) + (1\ (n)]$

(C) It contains only $[(1\ (P^+) + (1\ (n)]$ (D) D_2O is called the heavy water

8. Complete the following table :

	Symbol	No. of protons in nucleus	No. of neutrons in nucleus	No. of electrons	Netcharge
1	Y_{39}^{89}				
2	-	20	20		+2
3	-	23	28	20	
4	-	15	16		-3

Symbol		No. of protons in nucleus	No. of neutrons in nucleus	No. of electrons	Netcharge
1	Y_{39}^{89}	39	50	39	0
2	X^{+2}	20	20	18	+2
3	Z^{+3}	23	28	20	+3
4	A^{-3}	15	16	18	-3

Mole calculations

9. No. of atoms in 4.25 g of NH_3 is approx

(A) 1×10^{23} (B) 1.5×10^{23} (C) 2×10^{23} (D) 6×10^{23}

10. The volume occupied by 4.4 g of CO_2 at 273 K and (P = 1 atm) is

(A) 22.4 L (B) 2.24 L (C) 0.224 L (D) 0.1 L

11. The number of neutrons present in 9 mg of O^{18} is

(A) 10 (B) $5N_A$ (C) $0.005\ N_A$ (D) $0.0005\ N_A$

12. Rearrange the following (I to IV) in the order of increasing masses.

(I) 0.5 mole of O_3 (II) 0.5 gm molecule of Nitrogen

(III) 3.011×10^{23} molecule of O_2 (IV) 11.35 L of CO_2 at STP

(A) IV < III < II < I (B) II < III < IV < I (C) III < II < I < IV (D) I < II < III < IV

13. Total number of protons, neutrons and electrons present in 14 mg of ${}_6C^{14}$ is (Take $N_A = 6 \times 10^{23}$)

(A) 1.2×10^{22} (B) 1.2×10^{25} (C) 7.2×10^{21} (D) 1.08×10^{22}

14. Complete the following table : ($N_A = 6 \times 10^{22}$)

	Mass of sample	Moles of sample	Molecules in sample	Total atoms in sample
1	3.9g C_6H_6			
2		0.2 mole H_2O		
3			2.4×10^{22} molecules CO2	
4				3.6×10^{22} Total atoms in CH3OH sample

	Mass of sample	Moles of sample	Molecules in sample	Total atoms in sample
1	3.9g C_6H_6	0.05	$0.05N_A$	0.6 N_A
2	3.6 g	0.2 mole H_2O	0.2 N_A	0.6 N_A
3	1.76 g	0.04	2.4×10^{22} molecules CO_2	7.2×10^{22}
4	0.032g	0.001	6×10^{21}	3.6×10^{22} Total atoms in CH_3OH sample

15. Number of electrons in 36mg of ${}^{18}_{8}O^{-2}$ ions are (Take $N_A = 6 \times 10^{23}$)

(A) 1.2×10^{21} (B) 9.6×10^{21} (C) 1.2×10^{22} (D) 1.9×10^{22}

16. Molar mass of electron is nearly ($N_A = 6 \times 10^{23}$)

(A) 9.1×10^{-31} kg mol^{-1} (B) 9.1×10^{-31} gm mol^{-1}

(C) 54.6×10^{-8} gm mol^{-1} (D) 54.6×10^{-8} kg mol^{-1}

17. Which of the following contain highest number of molecules

(A) 2.8 g of CO (B) 3.2 g of CH_4 (C) 1.7 g of NH_3 (D) 3.2 g of SO_2

18. 5.6 L of oxygen at 273 K and 1 atm is equivalent to

(A) 1 mole (B) 1/2 mole (C) 1/4 mole (D) 1/8 mole

19. Which has maximum number of molecules of O_2

(A) 32 gm of O_2 (B) 1 mole of O_2

(C) 1 gram molecule of O_2 (D) All have same

20. 1 gm - atom of nitrogen does not represents

(A) 6.02×10^{23} N_2 molecules (B) 22.4 lit. of N_2 at N.T.P.

(C) 11.2 lit. of N_2 at N.T.P. (D) 28 g of nitrogen

21.

Column-I	Column-II
(A) 6.023×10^{23} molecules of CO_2	(P) 1 mol
(B) 6.023×10^{23} molecules of water	(Q) 22.4 L
(C) 96 g of O_2 gas	(R) 2 mol
(D) 88 g of CO_2 gas	(S) 3 mol

HOME WORK (NCERT : 1.10, 1.28, 1.30)

Mass percentage

1. The haemoglobin of most mammals contains approximately 0.33% of iron by mass. The molecular mass of haemoglobin is 67200. The number of iron atoms in each molecule of haemoglobin is

(A) 3 (B) 4 (C) 2 (D) 6

2. Percentage of Se in peroxidase anhydrous enzyme is 0.5% by weight (at.wt. = 78.4) then min.mol. wt. of peroxidase anhydrous enzymes is :-

(A) 1.568×10^4 (B) 1.568×10^3 (C) 15.68 (D) 2.136×10^4

Empirical formula

3. A compound contains 38.8% C, 16.0% H and 45.2% N. The empirical formula of the compound would be –

(A) CH_3NH_2 (B) CH_3CN C) C_2H_5CN (D) $CH_2(NH)_2$

4. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. Molecular formula of that compound (its mol. wt. is 120) could be -

(A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2

5. A hydrocarbon contains 80% of carbon, and its V.D. = 15 then the hydrocarbon is -

(A) CH_4 (B) C_2H_4 (C) C_2H_6 (D) C_2H_2

6. A carbon compound containing carbon and oxygen has molar mass equal to 288. On analysis it is found to contain 50% by mass of each element. Therefore molecular formula of the compound is

(A) $C_{12}O_9$ (B) C_4O_3 (C) C_3O_4 (D) C_9O_{12}

7. Two oxides of a metal contain 50% and 40% metal M respectively. If the formula of the first oxide is MO_2, the formula of the second oxide will be

(A) MO_2 (B) MO_3 (C) M_2O (D) M_2O_5

8. In a hydrocarbon, there is 3gm of carbon per gm of hydrogen present in the molecule. Therefore, molecular formula of the hydrocarbon is

(A) CH_4 (B) C_2H_6 (C) C_3H_8 (D) C_4H_{10}

9. On analysis, a certain compound was found to contain iodine and oxygen in the ratio of 254 gm of iodine (at. mass 127) and 80 gm oxygen (at. mass 16). What is the formula of the compound

(A) IO (B) I_2O (C) I_5O_3 (D) I_2O_5

10. The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is -

(A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) none

11. A given sample of pure compound contains 9.81 gm of Zn, 1.8×10^{23} atoms of chromium, and 0.60 mol of oxygen atoms. What is the simplest formula -

(A) $ZnCr_2O_7$ (B) $ZnCr_2O_4$ (C) $ZnCrO_4$ (D) $ZnCrO_6$

Laws of chemical combinatios

12. When 100 gm of C_2H_4 is polymerised to produce polyethylene according to the equation $nC_2H_4 \rightarrow (C_2H_4)_n$ Then, how many gm polyethylene (C_2H_4)n would be produced ?

At. wt : C-12 ; H - 1

(A) 100 gm (B) 100n gm (C) $\frac{100n}{2}$ gm (D) $\frac{100}{28} \times n$ gm.

13. 8.4 g $MgCO_3$ on heating leaves behind a residue weighing 4.0 g, then carbon dioxide released into the atmosphere at S.T.P. will be

(A) 2.24 L (B) 4.48 L (C) 1.12 L (D) 0.56 L

14. When 10 ml of propane (gas) is combusted completely, volume of $CO_2(g)$ obtained in similar condition is

(A) 10 ml (B) 20 ml (C) 30 ml (D) 40 ml

15. 2.76 g of silver carbonate on being strongly heated yields a residue weighing

(A) 2.16 g (B) 2.48 g (C) 2.32 g (D) 2.64 g

16. 0.54 gm of metal "M" yields 1.02 gm of its oxide M_2O_3. The at. wt. of metal "M" is

(A) 9 (B) 18 (C) 27 (D) 54

17. Suppose two elements X and Y combine to form two compounds XY_2 and X_2Y_3 when 0.05 mole of XY_2 weight 5 g while 3.011×10^{23} molecules of X_2Y_3 weighs 85 g. The atomic masses of X and Y are respectively

(A) 20, 30 (B) 30, 40 (C) 40, 30 (D) 80, 60

18. Chlorine is prepared in the laboratory by treating manganese dioxide (MnO_2) with aqueous hydrochloric acid according to the reaction

$$4HCl(aq) + MnO_2(s) \rightarrow 2H_2O(\ell) + MnCl_2(aq) + Cl_2(g)$$

How many gram of HCl react with 5.0 g of manganese dioxide ? (At. wt. of Mn = 55)

(A) 2.12 gm (B) 44.24 gm (C) 8.4 gm (D) 3.65 gm

19. One of the following combinations illustrate law of reciprocal proportions

(A) N_2O_3, N_2O_4, N_2O_5 (B) NaCl, NaBr, NaI (C) CS_2, CO_2, SO_2 (D) PH_3, P_2O_3, P_2O_5

20. The law of multiple proportions is illustrated by

(A) Carbon monoxide and carbon dioxide

(B) Potassium bromide and potassium chloride

(C) Water and heavy water

(D) Calcium hydroxide and barium hydroxide.

21. If law of conservation of mass was to hold true, then 20.8 gm of $BaCl_2$ on reaction with 9.8 gm of H_2SO_4 will produce 7.3 gm of HCl and $BaSO_4$ equal to

(A) 11.65 gm (B) 23.3 gm (C) 25.5 gm (D) 30.6 gm

22. 12 g carbon combines with 64 g sulphur to form CS_2. 12 g carbon also combines with 32 g oxygen to form CO_2. 10 g sulphur combines with 10 g oxygen to form SO_2. These data illustrate the

(A) Law of multiple proportions

(B) Law of definite proportions

(C) Law of reciprocal proportions

(D) Law of gaseous volumes.

HOME WORK NCERT 1.21, 1.23, 1.24, 1.8, 1.9

Quantum numbers

1. Principal, azimuthal and magnetic quantum numbers are respectively related to

(A) size, shape and orientation (B) shape, size and orientation

(C) size, orientation and shape (D) none of these

2. Which of the following sets of quantum numbers can be correct for an electron in 4f-orbital :

(A) $n = 4, \ell = 3, m = -2, s = 0$ (B) $n = 4, \ell = 3, m = +4, s = -\frac{1}{2}$

(C) $n = 4, \ell = 3, m = +1, s = +\frac{1}{2}$ (D) $n = 4, \ell = 2, m = -1, s = +\frac{1}{2}$

3. $\mathbf{S_1}$ **:** According to Bohr model, the angular momentum of revolving electron is directly proportional to the atomic number of H-like species bearing the electron.

$\mathbf{S_2}$ **:** An orbital cannot accomodate more than 2 electrons.

$\mathbf{S_3}$ **:** All orbitals have directional character.

(A) FTF (B) TFF (C) FFT (D) TTF

4. If an electron has spin quantum number of +1/2 and magnetic quantum number of –1 it cannot be present in :

(A) f-orbital (B) d-orbital (C) p-orbital (D) s-orbital

5. When the quantum number n,*l*,m,s are represented by 3,3,2,+1/2, the symbolism for the electron is -

(A) 3s (B) 3d

(C) 3f (D) Impossible set of quantum number

6. For a 6s electron the values of n,*l*,m,s respectively could be:

(A) 6,4,4,+1/2 (B) 1,0,0,+1/2 (C) 6,1,0,+1/2 (D) 6,0,0, +1/2

7. Any p-orbital can accomodate up to

(A) four electrons (B) Two electrons in parallel spin

(C) Six electrons (D) Two electrons with opposite spin

8. Which one of the following sets of quantum numbers (n,*l*,m,s) represents an impossible arrangement?

(A) 3,2,–2,+1/2 (B) 4,0,0,+1/2 (C) 3,2,–3,+1/2 (D) 5,3,0,–1/2

9. What type of orbital is designated $n = 2, \ell = 3, m_\ell = -2$?

(A) 4p (B) 4d

(C) 4f (D) Impossible set of quantum number

10. The maximum number of electrons that can be accomodated in s, p and d-subshells respectively are :

(A) 2 in each (B) 1, 3 and 5 (C) 2, 6 and 10 (D) 2, 6 and 14

11. Which of the following quantum numbers has not been derived from Schrodinger wave equation:

(A) Principal quantum number (n) (B) Subsidiary quantum number (l)

(C) Magnetic quantum number (m) (D) Spin quantum number (s)

12. The orbital angular momentum corresponding to n = 4 and m = –3 is :

(A) 0 (B) $\frac{h}{\sqrt{2}\pi}$ (C) $\frac{\sqrt{6}h}{2\pi}$ (D) $\frac{\sqrt{3}h}{\pi}$

13. Orbital angular momentum of an electron is $\sqrt{3}\frac{h}{\pi}$. Then, the number of orientations of this orbital in space are:

(A) 3 (B) 5 (C) 7 (D) 9

Electronic configurations

14. What is the maximum possible number of electrons in an atom with (n + l = 7) :

(A) 18 (B) 50 (C) 32 (D) 8

15. Consider the ground state of Cr (Z = 24). The numbers of electrons with the azimuthal quantum numbers l = 1 and 2 respectively are

(A) 16 and 4 (B) 12 and 5 (C) 12 and 4 (D) 16 and 5

16. Degenerate atomic orbitals have

(A) Equal energy (B) Nearly equal energy (C) Different energy (D) None of the above

17. What is a possible set of quantum numbers for the unpaired electron in the orbital box diagram below ?

[Ar] $\underbrace{\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}}_{3d}$ $\underbrace{\boxed{\uparrow\downarrow}}_{4s}$ $\underbrace{\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow}}_{4p}$

(A) n = 1, ℓ = 1, m_ℓ = –1, m_s = +1/2 (B) n = 4, ℓ = 1, m_ℓ = –1, m_s = +1/2

(C) n = 4, ℓ = 2, m_ℓ = –2, m_s = +1/2 (D) n = 4, ℓ = 0, m_ℓ = 0, m_s = +1/2

18. Which element has the following ground state electron configuration ?

[Ar] $\underbrace{\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}}_{3d}$ $\underbrace{\boxed{\uparrow\downarrow}}_{4s}$ $\underbrace{\boxed{\uparrow\downarrow}\boxed{\uparrow}\boxed{\uparrow}}_{4p}$

(A) Se (B) As (C) S (D) Ge

19. Hund's rule states that the most stable arrangement of electrons (for a ground state electron configuration)

(A) Has three electrons per orbital, each with identical spins

(B) Has m_ℓ values greater than or equal to +1

(C) Has the maximum number of unpaired electrons, all with the same spin in degenerate orbital

(D) Has two electrons per orbital, each with opposing spins

20. How many maximum electrons can be described by the quantum numbers n = 5, ℓ = 2 in a particular atom?

(A) 2 (B) 6 (C) 10 (D) 14

21. The total number of electrons in Cr atom for which m = 0

(A) 1 (B) 8 (C) 12 (D) 16

22. The Pauli exclusion principle states that

(A) no two electrons in an atom can have the same set of four quantum numbers

(B) electrons can have either ±1/2 spins

(C) electrons with opposing spins are attracted to each other

(D) no two electrons in an orbital can have the same spin

23. Which of the following statements regarding subshell filling order for a neutral atom is/are correct ?

(I) Electrons are assigned to the 4s subshell before they are assigned to the 3d subshell

(II) Electrons are assigned to the 4f subshell before they are assigned to the 6s subshell

(III) Electrons are assigned to the 4d subshell before they are assigned to the 5p subshell

(A) I only (B) II only (C) I and III (D) I, II and III

Home Work NCERT EXERCISE 2.23, 24, 27, 28, 29, 30, 311.

1. **Column-I** **Column-II**

Column-I	Column-II
(A) No. of electrons in Na(11) having m = 0	(P) 7
(B) No. of electrons in S(16) having $(n + \ell) = 3$	(Q) 15
(C) No. of maximum possible electrons having	(R) 8
s = +1/2 spin in Cr(24)	(S) 12

2. Imagine a universe in which the four quantum no. can have the same possible values as in our universe except that angular quantum no. (l) can have integral values from 0, 1, 2 .... n + 1.

Find the no. of electron n = 1 & 2 shell.

3. The total number of subshells in n^{th} main energy level are :

(A) n^2 (B) $2n^2$ (C) 2n + 1 (D) n.

4. Which of the following orbital does not make sense :

(A) 4d (B) 3f (C) 5p (D) 7s

5. The correct order of the maximum spin of $[_{25}Mn^{4+}, {}_{24}Cr^{3+}, {}_{26}Fe^{3+}]$ is :

(A) $Fe^{3+} > Cr^{3+} = Mn^{4+}$ (B) $Fe^{3+} = Cr^{3+} > Mn^{4+}$

(C) $Cr^{3+} = Mn^{4+} > Fe^{3+}$ (D) $Fe^{3+} > Mn^{4+} > Cr^{3+}$

6. A neutral atom of an element has 2K, 8L, 9M and 2N electrons. Which of the following is/are correctly matched

(A) Total number of s electrons - 8 (B) Total number of p electrons - 12

(C) Total number of d electrons -1 (D) Number of unpaired electrons in element - 3

7. Spin only magnetic moment of $_{25}Mn^{x+}$ ion is $\sqrt{15}$ B.M. Then, What is the value of x .

8. (a) If the value of Azimuthal Quantum Number ℓ for an electron in a particular subshell is 3, then the minimum value of shell number associated with this electron can be x

(b) Orbital angular momentum of an electron is $\sqrt{3}\frac{h}{\pi}$. Then, the number of orientations of this orbital in space is y :

Give the value of (y-x)

MATCH THE COLUMN

9.

Column-I	Column-II
(A) N_2	(P) 40% carbon by mass
(B) CO	(Q) Empirical formula CH_2O
(C) $C_6H_{12}O_6$	(R) Vapour density = 14
(D) CH_3COOH	(S) $14N_A$ ($N_A = 6.023 \times 10^{23}$) electrons in a mole

10. **Column-I** | **Column-II**

(A) Vapour density | (P) Unitless

(B) 1 mol | (Q) 6.023×10^{23} electrons

(C) 12 g carbon | (R) 6.023×10^{23} atoms

(D) 96500 C | (S) $\frac{1}{2} \times$ Molecular mass

11. **Column-I** | **Column-II**

(A) N^{3-} (1 mol) | (P) 10 mol electrons

(B) O^{2-} (1 mol) | (Q) 8 mol protons

(C) CH_4 (1 mol) | (R) 6.023×10^{24} electrons

(D) H_2O (1 mol) | (S) 10 mol protons

12. **Column-I** | **Column-II**

(A) 0.5 mol SO_2 (g) | (P) Occupy 11.2 L at NTP

(B) 1 g of H_2 (g) | (Q) Weighs 16 g

(C) 0.5 mol O_2 (g) | (R) Number of atoms = $2 \times 6.023 \times 10^{23}$

(D) One gram mole of O_2(g) | (S) Weighs 32 g

13. An unknown compound contains 8% sulphur by mass. Calculate

(a) Least molecular weight of the compound and

(b) Molecular weight if one molecule contains 4 atoms of "S"

(A) 200, 400 (B) 300, 400 (C) 400, 1600 (D) 400, 1200

Home Work NCERT EXERCISE 2.62, 63, 64, 65, 66, 67

Periodic classification

1. Which element's atomic weight had been corrected by Mendleeve :-

(A) Be (B) B (C) Br (D) Ba

2. Which element is a bridge element Acc. to Mendleeve :-

(A) Na (B) Li (C) K (D) Cu

3. Recently, a new element of Atomic No. 120 have been discovered. It will be placed in :-

(A) Inert gases (B) Alkali metal (C) Alkaline earth metal (D) Chalcogens

4. Which of the following is inner transition element :-

(A) Ca (B) Cu (C) Cm (D) Cd

5. What is the outermost electronic configuration of Pt :-

(A) $4d^{10}5s^0$ (B) $4d^9 5s^1$ (C) $5d^{10}\,6s^0$ (D) $5d^9 6s^1$

6. The long form of periodic table has

(A) Eight horizontal rows and seven vertical columns

(B) Seven horizontal rows and eighteen vertical columns

(C) Seven horizontal rows and seven vertical columns

(D) Eight horizontal rows and eight vertical columns

7. What is General electronic configuration of f-block element :-

(A) $(n-2)f^{0\text{ to }14}\,(n-1)d^{0\text{ to }4}\,ns^2$ (B) $(n-2)f^{0\text{ to }14}\,(n-1)d^{0\text{ to }1}\,ns^{0\text{ to }2}$

(C) $(n-2)f^{0\text{ to }14}\,(n-1)d^{0\text{ to }1}\,ns^0$ (D) $(n-2)f^{0\text{ to }14}\,(n-1)d^{0\text{ to }1}\,ns^2$

8. Which block element show allotropy :-

(A) s-block (B) p-block (C) d-block (D) f-block

9. Which group metal are non transition element

(A) 2^{nd} group (B) 12^{th} group (C) 16^{th} group (D) 18^{th} group

10. Which block contain gaseous element :-

(A) s-block (B) p-block (C) d-block (D) f-block

11. Rare earth metal are placed in :-

(A) s-block (B) p-block (C) Lanthanoides (D) Actinoids

12. Which contain radioactive element :-

(A) s-block (B) p-block (C) d-block (D) Actinoids

13. Which elements are called trans-uranium :-

(A) Element after uranium (B) Element after Lithium

(C) Element after Berrilium (D) Element after Boron

14. If an atom has electronic configuration

$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^3\,4s^2$, it will be placed in

(A) II A group (B) III A group (C) V B group (D) VI A group

15. The electronic configuration of an element is $1s^2\,2s^2\,2p^6\,3s^2\,3p^3$. What is the atomic number of the element which is just below the above element in the periodic table

(A) 33 (B) 34 (C) 31 (D) 49

16. Which one of the following belongs to representative group of elements in the periodic table

(A) Iron (B) Argon (C) Chromium (D) Aluminium

17. Which of the following pairs has both members from the same period of the periodic table

(A) Na,Ca (B) Na,Cl (C) Ca,Cl (D) Cl,Br

18. The elements having atomic number 72 belongs to

(A) s-block (B) p-block (C) d-block (D) f-block

19. An element has electronic configuration $1s^2\,2s^2\,2p^6\,3s^2\,3p^4$. Predict their period, group and block

(A) Period = 3rd, block = p , group = 16

(B) Period = 5th, block = s , group = 1

(C) Period = 3rd, block = p , group = 10

(D) Period = 4th, block = d , group = 12

20. Two friends Rohit and John, students of chemistry once discussing on periodic table, reach to a conclusion that because of Aufbau rule and other principles their thoughts are restricted for further discussion on electronic arrangements of atoms. They decided not to obey Aufbau rule and capacity of each orbital is increased to three electrons i.e. instead of two each orbital can take maximum of three electrons. Now on the basis of new arrangement, what is the number of elements in third period and fifth period respectively ? (Assume that total number of elements are 112)

(A) 12, 27 (B) 27, 22 (C) 12, 22 (D) 22, 27

21. Which atomic number represents a noble gas

(A) 56 (B) 59 (C) 86 (D) 72

22. The element with atomic number 35 will be placed in

(A) Noble gas family

(B) Alkali family

(C) Alkaline earth family

(D) Halogen family

23. Effective nuclear charge experienced by a valence electron in an atom, will be less than actual nuclear charge, due to :-

(A) Shielding effect

(B) Diagnonal relationship

(C) Inert pair effect

(D) Anamalous property

24. In a given shell the order of screening effect is :

(A) s > p > f > d (B) s > d > p > f (C) s > p > d > f (D) p > s > d > f

Atomic radius

25. Which of the following atom has smallest size

(A) He (B) F (C) H (D) None of these

26. Which of the alkali metals is smallest in size ?

(A) Rb (B) K (C) Na (D) Li

27. Which of the following has largest radius ?

(A) Mg^{2+} (B) Na^+ (C) O^{2-} (D) F^-

28. The radius of Au atom is known as

(A) covalent radius (B) molecular radius (C) metallic radius (D) ionic radius

Home Work NCERT EXERCISE 3.1, 2, 3, 4, 5, 6, 7, 8, 34, 35

1. The descending order in size of Al, $A1^{3+}$, Mg and Mg^{2+} would be

(A) Mg > Mg^{2+} > $A1^{3+}$ > Al (B) Mg > Al > Al^{3+} + > M^{2+}

(C) Mg > Mg^{2+} > Al > Al^{3+} (D) Mg > Al > Mg^{2+} > $A1^{3+}$

2. Ionic radii of

(A) Ti^{4+} < Mn^{7+} (B) $_{35}Cl^{-1}$ < $_{37}Cl^{-1}$ (C) K^{+} > Cl^{-1} (D) P^{3+} > P^{5+}

3. The atomic radius of each of the following element is given Which one has incorrect value of it's ionic radius Mg(1.6A°), Si(1.17 A°), P(l.1 A°), S(1.02°)

(A) Mg^{2+} (0.65 A°) (B) Si^{4+} (0.41 A°) (C) P^{3-} (2.12 °) (D) S^{2-} (1.0 A°)

4. Which radius order is correct :-

(A) V.W. radius > Covalent > Metallic (B) V.W. radius > Metallic > Covalent

(C) Metallic > V.W. radius > Covalent (D) Metallic > Covalent > V.W. radius

5. Size in lanthanoide element decreases from left to right due to :-

(A) Inert pair effect (B) Lanthanoid contraction

(C) Diagonal relationship (D) Absence of vacant orbital

6. The calculated atomic radius of Cl and Cu are 99 Pm and 128 Pm. These are :-

(A) Metallic and covalent respectively (B) Both metallic radius

(C) Covalent and metallic respectively (D) Both covalent radius

7. Which d-block metal has almost equal size :-

(A) Sc, Ti (B) Ti, V (C) Sc, Fe (D) Co, Ni

8. Which of the following has the maximum number of unpaired electrons -

(A) Mg^{2+} (B) Ti^{3+} (C) V^{3+} (D) Fe^{2+}

9. Which statement is correct

(A) For potassium, the atomic radius < ionic radius ; but for bromine, the atomic radius > ionic radius

(B) For potassium and bromine both, the atomic radii > ionic radii

(C) For potassium and bromine both, the atomic radii < ionic radii

(D) For potassium, the atomic radius > ionic radius but for bromine, the atomic radius < ionic radius

10. Al^{3+} has a lower ionic radius than Mg^{2+} because

(A) Mg atom has less number of neutrons than Al

(B) Al^{3+} has higher nuclear charge than Mg^{2+}

(C) Their electronegativities are different

(D) Al has a lower ionisation potential than Mg atom

11. In the isoelectronic species, the ionic radii (Å) of N^{3-}, O^{2-} and F^{-} are respectively given by :

(A) 1.36, 1.40, 1.71 (B) 1.36, 1.71, 1.40 (C) 1.71, 1.40, 1.36 (D) 1.71, 1.36, 1.40

12. The correct order of second ionization potential of carbon, nitrogen, oxygen and fluorine is :

(A) C > N > O > F (B) O > N > F > C (C) O > F > N > C (D) F > O > N > C

13. Which of the following is correct order of ionic radius.

(A) $Al^{+3} > Mg^{+2} > Na^{+}$ (B) $Na^{+} > Mg^{+2} > Al^{+3}$ (C) $Mg^{+2} > Na^{+} > Al^{+3}$ (D) $Mg^{+2} > Al/^{+3} > Na^{+}$

14. Which of the following atom has largest size

(A) Ba (B) Cs (C) K (D) Sr

15. From the given set of species, point out the species from each set having least atomic radius:-

(a) O^{-2}, F^{-}, Na^{+} (b) Ni, Cu, Zn (c) Li, Be, Mg (d) He, Li^{+}, H^{-}

Correct answer is

(A) O^{-2}, Cu, Li, H^{-} (B) Na^{+}, Ni, Be, Li^{+} (C) F^{-}, Zn, Mg, He (D) Na^{+}, Cu, Be, He

16. In the ions P^{3-} , S^{2-} and Cl^{-} the increasing order of size is:-

(A) $Cl^{-} < S^{2-} < P^{3-}$ (B) $P^{3-} < S^{2-} < Cl^{-}$ (C) $S^{2-} < Cl^{-} < P^{3-}$ (D) $S^{2-} < P^{3-} < Cl^{-}$

17. Which of the following order of atomic/ionic radius is not correct :–

(A) $I^{-} > I > I^{+}$ (B) $Mg^{+2} > Na^{+} > F^{-}$ (C) $P^{+5} < P^{+3}$ (D) Li > Be > B

18. Select correct order of size of A^{3+} , B^{3+} , C^{3+} :

(If atomic number of A = 58, B = 69 and C = 63)

(A) $A^{3+} > B^{3+} > C^{3+}$ (B) $C^{3+} > B^{3+} > A^{3+}$ (C) $A^{3+} > C^{3+} > B^{3+}$ (D) $B^{3+} > C^{3+} > A^{3+}$

19. If the difference in atomic size of :

Na – Li = x Rb – K = y Fr – Cs = z

Then correct order will be :-

(A) x = y = z (B) x > y > z (C) x < y < z (D) x < y << z

20. Match list I with list II and select the corrrect answer using the codes given below

List I	**List II**
Ion	Radius (in pm)
(A) Li^{+}	(a) 216
(B) Na^{+}	(b) 195
(C) Br^{-}	(c) 60
(D) I^{-}	(d) 95

Codes :

(A) a b d c (B) b c a d

(C) c d b a (D) d c b a

Subjectives

21. Mg^{2+}, O^{2-}, Na^{+}, F^{-}, N^{3-} (Arrange in decreasing order of ionic size)

22. Why Ca^{2+} has a smaller ionic radius than K^{+}.

23. Arrange in decreasing order of atomic size : Na, Cs, Mg, Si, Cl.

24. If internuclear distance between Cl atoms in Cl_2 is 10 Å & between H atoms in H_2 is 2 Å, then calculate internuclear distance between H & Cl (Electronegativity of H = 2.1 & Cl = 3.0).

NCERT EXERCISE 3.12, 13, 16, 19, 20, 25, 38

Ionisation energy

1. The first four ionisation energy values of an element are 191, 578, 872 and 5962 kcal. The number of valence electrons in the element is

(A) 1 (B) 2 (C) 3 (D) 4

2. The correct order of ionisation energy of C,N, O, F is

(A) F < N < C < O (B) C < N < O < F (C) C < O < N < F (D) F < O < N < C

3. The ionisation energy of nitrogen is more than that of oxygen because

(A) Nitrogen has half filled p-orbitals

(B) Nitrogen is left to the oxygen in the same period of the periodic table

(C) Nitrogen contains less number of electrons

(D) Nitrogen is less electronegative

4. Select correct about first ionization energy :

(A) Be > B (B) $Be^+ > B^+$ (C) $B^+ < C^+$ (D) B > C

5. Select correct about first ionization energy :

(A) Zn > Cu (B) Cu > Zn (C) Zn > Ga (D) Ga > Zn

6. Correct orders of I^{st} I.P. are :-

(a) Li < B < Be < C (b) O < N < F (c) Be < N < Ne

(A) a, b (B) b, c (C) a, c (D) a, b, c

7. IP_1 and IP_2 of Mg are 178 and 348 K. cal mol^{-1}. The enthalpy required for the reaction $Mg \rightarrow Mg^{2+} + 2e^-$ is :-

(A) + 170 K.cal (B) + 526 K.cal (C) – 170 K.cal (D) – 526 K.cal

8. The IP_1, IP_2, IP_3, IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be

(A) Na (B) Si (C) F (D) Ca

9. Which of the following has 2^{nd} IP < I^{st} IP

(A) Mg (B) Ne (C) C (D) None of these

10. The first (IE_1) and second (IE_2) ionization energies (kJ/mol) of a few elements designated by Roman numerals are given below. Which of these would be an alkali metal ?

	IE_1	IE_2		IE_1	IE_2
(A) I	2372	5251	(B) II	520	7300
(C) III	900	1760	(D) IV	1680	3380

11. Which of the following reaction correctly represent second ionization energy of atom magnesium :

(A) $Mg_{(s)} \longrightarrow Mg^{+2}_{(g)} + 2e^-$ (B) $Mg_{(g)} \longrightarrow Mg_{(g)}^{+2} + 2e^-$

(C) $Mg_{(g)}^{+} \longrightarrow Mg^{+2}_{(g)} + e^-$ (D) $Mg^{+2}_{(g)} \longrightarrow Mg^{+3}_{(g)} + e^-$

12. The ionisation energy of B and Al as compared to Be and Mg are

(A) Lower (B) Higher (C) Equal (D) None of these

13. Element X, Y and Z have atomic numbers 19, 37 and 55 respectively. Which of the following statements is true:-

(A) Their ionisation potential would increase with the increasing atomic number

(B) 'Y' would have an ionisation potential in between those of 'X' and 'Z'

(C) 'Z' would have the highest ionisation potential

(D) 'Y' would have the highest ionisation potential

14. Which of the following information is not specific for one element in periodic table :

(A) Atom in which one electron is present in outer most shell and helium gas configuration in penultimate shell.

(B) Atom which have maximum ionization energy (IE_1).

(C) Atom which have full filled 2^{nd} principal energy level but other higher energy levels are vacant.

(D) Atom which have higher value of IE_2 as compared to IE_1.

15. Consider the following changes :

1. $M(s) \longrightarrow M(g)$ 2. $M(s) \longrightarrow M^{2+}(g) + 2e^-$

3. $M(g) \longrightarrow M^+(g) + e^-$ 4. $M^+(g) \longrightarrow M^{2+}(g) + e^-$

5. $M(g) \longrightarrow M^{2+}(g) + 2e^-$

The second ionization energy of M could be calculated from the energy values associated with :

(A) 1 + 3 + 4 (B) 2 – 1 + 3 (C) 1 + 5 (D) 5 – 3

16. Incorrect order of ionisation energy is :-

(A) Pb (I.E.) > Sn (I.E.) (B) Na^+ (I.E.) > Mg^+ (I.E.)

(C) Li^+ (I.E.) < O^+ (I.E.) (D) Be^+ (I.E.) < C^+(I.E.)

17. The electronic configuration of some neutral atoms are given below :–

(A) $1s^2\ 2s^1$ (B) $1s^2\ 2s^2\ 2p^3$ (C) $1s^2\ 2s^2\ 2p^5$ (D) $1s^2\ 2s^2\ 2p^6\ 3s^1$

In which of these electronic configuration would you expect to have highest :–

(i) IE_1 (ii) IE_2

(A) C, A (B) B, A (C) C, B (D) B, D

SUBJECTIVES

18. The IE do not follow a regular trend in II & III periods with increasing atomic number. Why?

19. The IE values of Al (g) $\rightarrow Al^+(g) + e^-$ is 577.5 kJ mol^{-1} and ΔH for $Al(g) \rightarrow Al^{3+}(g) + 3e^-$ is 5140 kJ mol^{-1}. If second and third IE values are in the ratio 2 : 3. Calculate IE_2 and IE_3.

Similar questions beloNgs to NCERT Text Book

Problem - 3.5, 3.6

Excercise - 3.12, 3.17, 3.19, 3.16, 3.31

Electron affinity and Electronegativity

1. A compound AB whose electronegativity difference is 1.9. Atomic radius of A and B are 4 and 2 Å. The distance between A & B mean d_{A-B} is –

(A) 6.2Å (B) 5.82Å (C) 6.9Å (D) 7.5Å

2. Which of the following element has the lowest value of electron affinity -

(A) Carbon (B) Oxygen (C) Fluorine (D) Neon

3. In which case the energy released is minimum:-

(A) $Cl \rightarrow Cl^-$ (B) $P \rightarrow P^-$ (C) $N \rightarrow N^-$ (D) $C \rightarrow C^-$

4. Electron addition would be easier in :-

(A) O (B) O^+ (C) O^- (D) O^{+2}

5. Process in which maximum energy is released:-

(A) $O \rightarrow O^{-2}$ (B) $Mg^+ \rightarrow Mg^{+2}$ (C) $Cl \rightarrow Cl^-$ (D) $F \rightarrow F^-$

6. Select correct order of IE_3 :

(A) O > C > N > B (B) B > C > N > O (C) O > N > C > B (D) O > C > B > N

7. In the formation of a chloride ion, from an isolated gaseous chlorine atom, 3.8 eV energy is released, which would be equal to :-

(A) Electron affinity of Cl^- (B) Ionisation potential of Cl

(C) Electronegativity of Cl (D) Ionisation potential of Cl^-

8. The electron gain enthalpies of halogens are as given below.

F = – 332, Cl = – 349, Br = – 324, I = – 295 kJ mol^{-1}.

The less negative value for F as compared to that of Cl is due to :

(A) Strong electron-electron repulsions in the compact 2p sub shell of F.

(B) Weak electron-electron repulsions in the bigger 3p sub shell of Cl

(C) Smaller electronegativity value of F than Cl

(D) (A) & (B) both

9. Which of the following represent(s) the correct order of electron affinities ?

(A) F > Cl > Br > I (B) C < N < O < F (C) N < C < O < F (D) C < Si > P > N

10. The process(es) requiring the absroption of energy is/are :

(A) $Cl \rightarrow Cl^-$ (B) $S \rightarrow S^{2-}$ (C) $H \rightarrow H^-$ (D) $Ar \rightarrow Ar^-$

11. An element which have configuration ns^2np^5 of its outermost shell has highest electron affinity in its group of periodic table, what is the value of principle quantum number (n) of its penultimate shell :

(A) One (B) Two (C) Three (D) Four

12. Select correct order of electron affinity

(A) F > Cl > O > S (B) Cl > F > O > S (C) Cl > F > S > O (D) Cl > S > F > O

13. Highest electron affinity is shown by

(A) F^- (B) Cl^- (C) Li^+ (D) Na^+

14. Electron addition would be easier in

(A) S (B) S^+ (C) S^- (D) S^{+2}

15. Alkaline earth metals always form dipositive ions due to

(A) $IE_2 - IE_1 > 10$ eV (B) $IE_2 - IE_1 = 17$ eV (C) $IE_2 - IE_1 < 10$ eV (D) None of these

16. The element with least electronegative nature is –

(A) Cu (B) Cs (C) Cr (D) Ba

17. An element X have electronegativity on Paulings scale is 2.5, select correct about polarity of bond in :

(A) $\overset{\delta-}{H}—\overset{\delta+}{X}$ (B) $\overset{\delta+}{N}—\overset{\delta-}{X}$ (C) $\overset{\delta+}{Br}—\overset{\delta-}{X}$ (D) $\overset{\delta+}{B}—\overset{\delta-}{X}$

18. The nomenclature of ICl is iodine monochloride because of

(A) Size of I < Size of Cl (B) Atomic number of I > Atomic number of Cl

(C) E.N. of I < E.N. of Cl (D) E. A. of I < E. A. of Cl

19. The amount of energy released for the process $X_{(g)} + e^- \rightarrow X^-_{(g)}$ is minimum and maximum respectively for :–

(a) F (b) Cl (c) O (d) P

Correct answer is :–

(A) c & a (B) d & b (C) a & b (D) c & b

20. The ionization energy and electron affinity of an element are 17.42 and 3.42 eV respectively. Then the electronegativity of the element on Pauling scale is -

(A) 10.435 (B) 3.721 (C) 1.86 (D) 2.88

21. The correct order of electron affinity of B, C, N, O is :-

(A) O > C > N > B (B) B > N > C > O (C) O > C > B > N (D) O > B > C > N

22. Elements P, Q, R and S belong to the same group. The oxide of P is acidic, oxide of Q and R are amphoteric while the oxide of S is basic. Which of the following elements is the most electropositive?

(A) P (B) Q (C) R (D) S

23. For an element 'A', the first ionisation energy will be numerically equal to :

(A) EA of A^+ (B) EA of A^{2+} (C) IE of A^{2+} (D) None of these

24. Which is the correct order of electronegativity –

(A) Cl > S > P > Si (B) Si > Al > Mg > Na (C) F > Cl > Br > I (D) All

25. Electronegativity decreases in the order –

(A) F > O > N > Br (B) F > Br > N > O (C) F > O > Br > N (D) F > Br > O > N

SUBJECTIVES

26. Explain why a few elements such as Be, N & He have positive electron gain enthalpies while majority of elements do have negative values.

Similar questions belongs to NCERT Text Book

Excercise - 3.20, 3.22

Application of Electronegetivity

1. Arrange in the order of increasing acidic nature (NO_2, K_2O, ZnO) :-

(A) $NO_2 < ZnO < K_2O$ (B) $K_2O < ZnO < NO_2$ (C) $NO_2 < K_2O < ZnO$ (D) $K_2O < NO_2 < ZnO$

2. The basic character of MgO, SrO, K_2O and NiO increases in the order :-

(A) $K_2O < SrO < MgO < NiO$ (B) $NiO < MgO < SrO < K_2O$

(C) $MgO < NiO < SrO < K_2O$ (D) $K_2O < MgO < NiO < SrO$

3. The order in which the following oxides are arranged according to decreasing basic nature is :-

(A) $Na_2O > MgO > Al_2O_3 > SiO_2$ (B) $SiO_2 > Al_2O_3 > MgO > Na_2O$

(C) $Al_2O_3 > SiO_2 > MgO > Na_2O$ (D) $SiO_2 > MgO > Na_2O > Al_2O_3$

4. The correct order of acidic strength is

(A) $Cl_2O_7 > SO_3 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$

(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

5. Which of the following is an amphoteric oxide ?

(A) MgO (B) Al_2O_3 (C) SiO_2 (D) P_2O_5

6. Least basic oxide is :-

(A) Fe_2O_3 (B) FeO (C) BaO (D) Na_2O

7. Identify the correct order of acidic strengths :-

(A) $CaO < CuO < H_2O < CO_2$ (B) $H_2O < CuO < CaO < CO_2$

(C) $CaO < H_2O < CuO < CO_2$ (D) $H_2O < CO_2 < CaO < CuO$

8. Which of the following does not represent the correct order of the property indicated

(A) $Sc^{3+} > Cr^{3+} > Fe^{3+} > Mn^{3+}$ ionic radii (B) $Sc^{3+} < Y^{3+} < La^{3+}$ Ionic radii

(C) $FeO < CaO > MnO > CuO$ Basic nature (D) All

9. Which of the following is the most basic oxide?

(A) SeO_2 (B) Al_2O_3 (C) Sb_2O_3 (D) Bi_2O_3

10. Calculate individual and average Oxidation number (if required) of the marked element

(1) $H\underline{N}O_3$ (2) $\underline{Os}O_4$ (3) $\underline{P}H_3$ (4) $\underline{Cr}O_4^{2-}$

(5) $\underline{Cr_2}O_7^{2-}$ (6) $\underline{Cr}\,O_2Cl_2$ (7) $Na_2\,H\,\underline{P}O_4$ (8) $\underline{Fe}S_2$

(9) $\underline{C_6}H_{12}O_6$ (10) $\underline{Xe}\,O_2\,F_2$ (11) $Li\,\underline{Al}\,H_4$ (12) $Na_3\,\underline{Al}\,F_6$

(13) $\underline{P_4}$ (14) $\underline{O_3}$ (15) $\underline{I}\,(\underline{I}O_3)_3$ (16) $\underline{Fe_3}O_4$

(17) $Cs\underline{I_3}$ (18) $K\underline{O_3}$ (19) $\underline{O_2}\,F_2$ (20) $H_2\,\underline{Si}F_6$

(21) $\underline{P}(OH)_3$ (22) $\underline{P}OCl_3$ (23) $\underline{Si}\,(OH)_4$ (24) $Mg_2\,\underline{C_3}$

(25) $Ca\underline{C_2}$ (26) $Be_2\,\underline{C}$ (27) $Na\underline{B}H_4$ (28) $\underline{Fe}_{0.96}O$

11. Calculate the O.N. of all atoms in following compounds

(1) Fe_3O_4 (2) FeO (3) $Na_2S_4O_6$ (4) C_2H_5OH

(5) $FeSO_4.(NH_4)_2SO_4.6H_2O$ (6) CO_2 (7) FeS_2

(8) PbS (9) CS_2 (10) CrO_5 (11) $(N_2H_5)_2SO_4$

(12) N_2O_5 (13) HCN (14) HNC (15) $Ba[H_2PO_2]_2$

(16) OsO_4 (17) $H_2S_2O_3$ (18) $CH_3SO_3H_6$

Miscelleneous questions

12. Which are correct match :-

(a) O < C < S < Se — Atomic size

(b) Na < Al < Mg < Si — I^{st} I.P

(c) MgO < SrO < Cs_2O < K_2O — Basic character

(A) a, b (B) b, c (C) a, c (D) a, b, c

13. For the processes $K^+(g) \xrightarrow{I} K(g) \xrightarrow{II} K(s)$:-

(A) Energy is released in (I) and absorbed in (II)

(B) Energy is absorbed in (I) and released in (II)

(C) Energy is absorbed in both the processes

(D) Energy is released in both the processes

14. Which of the following option is/are correct :-

(A) IE_1 of Be > IE_1 of Be^- (B) IE_1 of Be^- > IE_1 of Be

(C) $|IE_1$ of $Be^-| = |EA$ of $Be|$ (D) IE_1 of Be > IE_1 of B

15. **Match the column :**

Column-I	Column-II
(A) F	(P) highest ionization energy
(B) Cl	(Q) highest electronegativity
(C) Br	(R) highest electron affinity
(D) I	(S) highest size
	(T) atom in which penultimate shell is full filled

SUBJECTIVES

16. From among the elements, choose the following: Cl, Br, F, Al, C, Li, Cs & Xe.

(i) The element with highest electron gain enthalpy

(ii) The element with lowest ionisation potential.

(iii) The element whose oxide is amphoteric.

(iv) The element which has smallest radii.

(v) The element whose atom has 8 electrons in the outermost shell.

17. Which oxide is more basic, MgO or BaO? Why?

18. Based on location in P.T., which of the following would you expect to be acidic & which basic.

(a) CsOH (b) IOH (c) $Sr(OH)_2$ (d) $Se(OH)_2$ (e) FrOH (f) BrOH

19. Compare the following giving reasons

Acidic nature of oxides : CaO, CO, CO_2, N_2O_5, SO_3

Ionic bond

1. State whether the following are ionic or covalent :

(i) Na_2S (ii) $SnCl_4$ (iii) diamond (iv) CaC_2 (v)NaH
(vi) C_2H_4 (vii) $CaCl_2$ (viii) HCl gas (ix) NH_4^+ (x) KBr

2. Indicate whether the following pairs of elements form ionic or covalent compounds and write down the molecular formula of the compound formed.

(i) Sodium and chlorine (ii) carbon and sulphur (iii) sulphur and oxygen (iv) calcium and hydrogen

3. What type of bonds are present in the following molecules ?

(i) MgF_2 (ii) BrCl (iii) CBr_4 (iv) H_2SO_4 (v) SO_2
(vi) HNO_3

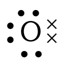
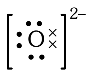
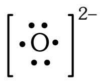
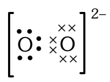
4. Most predominantly ionic compounds are obtained by the combination of elements of the groups :

(A) 1 and 7 (B) 2 and 6 (C) 4 and 8 (D) 3 and 5

5. Which set have strongest tendency to form anions :

(A) Ga, In, Te (B) Na, Mg, Al (C) N, O, F (D) V, Cr, Mn

6. Which lewis dot structure for O^{2-} ion is correct–

(A)  (B)  (C)  (D) 

7. Which of the following bonds is most polar ?

(A) O – H (B) P – H (C) C – F (D) S – Cl

8. Two elements X and Y have following electronic configuration. X : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ Y : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$,The expected compound formed by combination of X and Y will be expressed as-

(A) XY_2 (B) X_5Y_2 (C) X_2Y_5 (D) XY_5

9. The electronegativity of O,F,N, Cl and H are 3.5, 4.0, 3.0, 3.2 and 2.1 respectively.The strongest bond will be

(A) F – H (B) H – Cl (C) N – H (D) O – H

10. Ionic bonds are usually formed by combination of elements with

(A) high ionisation potential and low electron affinity (B) low ionisation potential and high electron affinity

(C) high ionisation potential and high electron affinity(D) low ionisation potential and low electron affinity

Lattice energy

11. For lattice energy the following statements are false :

(A) it increases with increase in charge on cation.

(B) it increases with increase in charge on anion.

(C) it increases with decrease in inter ionic distance

(D) it increases with increase in size of cations and anions.

12. Which of the following sequences represents the correct order of lattice energies ?

(A) LiI > LiBr < LiCl < LiF (B) KBr < KCl < KF < KI

(C) NaF < NaCl < NaBr > NaI (D) LiF > LiCl > LiBr > LiI

13. The correct expected order of decreasing lattice energy is

(A) CaO > $MgBr_2$ > CsI (B) $MgBr_2$ > CaO > CsI

(C) CsI > $MgBr_2$ > CaO (D) CsI > CaO > $MgBr_2$

14. Which of the following is the right order of lattice energy

(A) $Na_2O < Al_2O_3 < MgO$
(B) $MgO < Al_2O_3 < Na_2O$
(C) $Al_2O_3 < MgO < Na_2O$
(D) $Na_2O < MgO < Al_2O_3$

15. If it is known that on heating a ionic compound of a polyhalide with a cation it decomposese into more stable halide of that cation due to high lattice energy, for example $CsI_3 \xrightarrow{\Delta} CsI + I_2$

The complex compound Rb[IBrCl] after strong heating will

(A) RbI + BrCl
(B) RbCl + IBr
(C) RbBr + ICl
(D) None

Hydration energy

16. Find the correct ionic mobility order-

(A) F^- (aq) > Cl^- (aq)
(B) Li^+ (aq) > Be^{2+} (aq)
(C) Ca^{2+} (aq) > Ba^{2+}(aq)
(D) Li^+ (aq) < Al^{3+} (aq)

17. Choose the INCORRECT order of hydrated size of the ions -

(A) $F^{\ominus}_{(aq.)} > Cl^{\ominus}_{(aq.)} > Br^{\ominus}_{(aq.)} > I^{\ominus}_{(aq.)}$
(B) $Rb^{\oplus}_{(aq.)} > K^{\oplus}_{(aq.)} > Na^{\oplus}_{(aq.)} > Li^{\oplus}_{(aq.)}$
(C) $Na^{\oplus}_{(aq.)} > Mg^{2+}_{(aq.)} > Al^{3+}_{(aq.)}$
(D) $Be^{2+}_{(aq.)} > Mg^{2+}_{(aq.)} > Ca^{2+}_{(aq.)} > Sr^{2+}_{(aq.)}$

18. Find the INCORRECT ionic mobility order from the following options-

(A) $Be^{2+}_{(aq.)} < Li^{+}_{(aq.)}$
(B) $Mg^{2+}_{(aq.)} < Sr^{2+}_{(aq.)}$
(C) $Fe^{2+}_{(aq.)} < Fe^{3+}_{(aq.)}$
(D) $Br^{-}_{(aq.)} < I^{-}_{(aq.)}$

Polarisation (Fajan's rule)

19. Polarisibility of halide ions increases in the order

(A) F^-, I^-, Br^-, Cl^-
(B) Cl^-, Br^-, I^-, F^-
(C) I^-, Br^-, Cl^-, F^-
(D) F^-, Cl^-, Br^-, I^-

20. Which of the following is most covalent ?

(A) AlF_3
(B) $AlCl_3$
(C) $AlBr_3$
(D) AlI_3

21. Among LiCl, $BeCl_2$, BCl_3 and CCl_4, the covalent bond character follows the order -

(A) $LiCl < BeCl_2 > BCl_3 > CCl_4$
(B) $LiCl > BeCl_2 < BCl_3 < CCl_4$
(C) $LiCl < BeCl_2 < BCl_3 < CCl_4$
(D) $LiCl > BeCl_2 > BCl_3 > CCl_4$

22. Which has maximum covalent character ?

(A) NaCl
(B) $SiCl_4$
(C) $AlCl_3$
(D) $MgCl_2$

23. Choose the correct statement

(A) A cation with pseudo noble gas configuration is more polarising than the cation with noble gas configuration.
(B) Small cation has minimum capacity to polarise an anion.
(C) Small anion has maximum polarizability.
(D) None of these

24. Magnesium cation has plarisation power close to that of :-

(A) Li^+
(B) Na^+
(C) K^+
(D) Cs^+

25. Which of the following combination of ion will have highest polarisation :-

(A) Fe^{2+}, Br^-
(B) Ni^{4+}, Br^-
(C) Ni^{2+}, Br^-
(D) Fe, Br^-

26. An ion without pseudo-inert gas configuration is :

(A) Ag^+
(B) Cd^{2+}
(C) Zn^{2+}
(D) Fe^{3+}

Similar questions belongs to NCERT Text Book Problem - 4.1, 4.2, 4.3

Excercise - 4.4, 4.5, 4.6, 4.12, 4.19, 4.20

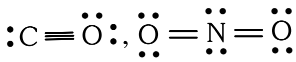
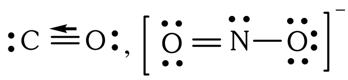
Lewis theory and formal charge

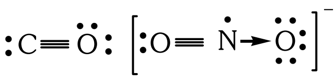
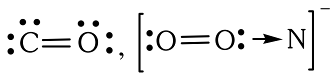
1. In ammonium ion, bond in between ammonia molecule and a proton is form by–

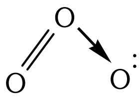
(A) Complete transfer of electron from NH_3 to H^+ (B) electrostatic attraction between NH_4^+ & H^+

(C) equal contribution of electrons by NH_3 & H^+ (D) One sided sharing of electrons

2. The correct structure of CO and NO_2^- are–

(A)  (B) 

(C)  (D) 

3. Lewis structure of O_3 is drawn as  : therefore formal charge on oxygen atoms are–

(A) 0, 0, 0 (B) 0, +1, –1 (C) 0, +1, +1 (D) –1, +1, –1

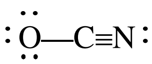
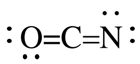
Paragraph for Q.4 to Q.5

The formal charge of an atom in a polyatomic molecule or ion may be defined as the difference between the number of valence electrons of that atom in an isolated or free state and the number of electrons assigned to that atom in the Lewis structure. It is

expressed as :

Formal charge [F.C.] on an atom in a Lewis structure = [total number of valence electron in the free atom] — [total number of non bonding (lone pair) electrons] — (1/2) [total number of bonding shared electrons]

4. Find the formal charge on "O" atom in given structure (I) & (II) respectively :

(I)  (II) 

(A) –1, –1 (B) –2, 0 (C) –1, 0 (D) 0, –1

5. Select correct about CO_3^{2-} carbonate ion in one of the lewis structure based on the presence of two single bonds and one double bond between carbon and oxygen atoms :

(A) Total number of lone pair = 8

(B) Formal charge on two oxygen = –1 and one oxygen = zero

(C) Oxidation number of C = +4 & Formal charge on C = zero

(D) All are correct

6. The Lewis theory does not account for the–

(A) cause of bond formation (B) Shape of molecules

(C) Strength of chemical bond (D) All

7. Draw the Lewis structure and find Formal charge of each atom:

1.	CO	2.	CO_2	3.	NO_2^-	4.	NO_3^-	5.	CCl_3^-	6.	$COCl_2$
7.	N_3^-	8.	O_3	9.	CH_3Cl	10.	NH_4^+	11.	NH_2Cl	12.	OCN^-
13.	CN^-	14.	SCN^-	15.	HCN	16.	HNC	17.	SiF_4	18.	$SnCl_3^-$

19.	BF_4^-	20.	BH_4^-	21.	BeF_4^{2-}	22.	H_3O^+	23.	SO_3	24.	SO_2
25.	CO_3^{2-}	26.	NO_2Cl	27.	NOCl	28.	F_2O	29.	SO_4^{2-}	30.	PO_4^{3-}
31.	SF_2	32.	CF_4	33.	PF_5	34.	PF_4^+	35.	PCl_3	36.	PCl_5
37.	SI_2	38.	SF_6	39.	SO_4^{2-}	40.	$POCl_3$	41.	ClO_4^-	42.	OF_2
44.	NO_3^-	45.	ClO_4^-	46.	PCl_4^+	47.	I_3^+	48.	ClO_3^-	49.	OCl_2
50.	$SnCl_3^-$	51.	HPO_3^{2-}	52.	SO_3^{2-}	53.	IO_3^-	54.	XeO_3	55.	NO_2^-

8. Which of the following species does not obey octet rule ?
(A) SiF_4 (B) PCl_5 (C) ICl (D) BF_4

9. Hypervalent compound is(are) :
(A) SO_3 (B) PO_4^{-3} (C) SO_4^{-2} (D) All of these

10. The octet rule is not valid for the molecule :
(A) CO_2 (B) H_2O (C) SF_2 (D) $Al_2(CH_3)_6$

11. In which of the excitation state of chlorine ClF_3 is formed
(A) In ground state (B) In triple excited state (C) In single excited state (D) In double excited state

12. Which of the following configuration shows second excitation state of Iodine

(A) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}\boxed{\uparrow}\boxed{\uparrow}$ $\boxed{\uparrow}\boxed{}\boxed{}\boxed{}\boxed{}$
(B) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{}$ $\boxed{\uparrow}\boxed{}\boxed{}\boxed{}\boxed{}$
(C) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}$ $\boxed{\uparrow}\boxed{\uparrow}\boxed{}\boxed{}\boxed{}$
(D) $\boxed{\uparrow}$ $\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}$ $\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}\boxed{}\boxed{}$

13. Highest extent of variable covalency is exhibited by.
(A) P and S (B) N and O (C) N and P (D) F and Cl

14. Which of the following species are hypervalent ?
(I) ClO_4^- (II) BF_3 (III) SO_4^{-2} (IV) CO_3^{-2}
(A) I, II, III (B) I, III (C) III, IV (D) I, III, IV

15. Find the number of isoelectronic species from the following having 14 electrons.
Mg^{2+}, Na^+, N^{3-}, S^{2-}, K^+, CN^-, N_2, NO^+, PH_3, P^+

VBT based questions

16. A sigma bond is formed by the overlapping of
(A) s-s orbital alone (B) s and p orbitals alone
(C) s–s, s–p or p–p orbitals along internuclear axis (D) p–p orbital along the sides

17. Which overlapping is involved in HCl molecule
(A) s–s overlap (B) p–p overlap (C) s–d overlap (D) s–p overlap

18. Which of the following bonds will have directional character
(A) Ionic bond (B) Metallic bond (C) Covalent bond (D) Both covalent & metallic

19. Which of the following compound is formed in the second excitation state of sulphur atom :
(A) SF_4 (B) SF_6 (C) SF_2 (D) None

Similar questions belongs to NCERT Text Book

Excercise - 4.13, 4.19, 4.22, 4.23, 4.25, 4.26

1. Which is not a characteristic of π-bond

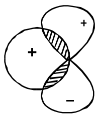
(A) π - bond is formed when a sigma bond already formed

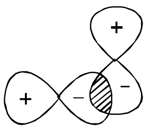
(B) π - bond are formed from hybrid orbitals

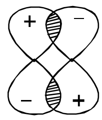
(C) π - bond may be formed by the overlapping of p-orbitals

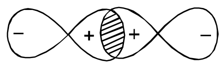
(D) π-bond results from lateral overlap of atomic orbitals

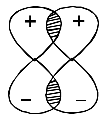
2. Which of the following atomic orbital overlappings would not lead to bond formation ?

(i) 

(ii) 

(iii) 

(iv) 

(v) 

(A) All (B) (i) (ii) (iii) (C) (i) (iii) (v) (D) (ii) only

3. Which of the following overlaps is **INCORRECT** [assuming z-axis to be the internuclear axis]

(a) $2\ p_y + 2\ p_y \rightarrow \pi\ 2\ p_y$ (b) $2\ p_z + 2\ p_z \rightarrow \sigma\ 2p_z$ (c) $2\ p_x + 2\ p_x \rightarrow \pi\ 2\ p_x$ (d) $2p_y + 3d_{xy} \rightarrow \pi(2p\text{-}3d)$

(A) 'a' & 'b' (B) 'b' & 'd' (C) only 'd' (D) None of these

4. Which of the following can lead to π-bond formation.

(A) d_{xy}-d_{yz} (B) d_{zx}-d_{zx} (C) d_{zx}-p_z (D) $d_{x^2-y^2} - p_x$

5. If the molecular axis is Z then which of the following overlapping is not possible.

(A) $p_z + p_z = \sigma$ bond (B) $p_x + p_y = \pi$ bond (C) $p_x + p_x = \pi$ bond (D) $p_y + p_y = \pi$ bond

6. Which of the following molecule(s) is/are having $2p\pi$ –$3d\pi$ bonding -

(A) SO_3 (B) NO_3^- (C) PO_4^{3-} (D) SO_4^{2-}

7. Strongest bond is formed by the head on overlapping of :

(A) 2s- and 2p- orbitals (B) 2p- and 2p- orbitals (C) 2s- and 2s- orbitals (D) All

8. π bond is formed

(A) By overlapping of hybridised orbitals (B) Overlapping of s - s orbitals

(C) Head on overlapping of p -p orbitals (D) By p - p collateral overlapping

9. Weakest bond is formed by the orbital overlapping of

(A) $sp^2 - s$ (B) $sp^3 - p$ (C) $s - s$ (D) p – p co-lateral

10. Overlapping of 2 hybrid orbitals can lead to the formation of

(A) Ionic bond (B) π –bond (C) σ –bond (D) (B) and (C) both

11. Which of the following bonds is most stable ?

(A) 1s – 1s (B) 2p – 2p (C) 2s – 2p (D) 1s – 2p

12. Which of the following statements is not correct ?

(A) Double bonds is shorter than a single bond (B) σ – bond is weaker than a π bond

(C) Double bond is stronger than a single bond (D) Covalent bond is stronger than a hydrogen bond

13. Which of the following compounds has an atom without stable duplet or octet configuration ?

(A) NaCl (B) LiH (C) B_2H_6 (D) HF

14. Answer the following :

(i) What is the co-valency of carbon in C_2H_4 and C_2H_2 ?

(ii) What types of bonds and how many of each are present in NH_4^+ ?

15. The bonds present in N_2O_5 (g) are :

(A) only ionic
(B) covalent and co-ordinate
(C) only covalent
(D) covalent and ionic

16. The compound which contains both ionic and covalent bonds is :

(A) CH_4 (B) H_2 (C) KCN (D) KCl

17. Which one of the following molecules are formed by p-p overlapping ?

(A) Cl_2 (B) HCl (C) H_2O (D) NH_3

18. Find σ and π bonds in the following molecules :

(i) CH_3CH_3 , (ii) $CH_2 = CH_2$, (iii) $HC \equiv CH$, (iv) $CH_2 = CHCOOH$, (v) $C_2(CN)_4$ (vi) $H_2S_2O_8$ (vii) $H_2S_4O_6$

19. A sigma bond may be formed by the overlap of two atomic orbitals of atoms Aand B. If the bond is formed along the x-axis, which of the following overlaps is acceptable ?

(A) s orbital of A and p_z orbital of B
(B) p_x orbital of A and p_y orbital of B
(C) p_z orbital of A and p_x orbital of B
(D) p_x orbital of A and s orbital of B

20. The pair of compounds which can form a co–ordinate bond is.

(A) $(C_2H_5)_3B$ and $(CH_3)_3N$
(B) HCl and HBr
(C) BF_3 and NH_3
(D) (A) & (C) both

VSEPR theory based question

21. The shape of sulphate ion is

(A) Hexagonal (B) Square planar (C) Trigonal bipyramidal (D) Tetrahedral

22. In which following set of compound/ion has linear shape ?

(A) CH_4, NH_4^+, BH_4^- (B) CO_3^{-2}, NO_4^-, BF_3 (C) NO_2^+, CO_2, XeF_2 (D) $BeCl_2$, BCl_3, CH_4

23. Shape of a molecule having 4 bond pairs and two lone pairs of electrons, will be

(A) Square planar (B) Tetra hedral (C) Linear (D) Octa hedral

24. Hybridisation in $XeOF_2$, XeO_2F_2 is sp^3d. But shape will be respectively

(A) T, 'V' shape
(B) T shape, distorted trigonal bipyramidal
(C) Both have T shape
(D) T shape, irregular octahedral

25. The shape of IF_4^+ will be

(A) Square planar
(B) See Saw
(C) Pentagonal bipyramidal
(D) Distorted tetrahedual

26. Which following compound, has four bond pair and one lone pair on central atom?

(A) NH_4^+ (B) ICl_4^- (C) SF_4 (D) XeF_4

27. A σ bonded molecule MX_3 is T–shaped. The number of non–bonding pairs of electrons is

(A) 0
(B) 2
(C) 1
(D) Can be predicted only if atomic number of M is known.

28. Incorrect code regarding shape is

(A) Linear : N_3^-, $(CN)_2$, ICl_2^-
(B) Pyramidal : CH_3^-, NH_3, XeO_3
(C) Trigonal planar : $CH_3^{\oplus}$, CH_3, $CH_3^{\ominus}$
(D) Tetrahedral : SiH_4, NH_4^+ , XeO_4

29. In molecules of the type AX_2L_n (where L represents lone pairs and n is its number) there exists a bond between element A and X. The $\angle$ X A X bond angle.

(A) Always decreases if n increases (B) Always increases if n increases

(C) Will be maximum for n = 3, 0 (D) No effect of value of n

30. Which of the following pairs of species have identical shapes ?

(A) NO_2^+ and NO_2^- (B) PCl_5 and BrF_5 (C) XeF_4 and ICl_4^- (D) $TeCl_4$ and XeO_4

31. The shape of CFClBrI will be

(A) Irregular tetrahedral (B) Octahedral

(C) See-saw (D) Trigonal bipyramidal

32. The structure of O_3 and N_3^- are -

(A) both linear (B) Linear and bent respectively.

(C) both bent (D) Bent and linear respectively.

33. Which of the following shape can not be obtained from sp^3d^2 hybridisation.

(A) Square planar (B) Square pyramidal (C) Tetrahedral (D) Octahedral

Hybridisation

34. Hybridization of orbitals of carbon in CH_4 is necessary to explain which of the following ?

(A) Equality of strength of all C-H bonds (B) Methane is non-polar

(C) Tetravalency of carbon (D) Carbon has complete octate

35. The d- orbitals involved in sp^3d hybridisation is

(A) $d_{x^2-y^2}$ (B) d_{z^2} (C) d_{xy} (D) d_{xz}

36. Hybridization of 2nd & 3rd carbon in $CH\equiv C–CH=CH_2$ are respectively :-

(A) sp & sp^2 (B) sp^3 & sp^2 (C) sp^2 & sp (D) sp^2 & sp^2

37. Which of the following contains Co-ordinate and covalent bonds.

(a) $N_2H_5^+$ (b) H_3O^+ (c) HCl (d) H_2O Correct answer is

(A) a & d (B) a & b (C) c & d (D) Only a

38. Predict the hybridisation on the central atom of following molecules :

1. ICl_4^-	2. BeF_2	3. CO_2	6. BF_3	36. ICl_2^-
7. $CH_2=CH_2$	9. HNO_3	10. HNO_2	11. SO_2	12. SO_3
13. HCO_3^-	14. CO_3^{-2}	15. NHO_4	16. $SnCl_2$	17. $AlCl_3$
18. CH_4	19. NH_4^+	20. BF_4^-	22.NF_3	23. PF_3
24. $AsCl_3$	25. NH_3	27. H_2O	28. OF_2	30. PCl_5
31. $SbCl_5$	32. SF_6	33. SeF_6	34. PF_6^-	37. ICl_5
38. XeF_4				

Similar questions belongs to NCERT Text Book

Excercise - 4.33, 4.34, 4.36, 4.40

1. Which starred carbon atom in the following molecules does not show sp^2 hybridisation :

(A) CH_3C^*HO (B) CH_3C^*OCl (C) $(C^*H_3)_3N \rightarrow O$ (D) $CH_3C^*OCH_2C^*OOC_2H_5$

2. In which of the following 'N' atom is sp^2 hybridised

(A) NH_3 (B) NH_4^+ (C) $\overset{\ominus}{N}H_2$ (D) $B_3N_3H_6$

3. Carbon atoms in $C_2(CN)_2$ are :

(A) All sp-hybridised (B) sp^3, sp^2 ,sp—hybridised

(C) sp^2, sp, sp^3—hybridised (D) sp, sp^3 , sp^2—hybridised.

4. $BF_3 + F^- \rightarrow BF_4^-$

What is the hybridiation state of B in BF_3 and BF_4^- ?

(A) sp^2, sp^3 (B) sp^3, sp^3 (C) sp^2, dsp^2 (D) sp^2d, sp^2

5. In a change from $PCl_3 \longrightarrow PCl_5$, the hybrid state of P changes from

(A) sp^2 to sp^3 (B) sp^3 to sp^2 (C) sp^3 to sp^3d (D) sp^3 to dsp^2

6. In which of the following process hybridisation of the central atom changes -

(A) $H_2O + H^+ \longrightarrow H_3O^+$ (B) $NF_3 + F^+ \longrightarrow NF_4^+$

(C) $BF_3 + F^- \longrightarrow BF_4^-$ (D) $NH_3 + H^+ \longrightarrow NH_4^+$

7. Match the species in column (I) with that geometry in column (II)

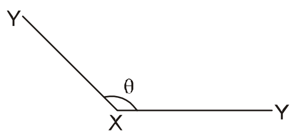
Column-I	**Column-II**
(P) BH_4^-	(1) 2 bond pair and 3 lone pair
(Q) ICl_2^+	(2) 4 bond pair and no lone pair
(R) ICl_2^-	(3) 3 bond pair and 1 lone pair
(S) ICl_4^-	(4) 2 bond pair and 2 lone pair
	(5) 4 bond pair and 2 lone pair

(A) P = 2; Q = 4; R = 3; S = 1 (B) P = 2; Q = 4; R = 1; S = 5

(C) P = 2; Q = 1; R = 5; S = 4 (D) P = 2; Q = 1; R = 3; S = 4

Dipole Moment.

10. Which bond angle θ would result in the maximum dipole moment for the triatomic molecule XY_2 shown below



(A) θ = 90° (B) θ = 120° (C) θ = 0° (D) θ = 180°

11. Which of the following molecule is/are non polar

(A) XeF_2 (B) PCl_3F_2 (C) XeF_4 (D) All

12. Which of the following molecule will not show zero dipole moment :

(A) CH_4 (B) CCl_4 (C) CO_2 (D) $CHCl_3$

13. The dipole moments of the given molecules are such that :

(A) $BF_3 > NF_3 > NH_3$ (B) $NF_3 > BF_3 > NH_3$ (C) $NH_3 > NF_3 > BF_3$ (D) $NH_3 > BF_3 > NF_3$.

14. Arrange in order of increasing dipole moment : BF_3, H_2S, H_2O.

15. In which type of molecule, the dipole moment may be non zero.

(A) AB_2L_2 (B) AB_2L_3 (C) AB_4L_2 (D) AB_4

Where A – Central atom, B – Bonded atom, L – Lone pair

16. A polar molecule AB have dipole moment 3.2 D (Debye) while the bond length is 1.6 Å. Find the percentage ionic character in the molecule.

(A) 31% (B) 41.6% (C) 39.6% (D) None of these

17.

Column–I	**Column–II**
(a) XeO_4^{2-}	(p) sp^3 with zero dipole moment
(b) PCl_2F_3	(q) sp^3d with nonzero dipole moment
(c) XeO_2F_2	(r) Shows resonance stability
(d) SO_4^{2-}	(s) No lone pair on central atom

18. Which of the following species is non polar with presence of polar bond and lone pair of electron on central atom.

(A) CO_2 (B) SF_4 (C) XeF_4 (D) CF_4

19. Which of the following molecule is planer as well as polar :

(A) PCl_3 (B) SF_4 (C) ClF_3 (D) None of thes

Hydrogen bond

20. The order of strength of hydrogen bond is :

(A) Cl–H....Cl > N–H....N > O–H....O > F–H....F (B) N–H....N > Cl–H....Cl > O–H....O > F–H....F

(C) O–H....O > N–H....N > Cl–H....Cl > F–H....F (D) F–H....F > O–H....O > N–H....N > Cl–H....Cl

21. Which one among the following does not have hydrogen bonds ?

(A) boric acid (solid) (B) N_2H_4 (liquid) (C) H_2O_2 (liquid) (D) C_6H_6 (liquid)

22. Which of the following substances does not exhibit H-bonding with water ?

(A) CH_3CH_2OH (B) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OH$ (C) $CH_3-CH_2-CH_3$ (D) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2$

23. I. When ice is melted, hydrogen bond starts breaking, molecule of water come closer by moving into vacant space. As a result, density of water decreases upto 4°C.

II. Due to open cage like structure, ice has a relatively large volume for a given mass of liquid water.

III. In ice, there are four water molecules attached tetrahedrally.

Which of the above statement is/are true.

(A) I, II and III (B) I and III (C) II and III (D) II only

24. Which of the following conditions is required for the formation of the hydrogen bond

(A) Hydrogen atom should be bonded to a highly electronegative atom

(B) The size of electronegative atom should be small

(C) There should be a lone pair of electron on the electronegative atom.

(D) All of the above

25. ***Assertion :-*** Acetylene is not soluble in H_2O but is highly soluble in acetone.

Reason :- Acetylene forms intermolecular H-bond with acetone easily but not with H_2O as water molecules themselves are highly associated through intermolecular H-bonding.

(A) A (B) B (C) C (D) D

26. Match the column :-

Column–I	**Column–II**
(a) Chloral hydrate	(P) Form Zig-zag chain
(b) HF	(Q) Form 2–D–sheet structure
(c) H_3BO_3	(R) Have low volatility
(d) H_2SO_4	(S) Intramolecular H-bond
	(T) Inter molecular H-bond

27. The maximum possible number of hydrogen bonds in which H_2O_2 molecule can participate:-

(A) 6 (B) 4 (C) 5 (D) 8

28. ***Statement 1 :-*** Boiling point of HF is lesser than water.

Statement 2 :- Hydrogen bond strength is stronger in water.

(A) Statement-1 and Statement-2 are true, Statement-2 is a correct explanation of Statement-1.

(B) Statement-1 and Statement-2 are true, Statement-2 is not the correct explanation of Statement-1.

(C) Statement-1 is true and Statement-2 is false.

(D) Statement-1 is false and Statement-2 is true.

29. Which of following statement is incorrect :-

(A) Boiling point of H_2O_2 is greater than that of H_2O

(B) Ethylene glycol is less viscous than glycerol

(C) o–nitrophenol can be separated from its meta and para isomer using its steam volatile property

(D) In ice each 'O' atom is tetrahedrally arranged by four H-atom which are all covalently bonded.

31. When two ice cubes are pressed over each other, they unit to form one cube. Which of the following force is responsible for holding them together

(A) Vander Waal's forces (B) Hydrogen bond

(C) Covalent attraction (D) Dipole-dipole attraction.

32. Arrange the following gases in the increasing order of their intermolecular forces of attraction (CO_2 , H_2O , H_2):

(A) $H_2 < CO_2 < H_2O$ (B) $H_2O < CO_2 < H_2$ (C) $CO_2 < H_2O < H_2$ (D) $H_2O < H_2 < CO_2$.

33. Which is <u>**incorrect**</u> order for net dipole moment -

(A) HF > HCl > HBr > HI (B) $CH_3 - F > CD_3 - F$

(C) $SO_3 > SO_2$ (D) $CH_3 - CH = CHCl$ (cis) > $CH_3 - CH = CHCl$ (trans)

34. Classsify the type of force of attraction existing in the sample of following compounds :

(i) $CH_3 — O — CH_3$ (ii) sugar (iii) ice (iv) $CH_3\ CO\ CH_3$

(v) $CH_3 — OH$ (vi) $N(CH_3)_3$ (vii) gold (viii) $CH_3 — NH_2$

(ix) H_2S (x) (aq.) Na^+ (xi) CCl_4 (xii) diamond

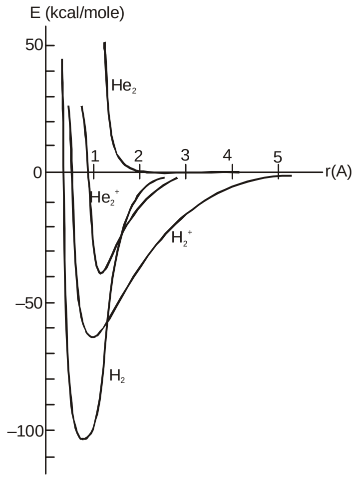
(xiii) Cl_2 (xiv) NH_4Cl (xv) HCl and Cl_2 (xvi) Ar

NCERT HOME WORK

Exercise Q 4.16,4.22,4.23. Q 4.39

MOT

1. The following graph is given, between total energy and distance between the two nuclei for species H_2^+, H_2, He_2^+ & He_2, which of the following statements is correct :



(A) He_2^+ is more stable than H_2^+.

(B) Bond dissociation energy of H_2^+ is more than bond dissociation energy of He_2^+.

(C) Since bond orders of He_2^+ and H_2^+ are equal hence both will have equal bond dissociation energy.

(D) Bond length of H_2^+ is less than bond length of H_2 .

2. **Match the following :**

Column	**Column**
(A) N_2^+ is stable than N_2^-	(p) due to one have higher electrons in antibonding than other
(B) NO can easily loose its electron than N_2	(q) one have B.O. 3 and other have 2.5
(C) NO have large bond length than NO^+	(r) It is easy to remove electron from higher energy level
(D) He_2^+ exist but less stable than H_2^+	(s) ABMO has more energy than corresponding BMO

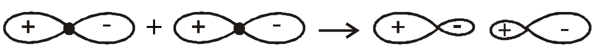
3. How many nodal plane is/are present in σ_{1s} bonding molecular orbital?

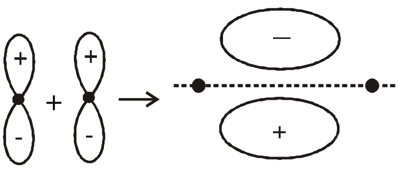
(A) zero (B) 1 (C) 2 (D) 3

4. Which of the following combination of orbitals is correct?

(A) 

(B) 

(C) 

(D) 

5. Which of the following statements is not correct regarding bonding molecular orbitals?

(A) Bonding molecular orbtials possess less energy than the atomic orbitals from which they are formed

(B) Bonding molecular orbtials have low electron density between the two nuclei

(C) Every electron in bonding molecular orbitals contributes to the attraction between atoms

(D) They are formed when the lobes of the combining atomic orbtials have the same sign

7. Fill in the blanks :

Molecule or ion	MO configuration	Bond order	Magnetic Behaviour
H_2	$(\sigma 1s^2)$	1	Diamagnetic
H_2^+	–	–	–
H_2^-	–	–	–
He_2	–	–	–
N_2	–	–	–
O_2	–	–	–
O_2^+	–	–	–
O_2^{2+}	–	–	–
F_2	–	–	–
Ne_2	–	–	–
CO	–	–	–
CN	–	–	–
CN^-	–	–	–

8. In the following which of the two are paramagnetic

(a) N_2 (b) CO (c) B_2 (d) NO_2, Then Correct answer is

(A) a and c (B) b and c (C) c and d (D) b and d

9. Of the following species, which has the highest bond order and shortest bond length ?NO, NO^+, NO^{2+}, NO^-

10. Which of the following pairs of species would you expect to have largest difference in magnetic moment ?

(A) O_2, O_2^+ (B) O_2, O_2^{2-} (C) O_2^+, O_2^{2-} (D) O_2^-, O_2^+

11. Order of stability of N_2, N_2^+ and N_2^- is

(A) $N_2 > N_2^+ > N_2^-$ (B) $N_2^+ > N_2 > N_2^-$ (C) $N_2^- > N_2 > N_2^+$ (D) $N_2^- = N_2^+ > N_2$

12. Which of the following forms only π-bond using M.O. theory :

(A) Li_2 (B) C_2 (C) N_2 (D) O_2

13. According to M.O. theory HOMO in O_2^- is :

(A) $\pi 2p_x = \pi 2p_y$ (B) $\pi 2p_x = \pi 2p_y$ (C) $\sigma 2p_z$ (D) $\sigma 2p_z$

Basic stoichiometry

1. What is the weight of oxygen required for the complete combustion of 2.8 kg of ethylene (C_2H_4)

(A) 2.8 kg (B) 6.4 kg (C) 6.72 kg (D) 9.6 kg

2. At 25°C for complete combustion of 5 mol propane (C_3H_8). The required volume of O_2 at STP will be.

(A) 5.6 L (B) 560 L (C) 360 L (D) 360 L

3. According to the following reaction the minimum quantity in gm of H_2S needed to precipitate 63.5 gm of Cu^{2+} ions will be nearly $Cu^{+2} + H_2S \rightarrow CuS + 2H^+$

(A) 63.5 gm (B) 31.75 gm (C) 34 gm (D) 20 gm

4. When 280 gm of ethylene polymerises to polyethylene according to the equation.

$$n(CH_2 = CH_2) \longrightarrow -(CH_2-CH_2)_n-$$

The weight and mole of polyethylene formed will be-

(A) 280, 10n (B) $\frac{280}{n}$, n (C) $\frac{280}{n}$, 280 (D) 280, $\frac{10}{n}$

5. 27 gm of Al with react completely with :

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

(A) 24 gm of O_2 (B) 0.75 moles of O_2

(C) 16.8 L of O_2 at 1atm,273K (D) 0.75 N_A molecules of O_2

6. 1.5 g of oxygen is produced by heating $KClO_3$. How much KCl is produced in the reaction :

$$2KClO_3(s) \longrightarrow 2KCl(s) + 3O_2(g)\uparrow$$

(A) 4.15×10^2 mol (B) 4.33 g (C) 3.12×10^{-2} mol (D) 2.33 g

Problem based on limiting reagent

7. For the reaction: $7A + 13B + 15C \longrightarrow 17P$

If 15 moles of A, 26 moles of B & 30.5 moles of C are taken initially then limiting reactant is

(A) A (B) B (C) C (D) None of these

8. For the reaction : $2A + 3B \rightarrow 4C + D$, 5 moles of A and 8 moles of B will produce

(A) 4 moles of C, 1 mole of D (B) 20 moles of C, 5 moles of D

(C) 10 moles of C, 2.5 moles of D (D) 32/3 moles of C, 8/3 moles of D

9. For reaction $A + 2B \longrightarrow C$. The amount of C formed by starting the reaction with 5 moles of A and 8 moles of B is:

(A) 5 mol (B) 8 mol (C) 16 mol (D) 4 mol

10. Calculate the number of moles of $SO_3(g)$ which are produced by the reaction of 5 moles of S and 6 moles of O_2 gas

(A) 2 moles (B) 6 moles (C) 4 moles (D) 8 moles

11. 446 g of PbO, 46 g of NO_2 and 16 g of O_2 are allowed to react according to the equation

$$PbO + 2NO_2 + \frac{1}{2}O_2 \rightarrow Pb(NO_3)_2$$

The amount of $Pb(NO_3)_2$ that can be produced is (At. wt. of Pb = 207)

(A) 331 gm (B) 662 gm (C) 165.5 gm (D) None of these

12. The mass of P_4O_{10} produced if 440 gm of P_4S_3 is mixed with 384 gm of O_2 is $P_4S_3 + O_2 \longrightarrow P_4O_{10} + SO_2$

(A) 568 gm (B) 426 gm (C) 284 gm (D) 396 gm

13. 0.6 mol of barium chloride in solution is mixed with 0.2 mol of sodium phosphate, the amount of barium phosphate produced is

(A) 0.1 mol (B) 0.3 mol (C) 0.4 mol (D) 0.5 mol

14. What is the number of moles of $Fe(OH)_3$ that can be produced by allowing 1 mole of Fe_2S_3, 2mole of H_2O and 3 mole of O_2 to react

$2Fe_2S_3 + 6H_2O + 3O_2 \longrightarrow 4Fe(OH)_3 + 6S$

(A) 2 (B) 1.33 (C) 3.52 (D) None

15. 28 gm lithium is mixed with 48 gm O_2 to react according to the following reaction.

$$Li + O_2 \rightarrow Li_2O$$

The mass of Li_2O formed is :

(A) 30 gm (B) 35 gm (C) 45 gm (D) 60 gm

16. Three substances A, B and C can react to form D and E as shwon :

$$2A + 3B + C \rightarrow 4D + 2E$$

If molar masses of A, B, C and D are 40, 30 , 20 and 15 respectively and 285 gm of mixture of A, B and C is reacted then maximum mass of E which can be obtained will be :

(A) 285 gm (B) 200 gm (C) 195 gm (D) 100 gm

17. How many moles of potassium chlorate need to be heated to produce 11.35 litre oxygen at STP ?

(A) $\frac{1}{2}$ mol (B) $\frac{1}{3}$ mol (C) $\frac{1}{4}$ mol (D) $\frac{2}{3}$ mol

18. In the reaction $4A + 2B + 3C \rightarrow A_4B_2C_3$ what will be the number of moles of product formed ? Starting from 2 moles of A, 1.2 moles of B and 1.44 moles of C.

(A) 0.5 (B) 0.6 (C) 0.48 (D) 4.64

19. 12 g of alkaline earth metal gives 14.8 g of its nitride, Atomic weight of metal is :

(A) 12 (B) 20 (C) 40 (D) 14.8

20. If 10 g of Ag reacts with 1 g of sulphur, the amount of Ag_2S formed will be :

[Atomic weight of Ag = 108, S = 32]

(A) 7.75 g (B) 0.775 g (C) 11 g (D) 10 g

21. According to following reaction :

$$A + BO_3 \rightarrow A_3O_4 + B_2O_3$$

The number of moles of A_3O_4 produced if 1 mole of A is mixed with 1 mole of BO_3 is :

(A) 3 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Problem based on mixtures

1. 3 litre of mixture of propane (C_3H_8) & butane (C_4H_{10}) on complete combustion gives10 litre CO_2. Find the composition of mixture.

(A)C_3H_8 2L and C_4H_{10} 1L (B) C_3H_8 3L and C_4H_{10} 0L(C)C_3H_8 1.5L and C_4H_{10} 1.5L(D) C_3H_8 0L and C_4H_{10} 3L

2. 0.01 mole of iodoform (CHI_3) reacts with Ag to produce a gas whose volume at NTP is

$2CHI_3 + 6Ag \longrightarrow 6AgI(s) + C_2H_2(g)$

(A) 224 ml (B) 112 ml (C) 336 ml (D) None of these

3. One mole mixture of CH_4 and air (containing 80% N_2 20% O_2 by volume) of a composition such that when underwent combustion gave maximum heat (assume combustion of only CH_4). Then which of the statements are correct, regarding composition of initial mixture. (X presents mole fraction).

(A) $X_{CH_4} = \frac{1}{11}, X_{O_2} = \frac{2}{11}, X_{N_2} = \frac{8}{11}$ (B) $X_{CH_4} = \frac{3}{8}, X_{O_2} = \frac{1}{8}, X_{N_2} = \frac{1}{2}$

(C) $X_{CH_4} = \frac{1}{6}, X_{O_2} = \frac{1}{6}, X_{N_2} = \frac{2}{3}$ (D) Data insufficient

4. A mixture of KBr and NaBr weighing 0.560 gm was treated with aqueous Ag^+ and all the bromide ion was recovered as 0.970 gm of pure AgBr. The fraction by weight of KBr in the sample is (approximately)

(A) 0.25 (B) 0.50 (C) 0.40 (D) 0.28

5. 40 gram of a carbonate of an alkali metal or alkaline earth metal containing some inert impurities was made to react with excess HCl solution. The liberated CO_2 occupied 12.315 lit. at 1 atm & 300 K. The correct option is

(A) Mass of impurity is 1 gm and metal is Be (B) Mass of impurity is 3 gm and metal is Li

(C) Mass of impurity is 5 gm and metal is Be (D) Mass of impurity is 2 gm and metal is Mg

Problem based on % yield and % purity

6. Calculate the weight of lime (CaO) obtained by heating 200 kg of 95% pure lime stone ($CaCO_3$).

(A) 104.4 kg (B) 105.4 kg (C) 212.8 kg (D) 106.4 kg

7. A silver coin weighing 11.34 g was dissolved in nitric acid. When sodium chloride was added to the solution all the silver (present as $AgNO_3$) was precipitated as silver chloride. The weight of the precipitated silver chloride was 14.35 g. Calculate the percentage of silver in the coin

(A) 4.8% (B) 95.2% (C) 90% (D) 80%

8. For the reaction

$2Fe(NO_3)_3 + 3Na_2CO_3 \rightarrow Fe_2(CO_3)_2 + 6NaNO_3$

initially 2.5 mol of $Fe(NO_3)_2$ and 3.6 mol of Na_2CO_3 is taken. If 6.3 mol of $NaNO_3$ is obtained then % yield of given reaction is

(A) 50% (B) 84% (C) 87.5% (D) 100%

9. For the reaction, $2x + 3y + 4z \longrightarrow 5w$ Initially if 1 mole of x, 3 mole of y and 4 mole of z is taken and 1.25 mole of w is obtained then % of this reaction is

(A) 25% (B) 50% (C) 75% (D) None of these

Problem based on sequential and parallel reaction

10. 120 g Mg was burnt in air to give a mixture of MgO and Mg_3N_2. The mixture is now dissolved in HCl to form $MgCl_2$ and NH_4Cl, if 107 gram NH_4Cl is produced. Then the moles of $MgCl_2$ formed is :

$$Mg + \frac{1}{2}O_2 \longrightarrow MgO \quad .... \text{(i)}$$

$$3Mg + N_2 \longrightarrow Mg_3N_2 \quad .... \text{(ii)}$$

$$MgO + 2HCl \longrightarrow MgCl_2 + H_2O \quad .... \text{(iii)}$$

$$Mg_3N_2 + 8HCl \longrightarrow 2NH_4Cl + 3MgCl_2 \quad .... \text{(iv)}$$

(A) 3 moles (B) 6 moles (C) 5 moles (D) 10 moles

Paragraph Question No. 11 to 13

NaBr, used to produced AgBr for use in photography can be self prepared as follows :

$Fe + Br_2 \longrightarrow FeBr_2$ ... (i)

$FeBr_2 + Br_2 \longrightarrow Fe_3Br_8$ ... (ii) (not balanced)

$Fe_3Br_8 + Na_2CO_3 \longrightarrow NaBr + CO_2 + Fe_3O_4$ ... (iii) (not balanced)

11. Mass of iron required to produce 4120 gm NaBr

(A) 420 gm (B) 840 kg (C) 840 gm (D) 420 kg

12. If the yield of (ii) is 50% and (iii) reaction is 60% then mass of iron required to produce 2060 gm NaBr

(A) 25 mol (B) 50 mol (C) 75 mol (D) 100 mol

13. If yield of (iii) reaction is 90% then mole of CO_2 formed when 1030 gm NaBr is formed

(A) 20 (B) 5 (C) 10 (D) 40

14. Two substance P_4 & O_2 are allowed to react completely to form mixture of P_4O_6 & P_4O_{10} leaving none of the reactants. Using this information calculate the composition of final mixture when mentioned amount of P_4 & O_2 are taken.

$$P_4 + 3O_2 \longrightarrow P_4O_6, \quad P_4 + 5O_2 \longrightarrow P_4O_{10}$$

(i) If 1 mole P_4 & 4 mole of O_2 (ii) If 3 mole P_4 & 11 mole of O_2

(iii) If 3 mole P_4 & 13 mole of O_2

15. Sulphur trioxide may be prepared by the following two reactions :

$S_8 + 8O_2(g) \rightarrow 8SO_2(g)$, $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

How many grams of SO_3 will be produced from 1 mol of S_8?

HOME WORK

NCERT 1.1

Concentration terms

1. 8 g NaOH is dissolved in one litre of solution, its molarity is

(A) 0.8 M (B) 0.4 M (C) 0.2 M (D) 0.1 M

2. For preparing 0.1 M solution of H_2SO_4 in one litre, we need H_2SO_4

(A) 0.98 g (B) 4.9 g (C) 49.0 g (D) 9.8 g

3. What is mass percent of the solute in the solution obtained by mixing 5 g of the solute in 50 g of water ?

(A) 10 % (B) 9.1 % (C) 91 % (D) 50 %

4. The number of moles of NaCl present in its 250 cm^3, 0.5 M solution are

(A) 0.5 mol (B) 0.25 mol (C) 0.125 mol (D) 12.5 mol

5. How many grams of NaOH are needed to prepare 250 cm^3 of 0.4 M NaOH solution ?

(A) 8 g (B) 40 g (C) 80 g (D) 4 g

6. The molarity of sugar ($C_{12}H_{22}O_{11}$) solution if its 20 g are dissolved in 2 litre solution, is

(A) 0.029 M (B) 0.29 M (C) 2.9 M (D) 0.0029 M

7. Determine mole fraction of CH_3OH in a solution obtained by mixing 1.2 mole CH_3OH with 4.8 mole H_2O

(A) 0.8 (B) 0.2 (C) 0.25 (D) 0.5

8. Calculate the volume in litre of 0.1 M solution of HCl which contains 0.365 g HCl ?

(A) 10^{-2} L (B) 0.1 L (C) 1 L (D) 10 L

9. The molarity of a HCl solution, which is 1.825 % (w/v) is :

(A) M/10 (B) M/2 (C) M/5 (D) M/20

10. What volume of a 0.8 M solution contains 100 millimoles of the solute ?

(A) 80 mL (B) 125 mL (C) 125 L (D) 80 L

11. What approximate volume of 0.40 M $Ba(OH)_2$ solution must be added to 50.0 mL of 0.30 M NaOH solution to get a solution in which the molarity of the OH^- ions is 0.50 M ?

(A) 33 mL (B) 66 mL (C) 133 mL (D) 100 mL

12. Equal moles of H_2O and NaCl are present in a solution. Hence, molality of NaCl solution is :

(A) 0.55 (2) 55.5 (C) 1.00 (D) 0.18

13. Calculate molality of a solution in which 5.6 g KOH is dissolved in 200 g water

(A) 0.5 m (B) 1.5 m (C) 1.5 m (D) 0.05 m

14. 1000 g aqueous solution of $CaCO_3$ contains 10 g of calcium carbonate. Concentration of solution is

(A) 10 ppm (B) 100 ppm (C) 1000 ppm (D) 10000 ppm

15. Calculate the molarity when

(a) 4.9 gm H_2SO_4 acid dissolved in water to result 500 ml solution

(b) 56 gm of KOH dissolved in water to result 500 ml solution

15. The mole fraction of I_2 in C_6H_6 is 0.02, then molality of solution approxmately will be:

(A) 0.16 (B) 0.26 (C) 2.6 (D) 1.6

Interconversions of different conentration terms

17. Arrange in increasing order of Molarity of solute in following solutions considering water as solvent. Show your calculations:

(i) 224 gm/lit. KOH (ii) 11.2% w/v KOH (iii) 5m KOH (d = 0.64 gm/ml)

(A) (ii) < (iii) < (i) (B) (iii) < (ii) < (i) (C) (iii) < (i) < (ii) (D) (i) < (ii) < (iii)

18. A solution of A(mol. wt. = 20) and B(mol. wt. = 10), [Mole fraction $X_B = 0.6$] having density 0.7 gm/ml then molarity and molality of B in this solution will be _________ and ________ respectively.

(A) 30 M, 75 m (B) 75 m, 30 M (C) 7.5 m, 30 M (D) None of these

19. **Match the column :**

Column I	**Column II**
(A) 16% w/v. $H_2C_2O_4$ (d = 1.1602 g/ml.)	(P) 1.78 M
(B) 17.45 % w/v H_2SO_4 (d = 1.1745 g/ml)	(Q) 1.78 m
(C) Pure water	(R) 1.5 M
(D) 5 % w/w NaOH (d = 1.2 gm/ml)	(S) 55.5 M

20.

Column I	**Column II**
(A) 10 M MgO ($d_{solution}$= 1.20 gm/ml) Solute : MgO, Solvent: H_2O	(P) $W_{solvent}$ = 120 gm per 100 ml of solution
(B) 40% w/v NaOH ($d_{solution}$ = 1.6 gm/ml) Solute : NaOH, Solvent: H_2O	(Q) $W_{solution}$ = 150 gm per 100 gm solvent
(C) 8 m $CaCO_3$ Solute : $CaCO_3$, Solvent: H_2O	(R) W_{solute} = 120 gm per 100 gm of solvent
(D) 0.6 mol fraction of 'X' (molecular mass = 20) in 'Y' (molecular mass 25) Solute : X, Solvent : Y	(S) $W_{solvent}$ = 125 gm per 100 gm of solute

Mixing of solutions

1. 20 mL of 0.2M $Al_2(SO_4)_3$ is mixed with 30 mL of 0.6 M $BaCl_2$. Calculate the mass of $BaSO_4$ formed in solution.
$BaCl_2 + Al_2(SO_4)_3 \rightarrow BaSO_4 + AlCl_3$

2. 300 ml of 3.0 M NaCl solution is added to 200 ml of 4.0 M $BaCl_2$ solution. The concentration of Cl^- ions in the resulting solution is

(A) 7 M (B) 6 M (C) 5.5 M (D) 5 M

3. What volumes should you mix of 0.2 M NaCl and 0.1 M $CaCl_2$ solution so that in resulting solution the concentration of positive ion is 40% lesser than concentration of negative ion. Assuming total volume of solution 1000 ml.

(A) 400 ml NaCl, 600 ml $CaCl_2$ (B) 600 ml NaCl, 400 ml $CaCl_2$
(C) 800 ml NaCl, 200 ml $CaCl_2$ (D) None of these

4. Assuming complete precipitation of AgCl, calculate the sum of the molar concentration of all the ions if 2 L of 2 MAg_2SO_4 is mixed with 4 L of 1 M NaCl solution is

(A) 4 M (B) 2 M (C) 3 M (D) 2.5M

5. What approximate volume of 0.40 M $Ba(OH)_2$ solution must be added to 50.0 mL of 0.30 M NaOH solution to get a solution in which the molarity of the OH^- ions is 0.50 M ?

(A) 33 mL (B) 66 mL (C) 133 mL (D) 100 mL

6. How many grams of sodium dichromate, $Na_2Cr_2O_7$, should be added to a 50.0mL volumetric flask to prepare 0.025 M $Na_2Cr_2O_7$ when the flask is filled to the mark with water ?

7. Calculate molarity of NaOH in a solution made by mixing 2 lit. of 1.5 M NaOH, 3 lit. of 2M NaOH and 1 lit. water.

8. How would you prepare exactly 3.0 litre of 1.0 M NaOH by mixing proportions of stock solution of 2.50 M NaOH and 0.40 M NaOH. No water is to be used. Find the ratio of the volume (v_1/v_2).

9. The concentration of H_2SO_4 in a solution which has a density 1.2 g /ml. and mass percent of H_2SO_4 is 9.8%, is

(A) 9.8 M (B) 1.2 M (C) 0.6 M (D) 1.8 M

10. What volume of 0.250 MHNO_3 (nitric acid) reacts with 50mL of 0.150MNa_2CO_3 (sodium carbonate) in the following reaction ?

$$2HNO_3(aq) + Na_2CO_3(aq) \rightarrow 2NaNO_3(aq) + H_2O(l) + CO_2(g)$$

11. 20 ml of 0.2 M $Al_2(SO_4)_3$ is mixed with 20 ml of 0.6 M $BaCl_2$. Concentration of Al^{3+} ion in the solution will be

(A) 0.2 M (B) 10.3 M (C) 0.1 M (D) 0.25 M

12. 5 g of K_2SO_4 was dissolved in water to prepare 250 mL of solution. What volume of this solution should be used so that 2.33 g of $BaSO_4$ may be precipitated from $BaCI_2$ solution.

$$K_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2KCl$$

(A) 87 mL (B) 174 mL (C) 8.7 mL (D) 17.4 mL

EUDIOMETRY

13. $C_6H_5OH\ (g) + O_2\ (g) \longrightarrow CO_2\ (g) + H_2O\ (l)$

Magnitude of volume change if 30 ml of C_6H_5OH (g) is burnt with excess amount of oxygen, is

(A) 30 ml (B) 60 ml (C) 20 ml (D) 10 ml

14. 10 ml of a compound containing 'N' and 'O' is mixed with 30 ml of H_2 to produce H_2O (*l*) and 10 ml of N_2(g). Molecular formula of compound if both reactants reacts completely, is

(A) N_2O (B) NO_2 (C) N_2O_3 (D) N_2O_5

15. When 20 ml of mixture of O_2 and O_3 is heated, the volume becomes 29 ml and disappears in alkaline pyragallol solution. What is the volume precent of O_2 in the original mixture?

(A) 90% (B) 10% (C) 18% (D) 2%

16. A mixture of C_2H_2 and C_3H_8 occupied a certain volume at 80 mm Hg. The mixture was completely burnt to CO_2 and $H_2O(l)$. When the pressure of CO_2 was found to be 230 mm Hg at the same temperature and volume, the fraction of C_2H_2 in mixture is

(A) 0.125 (B) 0.5 (C) 0.87 (D) 0.25

17. 20 mL of a mixture of CO and H_2 were mixed with excess of O_2 and exploded & cooled. There was a volume contraction of 23 mL. All volume measurements corresponds to room temperature (27°C) and one atmospheric pressure. Determine the volume ratio $V_1 : V_2$ of CO and H_2 in the original mixture

(A) 6.5 : 13.5 (B) 5 : 15 (C) 9 : 11 (D) 7 : 13

18. The % by volume of C_4H_{10} in a gaseous mixture of C_4H_{10}, CH_4 and CO is 40. When 200 ml of the mixture is burnt in excess of O_2. Find volume (in ml) of CO_2 produced.

(A) 220 (B) 340 (C) 440 (D) 560

COMPREHENSION

A 10 ml mixture of N_2, a alkane & O_2 undergo combustion in Eudiometry tube. There was contraction of 2 ml, when residual gases are passed through KOH. To the remaining mixture comprising of only one gas excess H_2 was added & after combustion the gas produced is absorbed by water, causing a reduction in volume of 8 ml.

19. Gas produced after introduction of H_2 in the mixture ?

(A) H_2O (B) CH_4 (C) CO_2 (D) NH_3

20. Volume of N_2 present in the mixture?

(A) 2 ml (B) 4 ml (C) 6 ml (D) 8 ml

21. Volume of O_2 remained after the first combustion?

(A) 4 ml (B) 2 ml (C) 0 ml (D) 8 ml

22. Identify the hydrocarbon.

(A) CH_4 (B) C_2H_6 (C) C_3H_8 (D) C_4H_{10}

Types of redox reaction and oxidation number

1. Calculate individual and average Oxidation number (if required) of the marked element and also draw the structure of the following compounds or molecules.

(1) $Na_2\underline{S_2}O_3$ (2) $Na_2\ \underline{S_4}O_6$ (3) $H_2\underline{S}\,O_5$ (4) $H_2\ \underline{S_2}O_8$
(5) $H_2\underline{S_2}O_7$ (6) $\underline{S_8}$ (7) $(CH_3)_2\underline{S}O$ (8) $H\underline{N}O_4$
(9) $\underline{C_3}O_2$ (10) $\underline{Os}O_4$ (11) $\underline{P}H_3$ (12) $\underline{Cr}O_4^{2-}$
(13) $\underline{Cr_2}O_7^{2-}$ (14) $\underline{Cr}\ O_2Cl_2$ (15) $\underline{Cr}O_5$ (16) $Na_2H\ \underline{P}O_4$
(17) $\underline{Fe}\underline{S}_2$ (18) $\underline{C_6}H_{12}O_6$ (19) $\underline{N}H_4\ \underline{N}O_3$

2. Calculate individual and average Oxidation number (if required) of the marked element and also draw the structure of the following compounds or molecules.

(1) $F\underline{N}O_3$
(2) $CaO\underline{Cl}_2$ (3) $\underline{Xe}O_3F_2$ (4) $Li\ \underline{Al}\ H_4$ (5) $Na_3\underline{Al}F_6$
(6) $\underline{P_4}$ (7) $\underline{O_3}$ (8) $\underline{I}(\underline{I}O_3)_3$ (9) $\underline{Fe_3}O_4$
(10) $Cs\underline{I_3}$ (11) $K\underline{O_3}$ (12) $\underline{O_2}F_2$ (13) $H_2\underline{Si}F_6$
(14) $\underline{P}(OH)_3$ (15) $\underline{P}OCl_3$ (16) $\underline{Si}(OH)_4$ (17) $Mg_2\underline{C_3}$
(18) $Ca\underline{C_2}$ (19) $Be_2\underline{C}$

3. The reaction $3ClO^-(aq.) \rightarrow ClO_3^-(aq.) + 2Cl^-(aq.)$ is an example of
(A) oxidation (B) reduction (C) disproportionation (D) decomposition reaction

4. White phosphorus reacts with caustic soda, the products are PH_3 and NaH_2PO_2. This reaction is an example of
(A) Oxidation (B) Reduction (C) Disproportionation (D) Neutralisation

5. In the reaction $4P + 3KOH + 3H_2O \rightarrow 3KH_2PO_2 + PH_3$
(A) P undergoes reduction only (B) P undergoes oxidation only
(C) P undergoes both oxidation and reduction (D) neither undergoes oxidation nor reduction

6. Which of the following species does not show disproportionation :-
(A) ClO^- (B) ClO_2^- (C) ClO_3^- (D) ClO_4^-

7. Which of the following reagent can act as reducing agent with SO_2 :-
(A) Cl_2 (B) $KMnO_4$ (C) H_2O (D) H_2S

8. Which of the following can only acts as oxidising agent ?
(A) $KMnO_4$ (B) K_2MnO_4 (C) H_2O_2 (D) SO_2

9. Which will be the proper alternative in place of A in the following equation
$2Fe^{3+}(aq) + Sn^{2+}(aq) \rightarrow 2Fe^{2+}(aq) + A$
(A) Sn^{4+} (B) Sn^{3+} (C) Sn^{2+} (D) Sn^0

10. Which of the following reactions does not involve either oxidation or reduction ?
(A) $VO^{2+} \rightarrow V_2O_3$ (B) $Na \rightarrow Na^+$ (C) $Zn^{+2} \rightarrow Zn$ (D) $CrO_4^{-2} \rightarrow Cr_2O_7^{-2}$

11. Identify the oxidant and the reductant in the following reactions :

(A) $KMnO_4 + KCl + H_2SO_4 \longrightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$

(B) $FeCl_2 + H_2O_2 + HCl \longrightarrow FeCl_3 + H_2O$

(C) $Cu + HNO_3$ (dil) $\longrightarrow Cu(NO_3)_2 + H_2O + NO$

(D) $Na_2HAsO_3 + KBrO_3 + HCl \longrightarrow NaCl + KBr + H_3AsO_4$

(E) $I_2 + Na_2S_2O_3 \longrightarrow Na_2S_4O_6 + NaI$

n-factor calculation

12. Find the **n** factor in following non-redox interaction.

(a) Of base

(i) $Ba(OH)_2 + HCl \rightarrow BaCl_2 + H_2O$ (ii) $Al(OH)_3 + H_2SO_4 \rightarrow Al(OH)(HSO_4)_2 + H_2O$

(b) Of acid

(i) $H_3SbO_4 \xrightarrow{KOH} KH_2SbO_4 + H_2O$ (ii) $H_3SbO_4 + KOH \rightarrow K_2HSbO_4 + H_2O$

13. Find the **n** factor of underlined compound in following interaction

(i) $\underline{Pb(NO_3)_2} + \underline{Cr_2(SO_4)_3} \longrightarrow \underline{PbSO_4} + \underline{Cr(NO_3)_3}$ (ii) $\underline{KMnO_4} + \underline{MnSO_4} \longrightarrow \underline{MnO_2}$

(iii) $\underline{P_4} \longrightarrow \underline{H_2PO_2^-} + \underline{PH_3}$

14. In the reaction, $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$, the eq. wt. of $S_4O_6^{-2}$ is equal to its -

(A) Mol. wt. (B) Mol. wt./2 (C) 2 × mol. wt. (D) Mol. wt./6

15. Equivalent weight of NH_3 in the change $N_2 \rightarrow NH_3$ is :

(A) 17/6 (B) 17 (C) 17/2 (D) 17/3

16. The molecular weight of the compounds (a) Na_2SO_4, (b) $Na_3PO_4 . 12H_2O$ and (c) $Ca_3(PO_4)_2$ respectively are X, Y and Z. the correct set of their equivalent weights will be -

(A) (a) $\frac{X}{2}$ (b) $\frac{Y}{3}$ (c) $\frac{Z}{6}$ (B) (a) X (b) $\frac{Y}{3}$ (c) $\frac{Z}{3}$ (C) (a) $\frac{X}{2}$ (b) Y (c) $\frac{Z}{3}$ (D) (a) X (b) Y (c) Z

17. In the following change -

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. If the atomic weight of iron is 56, then its equivalent weight of Fe will be

(A) 42 (B) 21 (C) 63 (D) 84

18. When one mole NO_3^- is converted into 1 mole NO_2, 0.5 mole N_2 and 0.5 mole N_2O respectively. It accepts x, y and z mole of electrons - x , y and z are respectively -

(A) 1, 5, 4 (B) 1 ,2 , 3 (C) 2 , 1 , 3 (D) 2, 3, 4

19. In the reaction $2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + I_2 + 2K_2SO_4$ the equivalent weight of Cu in $CuSO_4$

(A) 31.75 (B) 63.5 (C) 127 (D) 15.88

20. In the following reaction hydrazine is oxidized to N_2.

$N_2H_4 + OH^- \longrightarrow N_2 + H_2O + e$

The equivalent weight of N_2H_4 (hydrazine) is

(A) 8 (B) 16 (C) 32 (D) 64

21. Calculate the equivalent mass of each oxidant and reductant in:

(A) $FeSO_4 + KClO_3 \rightarrow KCl + Fe_2(SO_4)_3$ (B) $Na_2SO_3 + Na_2CrO_4 \rightarrow Na_2SO_4 + Cr(OH)_3$

(C) $Fe_3O_4 + KMnO_4 \rightarrow Fe_2O_3 + MnO_2$ (D) $KI + K_2Cr_2O_7 \rightarrow Cr^{3+} + 3I_2$

Balancing of redox reaction

1. Balance following redox reactions.

(i) $MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2$ (acidic medium)

(ii) $Cl_2 + O\overline{H} \rightarrow C\overline{l} + ClO_3^- + H_2O$ (Basic medium)

(iii) $AsO_3^{3-} + MnO_4^- \rightarrow AsO_4^{3-} + MnO_2$ (Acidic medium)

(iv) $FeC_2O_4 \rightarrow Fe^{3+} + 2CO_2$ (acidic medium)

(v) $P_2H_4 \rightarrow P_4H_2 + PH_3$ (acidic medium)

2. Balance the following redox equations and identify the valency factor (n-factor) for different compunds or molecule involved in the reactions at reactant and product side.

(i) $K_2Cr_2O_7 + H_2O_2 + H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O + O_2$

(ii) $Zn + NaNO_3 + NaOH \longrightarrow Na_2ZnO_2 + H_2O + NH_3$

(iii) $ClO_3^- + Fe^{2+} + H^+ \longrightarrow Cl^- + Fe^{3+} + H_2O$

(iv) $N_2O_4 + BrO_3^- \longrightarrow NO_3^- + Br^-$ (Basic)

(v) $S_2O_3^{-2} + Sb_2O_5 \longrightarrow SbO + H_2SO_3$ (Acidic)

(vi) $Cr_2O_7^{-2} + I^- + H^+ \longrightarrow Cr^{+3} + I_2 + H_2O$

(vii) $IO_3^- + I^- + H^+ \longrightarrow I_2 + H_2O$

3. Choose the set of coefficients that correctly balances the following equation :

$x\, Cr_2O_7^{2-} + yH^+ + z\, e^- \rightarrow a\, Cr^{+3} + bH_2O$

	x	y	z	a	b		x	y	z	a	b
(A)	2	14	6	2	7	(B)	1	14	6	2	7
(C)	2	7	6	2	7	(D)	2	7	6	1	7

Acid Base titrations

4. How many millilitres of a 9 N H_2SO_4 solution will be required to neutralize completely 20 mL of a 3.6 N NaOH solution ?

(A) 18.0 mL (B) 8.0 mL (C) 16.0 mL (D) 80.0 mL

5. What is the normality of the H_2SO_4 solution, 18.6 mL of which neutralizes 30.0 mL of a 1.55 N KOH solution?

(A) 5.0 N (B) 1.25 N (C) 2.5 N (D) 3.5 N

6. A sample of 1.0 g of calcite (pure $CaCO_3$) required 40 mL of HCl complete reaction. Calculate the normality of the acid.

(A) 0.40 N (B) 0.60 N (C) 0.50 N (D) 0.15 N

7. An excess of NaOH was added to 100 mL of a ferric chloride solution. This caused the precipitation of 1.425 g of $Fe(OH)_3$. Calculate the normality of the ferric chloride solution

(A) 0.20 N (B) 0.50 N (C) 0.25 N (D) 0.40 N

8. One mole of H_2SO_4 will exactly neutralize

(A) 4 moles of an ammonia solution
(B) 1 mole of $Ba(OH)_2$
(C) 3 moles of Al $(OH)_3$
(D) 0.5 mole of $Ba(OH)_2$

9. Hydrochloric acid solutions A and B have concentrations 0.5 N and 0.1 N respectively. The volumes of solution A and solution B required to make a 2-litre solution of 0.2 N HCI are

(A) 0.5 L of A and 1.5 L of B
(B) 3.5 L of A and 0.5 L of B
(C) 1.0 L of A and 1.0 L of B
(D) 0.75 L of A and 1.25 L of B

10. Calculate the normality of an NaOH solution, 21.5 mL of which is required to convert 0.240 g of NaH_2PO_4 in a solution to monohydrogen phosphate.

(A) 1.093 N
(B) 0.093 N
(C) 0.048 N
(D) 0.93 N

11. 2.00 g of a mixture of $NaHCO_3$ and $KClO_3$ requires 100 mL of 0.1 N HCl for complete neutralization. The weight of $KClO_3$ present in the mixture is

(A) 0.84 g
(B) 1.84 g
(C) 1.16 g
(D) 0.16 g

12. A 25-mL HCl solution containing 3.65 g HCl/L is neutralized by 50 mL of NaOH solution. Again, 25 mL of the same NaOH solution neutralized by 50 mL of an H_2SO_4 solution of unknown strength.The normality of the H_2SO_4 solution is

(A) 0.25 N
(B) 0.025 N
(C) 0.05 N
(D) 0.50 N

13. 5 ml of N–HCl, 20 ml of $\frac{N}{2}$ H_2SO_4 and 30 ml of $\frac{N}{3}$ HNO_3 are mixed together and the volume made of 1 litre. What is the weight of pure NaOH required to neutralize the solution?

(A) 1 gm
(B) 0.5 gm
(C) 0.1 gm
(D) 2 gm

14. 10 ml of 0.10 M solution of a compound ($Na_2CO_3.NaHCO_3 . 2H_2O$) is titrated against 0.05 M HCl. X ml of HCl is used when phenolphthalein is the indicator and Y ml of HCl is used when methyl orange is the indicator in two separate titration, hence (Y – X) is

(A) 20 ml
(B) 40 ml
(C) 60 ml
(D) None of these

Redox titration

1. The amount of $KMnO_4$ required to prepare 100 mL of a 0.1 N solution in an acidic medium is

(A) 3.16 g (B) 1.58 g (C) 0.316 g (D) 31.6 g

2. 0.185 g of an iron wire containing 99.8% iron is dissolved in an acid to form ferrous ions. The solution requires 29.3 mL of $K_2Cr_2O_7$ solution for complete reaction. The normality of the $K_2Cr_2O_7$ solution is

(A) 0.05 (B) 0.02 (C) 0.22 (D) 0.11

3. 25 mL of a solution of $KMnO_4$ containing 3.16 g potassium permanganate per litre of the solution oxidizes 20 mL of a solution of $FeSO_4$. in acidic medium. Calculate the weight of the crystalline $FeSO_4$ in 500 mL of the solution.

(A) 9.5 g (B) 7.5 g (C) 17.5 g (D) 1.7 g

4. 4 mL of concentrated H_2SO_4 having a density of 1.8 g / mL was diluted to one litre. 25.0 mL of this dilute solution needed 35.0 mL of an N/10 Na_2CO_3 solution for complete neutralization. The percentage purity of H_2SO_4 is

(A) 95.28% (B) 98.48% (C) 87.78% (D) 90.76%

5. As_2O_3 is oxidised to H_3AsO_4 by $KMnO_4$ in acidic medium. Volume of 0.02M $KMnO_4$ required for this purpose by 1mmol of As_2O_3 will be

(A) 10 mL (B) 20 mL (C) 40 mL (D) 80 mL

6. 25 ml of a solution of Fe^{2+} ions was titrated with a solution of the oxidizing agent $Cr_2O_7^{2-}$. 32.0 ml of 0.025M $K_2Cr_2O_7$ solution was required.What is the molarity of the Fe^{2+} solution.

(A) 0.189M (B) 0.92M (C) 0.192M (D) 0.190M

7. When ferrous oxalate is titrated against $K_2Cr_2O_7$, meq of Fe^{2+}, $C_2O_4^{2-}$ and $Cr_2O_7^{2-}$ in this redox reaction are x, y and z respectively. Then

(A) x = y (B) x + y = z (C) x + 2y = z (D) 2x + 6y = 6z

8. What mass of MnO_2 is reduced by 35 ml of 0.16 N oxalic acid in acid solution? The skeleton reaction is

$MnO_2 + H^+ + H_2C_2O_4 \rightarrow CO_2 + H_2O + Mn^{2+}$

(A) 8.7 g (B) 0.24 g (C) 0.84 g (D) 43.5 g

9. How many grams of I_2 are present in a solution which requires 40 ml of 0.11 N $Na_2S_2O_3$ to react with it?

$S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

(A) 12.7 g (B) 0.558 g (C) 25.4 g (D) 11.4 g

10. What volume of 2N $K_2Cr_2O_7$ solution is required to oxidise 0.81 g of H_2S in acid medium?

$Cr_2O_7^{2-} + H_2S \rightarrow Cr^{+3} + S$

(A) 47.8 ml (B) 23.8 ml (C) 40 ml (D) 72 ml

11. What volume of 0.1 N oxalic acid solution can be oxidized by 250 gram of an 8 percent $KMnO_4$ solution?

(A) 6.3 L (B) 12.6 L (C) 25.2 L (D) 0.63 L

12. How many gram of $KMnO_4$ are contained in 4 litres of 0.05 N solution. The $KMnO_4$ is to be used as an oxidant in acid medium?

(A) 1.58 g (B) 15.8 g (C) 6.32 g (D) 31.6 g

13. The eq. wt. of $Fe_2(SO_4)_3$, the salt to be used as an oxidant in an acid solution is

(A) (mol. wt)/1 (B) (mol. wt./2) (C) (mol. wt.)/3 (D) (mol.wt.)/5

14. In basic medium CrO_4^{2-} oxidises $S_2O_3^{2-}$ to form $Cr(OH)_4^-$ & SO_4^{2-}. How many ml of 0.25 M Na_2CrO_4 are required to react with 40 ml of 0.3 M $Na_2S_2O_3$.

(A) 16 ml (B) 32 ml (C) 128 ml (D) 42 ml

15. What is the molarity of H_2O_2 solution whose 100 ml produce the 0.5 mole of I_2 when reacted with excess KI solution.

(A) 0.5 M (B) 1 M (C) 2.5 M (D) 5 M

16. 2 mole, equimolar mixture of $Na_2C_2O_4$ and $H_2C_2O_4$ required V_1L of 0.1 M $KMnO_4$ in acidic medium for complete oxidation. The same amount of the mixture required V_2L of 0.2 M NaOH for neutralization. The ratio of V_1 to V_2 is :-

(A) 1 : 2 (B) 2 : 1 (C) 4 : 5 (D) 5 : 4

17. The number of mole of oxalate ions oxidised by one mole of MnO_4^- is :

(A) 1/5 (B) 2/5 (C) 5/2 (D) 5

18. 10 mole of ferric oxalate is oxidised by x mole of MnO_4^- in acidic medium. The value of 'x' is-

(A) 12 (B) 4 (C) 40 (D) 18

19. A solution containing 2.10 gm of $Fe(NH_4)_2(SO_4)_2\ 6H_2O$($M = 392g\ mol^{-1}$) was titrated with acidic 23 mL of $Na_2Cr_2O_7$ solution. What is molarity of $Na_2Cr_2O_7$ -

(A) 0.0215 M (B) 0.0388 M (C) 0.0644 M (D) 0.0744 M

20. Consider the redox reactions in column-I and molar ratios of oxidizing to reducing agents in column-II respectively. Match the items in the column appropriately.

Column-I	**Column-II**
(A) $MnO_4^- + C_2O_4^{2-} \rightarrow MnO_2 + CO_2$	(P) 2 : 1
(B) $ClO^- + Fe(OH)_3 \rightarrow Cl^- + FeO_4^{2-}$	(Q) 3 : 1
(C) $HO_2^- + Cr(OH)_3^- \rightarrow CrO_4^{2-} + HO^-$	(R) 2 : 3
(D) $N_2H_4 + Cu(OH)_2 \rightarrow N_2O + Cu$	(S) 3 : 2

Volume strength of H_2O_2

21. What is the volume strength of 1.5 N H_2O_2 (aq.)

(A) 4.8 (B) 8.4 (C) 3.0 (D) 8.0

22. A 10-volume H_2O_2 solution is equal to

(A) 3% (w/w) H_2O_2 (B) 30 g/L H_2O_2 (C) 1.76 N (D) all of these

23. 10 mL of H_2O_2 solution (vol strength x) required 10 mL of (1/0.56) N MnO_4^- solution in acidic medium. Hence, x is -

(A) 0.56 (B) 5.6 (C) 0.1 (D) 10

1. A 5.0 mL solution of H_2O_2 liberates 0.508 g of iodine from an acidified Kl solution. The volume strength of the H_2O_2 solution at STP is approximately

(A) 4.00 (B) 4.5 (C) 6.05 (D) 5.5

2. A fresh H_2O_2 solution is labeled 11.2 V. This solution has the same concentration as a solution which is

(A) 3.4% (wt/wt) (B) 3.4%(vol/vol) (C) 3.4%(wt/vol) (D) None of these

3. 20 ml of H_2O_2 after acidification with dil H_2SO_4 required 30 ml of $\frac{N}{12}$ $KMnO_4$ for complete oxidation. The strength of H_2O_2 solution is nearly. [Molar mass of H_2O_2 = 34]

(A) 2 g/L (B) 4 g/L (C) 8 g/L (D) 6 g/L

4. 10 ml of a solution of H_2O_2 labelled .10 volume. just decolorises 100 ml of potassium permangante solution acidified with dilute H_2SO_4. Calculate the amount of the potassium permanganate in the given solution.

(A) 0.1563 gm (B) 0.563 gm (C) 5.63 gm (D) 0.256 gm

5. 50 ml of an aqueous solution of H_2O_2 was added into excess of a solution of KI in dil. H_2SO_4. The liberated iodine required 20 ml of 0.1 M $Na_2S_2O_3$ solution for complete reaction. The same amount of H_2O_2 with exactly titrate with

(A) 25 mL of 0.016M $KMnO_4$ in acidic medium

(B) 20 mL of $\frac{M}{30}$ $K_2Cr_2O_7$ solution in acidic medium.

(C) 100 mL of 0.008M $KMnO_4$ in acidic medium.

(D) 10 mL of $\frac{M}{30}$ $K_2Cr_2O_7$ solution in acidic medium.

% Labelling of Oleum

6. What is the % of free SO_3 in an oleum sample that is labelled as '104.5% H_2SO_4' ?

(A) 30% (B) 15% (C) 20% (D) 25%

7. Maximum labelling of oleum is :

(A) 109% (B) 100% (C) 122.5% (D) 112%

8. What is the % of free SO_3 in an oleum sample that is labelled as '109% H_2SO_4' ?

(A) 30% (B) 15% (C) 20% (D) 40%

9. Similar to the % labelling of oleum, a mixture of H_3PO_4 and P_4O_{10} is labelled as (100 + x)% where x is the maximum amount of water which can react with P_4O_{10} present in the mixture. If such a mixture is labelled as 127%. The mass of P_4O_{10} in 100 gm of mixture is :

(A) 71 g (B) 47 g (C) 83 (D) 35 g

10. A sample of oleum is labelled as 104.5%. What amount of pure NaOH is required to neutralize the 100 g of the sample of oleum.

(A) 172.2 g (B) 80 g (C) 85.3 g (D) 62 g

11*. Fuming H_2SO_4 (oleum) is a homogenous mixture of H_2SO_4 and SO_3. Then which of the following statement(s) are correct :

(A) If H_2SO_4 and SO_3 are equimolar in an oleum sample, then strength of oleum is 110.11%

(B) If H_2SO_4 and SO_3 are having equal masses in an oleum sample, then strength of oleum is 111.25%

(C) Strength of an oleum sample may be less than 100%.

(D) If strength of oleum is (100 + x) %, then x g of water is to be added to 100 g oleum sample to convert whole of SO_3 to H_2SO_4.

12. An oleum sample is labelled as 118 %, Calculate

(i) Mass of H_2SO_4 in 100 gm oleum sample.

(ii) Maximum mass of H_2SO_4 that can be obtained if 30 gm sample is taken.

(iii) Composition of mixture (mass of components) if 40 gm water is added to 30 gm given oleum sample.

13. A mixture is prepared by mixing 10 gm H_2SO_4 and 40 gm SO_3 calculate,

(a) mole fraction of H_2SO_4 (b) % labelling of oleum

%Availiblity of chlorine in bleaching powder

14. 10 gm sample of bleaching powder was dissolved into water to make the solution one litre. To this solution 35 mL of 1.0 M Mohr salt solution was added containing enough H_2SO_4. After the reaction was complete, the excess Mohr salt required 30 mL of 0.1 M $KMnO_4$ for oxidation. The % of available Cl_2 approximately is

(A) 10% (B) 12.7% (C) 7.1 % (D) 22%

15. 3.55 gm sample of bleaching powder suspended in H_2O was treated with enough acetic acid and KI solution. Iodine thus liberated required 80ml of 0.2M hypo for titration. Calculate the % of available chlorine.

[Available Chlorine = mass of chlorine liberated/ mass of bleaching powder × 100]

(A) 20% (B) 16 % (C) 30% (D) 10 %

Hardness of water

16. One litre of a sample of hard water contain 4.44 mg $CaCl_2$ and 1.9 mg of $MgCl_2$. What is the total hardness in terms of ppm of $CaCO_3$?

(A) 6 (B) 3 (C) 1 (D) 10

17. One litre hard water contains 1 mg $CaCl_2$ and 1 mg $MgSO_4$. Find hardness of water sample (in ppm).

(A) 1.734 ppm (B) 1.934 ppm (C) 1.534 ppm (D) 1.334 ppm

18. Calculate the amount of $Ca(OH)_2$ required to remove the hardness in 60 litre of pond water,containing 1.62mg of calcium bicarbonate per 100mL of water.

(A) 0.246g (B) 0.897g (C) 0.444g (D) 1.286g

19. If 100 Kg of a hard water sample contains 5g $MgSO_4$, find hardness of water (in ppm).

(A) 33.46ppm (B) 52.24ppm (C) 64.26ppm (D) 41.66ppm

Basic definitions

1. Which of the following are isotopes :

(i) Atom, whose nucleus contains 20p + 15n
(ii) Atom, whose nucleus contains 20p + 17n
(iii) Atom, whose nucleus contains 18p + 22n
(iv) Atom, whose nucleus contains 18p + 21n

(A) (i) and (iii) (B) (i) and (iv) (C) (ii) and (iii) (D) (iii) and (iv)

2. Which of the following are isobars :

(i) Atom, whose nucleus contains 20p + 15n
(ii) Atom, whose nucleus contains 20p + 20n
(iii) Atom, whose nucleus contains 18p + 17n
(iv) Atom, whose nucleus contains 18p + 22n

(A) (i) and (iv) (B) (ii) and (iii) (C) (iii) and (iv) (D) (i) and (iii)

3. Which of the following pairs contain isoelectronic species :

(A) CO_3^{2-}, NO_3^- (B) SO_4^{2-}, PO_4^{3-} (C) CO_2, N_2O (D) N^{3-}, Al^{3+}

4. The atom A, B, C have the configuration

A $\rightarrow$ [Z(90) + n(146)], B $\rightarrow$ [Z(92) + n(146)], C $\rightarrow$ [Z(90) + n(148)] So that :-

(a) A and C - Isotones (b) A and C - Isotopes (c) A and B - Isobars (d) B and C - Isobars
(e) B and C - Isotopes; The wrong statement's are:-

(A) a and c (B) c, d and e (C) a, c, and d (D) a, c and e

5. Atoms $_6C^{13}$ and $_8O^{17}$ are related to each other as

(A) Isotone's (B) Isoelectronic (C) Isodiapher's (D) Isoster's

6. Which one of the following sets of ions represents the collection of isoelectronic species ?

(A) Na^+, Mg^{2+}, Al^{3+}, Cl^- (B) Na^+, Ca^{2+}, Sc^{3+}, F^- (C) K^+, Cl^-, Mg^{2+}, Sc^{3+} (D) K^+, Ca^{2+}, Sc^{3+}, Cl^-

7.

(i) $_{26}Fe^{54}$, $_{26}Fe^{56}$, $_{26}Fe^{57}$, $_{26}Fe^{58}$	(a) Isotopes
(ii) $_1H^3$, $_2He^3$	(b) Isotones
(iii) $_{32}Ge^{76}$, $_{33}As^{77}$	(c) Isodiaphers
(iv) $_{92}U^{235}$, $_{90}Th^{231}$	(d) Isobars
(v) $_1H^1$, $_1D^2$, $_1T^3$	

Match the above correct terms:-

(A) [(i), - a], [(ii) - d], [(iii) - b], [(iv) - c], [(v) - a]
(B) [(i) - a] [(ii) - d], [(iii) - d] [(iv) - c] [v - a]
(C) [v -a] [(iv) - c]. [(iii) - d] [(ii) - b] [(i) - a]
(D) None of them

8. If the atomic number (Z) and mass number (A) of an element X are related by the equation : A + Z = 46 and the total number of neutrons in one atom of X is 16, then the total number of protons and electrons in one atom of element X = ‘a’. Find value of ‘a’.

9. If an atom of an element Y contains equal number of protons, neutrons and electrons, and its atomic number (Z) and mass number (A) are related as : 2A + 3Z = 140, then the total number of nucleons present in one atom of element Y = ‘b’. Find value of ‘b’

10. Match the following : (Mass numbers : H = 1, C = 12, N = 14, O = 16)

Column-I	**Column-II**
(A) CH_4	(p) Species contain a total of 10 electrons.
(B) NO^+	(q) Total number of protons is greater than or equal to 13.
(C) CN^-	(r) Total number of neutrons is less than or equal to 9.
(D) H_2O	(s) Species is isoelectronic with CO.

Cathode and Anode Rays

11. Which is not true with respect to cathode rays :

(A) Cathode rays consist of fast moving electrons.

(B) For production of Cathode rays in a discharge tube, the gas filled is at a low pressure.

(C) For production of Cathode rays in a discharge tube, the voltage applied across the electrodes should be high.

(D) None of these

12. Select the correct statement :

(A) Cathode rays have charge only, no mass.

(B) Cathode move with same speed as that of light.

(C) The magnitude of e/m ratio for Cathode rays is 1.76×10^{11} C/g.

(D) Cathode rays are deflected by electric and magnetic field.

13. Which of the following statements is/are INCORRECT regarding anode rays :

(A) Anode rays consist of fast moving protons.

(B) Anode rays are produced by the ejection of protons from the anode material.

(C) Both (A) & (B)

(D) None of these.

14. Select the correct statement(s) :

(A) Anode rays have charge as well as mass.

(B) Anode rays are deflected by electric and magnetic field.

(C) Anode rays are also known as Positive rays or Canal rays.

(D) All of these.

15. The e/m ratio for Anode rays :

(A) varies with the element forming the anode in the discharge tube.

(B) varies with the gas in the discharge tube.

(C) is constant.

(D) Both (A) & (B).

16. The highest value for e/m for anode rays has been observed when the discharge tube is filled with ____ gas:

(A) nitrogen (B) oxygen (C) hydrogen (D) helium

17. Which of the following pairs do not have identical values of e/m :

(A) A proton and a neutron (B) A proton and a Deuterium nucleus

(C) A Deuterium nucleus and an α-particle (D) An electron and α-particle

18. How many moles of protons will have total charge equal to about 4825 Coulombs :

(A) 0.02 mole (B) 0.5 mole (C) 0.05 mole (D) 0.2 mole

19. The ratio of the e/m values of a proton and an α-particle is :

(A) 2 : 1 (B) 1 : 1 (C) 1 : 2 (D) 1 : 4

20. Which of the following is an arrangement of increasing value of e/m (wrt magnitude only) :

(A) $n < \alpha < p < e$ (B) $e < p < \alpha < n$ (C) $n < p < e < \alpha$ (D) $n < p < \alpha < e$

***Home Work : NCERT Exercise* 2.1 To 2.4 ; 2.22 & 2.26**

Rutherford model

1. Atomic radius is of the order of 10^{-8} cm and nuclear radius is of the order of 10^{-13} cm. The fraction of atom that is occupied by nucleus is :

(A) 10^{-5} (B) 10^{5} (C) 10^{-15} (D) None of these

2. Rutherford's experiment, which established the nuclear model of the atom, used a beam of :

(A) β-particles, which impinged on a metal foil & got absorbed.

(B) γ-rays, which impinged on a metal foil & ejected electrons.

(C) helium atoms, which impinged on a metal foil & got scattered.

(D) helium nuclei, which impinged on a metal foil & got scattered.

3. The potential energy of the electron present in the ground state of Li^{2+} ion is represented by :
(r = Radius of ground state)

(A) $+\dfrac{3e^2}{4\pi \epsilon_0 r}$ (B) $-\dfrac{3e}{4\pi \epsilon_0 r}$ (C) $-\dfrac{3e^2}{4\pi \epsilon_0 r^2}$ (D) $-\dfrac{3e^2}{4\pi \epsilon_0 r}$

4. As an electron is brought from an infinite distance close to the nucleus of an atom, the potential energy of the electron-nucleus system :

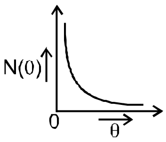
(A) Increases to a greater positive value (B) Decreases to a smaller positive value

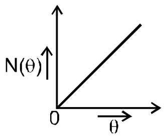
(C) Decrease to a greater negative value (D) Increases to a smaller negative value

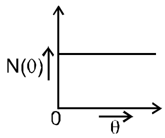
5. If the diameter of two different nuclei are in the ratio 1 : 2, then their mass number are in the ratio :

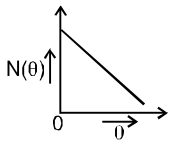
(A) 1 : 2 (B) 8 : 1 (C) 1 : 8 (D) 1 : 4

6. Select the correct graph :

(A) 

(B) 

(C) 

(D) 

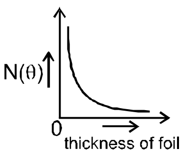
7. In Rutherford experiment, minimum number of α-particles will be deflected on using the thin foil of same thickness of which of the following metals?

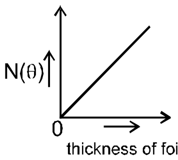
(A) Ag (B) Au (C) Pt (D) Al

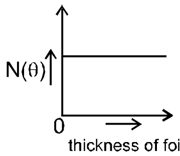
8. In Rutherford formula, maximum number of α-particles deflecting is for which among the following angles

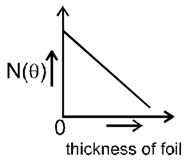
(A) 30° (B) 45° (C) 60° (D) 90°

9. Select the correct graph :

(A) 

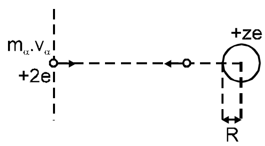
(B) 

(C) 

(D) 

COMPREHENSION : Read the following passage carefully and answer the questions :

RUTHERFORD MODEL :



The approximate size of the nucleus can be calculated by using energy conservation theorem in Rutherford's α-scattering experiment. If an α-particle is projected from infinity with speed v, towards the nucleus having Z protons, then the α-particle which is reflected back or which is deflected by 180^0 must have approached closest to the nucleus. It can be approximated that α-particle collides with the nucleus and gets back. Now, if we apply the energy conservation equation at initial point and collision point, then :

$$(\text{Total Energy})_{\text{initial}} = (\text{Total Energy})_{\text{final}}$$

$$(K.E.)_i + (P.E.)_i = (K.E.)_f + (P.E.)_f$$

$(P.E.)_i = 0$, since P.E. of two charge system separated by infinite distance is zero. Finally the particle stops and then starts coming back.

$$\Rightarrow \frac{1}{2} m_\alpha v_\alpha^2 + 0 = 0 + \frac{Kq_1q_2}{R}$$

$$\Rightarrow \frac{1}{2} m_\alpha v_\alpha^2 = K\frac{2e \times Ze}{R}$$

$$\Rightarrow R = \frac{4KZe^2}{m_\alpha v_\alpha^2}$$

Thus the radius of nucleus can be calculated using above equation. The nucleus is so small a particle that we can't define a sharp boundary for it.

10. An α-particle with initial speed v_0 is projected from infinity and it approaches up to r_0 distance from a nuclie. Then, the initial speed of α-particle, which approaches upto $2r_0$ distance from the nucleus, is :

(A) $\sqrt{2}v_0$ (B) $\frac{v_0}{\sqrt{2}}$ (C) $2v_0$ (D) $\frac{v_0}{2}$

11. Radius of a particular nucleus is calculated by the projection of α-particle from infinity at a particular speed. Let this radius is the true radius. If the radius calculation for the same nucleus is made by another α-particle with half of the earlier speed, then the percentage error involved in the radius calculation is :

(A) 75% (B) 100% (C) 300% (D) 400%

Electromagnetic wave

12. Match the following with respect to waves :

Column-I	**Column-II**
(1) Frequency	(p) Linear distance travelled by a wave per unit time.
(2) Wavelength	(q) Number of waves passing through a point per unit time.
(3) Wave number	(r) Linear distance between starting and end point of one complete wave.
(4) Speed	(s) Number of waves contained in a unit length.

(A) [1 – q] ; [2 – p] ; [3 – s] ; [4 – r] (B) [1 – s] ; [2 – r] ; [3 – q] ; [4 – p]
(C) [1 – q] ; [2 – r] ; [3 – s] ; [4 – p] (D) [1 – s] ; [2 – p] ; [3 – q] ; [4 – r]

13. If the frequency of violet radiation is 7.5×10^{14} Hz, find the value of wavenumber ($\bar{v}$) (in m^{-1}) for it :

(A) 2.5×10^6 (B) 4×10^{-7} (C) 1.33×10^{-15} (D) Data insufficient

14. Visible spectrum contains light of following colours "Violet - Indigo - Blue - Green - Yellow - Orange - Red" (VIBGYOR).
Its frequency ranges from Violet (7.5×10^{14} Hz) to Red (4×10^{14} Hz). Find out the maximum wavelength in this range :

(A) 400 Å (B) 750 Å (C) 4000 Å (D) 7500 Å

15. Select the incorrect statement :

(A) Cathode rays are electromagnetic waves, but anode rays are not.
(B) Electromagnetic waves need a material medium for their propagation.
(C) Electromagnetic waves may have different frequencies.
(D) Electromagnetic waves consist of some material particles travelling with the speed of light.

16. A wavelength of 400 nm of an electromagnetic radiation corresponds to :

(A) frequency = 7.5×10^{14} Hz (B) wave number = 2.5×10^6 m^{-1}.
(C) velocity = 3×10^8 ms^{-1} (D) λ = 40 Å

17. For a wave, frequency is 10 Hz and wavelength is 2.5 m. How much linear distance will it travel in 40 seconds:

Home Work (NCERT EXERCISE 2.5 To 2.8 ; 2.39)

Planck's Quantum theory and photoelectric effect

1. The ratio of the energy of a photon of wavelength 3000 Å to that of a photon of wavelength 6000Å respectively is:

(A) 1 : 2 (B) 2 : 1 (C) 3 : 1 (D) 1 : 3

2. Calculate the approximate molar energy (in KJ) corresponding to photons of energy 2 eV :

(A) 386 KJ (B) 192 KJ (C) 289.5 KJ (D) 144.75 KJ

3. What approximate change in molar energy would be associated with an atomic transition giving rise to a radiation of 1 Hz frequency : ($h = 6.6 \times 10^{-34}$ Js)

(A) 4×10^{-10} J (B) 2.5×10^{10} J (C) 4×10^{-11} J (D) 2.5×10^{9} J

4. Find out the number of photons emitted by a 60 watt bulb in one minute, if wavelength of an emitted photon is 620 nm:

(A) 1.125×10^{22} (B) 1.875×10^{20} (C) 1.5×10^{21} (D) Data insufficient

5. A photon of 300 nm is absorbed by a gas and then, it re-emits two photons and attains the same initial energy level. One re-emitted photon has wavelength 500 nm. Calculate the wavelength of other photon re-emitted out:

(A) 450 nm (B) 800 nm (C) 200 nm (D) 750 nm

6. Assume that 10^{-17} J of light energy is needed by the interior of the human eye to see an object. How many minimum photons of green light (λ = 310 nm) are needed to generate this energy :

(A) 14 (B) 15 (C) 16 (D) 17

7. Infrared lamps are used in restaurants to keep the food warm. The infrared radiation is strongly absorbed by water, raising its temperature and that of the food. If the wavelength of infrared radiation is assumed to be 1500 nm, then the number of photons per second of infrared radiation produced by an infrared lamp that consumes energy at the rate of 100 W and is 12% efficient only is $y \times 10^{19}$. Find the integer value of y.

8. The wavelength (λ) of monochromatic light coming from some light sources is listed below. How many of these sources will be able to exhibit photoelectric effect if incident upon surface of Li metal (work function, ϕ = 2.4 eV).

Light Source	A	B	C	D	E	F	G	H	I
λ(nm)	100	200	300	400	500	600	700	800	900

9. If a photon having wavelength 620 nm is used to break the bond of A_2 molecule having bond energy 144 KJ mol^{-1}, then find the % of energy of photon that is converted into kinetic energy of A atoms.

[hc = 12400 eVÅ ,1 eV/atom = 96 KJ/mol]

(A) 75 % (B) 50 % (C) 12.5 % (D) 25 %

10. In I experiment, electromagnetic radiations of a certain frequency are irradiated on a metal surface ejecting photoelectrons having a certain value of maximum kinetic energy. However, in II experiment,on doubling the frequency of incident electromagnetic radiations, the maximum kinetic energy of ejected photoelectrons becomes three times. What percentage of incident energy is converted into maximum kinetic energy of photoelectrons in II experiment ?

(A) 75 % (B) 50 % (C) 12.5 % (D) 25 %

Bohr atomic model.

11. If the radius of the first Bohr orbit of the H atom is r, then for Li^{2+} ion, it will be :

(A) 3r (B) 9r (C) r/3 (D) r/9

12. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom.

(A) He^{+} (n = 2) (B) Li^{2+} (n = 2) (C) Li^{2+} (n = 3) (D) Be^{3+} (n = 2)

13. The ratio of the radii of two Bohr's orbit of H-atoms is 4:1. What would be their nomenclature ?

(A) K and L (B) L and K (C) N and L (D) (b) and (c) both

14. The ratio of radius of first orbit in hydrogen to the radius of first orbit in deuterium will be :

(A) 1:1 (B) 1: 2 (C) 2 :1 (D) 4:1

15. If radius of ground state of H-atom (according to Bohr's model) is 'R' then radius of 3rd energy state of Li^{+2} is:

(A) R (B) 3R (C) $\frac{R}{3}$ (D) 9R

16. For Li^{+2}, $r_2 : r_5$ will be :

(A) 9 : 25 (B) 4 : 25 (C) 25 : 4 (D) 25 : 9

17. If the velocity of an electron in 1st orbit of H-atom is v, what will be the velocity of electron in 3rd orbit Li^{+2}?

(A) v (B) $\frac{v}{3}$ (C) 3v (D) 9v

18. The velocity of electron in the ground state of H-atom is 2.185×10^8 cm/sec. The velocity of electron in the first excited state of Li^{2+} ion in cm/sec would be:

(A) 3.277×10^8 (B) 2.185×10^8 (C) 4.91×10^8 (D) 1.638×10^8

19. The time period of revolution in the 3rd orbit of Li^{2+} ion is x sec. The time period of revolution in the 2nd orbit of He^+ ion, should be :

(A) x sec (B) $\frac{3}{2}$ x sec (C) $\frac{2}{3}$ x sec (D) $\frac{8}{27}$ x sec

20. Ratio of frequency of revolution of electron in the second excited state of He^+ and second energy state of hydrogen is

(A) $\frac{32}{27}$ (B) $\frac{27}{32}$ (C) $\frac{1}{54}$ (D) $\frac{27}{2}$

21. When an electron makes a transition from (n +1) to n state, the frequency of emitted radiations (v) is related to n according to (n > > 1) ;

(A) $v \propto n^{-3}$ (B) $v \propto n^2$ (C) $v \propto n^3$ (D) $v \propto n^{2/3}$

22. If the angular momentum of an electron in a Bohr orbit is $\frac{2h}{\pi}$, then the value of potential energy of this electron present in He^+ ion is :

(A) – 13.6 eV (B) – 3.4 eV (C) – 6.8 eV (D) – 27.2 eV.

23. The ratio of time periods in first and third orbits of hydrogen atom is :-

(A) 1 : 3 (B) 1 : 4 (C) 1 : 27 (D) 1 : 9

Home Work NCERT EXERCISE 2.10, 2.11, 2.12, 2.51 To 2.54

BOHR ATOMIC MODEL

1. The ratio between kinetic energy and the total energy of the electrons of hydrogen atom according to Bohr's model is :

(A) 2 : 1 (B) 1 : 1 (C) 1 : – 1 (D) 1 : 2

2. The kinetic energy of electron in n = 2 of Li^{2+} ion is:

(A) 61.2 eV (B) – 61.2 eV (C) 15.3 eV (D) 30.6 eV

3. The radii of two of the first four Bohr's orbits of the hydrogen atom are in the ratio 1 : 4. The energy difference between them may be :

(A) Either 12.09 eV or 3.4 eV (B) Either 2.55 eV or 10.2 eV

(C) Either 13.6 eV or 3.4 eV (D) Either 3.4 eV or 0.85 eV

4. The ratio of of (E_2 - E_t) to (E_4 - E_3) for He^+ ion is approximately equal to (where E_n is the energy of nth orbit):

(A) 10 (B) 15 (C) 17 (D) 12

5. If the binding energy of 2nd excited state of hydrogen like sample is 24 eV approximately, then the ionization energy of the sample is approximately:

(A) 54.4 eV (B) 24 eV (C) 122.4 eV (D) 216 eV

6. Tne ionization energy of H atom is 21.79 x 10^{-19} J. Then the value of binding energy of second excited state of Li^{2+} ion:

(A) $3^2 \times 21.7 \times 10^{-19}$ J (B) 21.79×10^{-19} J (C) $\frac{1}{3} \times 21.79 \times 10^{-19}$ J (D) $\frac{1}{3^2} \times 21.79 \times 10^{-19}$

7. The binding energy of e^- in ground state of hydrogen atom is 13.6 eV. The energies required to eject out an electron from three lowest states of He^+ atom will be (in eV):

(A) 13.6,10.2,3.4 (B) 13.6,3.4,1.5 (C) 13.6, 27.2, 40.8 (D) 54.4,13.6, 6

8. The species which has its fifth ionization potential equal to 340 eV is :

(A) B^+ (B) C^+ (C) B (D) C

9. The radii of two Bohr's orbits of hydrogen atom are in the ratio of 4 : 9. Which of the following value of energy difference is not possible between the two orbits? [IE. = 13.6 eV]

(A) 1.9 eV (B) 0.472 eV (C) 0.66 eV (D) 0.21 eV

10. A hydrogen like species with atomic number Z is present in a higher excited state(n). This electron can make transition to the first excited level by successively emitting two photons of energy 2.64 eV and 48.36 eV. This electron can also make transition to third excited state by emitting three photons of energy 2.64 eV, 2.66. eV and 4.9 eV. Identify the hydrogen like species involved.

(A) He^+ (B) Li^{2+} (C) Be^{+3} (D) B^{+4}

11. The energy of hydrogen atom in its ground state is -13.6 eV. The energy of the level corresponding to n = 5 is:

(A) –0.54 eV (B) –5.40 eV (C) –0.85 eV (D) –2.72 eV

12. If first ionization potential of an atom is 16V, then the first excitation potential will be :

(A) 10.2 V (B) 12 V (C) 14 V (D) 16 V

13. The ionisation enthalpy of hydrogen atom is 1.312×10^6 J mol^{-1}. The energy required to excite the electron in the atom from n = 1 to n = 2is :

(A) 8.51×10^5 $Jmol^{-1}$ (B) 6.56×10^5 $Jmol^{-1}$

(C) 7.56×10^5 $Jmol^{-1}$ (D) 9.84×10^5 $Jmol^{-1}$

14. In a single isolated atom of hydrogen, electrons make transition from 4^{th} excited state to ground state producing maximum possible number of wavelengths. If the 2^{nd} lowest energy photon is used to further excite an already excited sample of Li^{2+} ion, then transition will be :

(A) $12 \rightarrow 15$ (B) $9 \rightarrow 12$ (C) $6 \rightarrow 9$ (D) $3 \rightarrow 6$

15. In a mixture of sample of H-atoms and He^+ ions, electrons in all the H-atoms and He^+ ions are present in $n = 4^{th}$ state. Then, find maximum number of different spectral lines obtained when all the electrons make transition from n = 4 upto ground state :

(A) 12 (B) 6 (C) 11 (D) 16

16. If the binding energy of 2nd excited state of a hypothetical H-like atom is 12 eV, then the CORRECT option is/are :

(A) I excitation potential = 81 V (B) II Excitation energy = 96 eV

(C) Ionisation potential = 192 V (D) Binding energy of 2^{nd} state = 27 eV

17. If electron is present in n_1 orbit of hydrogen atom and n_2 orbit of He^+ ions then which of the following relation is correct for the same value of potential energy of electron in both the chemcial specie

(A) $4n_2^2 = n_1^2$ (B) $n_1 = 2n_2$ (C) $2n_1 = n_2$ (D) $n_2 = 4n_1$

18. Select the correct expression(s) according to Bohr model -

(A) $TE = \dfrac{-2\pi^2k^2z^2e^4m}{n^2h^2}$ (B) $PE = \dfrac{-4\pi^2k^2z^2e^4m}{n^2h^2}$

(C) $v = \dfrac{2\pi kze^2}{nh}$ (D) $\dfrac{hc}{\lambda} = 13.6z^2\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right)$

SPECTRUM

19. Wavelength of radiation emitted when an electron jumps from a state A to C is 3000A and it is 6000A when the electron jumps from state B to C. Wavelength of the radiation emitted when an electron jumps from state A to B will be :

(A) 2000Å (B) 3000Å (C) 4000Å (D) 6000Å

20. Difference in wavelength of two extreme lines of Lyman series in emission spectrum of He+ would be:

(A) $\dfrac{1}{12R_H}$ (B) $\dfrac{12}{R_H}$ (C) $\dfrac{1}{4R_H}$ (D) $\dfrac{1}{3R_H}$

21. A collection of H-atoms in 9th excited state returns to ground state. Calculate ratio of total number of spectral lines emitted without emitting any line in Brackett series to number of Brackett series lines.

(A) $\dfrac{39}{6}$ (B) $\dfrac{45}{6}$ (C) $\dfrac{45}{39}$ (D) 6

22. Which transition emits photon of maximum frequency:

(A) 2nd spectral line of Balmer series (B) 2nd spectral line of Paschen series

(C) 5th spectral line of Humphery series (D) Ist spectral line of Lymen series

Home Work : NCERT EXERCISE 2.13, 14, 15, 18, 19, 20

1. The number of possible lines of Paschen series when electron jumps from 7th excited state to ground state (in hydrogen like atom) is :

(A) 2 (B) 5 (C) 4 (D) 3

2. The wavenumber of the first line in the Balmer series of hydrogen is 15200 cm^{-1}. What would be the wave number of the first line in the Balmer series of the Be^{3+} ion ?

(A) $2.4 \times 10^5\ cm^{-1}$ (B) $24.3 \times 10^5\ cm^{-1}$ (C) $6.08 \times 10^5\ cm^{-1}$ (D) $60.8 \times 10^5\ cm^{-1}$

3. The shortest wavelength in Lyman series of Li^{2+} ion is :

(A) 10.13 A (B) 135 A (C) 13.5 A (D) 101.3 A

4. A hydrogen like ion having wavelength difference between first Balmer and Lyman series equals to 593 Å, has Z equal to :

(A) 2 (B) 3 (C) 4 (D) 1

5. Total number of lines in Lyman series of H spectrum will be (where n = no. of orbits):

(A) n (B) n - 1 (C) n -2 (D) n(n + 1)

6. Number of visible lines when an electron returns from 5th orbit to ground state in H spectrum :

(A) 5 (B) 4 (C) 3 (D) 10

7. The difference between the wave number of 1st line of Balmer series and last line of Paschen series for Li^{2+} ion is :

(A) $\frac{R}{36}$ (B) $\frac{5R}{36}$ (C) 4R (D) $\frac{R}{4}$

8. Which electronic transition in atomic hydrogen corresponds to the emission of visible light?

(A) $n = 5 \to n = 2$ (B) $n = 1 \to n = 2$ (C) $n = 3 \to n = 4$ (D) $n = 3 \to n = 1$

9. Wave number of a spectral line for a given transition is x cm^{-1} for He^+, then its value for Be^{3+} (isoelectronic with He^+) for same transition is :

(A) x cm^{-1} (B) 4x cm^{-1} (C) x/4 cm^{-1} (D) 2x cm^{-1}

10. The ratio of wavelength of photon corresponding to the beta line of Lyman series in H-atom and β-line of Balmer series in He+ is:

(A) 1: 1 (B) 1: 2 (C) 1: 4 (D) 3 : 16

11. Match the following : [Atomic structure]

List-I	**List-II**
(A) From n = 6 upto n = 3 (In H-atom sample)	(P) 10 lines in the spectrum
(B) From n = 7 upto n = 3 (In H-atom sample)	(Q) Spectral lines in visible region
(C) From n = 5 upto n = 2 (In H-atom sample)	(R) 6 lines in the spectrum
(D) From n = 6 upto n = 2 (In H-atom sample)	(S) Spectral lines in infrared region

12. Suppose a hypothetical H-like atom produces a blue, yellow, red and violet line in emission spectrum. Match the above lines with their corresponding possible electronic transition :

Colour of spectral lines	**Possible corresponding transitions**
(A) Blue	(P) $6 \to 3$
(B) Yellow	(Q) $2 \to 1$
(C) Red	(r) $5 \to 2$
(D) Violet	(s) $4 \to 3$

(A) (A) $\to$ r , (B) $\to$ p , (C) $\to$ s , (D) $\to$ q (B) (A) $\to$ r , (B) $\to$ s , (C) $\to$ q, (D) $\to$ p

(C) (A) $\to$ p , (B) $\to$ r , (C) $\to$ s , (D) $\to$ q (D) (A) $\to$ p , (B) $\to$ r , (C) $\to$ q, (D) $\to$ s

13. If the shortest wavelength of Lyman series of H atom is x, then the wavelength of the first line of Balmer series of H-atom will be :

(A) $\frac{9x}{5}$ (B) $\frac{36x}{5}$ (C) $\frac{5x}{9}$ (D) $\frac{5x}{36}$

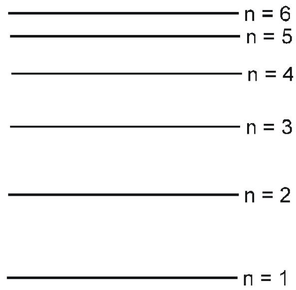
14. Which spectral series for hydrogen contains lines in the visible region of the spectrum?

(A) Lyman series (B) Balmer series (C) Paschen series (D) Brackett series

15. The ratio of wavelength of 1st line of Balmer series and 2^{nd} line of Lyman series is

(A) 32/5 (B) 2 (C) 3 (D) 16

16. What would be the maximum number of emission lines for atomic hydrogen that you would expect to see with the naked eye, if the only electronic energy levels involved are those shown in the Figure :



(A) 4 (B) 6 (C) 5 (D) 15

17. The radius ratio of two Bohr's orbit n_1 & n_2 are in the ratio of 4 : 9. If $n_1 + n_2 = 5$ & an electron make transition between these orbits then

(A) longest wavelength of lyman series will be emitted

(B) longest wavelength of paschen series will be emitted

(C) shortest wavelength of lyman series will be emitted

(D) Radiation emitted will have minimum frequency of Balmer series

18. An ion (atomic number Z), isoelectronic with Hydrogen, is in n^{th} excited state. This ion emits two photons of energies 10.2 eV and 17eV successively to return to first exited state. It can also emit two photons of energies 4.25 and 5.95 eV successively to return to second excited state. What is the sum of values of n and Z ?

19. A sample of H–like ion is in a particular excited state n_2. The electron in it makes back transition upto a lower excited state n_1 producing a maximum of 10 different spectral lines. The change in angular momentum of electron corresponding to maximum frequency line is expressed as y $\frac{h}{4\pi}$ J-s. Then, find the value of y.

Home Work : NCERT Exercise 2.16, 17, 21, 56

Heignberg Uncertainity principal and De-broglie

1. The wavelength associated with a gold ball weighing 200 g and moving at a speed of 5 m/h is of the order of

(A) 10^{-1} m (B) 10^{-20} m (C) 10^{-30} m (D) 10^{-40} m

2. If kinetic energy of a proton is increased nine times the wavelength of the de-Broglie wave associated with it would become

(A) 3 times (B) 9 times (C) $\frac{1}{3}$ times (D) $\frac{1}{9}$ times

3. If numerical value of mass and velocity are equal for a particle, then its de-Broglie wavelength in terms of K.E. is :

(A) $\frac{mh}{2K.E.}$ (B) $\frac{h}{2mK.E.}$ (C) both are correct (D) none is correct.

4. In which transition, the change in de-Broglie wavelength of electron is maximum :

(A) $n = 8 \rightarrow n = 6$ (B) $n = 5 \rightarrow n = 4$ (C) $n = 3 \rightarrow n = 2$ (D) $n = 2 \rightarrow n = 1$

5. Calculate ratio of de-Broglie wavelength for a proton and an α-particle, if they are accelerated from rest through same potential difference V :

(A) $1:2\sqrt{2}$ (B) 2 : 1 (C) $2\sqrt{2}:1$ (D) 1 : 2

6. If the de-Broglie wavelength of an electron revolving in 2^{nd} orbit of H-atom is x, then radius of that orbit is given by

(A) $\frac{x}{\pi}$ (B) $\frac{2x}{\pi}$ (C) $\frac{x}{2\pi}$ (D) Cannot be determined

7. If the radius of first Bohr's orbit of H-atom is x, which of the following is the INCORRECT conclusion :

(A) The de-Broglie wavelength of electron in the third Bohr orbit of H-atom = $6\pi x$

(B) The fourth Bohr's radius of He^{+} ion = 8x

(C) The de-Broglie wavelength of electron in third Bohr's orbit of $Li^{2+} = 2\pi x$

(D) The second Bohr's radius of Be^{2+} = x

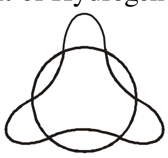
8. In one experiment, a proton having initial kinetic energy of 1 eV is accelerated through a potential difference of 3 V. In another experiment, an α-particle having initial kinetic energy 20 eV is retarded by a potential difference of 2 V. The ratio of de-Broglie wavelengths of proton and α-particle is :

(A) $2\sqrt{6}$: 1 (B) 8 : 1 (C) 4 : 1 (D) $2\sqrt{2}$: 1

9. Determine the de-Broglie wavelength associated with an electron in the 3^{rd} Bohr's orbit of He^{+} ion :

(A) 10 Å (B) 2 Å (C) 5 Å (D) 1 Å

10. The wave motion of an electron in a Bohr's orbit of Hydrogen atom is as shown in diagram. The orbit number is:



(A) 2 (B) 3 (C) 4 (D) 6

11. **Statement-1 :** It is possible to measure simultaneously both the position and momentum of a moving microscopic particle with absolute accuracy.

Statement-2 : In case of moving microscopic particle, if uncertainty in position is reduced, then uncertainty in velocity is very high and vice versa.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

12. If uncertainty in position of an electron is zero, then the uncertainty in its momentum would be :

(A) Zero (B) $h/2\pi$ (C) $h/4\pi$ (D) Infinity

13. Calculate the uncertainty in the velocity of a cricket ball having a mass 150 g, if the uncertainty in its position is the order of 1Å :

(A) 3.50×10^{-24} ms^{-1}. (B) 7.20×10^{-24} ms^{-1}.

(C) 9.30×10^{-22} ms^{-1}. (D) None of these

14. Alveoli are the tiny sacs of air in the lungs, whose average diameter is 50 pm. Consider an oxygen molecule trapped within a sac. Calculate uncertainty in the velocity of oxygen molecule :

(A) 1.98×10^{-2} ms^{-1} (B) 19.8 ms^{-1} (C) 198×10^{-4} ms^{-1} (D) 19.8×10^{-6} ms^{-1}

15. Which of the following have the maximum frequency :

(A) Cosmic rays (B) Radiowaves (C) X-rays (D) Infra red rays

16. Photon of which light has minimum energy :

(A) green (B) blue (C) violet (D) red

17. What is the potential difference through which an electron, with a de-Broglie wavelength of 1.5 Å should be accelerated, if its de-Broglie wavelength has to be reduced to 1 Å :

(A) 110 volts (B) 168 volts (C) 84 volts (D) 55 volts

18. According to Heisenberg's uncertainty principle, on increasing the wavelength of light used to locate a microscopic particle, uncertainty in measurement of position of particle :

(A) Increases (B) Decreases

(C) Remains unchanged (D) Cannot be predicted

19. If uncertainty in momentum is twice the uncertainty in position of an electron, then uncertainty in velocity of electron is : $[\hbar = \frac{h}{2\pi}]$

(A) $\frac{1}{2m}\sqrt{\hbar}$ (B) $\frac{h}{4\pi m}$ (C) $\frac{1}{4m}$ (D) $\frac{1}{m}\sqrt{\hbar}$

20. Debroglie wavelength associated with α-particle & proton are in the ratio 2 : 1 then ratio of kinetic energy associated will be :

(A) 1 : 4 (B) 4 : 1 (C) 1 : 16 (D) 16 : 1

Home Work : NCERT EXERCISE 2.32, 57, 59, 60, 61

SCHRODINGER WAVE EQUATION

1. The radial probability distribution curve of an orbital of H has '4' local maxima. If orbital has 3 angular node then orbital will be :

(A) 7f (B) 8f (C) 7d (D) 8d

2. Which orbital is represented by the complete wave function, ψ_{420}?

(A) 4s (B) 4p (c) 4d (D) 4f

3. The orbitals amongest the following having three nodal sufaces:

(A) 1s (B) 2s (C) 3s (D) 4s

4. The number of radial nodes of 3s and 2p orbitals are respectively:

(A) 2, 0 (B) 0,2 (C) 1, 2 (D) 2 , 1

5. For an electron, with n = 3 has only one radial node. The orbital angular momentum of the electron Will be:

(A) 0 (B) $\sqrt{6}\frac{h}{2\pi}$ (C) $\sqrt{2}\frac{h}{2\pi}$ (D) $3\left(\frac{h}{2\pi}\right)$

6. How many angular nodes does a d-orbital possess?

(A) 1 (B) 2 (C) 3 (D) 4

7. How many radial nodes does a 3d-orbital possess?

(A) 0 (B) 1 (C) 2 (D) 3

8. The radial part of schrodinger wave equation for hydrogen atom is

$$\psi(r) = \frac{1}{16\sqrt{4}\,a_0^{3/2}} (\sigma - 1)(\sigma^2 - 8\sigma + 12)\, e^{-\sigma/2}$$

Where a_0 = constant & $\sigma = 2r/na_0$; n = principle quantum number

Select the correct statements :

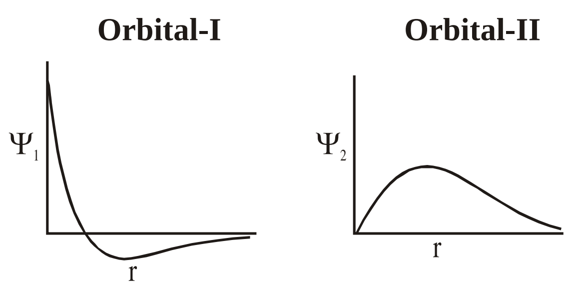
(A) Distance of nearest radial node from the nucleus is $2a_0$.

(B) Distance of farthest radial node from the nucleus is $12a_0$.

(C) Number of maxima in the curve $4\pi r^2\psi^2(r)$ vs r are 4.

(D) $\psi(r)$ is for 4p orbital.

9. Observe the following plots carefully :



Identify the correct option (s) -

(A) Ψ_2 may represent a 2p-orbital

(B) Ψ_1 should represent 2s-orbital

(C) Variation of probability with r in each direction is same for orbital-I

(D) Variation of probability with r in each direction is same for orbital-II

10. The schrodinger wave equation for hydrogen atom is

$\psi(r)=\frac{1}{16\sqrt{4}}\left(\frac{Z}{a_0}\right)^{3/2}\left\{\left[(\sigma-1)(\sigma^2-8\sigma+12)\right]e^{-\sigma/2}\right\}$ where a_0 & Z are constant, $\sigma=\frac{2Zr}{a_0}$ then select correct statement.

(A) Minimum distance of radial node from nucleus is $\frac{a_0}{Z}$

(B) Maximum distance of radial node from nucleus is $\frac{a_0}{Z}$

(C) $\psi(r)$ is for 4s-orbital

(D) $\psi(r)$ is for 5p-orbital

11. Which of the following is the nodal plane of d_{xy} orbital ?

(A) XY (B) YZ (C) ZX (D) All

Paragraph for Question 12 to 14

For an orbital of H-like species radial function is :

$$R(r)=\frac{1}{9\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}(4-\sigma)\,\sigma\,e^{-\sigma/2}$$

where $\sigma=\frac{2Zr}{na_0}$; $a_0 = 0.5$ Å , Z = atomic number of species , r = radial distance from nucleus.

12. Which of the following statements about the orbital whose radial function is given above is **CORRECT**?

(A) This is an s-orbital

(B) This cannot be p-orbital

(C) Principal quantum number (n) of this orbital is 4

(D) This orbital has one radial node.

13. At what distance from the nucleus will this orbital have a radial node in Be^{+3} ion ?

(A) 0.5 Å (B) 2 Å (C) 1 Å (D) 0.75 Å

14. Choose the correct statement among the following

(A) Radial distribution function ($\Psi^2\cdot 4\pi r^2 dr$) give probability at a particular distance along one chosen direction

(B) $\Psi^2(r)$ give probability density at a particular distance over a spherical surface

(C) For 's' orbitals $\Psi(r,\theta,\phi)$ is dependent on θ and ϕ

(D) '2p' orbital with quantum numbers. n = 2, ℓ = 1, m = 0, also shows angular dependence

Ideal Gas Laws

1. Which of the following expression at constant pressure represents Charle's law

(A) $V \propto \frac{1}{T}$ (B) $V \propto \frac{1}{T^2}$ (C) $V \propto T$ (D) $V \propto d$

2. A 10 g of a gas at atmospheric pressure is cooled from 273^oC to 0^oC keeping the volume constant, its pressure would become

(A) 1/2 atm (B) 1/273 atm (C) 2 atm (D) 273 atm

***3.** Pressure remaining the same, the volume of a given mass of an ideal gas increases for every degree centigrade rise in temperature by definite fraction of its volume at

(A) 0^oC (B) Its critical temperature
(C) Absolute zero (D) Its Boyle temperature

4. A certain sample of gas has a volume of 0.2 litre measured at 1 atm. pressure and 0°C. At the same pressure but at 273°C, its volume will be

(A) 0.4 litres (B) 0.8 litres (C) 27.8 litres (D) 55.6 litres

5. Correct gas equation is

(A) $\frac{V_1T_2}{P_1} = \frac{V_2T_1}{P_2}$ (B) $\frac{P_1V_1}{P_2V_2} = \frac{T_1}{T_2}$ (C) $\frac{P_1T_2}{V_1} = \frac{P_2V_2}{T_2}$ (D) $\frac{V_1V_2}{T_1T_2} = P_1P_2$

6. The densities of two gases are in the ratio of 1 : 16. The ratio of their rates of diffusion is

(A) 16 : 1 (B) 4 : 1 (C) 1 : 4 (D) 1 : 16

7. Densities of two gases are in the ratio 1 : 2 and their temperatures are in the ratio 2 : 1, then the ratio of their respective pressures is given molar masses of two gasses are same

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1

***8.** Two separate bulbs contain ideal gases A and B. The density of gas A is twice that of gas B. The molecular mass of A is half that of gas B. The two gases are at the same temperature. The ratio of the pressure of A to that of gas B is

(A) 2 (B) 1/2 (C) 4 (D) 1/4

9. If P, V, T represents the pressure, volume and temperature of gas respectively, then according to Boyle's law, which is correct for a fixed amount of ideal gas :

(A) V $\propto \frac{1}{T}$ (At constant P) (B) V $\propto$ P (At constant T)

(C) V $\propto \frac{1}{P}$ (At constant T) (D) PV = nRT

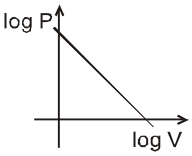
10. A vessel of 120 mL capacity contains a certain mass of an ideal gas at 20ºC and 750 mm pressure. The gas was transferred to another vessel, whose volume is 180 mL. Then the pressure of gas at 20ºC is :

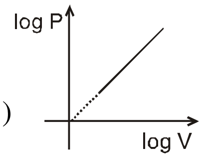
(A) 500 mm (B) 250 mm (C) 1000 mm (D) None of these

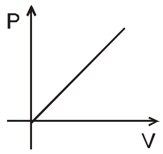
***11.** 5 L of a sample of a gas at 27°C and 1 bar pressure is compressed to a volume of 1000 mL keeping the temperature constant. The percentage increase in pressure is :

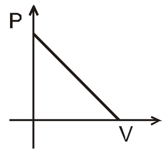
(A) 100 % (B) 400 % (C) 500% (D) 80%

12. For a fixed amount of ideal gas at constant temperature, which of the following plots is correct :

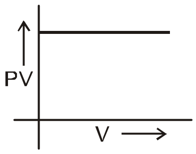
(A) 

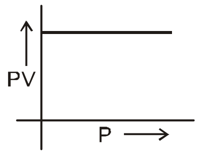
(B) 

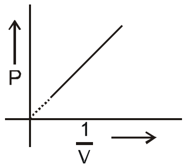
(C) 

(D) 

13. For a fixed amount of ideal gas at constant temperature, which of the following plots is/are correct

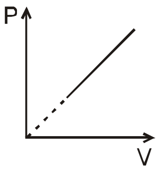
(A) 

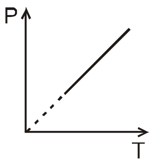
(B) 

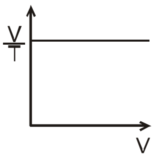
(C) 

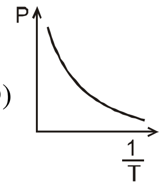
(D) All of these

14. Which of the following graphs is /are possible for a fixed amount of ideal gas :

(A) 

(B) 

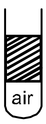
(C) 

(D) 

15. What should be the percentage increase in pressure for a 5% decrease in volume of an ideal gas at constant temperature ?

(A) 5 % (B) 1/5% (C) 5.26 % (D) 4.76 %

16. In which of the following cases is the pressure of air in air column maximum : (Assume same length of Hg column in each case) :

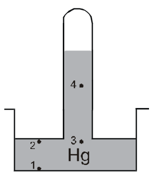
(A) 

(B) 

(C) 

(D) 

***17.** Compare the values of pressure at different points in the given diagram :



(A) $P_1 > P_2 > P_3 > P_4$ (B) $P_1 < P_2 < P_3 < P_4$ (C) $P_1 > P_2 = P_3 > P_4$ (D) $P_1 < P_2 = P_3 < P_4$

18. If some gas is trapped above the mercury column in a Barometer during measurement of atmospheric pressure, the height of Hg column is observed to be h. Then :

(A) h > 76 cm (B) h < 76 cm (C) h = 76 cm (D) cannot be predicted.

19. If another liquid L (ρ = 10.2 g/cm^3) is used in place of mercury, then what should be the minimum length of Barometer tube to measure normal atmospheric pressure ? (Take atmoshperic pressure to be 76 cm of Hg).

(A) ≈ 101.3 cm (B) 57 cm (C) 1 m (D) Cannot be determined

20. The density of neon will be highest at:

(A) S.T.P. (B) 0°C, 2 atm (C) 273°C, 1 atm (D) 273°C, 2 atm

NCERT HOME WORK: Exercise Q. 5.1, 5.2, 5.3, 5.4, 5.6, 5.9, 5.10, 5.15.

1. 1.0 litre of N_2 and 7/8 litre of O_2 at the same temperature and pressure were mixed together. What is the relation between the masses of the gases in the mixture ?

(A) $M_{N_2} = 3M_{O_2}$ (B) $M_{N_2} = 8M_{O_2}$ (C) $M_{N_2} = M_{O_2}$ (D) $M_{N_2} = 16M_{O_2}$

2. Equal masses of CH_4 and O_2 are mixed in an empty container at 25°C. The fraction of the total pressure exerted by O_2 is:

(A) 1/3 (B) 1/2 (C) (1/3) × 273/298 (D) 2/3

***3.** 2 g of a gas A is introduced into an evacuated flask kept at 27°C. The pressure is found to be 1 atm. If 3 g of another gas B is added to the same flask, the total pressure becomes 1.5 atm. Assuming constant temperature and ideal gas behaviour, calculate :

(A) the ratio of mol. weight of gases, M_A and M_B.

(B) the volume of the vessel, if gas A is He

(A) (A) 1 : 1 ; (B) 12.3 litre (B) (A) 1 : 3 ; (B) 12.3 litre

(C) (A) 1 : 1 ; (B) 24.6 litre (D) (A) 1 : 3 ; (B) 24.6 litre

4. 4 g of oxygen occupy 10 L at a particular pressure and temperature. If the pressure of gas is doubled and absolute temperature is halved, in order to maintain constant volume :

(A) 3 g gas should be removed from container. (B) 8 g gas should be added in the container.

(C) 16 g gas should be added in the container. (D) 12 g gas should be added in the container

***5.** Equal volumes of two gases, which do not react together, are enclosed in separate vessels. Their pressures are 100 mm and 400 mm respectively. If the two vessels are joined together, then what will be the pressure of the resulting mixture (temperature remaining constant) :

(A) 150 mm (B) 500 mm (C) 1000 mm (D) 250 mm

***6.** Two flasks of equal volume connected by a narrow tube (of negligible volume) contains a certain amount of N_2 gas at 2 atm and 27°C. The I^{st} flask is then immersed into a bath kept at 47°C while the II^{nd} flask is immersed into a bath kept at 127°C. The ratio of the number of moles of N_2 in I^{st} flask and II flask respectively after sometime will be :

(A) 5 : 4 (B) 2 : 3 (C) 3 : 2 (D) 4 : 5

7. A flask of 4.48 L capacity contains a mixture of N_2 and H_2 at 0ºC and 1 atm pressure. If the mixture is made to react to form NH_3 gas at the same temperature, the pressure in the flask reduces to 0.75 atm. The partial pressure of NH_3 gas in the final mixture is :

(A) 0.33 atm (B) 0.50 atm (C) 0.66 atm (D) 0.25 atm

8. For a certain gas which deviates a little from ideal behaviour, the values of density, ρ were measured at different values of pressure, P. The plot of P/ρ on the Y–axis versus P on the X–axis was nonlinear and had an intercept on the Y–axis, which was equal to

(A) $\frac{RT}{M}$ (B) $\frac{M}{RT}$ (C) RT (D) $\frac{RT}{V}$ (where M = molar mass.)

Kinetic Theory Of Gasses

9. According to the kinetic theory of gases:

(A) the pressure exerted by a gas is proportional to the mean velocity of the molecules

(B) the pressure exerted by the gas is proportional to the root mean square velocity of the molecules

(C) the root mean square velocity is inversely proportional to the temperature

(D) the mean translational kinetic energy of the molecules is directly proportional to the absolute temperature

***10.** According to kinetic theory of gases :

(A) collisions are always elastic

(B) heavier molecules transfer more momentum to the wall of the container

(C) only a small number of molecules have very high velocity

(D) between collisions, the molecules move in straight lines with constant velocities

11. At constant volume for a fixed number of moles of a gas, the pressure of the gas increases with rise in temperature due to :

(A) increase in average molecular speed (B) increase in rate of collisions

(C) increase in molecular attraction (D) increase in mean free path

12. At same temperature and pressure, which of the following gases will have same average translational kinetic energy per mole as N_2O :

(A) He (B) H_2S (C) CO_2 (D) All of these

***13.** At what temperature, will hydrogen molecules have the same average translational kinetic energy as nitrogen molecules have, at 35°C?

(A) $\left(\frac{28\times35}{2}\right)$°C (B) $\left(\frac{2\times35}{28}\right)$°C (C) $\left(\frac{2\times28}{35}\right)$°C (D) 35 °C

14. Average translational K.E. of one mole of helium gas at 273 K in calories is : (Use: R = 2 cal/mollk)

(A) 819 (B) 81.9 (C) 3412.5 (D) 34.125

15. A sample of gas contains N_1 molecules and the average translational kinetic energy at –123°C is E_1 ergs. Another sample of gas at 27°C has average translational kinetic energy as $2E_1$ ergs. Assuming gases to be ideal, the number of gas molecule in the second sample will be

(A) N_1 (B) $\frac{N_1}{4}$ (C) $2N_1$ (D) $4N_1$

16. If average kinetic energy of the molecules in 5 L of He at 2 atm is E, then average kinetic energy of the same number of molecules in 15 L of Ne at 3 atm in terms of E is .... :

(A) 4.5 E (B) E (C) 0.9 E (D) 22.5 E

17. Longest mean free path stands for:

(A) H_2 (B) N_2 (C) O_2 (D) Cl_2

18. The ratio between rms velocity of H_2 at 50 K and that of O_2 at 800 K is:

(A) 4 (B) 2 (C) 1 (D) 1/4

19. The ratio of root mean square velocity to average velocity of a gas molecule at a given temperature is:

(A) 1.086 : 1 (B) 1 : 1.086 (C) 2 : 1.086 (D) 1.086 : 2

NCERT HOME WORK: Exercise Q. 5.11, 5.12, 5.13, 5.16, 5.17, 5.18, 5.21.,

1. The average, root mean square and most probable speeds of gas molecules at STP increase in the order

(A) Root mean square < average < most probable

(B) Most probable < root mean square < average velocity

(C) Average < most probable < root mean square

(D) Most probable < average < root mean square

2. Calculate the temperature at which the R.M.S. velocity of sulphur dioxide molecules is the same as that of oxygen at 300 K -

(A) 600°C (B) 600 K (C) 300 K (D) 300°C

3. Total kinetic energy of 14 grams of dinitrogen gas at 127°C is nearly :

(A) 4980 J (B) 25 J (C) 2490 J (D) 50 J

4. If the density of a gas sample is 4 g/L at pressure 1.2×10^5 Pa, the value of v_{rms} will be :

(A) 600 m/s (B) 300 m/s (C) 150 m/s (D) 450 m/s

5. The total KE of 1 mole a monoatomic ideal gas at 27°C is:

(A) 300 cal (B) 900 cal (C) 1800 cal (D) none of these

6. The average velocity of an ideal gas molecule at 27°C is 0.3 m/s. The average velocity at 927°C will be:

(A) 0.6 m/s (B) 0.8 m/s (C) 0.9 m/s (D) 3.0 m/s

7. Let the most probable velocity of hydrogen molecule at a temperature t°C is V_0. Suppose all the molecules dissociate into atoms when temp is raised to (2t + 273)ºC, then the new r.m.s velocity is -

(A) $\sqrt{2/3}\,V_0$ (B) $2V_0$ (C) $\sqrt{2/3}\,V_0$ (D) $\sqrt{6}\,V_0$

8. Collision diameter and collision cross section for molecules each having radius 'r' are respectively:

(A) πr^2, r (B) $4\pi r^2$, 2r (C) 2r, $4\pi r^2$ (D) r, πr^2

9. Correct expression for collision number and collison frequency would be given as

(A) $\sqrt{2}\pi\sigma^2 U_{avg} N^*, \frac{1}{\sqrt{2}}\pi\sigma^2 U_{avg} N^{*2}$ (B) $\pi\sigma^2 U_{avg} N^*, \frac{1}{\sqrt{2}}\pi\sigma^2 U_{avg} N^{*2}$

(C) $\sqrt{2}\pi\sigma^2 U_{avg} N^*, \frac{1}{\sqrt{2}}\pi\sigma^2 U_{avg} N^*$ (D) $\pi\sigma^2 U_{avg} N^*, \frac{1}{\sqrt{2}}\pi\sigma^2 U_{avg} N^*$

(σ ; collision diameter U_{avg} ; average speed of particles N^* ; number density)

10. Units of collision number & collision frequency respectively

(A) mol, mol sec^{-1} (B) sec^{-1}, sec^{-1} m^{-3}

(C) mol sec^{-1}, mol sec^{-1} m^3 (D) sec^{-1}, sec^{-1} m^3

Effusion and Diffusion

11. The rms velocity of hydrogen is $\sqrt{7}$ times the rms velocity of nitrogen. If T is the temperature of the gas, then :

(A) $T_{(H_2)} = T_{(N_2)}$ (B) $T_{(H_2)} > T_{(N_2)}$ (C) $T_{(H_2)} < T_{(N_2)}$ (D) $T_{(H_2)} = \sqrt{7}\,T_{(N_2)}$

12. The ratio of rates of diffusion of SO_2, O_2 and CH_4 under identical conditions is :

(A) $1 : \sqrt{2} : 2$ (B) 1 : 2 : 4 (C) $2 : \sqrt{2} : 1$ (D) $1 : 2 : \sqrt{2}$

13. The molecular weight of a gas, which diffuse through a porous plug at 1/6th of the speed of hydrogen under identical conditions, is :

(A) 12 u (B) 72 u (C) 36 u (D) 24 u

14. The rate of diffusion of two gases A and B is in the ratio of 1 : 4 and that of B and C in the ratio of 1 : 3 the rate of diffusion of C with respect to A is -

(A) $\frac{1}{12}$ (B) 12 (C) 6 (D) 4

15. The rate of effusion of helium gas at a pressure of 1000 torr is 10 torr min-1. What will be the rate of effusion of hydrogen gas at a pressure of 2000 torr at the same temperature ?

(A) 20 torr min-1 (B) 40 torr min-1 (C) 20 $\sqrt{2}$ torr min-1 (D) 10 torr min-I

***16.** The time taken for effusion of 32 mL of oxygen gas will be the same as the time taken for effusion of which gas sample under identical conditions : (Take $\sqrt{2}$ = 1.4, $\sqrt{3}$ = 1.7)

(A) 64 mL H_2 (B) 50 mL N_2 (C) 44.8 mL CH_4 (D) 22.4 mL SO_2

17. A bottle of dry NH_3 & a bottle of dry HCl connected through a long tube are opened simultaneously under identical conditions at both ends. The white ammonium chloride ring first formed will be :

(A) at the centre of the tube (B) near the HCl bottle

(C) near the NH_3 bottle (D) throughout the length of tube

18. A straight glass tube has 2 inlets X & Y at the two ends of 200 cm long tube. HCl gas through inlet X and NH_3 gas through inlet Y are allowed to enter in the tube at the same time and white fumes form at a point P inside the tube. The distance of point P from X is :

(A) 118.9 cm (B) 81.1 cm (C) 91.1 cm (D) 108.9 cm

***19.** At room temperature, A_2 gas (vapour density = 40) and B_2 gas are allowed to diffuse through identical pinholes from opposite ends of a glass tube of 1m length and of uniform cross-section. The two gases first meet at a distance of 60 cm from the A_2 end. The molecular mass of B_2 gas is :

(A) 90 u (B) 180 u (C) 45 u (D) 35.5 u

20. A mixture containing 2 moles of He and 1 mole of CH_4 is taken in a closed container and made to effuse through a small orifice of container. Then, which is the correct effused volume percentage of He and CH_4 initially, respectively :

(A) 40%, 60% (B) 20%, 80% (C) 80%, 20% (D) 60%, 40%

21. A 4.0 dm^3 flask containing N_2 at 4.0 bar was connected to a 6.0 dm^3 flask containing helium at 6.0 bar, and the gases were allowed to mix isothermally. Then the total pressure of the resulting mixture will be

(A) 4.8 bar (B) 5.2 bar (C) 5.6 bar (D) 5.4 bar

NCERT HOME WORK: Exercise Q 5.5, 5.7, 5.8, 5.14, 5.19.

Real gas

1. The compressibility factor for an ideal gas is :

(A) 0 (B) 1 (C) 3/8 (D) ∞

2. The compressibility of a gas is less than unity at S.T.P. Therefore :

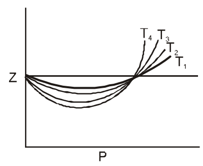
(A) $V_m > 22.4$ litres (B) $V_m < 22.4$ litres (C) $V_m = 22.4$ litres (D) $V_m = 44.8$ litres

3. Gases deviate from the ideal gas behavior because their molecules :

(A) Possess negligible volume (B) Have forces of attraction between them

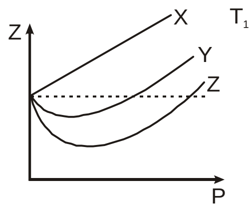
(C) Are polyatomic (D) Are not attracted to one another

***4.** Which of the following is correct order of temperature shown in the following Z Vs P graph for the same gas:

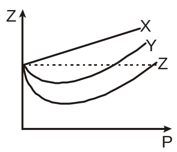


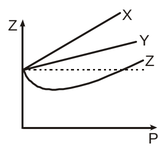
(A) $T_1 < T_2 < T_3 < T_4$ (B) $T_4 < T_3 < T_2 < T_1$ (C) $T_1 < T_3 < T_2 < T_4$ (D) $T_4 < T_2 < T_3 < T_1$

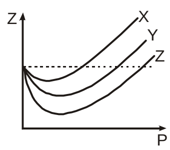
***5.** Z vs P graph is plotted for 1 mole of three different gases X, Y and Z at temperature T_1.

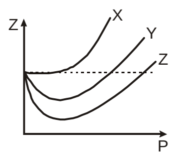


Then, which of the following graph is incorrect if the above plot is made for 1 mole of each gas at T_2 temperature ($T_2 < T_1$)

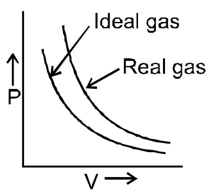
(A)


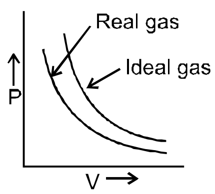
(B)


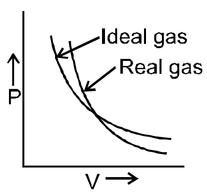
(C)


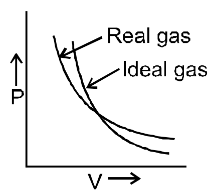
(D)


6. Which of the following graph is commonly correct for real gases other than hydrogen and helium ?

(A)


(B)


(C)


(D)


7. A real gas most closely approaches the behaviour of an ideal gas at :

(A) low pressure & low temperature (B) high pressure & high temperature

(C) low pressure & high temperature (D) high pressure & low temperature

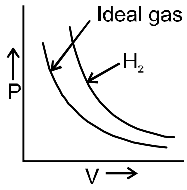
8. At the critical point for H_2 gas, value of Z = 3/8. Then, the value of Z under the similar conditions for CO_2, O_2, SO_2 at their respective critical points will be :

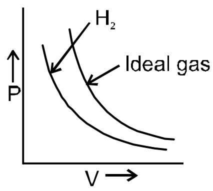
(A) greater than 3/8 (B) smaller than 3/8 (C) equal to 3/8 (D) nothing can be said

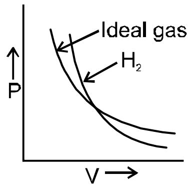
9. For the four gases A, B, E and D, the value of the excluded volume per mole is same. If the order of the critical temperature is $T_B > T_D > T_A > T_E$, then the order of their liquefaction pressure at a temperature T ($T < T_E$) will be

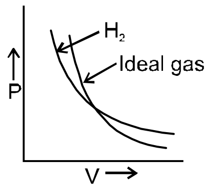
(A) $P_A < P_B < P_E < P_D$ (B) $P_B < P_D < P_A < P_E$ (C) $P_E < P_A < P_D < P_B$ (D) $P_D < P_E < P_A < P_B$

10. Which of the following graph is commonly correct for hydrogen gas ?

(A) 

(B) 

(C) 

(D) 

11. Compressibility factor (Z) for N_2 at – 23°C and 820 atm pressure is 1.9. Find the number of moles of N_2 gas required to fill a gas cylinder of 95 L capacity under the given conditions.

(A) 2000 (B) 200 (C) 2×10^4 (D) Cannot be determined

12. Which of the following statements regarding compressibility factor (Z) is/are correct :

(A) In the lower pressure region, generally value of Z initially decreases on increasing pressure and then increases, however H_2 and He gases are exception to this.

(B) Z for a non-ideal gas can be greater than or less than unity depending on temperature and pressure.

(C) When Z < 1, intermolecular attraction dominates over intermolecular repulsion.

(D) All of these

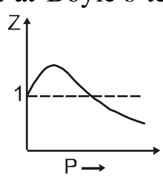
***13.** Match the following forms of Vander waal's equation for 1 mole of a real gas with given conditions

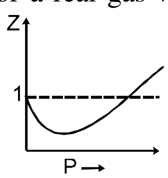
(a) high pressure	(i) PV = RT + Pb
(b) low pressure	(ii) PV = RT – a/V
(c) force of attraction between gas molecules is negligible	(iii) PV = RT + a/V
(d) volume of gas molecules is negligible	(iv) PV = RT – Pb

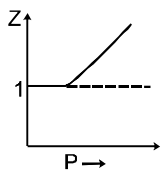
(A) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii) (B) (a)-(i), (b)-(ii), (c)-(i), (d)-(ii)

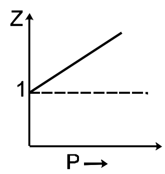
(C) (a)-(iii), (b)-(iv), (c)-(iv), (d)-(iii) (D) (a)-(iii), (b)-(iv), (c)-(i). (d)-(ii).

14. Plot at Boyle's temperature for a real gas will be :

(A) 

(B) 

(C) 

(D) 

15. Critical temperature of a gas is _______ Boyle temperature

(A) higher than (B) equal to (C) lower than (D) no relation between them

16. The critical pressure P_C and critical temperature T_C for a gas obeying Vander Waal's equation are 80 atm and 87°C. Molar mass of the gas is 130 g/mole. The compressibility factor for the above gas will be smaller than unity under the following conditions :

(A) 1 atm and 800 °C (B) 1 atm and 1200 °C (C) 1 atm and 1000 °C (D) 1 atm and 1100 °C

17. Four different identical vessels at same temperature contains one mole each of C_2H_6, CO_2, Cl_2 and H_2S at pressures P_1, P_2, P_3 and P_4 respectively. The value of Vander waal's constant 'a' for C_2H_6, CO_2, Cl_2 and H_2S is 5.562, 3.640, 6.579 and 4.490 atm.L^2.mol^{-2} respectively. If value of Vander waal's constant 'b' is taken to be same for all gases, then :

(A) $P_2 < P_4 < P_1 < P_3$ (B) $P_1 < P_3 < P_4 < P_2$ (C) $P_3 < P_1 < P_4 < P_2$ (D) $P_1 = P_2 = P_3 = P_4$

1. Which of the following reactions can reach equilibrium in open vessel ?

(A) $NH_4HS\ (s) \rightleftharpoons NH_{3\ (g)} + H_2S_{\ (g)}$

(B) $N_{2\ (g)} + 3H_{2\ (g)} \rightleftharpoons 2NH_{3\ (g)}$

(C) $BaCl_{2\ (s)} + H_2SO_{4\ (aq)} \rightleftharpoons BaSO_{4\ (s)} + HCl_{\ (aq)}$

(D) None of these

2. Which of the following represents heterogeneous equilibrium ?

(A) $KClO_3(s) \rightleftharpoons KCl(s) + O_2(g)$

(B) $C_{\ graphite(s)} \rightleftharpoons C_{\ diamond(s)}$

(C) $C_2H_5OH_{\ (l)} + CH_3COOH_{\ (l)} \rightleftharpoons CH_3COOC_2H_{5\ (l)} + H_2O_{\ (l)}$

(D) $N_{2\ (g)} + O_{2\ (g)} \rightleftharpoons 2NO_{\ (g)}$

3. Which of the following represent physical equlibrium ?

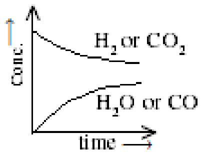
(A) Dissolution of solid KI in water (B) Sublimation of NH_4Cl in a closed container

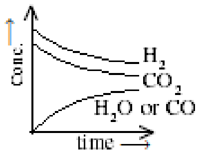
(C) Evaporation of dry ice in a closed container (D) Condesation of water vapours

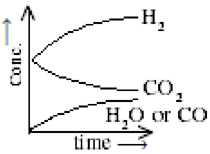
4. Which of the following graphs are correct for the given reaction ,

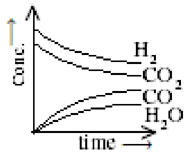
$H_2\ (g) + CO_2\ (g) \rightleftharpoons H_2O\ (g) + CO\ (g)$.

Assume initially only H_2 & CO_2 are present .

(A) 

(B) 

(C) 

(D) 

5. For the homogeneous reaction $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$ the equilibrium constant K_c has the units of

(A) $(Conc.)^{-10}$ (B) $(conc.)^{1}$ (C) $(Conc.)^{-1}$ (D) It is dimensionless.

6. Select the reaction for which the equilibrium constant is written as $[MX_3]^2 = K[MX_2]^2\ [X_2]$

(A) $MX_3 \rightleftharpoons MX_2 + \frac{1}{2}X_2$ (B) $2MX_3 \rightleftharpoons 2MX_2 + X_2$

(C) $2MX_2 + X_2 \rightleftharpoons 2MX_3$ (D) $MX_2 + \frac{1}{2}X_2 \rightleftharpoons MX_3$.

7. The active mass of 64 g of HI in a two litre flask would be

(A) 2 (B) 1 (C) 5 (D) 0.25

8. In order to increase the forward rate of the reaction : $2A + 3B \rightleftharpoons$ Product, 32 times, it is necessary to

(A) Make the conc. of A and B three times (B) Make the conc. of A and B two times

(C) Make the conc. of A and B half (D) Make the conc. of A and B four times

9. In a reaction $A + 2B \rightleftharpoons 2C$, 2.0 mole of 'A', 3.0 mole of 'B' and 1.0 mole of 'C' are placed in a 2.0 L flask and the equilibrium concentration of 'C' is 1.0 mole/L. The equilibrium constant (K) for the reaction is

(A) 0.33 (B) 1.33 (C) 1.66 (D) 0.66.

10. Which of the following reaction(s) is /are follow the relationship, $\log\frac{K_p}{K_c} - \log RT = 0$

(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (B) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(C) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$ (D) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

11. For the equilibrium in a closed vessel $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

K_p is found to be double of K_c. This is attained when:

(A) T = 2 K (B) T = 12.18 K (C) T = 24.36 K (D) T = 27.3 K

12. K_p/K_c for the reaction $CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$ will be

(A) 1 (B) $\sqrt{RT}$ (C) $\frac{1}{\sqrt{RT}}$ (D) R T

13. For the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the equilibrium constant K_p changes with

(A) total pressure (B) catalyst

(C) the amounts of H_2 and I_2 present (D) temperature

14. In a reversible chemical reaction having two reactants in equilibrium, if the concentration of the reactants are doubled then the equilibrium constant will

(A) also be doubled (B) be halved (C) become one fourth (D) remain the same.

15. For which of the following equilibria, is $K_p = K_c$.

(A) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$ (B) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$

(C) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ (D) $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$

16. For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, at 350 K, the value of $K_c = 0.4$. The value of K_p for the reaction at the same temperature would be

(A) 11.48 atm (B) 1.148 atm (C) 1.4×10^{-2} atm (D) 1.4×10^{-3} atm

17. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ Starting with one mole of nitrogen and 3 moles of hydrogen, at equilibrium 50% of each had reacted. If the equilibrium pressure is P, the partial pressure of hydrogen at equilibrium would be

(A) P/2 (B) P/3 (C) P/4 (D) P/6.

18. For a reaction $H_2 + I_2 \rightleftharpoons 2HI$ at 721 K, the value of equilibrium constant is 50. If 0.5 moles each of H_2 and I_2 is added to the system the value of equilibrium constant will be

(A) 0.02 (B) 0.2 (C) 50 (D) 25

19. For the reaction : $P \rightleftharpoons Q + R$. Initially 2 moles of P was taken. Up to equilibrium 0.5 moles of P was dissociated. What would be the degree of dissociation :-

(A) 0.5 (B) 1 (C) 0.25 (D) 4.2

20. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

In above reaction, at equilibrium condition mole fraction of PCl_5 is 0.4 and mole fraction of Cl_2 is 0.3. Then find out mole fraction of PCl_3

(A) 0.3 (B) 0.7 (C) 0.4 (D) 0.6

21. 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium :-

(A) 4.5 (B) 6 (C) 3 (D) 4

22. The dissociation of CO_2 can be expressed as $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$. If the 2 moles of CO_2 is taken initially and 40% of the CO_2 is dissociatated completely. What is the total number of moles at equilibrium:-

(A) 2.4 (B) 2.0 (C) 1.2 (D) 5

23. In the reaction $2P(g) + Q(g) \rightleftharpoons 3R(g) + S(g)$. If 2 moles each of P and Q taken initially in a 1 litre flask. At equilibrium which is true:-

(A) [P] < [Q] (B) [P] = [Q] (C) [Q] = [R] (D) None of these

Home Work NCERT EXERCISE 7.1, 7.2, 7.4 To 7.6

CHEMICAL EQUILIBRIUM AND ACTIVE MASS

1. The active mass of 64 g of HI in a two litre flask would be
(A) 2 (B) 1 (C) 5 (D) 0.25

2. Ratio of active masses of 22g CO_2, 3g H_2 and7g N_2 in a gaseous mixture :-
(A) 22 : 3 : 7 (B) 0.5: 3 : 7 (C) 1 : 3 : 1 (D) 1 : 3 : 0.5

3. When rate of forward reaction becomes equal to backward reaction, this state is termed as:-
(A) Irreversible state (B) Reversible state (C) Equilibrium (D) All of these

4. A reversible reaction is one which:-
(A) Proceeds in one direction (B) Proceeds in both direction
(C) Proceeds spontaneously (D) All the statements are wrong

5. Amongst the following chemical reactions the irreversible reaction is :-
(A) $H_2 + I_2 \rightleftharpoons 2HI$ (B) $AgNO_3 + NaCl \rightleftharpoons AgCl + NaNO_3$
(C) $CaCO_3 \rightleftharpoons CaO + CO_2$ (D) $O_2 + 2SO_2 \rightleftharpoons 2SO_3$

LAW OF EQUILIBRIUM AND EQUILIBRIUM CONSTANT

6. The equilibrium concentration of B $[(B)_e]$ for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:-
(A) $K_C[A]_e^{-1}$ (B) $\frac{k_f}{k_b}[A]_e^{-1}$ (C) $k_f k_b^{-1}[A]_e$ (D) $k_f\ k_b\ [A]^{-1}$

7. For the reaction $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$
Which one is correct representation :–
(A) $K_p = (P_{(H_2O)})^2$ (B) $K_c = [H_2O]^2$ (C) $K_p = K_c(RT)^2$ (D) All

8. For the equilibrium $2H_2O(g) \rightleftharpoons 2H_2\ (g) + O_2(g)$ equilibrium constant is K_1.
For the equilibrium $2CO_2 \rightleftharpoons 2CO(g) + O_2\ (g)$, equilibrium constant is K_2.
The equilibrium constant for $CO_2(g) + H_2(g) \rightleftharpoons CO(s) + H_2O(g)$ is:
(A) $K_1\ K_2$ (B) $\frac{K_1}{K_2}$ (C) $\sqrt{\frac{K_1}{K_2}}$ (D) $\sqrt{\frac{K_2}{K_1}}$

9. Which of the following statements is (are) correct?
(A) The value of equilibrium constant for a particular reaction is constant under all conditions of temperature and pressure
(B) The units of K_c for the reaction $H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$ are mol L^{-1}
(C) In the reaction $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$, $[CaCO_3]=[CaO]=1$
(D) K_p is always greater than K_c for a particular reaction

10. In which of the following gaseous equilibria, K_p is less than K_c?
(A) $N_2O_4 \rightleftharpoons 2NO_2$ (B) $2HI \rightleftharpoons H_2 + I_2$
(C) $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (D) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

11. The value of K_p, for the reaction, $A(g) + 2B(g) \rightleftharpoons C(g)$ is 25 atm^{-2} at a certain temperature. The value of K_p for the reaction, $\frac{1}{2}C(g) \rightleftharpoons \frac{1}{2}A(g) + B(g)$ at the same temperature would be
(A) 25 atm^{-1} (B) 1/25 atm^{-1} (C) 1/5 atm (D) 5 atm.

12. For the reaction, $A + 2B \rightleftharpoons 2C$, the rate constants for the forward and the backward reactions are 1×10^{-4} and 2.5×10^{-2} respectively. The value of equilibrium constant, K for the reaction would be

(A) 1×10^{-4} (B) 2.5×10^{-2} (C) 4×10^{-3} (D) 2.5×10^{2}

13. The equilibrium constant for the reaction $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ is 20 at 500 K.

The equilibrium constant for the reaction, $2AB(g) \rightleftharpoons A_2(g) + B_2(g)$, would be-

(A) 20 (B) 0.5 (C) 0.05 (D) 10.

14. For the reaction, $A + B \longleftrightarrow 3\,C$, if 'a' moles/litre of each 'A' & 'B' are taken initially, then the incorrect relation is

(A) $[A] - [B] = 0$ (B) $3\,[B] + [C] = 3\,a$ (C) $3\,[A] + [C] = 3\,a$ (D) $[A] + [B] = 3\,[C]$

DEGREE OF DISSOCIATION

15. For the reaction : $A \rightleftharpoons nB$, if 'a' mole of A is taken initially and x moles of A get dissociated at equilibrium. What is the value of degree of dissociation -

(A) $\left(\frac{x}{a^n}\right)$ (B) ax (C) $\left(\frac{x}{a}\right)^n$ (D) $\left(\frac{x}{a}\right)$

16. An equilibrium system for the reaction between hydrogen and iodine to give hydrogen iodide at 765 K in a 5 litre volume contains 0.4 mole of hydrogen, 0.4 mole of iodine and 2.4 moles of hydrogen iodide. The equilibrium constant for the reaction is :

$H_2 + I_2 \rightleftharpoons 2HI$, is

(A) 36.0 (B) 15.0 (C) 0.067 (D) 0.28.

17. For a gasesous reaction, $2A + B \rightleftharpoons 2C$, the partial pressures of A, B and C at equilibrium are 0.3 atm, 0.4 atm and 0.6 atm respectively. The value of K_p for the reaction would be :

(A) 10 atm^{-1} (B) 1/10 atm^{-1} (C) 0.2 atm^{-1} (D) 5 atm^{-1} .

18. For the reaction $A(g) + B(g) \rightleftharpoons C(g)$ at equilibrium the partial pressure of the species are $P_A = 0.15$ atm, $P_C = P_B = 0.30$ atm. If the capacity of reaction vessel is reduced, the equilibrium is re-established. In the new situation partial pressure A and B become twice. What is the partial pressure of C?

(A) 0.30 (B) 0.60 (C) 1.20 (D) 1.80

19. For the reaction : $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = 1.16$ atm at 800°C. If 20g of $CaCO_3$ were kept in a 10 litre vessel at 800°C, the amount of $CaCO_3$ remained at equilibrium is

(A) 34% (B) 64% (C) 46% (D) none of these

20. The equilibrium $N_2 + O_2$ estabilished in a reaction vessel of 2.5 L capacity. The amounts of N_2 and O_2 taken at the start were respectively 2 moles and 4 moles. Half a mole of nitrogen has been used up at equilibrium. The molar concentration of nitric oxide is

(A) 0.2 (B) 0.4 (C) 0.6 (D) 0.1

21. Two moles of HI were heated in a sealed tube at 440°C till the equilibrium was reached. HI was found to be 22% decomposed. The equilibrium constant for dissociation is

(A) 0.282 (B) 0.0796 (C) 0.0199 (D) 1.99

22. In an evacuated closed isolated chamber at 250^0 C .02 mole PCl_5 and .01 mole Cl_2 are mixed ($PCl_5 \rightleftharpoons PCl_3 + Cl_2$) At equilibrium density of mixture was 2.48 g/L and pressure was 1 atm . The number of total moles at equilibrium will be approximately

(A) 0.012 (B) 0.022 (C) 0.032 (D) 0.045

Home Work NCERT EXERCISE 7.8, 7.9, 7.10, 7.11 & 7.13

1. The equilibrium constant for the following reaction , H_2 (g) + Br_2 (g) $\rightleftharpoons$ 2 HBr (g) is 1.6×10^5 at 1024 K. Find the equilibrium pressure of all gases if 10 bar of HBr is introduced into a sealed container at 1024 K.

2. The degree of dissociation of PCl_5 (α) obeying the equilibrium, $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, is approximately related to the presure at equilibrium by -

(A) $\alpha \propto P$ (B) $\alpha \propto \frac{1}{\sqrt{P}}$ (C) $\alpha \propto \frac{1}{P^2}$ (D) $\alpha \propto \frac{1}{P^4}$

3. At temperature, T, a compound AB_2(g) dissociates according to the reaction $2AB_2(g) \Leftrightarrow 2AB\ (g) + B_2(g)$ with a degree of dissociation, x, which is small compared with unity. Deduce the expression for K_p, in terms of x and the total pressure, P.

(A) $\frac{Px^3}{2}$ (B) $\frac{Px^2}{3}$ (C) $\frac{Px^3}{3}$ (D) $\frac{Px^2}{2}$

4. For $A \longleftrightarrow 2B$ equilibrium constant at total pressure p_1 is k_{p1} & for $C \rightleftharpoons D + E$ equilibrium constant at total pressure p_2 is k_{p2}. If both reaction are gaseous homogeneous reactions and degree of dissociation of A & C are same, then the ratio of p_1/p_2 if $k_{p1} = 2\ k_{p2}$ is

(A) 1/2 (B) 1/3 (C) 1/4 (D) 2

5. Match the following .

Reaction (Homogeneous gaseous phase)	**Degree of dissociation in terms of equilibrium constant**
1. $A + B \longleftrightarrow 2C$	(a) $\left(\sqrt{k}\right)/\left(1+\sqrt{k}\right)$
2. $2A \longleftrightarrow B + C$	(b) $\left(\sqrt{k}\right)/\left(2+\sqrt{k}\right)$
3. $A + B \longleftrightarrow C + D$	(c) $2k/\left(1+2k\right)$
4. $AB \longleftrightarrow \frac{A}{2} + \frac{B}{2}$	(d) $\frac{2\sqrt{k}}{1+2\sqrt{k}}$

(A) 1 – d , 2 – c , 3 – b , 4 – a (B) 1 – a , 2 – c , 3 – b , 4 – d
(C) 1 – b , 2 – d , 3 – a , 4 – c (D) 1 – b , 2 – a , 3 – d , 4 – c

6. In the following reaction, $3A + B \rightleftharpoons 2C + D$

Initial mol of B is double of A. At equilibrium mol of A and C are equal. Hence % dissoication of B is

(A) 10% (B) 20% (C) 40% (D) 5%

7. $CH_3\text{–}CO\text{–}CH_3(g) \rightleftharpoons CH_3\text{–}CH_3(g) + CO(g)$

Initial pressure of CH_3COCH_3 is 100 mm. When equilibrium set up and mol fraction of CO(g) is 1/3 hence K_p is

(A) 100 mm (B) 50 mm (C) 25 mm (D) 0.6 mm

8. For the following equilibrium $N_2O_4 \rightleftharpoons 2NO_2$ in gaseous phase, NO_2 is 50% of the total volume when equilibrium is set up. Hence percent of dissoicaiton of N_2O_4 is

(A) 50% (B) 25% (C) 66.66% (D) 33.33%

9. PCl_5 is 40% dissociated when pressure is 2 atm. It will be 80% dissociated when pressure is approximately

(A) 0.2 atm (B) 0.5 atm (C) 0.3 atm (D) 0.6 atm

10. A quantity of PCl_5 was heated in a 10 dm^3 vessel at 250°C : PCl_5 (g) $\rightleftharpoons$ $PCl_3(g) + Cl_2(g)$. At equilibrium, the vessel contains 0.1 mole of PCl_5 and 0.2 mole of Cl_2. The equilibrium constant of the reaction is :

(A) 0.04 mol/L (B) 0.4 mol/L (C) 4 mol/L (D) cannot be determined

11. The equilibrium constant (K_p) for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 16 atm. If the volume of the container is reduced to one half its original volume, the value of K_p for the reaction at the same temperature will be :

(A) 32 atm (B) 64 atm (C) 16 atm (D) 8 atm

12. The vapour density of N_2O_4 at a certain temperature is 40. What is the percentage dissociation of N_2O_4 at this temperature according to the given reaction ? $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(A) 7.5% (B) 95% (C) 15% (D) 1.5%

LE-CHATELIER'S PRINCIPALE

13. The exothermic formation of NH_3 is represented by the equation : N_2 (g) + $3H_2$ (g) $\rightleftharpoons$ $2NH_3$ (g)

Which of the following will increase the quantity of NH_3 in an equilibrium mixture of N_2 , H_2 and NH_3 :

(A) Increasing the temperature (B) Increasing the volume of container

(C) Removing N_2 (D) Adding H_2

14. $aA + bB \rightleftharpoons cC + dD$

In above gaseous phase reaction, low pressure and high temperature shift the equilibrium in backward direction. So correct set is :$aA + bB \rightleftharpoons cC + dD$

(A) $(a + b) > (c + d), \Delta H > 0$ (B) $(a + b) < (c + d), \Delta H > 0$

(C) $(a + b) < (c + d), \Delta H < 0$ (D) $(a + b) > (c + d), \Delta H < 0$

15. Consider the following reactions :

(i) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (ii) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

The addition of an inert gas at constant volume :

(A) will increase the dissociation of PCl_5 as well as HI.

(B) will reduce the dissociation of PCl_5 and not affect the dissociation of HI.

(C) will increase the dissociation of PCl_5 and not affect the dissociation of HI.

(D) will not disturb the equilibrium of the reactions.

16. For a chemical reaction $X(g) + 2Y(g) \rightleftharpoons XY_2(g)$, the amount of XY_2 at equilibrium is affected by :

(A) temperature and pressure (B) temperature only

(C) pressure only (D) temperature, pressure and catalyst

17. For $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$,for a reaction started only with $NH_4HS(s)$ the observed pressure for reaction mixture in equilibrium is 1.12 atm at 106°C. What is the value of K_p for the reaction

18. Consider the equilibrium: $P(g) + 2Q(g) \rightleftharpoons R(g)$. When the reaction is carried out at a certain temperature, the equilibrium conceentration of P and Q are 3M and 4M respectively. When the volume of the vessel is doubled and the equilibrium is allowed to be reestablished, the concentration of Q is found to be 3M. Find (A) K_c (B) concentration of R at two equilibrium stages.

19. The dissociation of ammonium carbamate may be represented by the reaction $NH_2COONH_{4(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$

$\Delta H°$ for the forward reaction is negative. The equilibrium will shift from right to left if there is

(A) A decrease in pressure (B) An increase in temperature

(C) An increase in the concentration of ammonia (D) An increase in the concentration of carbondioxide

20. $N_{2(g)} + 3H_{2(g)} \underset{500°C}{\overset{Catalyst}{\rightleftharpoons}} 2NH_3 + heat$

In the above reaction, the direction of equilibrium will be shifted to the right by

(A) Increasing the concentration of nitrogen (B) Compressing the reaction mixture

(C) Removing the catalyst (D) Decreasing the concentration of ammonia

Home Work NCERT Exercise 7.12, 7.14, 7.15 To 7.20

1. The equilibrium of which of the following reactions will not be disturbed by the addition of an inert gas at constant volume

(A) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ (B) $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

(C) $CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$ (D) $C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$

2. Which of the following factors will increase solubility of $NH_{3(g)}$ in H_2O

$NH_{3(g)} + H_2O_{(aq)} \rightleftharpoons NH_4OH_{(aq)}$

(A) Increase in pressure (B) Addition of water (C) Liquefaction of NH_3 (D) Decrease in pressure

3. For the gas phase reaction $C_2H_4(g) + H_2(g) \rightleftharpoons C_2H_6(g)$; $\Delta H = -32.7$ kcal carried out in a vessel, the equilibrium concentration of C_2H_4 can be increased by

(A) Increasing temperature (B) Decreasing pressure

(C) Removing some H_2 (D) Adding some C_2H_6

4. When $NaNO_3$ is heated in a closed vessel, O_2 is liberated and $NaNO_2$ is left behind. At equilibrium.

(A) Addition of $NaNO_2$ favours reverse reaction (B) Addition of $NaNO_3$ favours forward reaction

(C) Increasing temperature favours forward reaction

(D) Increasing pressure favours forward reaction

5. For which of the following statements about the reaction quotient, Q, are correct?

(A) the reaction quotient, Q, and the equilibrium constant always have the same numerical value

(B) Q may be $>< = K_{eq}$

(C) Q (numerical value) varies as reaction proceeds

(D) Q = 1 at equilibrium

6. For the following equilibrium $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$ partial pressure of NH_3 will increase:

(A) if NH_3 is added after equilibrium is established (B) if H_2S is added after equilibrium is established

(C) temperature is increase (D) volume of the flask is decreased

7. Which are true statements for the following equilibrium?

$H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$

(A) increase in pressure will result in the formation of more liquid water

(B) increase in pressure will increase boiling point

(C) decrease in pressure will vaporize $H_2O_{(l)}$ to a greater extent

(D) increase in pressure will liquefy steam.

8. Volume of the flask in which species are transferred is double of the earlier flask. In which of the following cases, equilibrium concentration is affected?

(A) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ (B) $N_{2(g)} + O_{2\,(g)} \rightleftharpoons 2NO_{(g)}$

(C) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ (D) $2NOBr_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$

9. Which of the following statements is (are) correct?

(A) An irreversible reaction goes to almost completion

(B) A reversible reaction always goes to completion if carried out in a closed vessel

(C) At equilibrium, the rate of forward reaction becomes equal to that of backward reaction

(D) In the beginning, the rate of backward reaction is much greater than that of forward reaction.

10. The value of equilibrium constant of a reversible reaction at a given temperature

(A) depends on the initial concentration of reactants

(B) depends on the concentration of products at equilibrium

(C) gets reversed when the mode of representation of the reaction is reversed

(D) changes when the unit of active mass is changed

11. In the dissociation of $2HI \rightleftharpoons H_2 + I_2$, the degree of dissociation will be affected by the

(A) Addition of inert gas (B) Addition of H_2 and I_2

(C) Increase of temperature (D) Increase of pressure

12. Which of the following equilibria is (are) not affected by pressure?

(A) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ (B) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

(C) $2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$ (D) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$

13. Which of the following will favour the formation of products in the following reaction?

$$A_{(g)} + 2B_{(g)} \longrightarrow 3C_{(g)} + D_{(g)} - \text{heat}$$

(A) An increase in temperature (B) An increase in pressure

(C) Addition of B (D) Removal of C

14. In the reaction $2A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)}$ + 362 Kcal. Which combination of pressure and temperature gives the highest yield of C at equilibrium

(A) 1000 atm and 500°C (B) 500 atm and 500°C

(C) 1000 atm and 50°C (D) 500 atm and 100°C

15. For the reaction, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ the forward reaction at constant temperature is favoured by :-

(a) Introducing an inert gas at constant volume (b) Introducing chlorine gas at constant volume

(c) Introducing an inert gas at constant pressure (d) Increasing volume of the container

(e) Introducing PCl_5 at constant volume

(A) a, b, c (B) b, c, d (C) c, d, e (D) a, c, d, e

16. Following equilibrium is present in a closed container at the temperature of 25° C.

$$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$$

When Cl_2 is added to the equlibrium mixture, the following statements will be correct for the system.

(a) Concentrations of SO_2, Cl_2 and SO_2Cl_2 change.(b) Cl_2 is formed in more amount.

(c) Concentration of SO_2 decreases and that of SO_2Cl_2 increases.

(A) a, c (B) a, b (C) b, c (D) a, b, c

17. Calculate $\Delta G°$ for conversion of oxygen to ozone $3/2\ O_2(g) \rightleftharpoons O_3(g)$ at 298 K, if K_p for this conversion is 3×10^{-29}

(A) 162.74 kJ mol^{-1} (B) 2.4×10^2 kJ mol^{-1} (C) 1.63 kJ mol^{-1} (D) 2.38×10^6 kJ mol^{-1}

18. At 250° C the vapour density of PCl_5 is 100. Calculate the degree of dissociation at this tempeature

(A) 4 (B) 0.04 (C) 0.02 (D) 2

19. The vapour density of N_2O_4 at a certain temperature is 30. What is the percentage dissociation of N_2O_4 at this temperature ?

(A) 53.3% (B) 106.6% (C) 26.7% (D) None of these

Home Work NCERT EXERCISE 7.21 To 7.34

THEORIES OF ACIDS & BASES

1. Which of the following acids is monoprotic :

(A) H_2S (B) H_3PO_2 (C) H_3PO_3 (D) H_3PO_4

2. The weakest Bronsted base among the following is :

(A) Cl^- (B) HS^- (C) ClO_4^- (D) NH_3

3. Several acids are listed below with their respective equilibrium constants :

$HF(aq) + H_2O(\ell) \rightleftharpoons H_3O^+ (aq) + F^- (aq)$ $K_a = 7.2 \times 10^{-4}$

$HS^- (aq) + H_2O(\ell) \rightleftharpoons H_3O+ (aq) + S^{2-} (aq)$ $K_a = 1.3 \times 10^{-11}$

$CH_3COOH(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + CH_3COO (aq)$ $K_a = 1.8 \times 10^{-5}$

Which is the strongest acid and which acid has the strongest conjugate base ?

(A) HF and HF (B) HF and HS^- (C) HS^- and HF (D) HS^- and CH_3COOH

4. Several bases are listed below with their respective K_b values :

$NH_3(aq) + H_2O (\ell) \rightleftharpoons NH_4^+(aq) + OH^- (aq).$ $K_b = 1.8 \times 10^{-5}$

$C_5H_5N(aq) + H_2O (\ell) \rightleftharpoons C_5H_5NH^+(aq) + OH^- (aq).$ $K_b = 1.5 \times 10^{-9}$

$N_2H_4(aq) + H_2O(\ell) \rightleftharpoons N_2H_5^+(aq) + OH^- (aq).$ $K_b = 8.5 \times 10^{-7}$

Which is the weakest base and which base has the weakest conjugate acid ?

(A) C_5H_5N and C_5H_5N (B) NH_3 and NH_3 (C) C_5H_5N and NH_3 (D) NH_3 and N_2H_4

5. Which of the following can act both as a Bronsted acid & a Bronsted base (amphiprotic species) in aq. solution

(A) NH_3 (B) $H_2PO_3^-$ (C) HCO_3^- (D) OH^-

6. How many of the following species behave as a strong acid or as a strong base in aqueous solution ?

(a) HNO_2 (b) HNO_3 (c) NH_4^+ (d) Cl^- (e) H^-(f) O^{2-} (g) H_2SO_4 (h) 3

(i) 4 (j) 5 (k) 6

7. Consider following reactions

(a) H_2CO_3 (aq) + HSO_4^-(aq) $\rightleftharpoons$ H_2SO_4(aq) + HCO_3^- (aq)

(b) HF (aq) + Cl^- (aq) $\rightleftharpoons$ HCl (aq) + F^- (aq)

(c) HF (aq) + NH_3 (aq) $\rightleftharpoons$ NH_4^+ (aq) + F^- (aq)

(d) HSO_4^- (aq) + CN^- (aq) $\rightleftharpoons$ HCN (aq) + SO_4^{2-} (aq)

Reactions proceeding to the right are :

(A) a, b (B) c, d (C) a, c (D) b, d

8. What are the conjugate bases of each of the following Bronsted Lowry acids ?

(a) HOCl (b) HPO_4^{2-} (c) H_2O (d) $CH_3NH_3^+$

(e) H_2CO_3 (f) H_2 (g) H_2O_2 (h) HO_2^-

9. Given : $HF + H_2O \xrightarrow{K_a} H_3O^+ + F^-$; $F^- + H_2O \xrightarrow{K_b} HF + OH^-$.

Which relation is correct :

(A) $\frac{K_b}{K_a} = K_w$ (B) $K_a \times K_b \times K_w = 1$ (C) $K_b \times K_w = K_a$ (D) $\frac{K_w}{K_a} = K_b$

10. For the following equilibrium at 25°C : $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, equilibrium constant is 1.8×10^{-5}.

Calculate equilibrium constant for the equilibrium at 25°C : $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

(A) 1.8×10^{-19} (B) 5.55×10^{-10} (C) 1.8×10^{-9} (D) data insufficient

11. If equilibrium constant of $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$ is 5.55×10^{-10} at 25°C, calculate equilibrium constant of $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$ at 25°C.
(A) 1.8×10^{-4} (B) 1.8×10^{9} (C) 5.55×10^{4} (D) 1.8×10^{-5}

PH OF SOLUTION

12. K_b for trimethylamine at 25°C is 6.667×10^{-5}. Calculate pK_a for trimethyl ammonium ion $(CH_3)_3NH^+$ at 25°C.
(A) 9.82 (B) 10.18 (C) 8.82 (D) – 9.82

13. For which temperature, the pOH of pure water can be greater than 7 :
(A) 45ºC (B) 25ºC (C) 15ºC (D) 35ºC

14. For pure water at 25ºC and 50ºC, the correct relation is :
(A) $pH_{25^\circ C} = pH_{50^\circ C}$ (B) $pH_{25^\circ C} > pH_{50^\circ C}$ (C) $pH_{50^\circ C} > pH_{25^\circ C}$ (D) Can't say

15. The ionic product of water at 65ºC is 16×10^{-14}. What is pH of pure water at this temperature : ?
(A) 6.7 (B) 7 (C) 7.6 (D) 6.4

16. For pure water at 10ºC and 60ºC, the correct relation is :
(A) $pOH_{10^\circ C} = pOH_{60^\circ C}$ (B) $pOH_{10^\circ C} > pOH_{60^\circ C}$ (C) $pOH_{60^\circ C} > pOH_{10^\circ C}$ (D) Can't say

17. At – 50ºC, autoprotolysis of NH_3 gives $[NH_4^+] = 1 \times 10^{-15}$ M. Hence, autoprotolysis constant of NH_3 is :
(A) $\sqrt{1\times10^{-15}}$ (B) 1×10^{-30} (C) 2×10^{-30} (D) 2×10^{-15}

18. For a 10^{-3} M solution of a weak monoacidic base ($K_b = 5 \times 10^{-4}$), calculate the % ionisation of base. Report your answer rounding it off to the nearest whole number.

19. What is the K_b of a weak base that can produce one OH^- ion per molecule, if its 0.04 M solution is 2.5% ionized:
(A) 2.5×10^{-4} (B) 2.5×10^{-6} (C) 2.5×10^{-5} (D) 2.5×10^{-7}

20. The degree of dissociation of 0.04 M HA solution is 0.01. What would be the degree of dissociation of 0.01 M solution of the acid at the same temperature ?
(A) 0.02 (B) 0.16 (C) 0.005 (D) 0.04

21. CO_2 in aqueous solution shows following ionic equilibrium : $2H_2O + CO_2 \rightleftharpoons HCO_3^- + H_3O^+$
If hydronium ion (H_3O^+) concentration at 25°C is 2 x 10^{-6} M, what is hydroxide ion (OH^-) concentration ?
(A) 5×10^{-8} M (B) 2×10^{8} M (C) 5×10^{-9} M (D) 0.05 M

22. What is the percent ionization of a 0.01 M HCN solution : Given $K_a = 6.4 \times 10^{-9}$.
(A) 8×10^{-4} % (B) 0.08 % (C) 8×10^{-3} % (D) 0.8 %

23. Which solution has maximum pH :
(A) 0.01 M H_2SO_4 (B) 0.01 M HCl (C) 0.01 M $Ba(OH)_2$ (D) 0.01 M NaOH

24. (a) pH of a NaOH solution is 10.3. What is concentration of NaOH solution ?
(b) What is molar concentration of $Sr(OH)_2$ if its solution has pH of 12 ?
(A) $[NaOH] = 2 \times 10^{-4}$M (B) $[Sr(OH)_2] = 5 \times 10^{-3}$ M
(C) Both (A) and (B) (D) None of these

25. How many moles of sulphuric acid must be dissolved to produce 250 ml of an aqueous solution of pH = 3.52? Assume complete dissociation.
(A) 3.75×10^{-5} (B) 7.5×10^{-4} (C) 3.75×10^{-4} (D) 7.5×10^{-5}

26. pH of 10^{-8} N NaOH solution is
(A) 7.2 (B) 6.8 (C) 6.98 (D) 7.02

Home Work NCERT EXERCISE 7.35 To 7.43

1. [Cl^-] in a mixture of 0.01 M HCl and 0.01 M $BaCl_2$ solutions in volume ratio 2 : 1 is

(A) 0.01 M (B) 0.0133 M (C) 0.015 M (D) 0.014 M

2. 100 mL of 1 M HCl solution is mixed with 50 mL of 2 M HCl solution. Hence, [H_3O^+] in resulting solution is

(A) 1.25 M (B) 1.30 M (C) 1.33 M (D) 1.40 M

3. What will be the pOH of a solution obtained by mixing 800 ml of 0.05 N NaOH solution and 200 ml of 0.1 N HCl solution ?

(A) 2.7 (B) 1.7 (C) 11.3 (D) 12.3

4. At 25ºC, the dissociation constants of CH_3COOH and NH_4OH in an aqueous solution are almost the same. The pH of a solution of 0.01 N CH_3COOH is 4 at 25ºC. The pH of 0.01 N NH_4OH solution at the same temperature would be

(A) 8 (B) 9 (C) 10 (D) 4

5. K_a of CH_3COOH is 1.8×10^{-5}. For 0.02 M CH_3COOH solution

(A) [H_3O^+] = 6×10^{-4} M (B) % ionisation = 3 %

(C) pH = 3.22 (D) All of these

6. The pH of 0.05 M aqueous solution of diethyl amine is 12. Calculate its K_b.

(A) 2×10^{-3} (B) 2.5×10^{-4} (C) 2.5×10^{-3} (D) 2×10^{-4}

7. A 0.02 M monobasic acid solution is 20% dissociated. The equilibrium constant K_a for the acid is

(A) 10^{-2} (B) 10^{-3} (C) 8×10^{-4} (D) 8.33×10^{-4}

8. The dissociation constants of HCOOH & CH_3COOH are 2×10^{-4} & 1.8×10^{-5} respectively. Calculate the relative strengths of the acids.

(A) $\frac{3}{10}$ (B) $\frac{10}{3}$ (C) $\frac{9}{100}$ (D) $\frac{100}{9}$

9. Calculate the change in pH when a 0.1 M solution of CH_3COOH in water at 25ºC is diluted to a final concentration of 0.01 M. [K_a of acetic acid = 1.85×10^{-5}]

(A) +0.5 (B) + 1.5 (C) +1.0 (D) –0.5

10. $HCOOH \rightleftharpoons H^+ + HCOO^-$, $K_a = 1.7 \times 10^{-4}$. Then [H^+] concentration in a solution containing 0.1 M HCOOH & 0.05 M HCOONa is nearly equal to :

(A) 8.5×10^{-5} (B) 3.4×10^{-4} (C) 3.4×10^{-3} (D) 8.5×10^{-4}

11. Blue litmus turns red in which of the following mixtures of acid and base solutions :

(A) 100 mL of 1×10^{-2} M H_2SO_4 + 100 mL of 1×10^{-2} M $Sr(OH)_2$.

(B) 100 mL of 1×10^{-2} M HCl + 100 mL of 1×10^{-2} M $Ba(OH)_2$.

(C) 100 mL of 1×10^{-2} M H_2SO_4 + 10 mL of 1×10^{-2} M NaOH.

(D) 100 mL of 1×10^{-2} M HCl + 100 mL of 1×10^{-2} M NaOH.

12. Which of the following statements is/are incorrect at 25°C :

(A) pK_a for H_3O^+ is 15.74

(B) Percentage dissociation of water is 1.8×10^{-9} %

(C) $pK_a + pK_b = pK_w$ for HCl & ClOH

(D) pK_b for OH^- is – 1.74

13. Which of the following increases with dilution at a given temperature :

(A) pH of 10^{-3} M acetic acid solution

(B) pH of 10^{-3} M aniline solution

(C) degree of dissociation of 10^{-3} M acetic acid solution

(D) degree of dissociation of 10^{-3} M aniline solution

14. When 0.1 mole of NH_3 is dissolved in sufficient water to make 1 L of solution, the solution is found to have a hydroxide ion concentration of 1.2×10^{-3} M. Then :

(A) pH of the solution is 2.92.

(B) pH of the solution after 0.1 mole of NaOH is added to it is 13.

(C) K_b for ammonia is 1.44×10^{-5}.

(D) NaOH added to the solution decreases the extent of dissociation of ammonia.

15. The first and second dissociation constants of an acid, H_2A, are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be :

(A) 0.2×10^{5} (B) 5.0×10^{-5} (C) 5.0×10^{-15} (D) 5.0×10^{15}

16. A 50 mL solution of pH 1 is mixed with a 50 mL solution of pH = 2. The pH of the mixture will be nearly :

(A) 0.76 (B) 1.26 (C) 1.76 (D) 2.26

17. The pH of a solution is 5.0. To this solution sufficient acid is added to decrease the pH to 2.0. The increase in hydrogen ions concentration is :

(A) 1000 times (B) 5/2 times (C) 100 times (D) 5 times

18. 10^{-6} M HCl is diluted to 100 times. Its pH is :

(A) 6.0 (B) 8.0 (C) 6.95 (D) 9.5

19. Calculate (i) $[H^+]$ and (ii) $[PO_4^{3-}]$ in a 0.15 M solution of phosphoric acid, H_3PO_4. Given that for H_3PO_4: $K_{a1} = 7.5 \times 10^{-3}$, $K_{a2} = 6.2 \times 10^{-8}$, $K_{a3} = 3.6 \times 10^{-13}$.

(A) $[H^+] = 0.03$ M (B) $[PO_4^{3-}] = 7.44 \times 10^{-19}$ M

(C) Both (A) and (B) (D) None of these

20. 100 ml of 0.5M hydrazoic acid (HN_3, $K_a = 3.6 \text{ x } 10^{-4}$) and 400ml of 0.1M cyanic acid (HOCN, $K_a = 8 \times 10^{-4}$) are mixed. Which of the following options is false for the final solution :

(A) $[H^+] = 10^{-2}$ M (B) $[N_3^-] = 3.6 \times 10^{-3}$ M

(C) $[OCN^-] = 6.4 \times 10^{-3}$ M (D) $[H^+] = 1.4 \times 10^{-2}$ M

21. Aqueous solution of $CuSO_4$ is ______ due to the hydrolysis of ______ ions.

(A) acidic, SO_4^{2-} (B) acidic, Cu^{2+} (C) basic, Cu^{2+} (D) basic, SO_4^{2-}

Home Work NCERT EXERCISE 7.45 To 7.53

Salt Hydrolysis :

1. In which of the following cases, the aqueous solution is not basic :

I . $BeCl_2$ II. LiCN III. $(NH_4)_2C_2O_4$ IV. C_5H_6NBr V. NaF

(A) II & V only (B) I & IV only (C) I, III & IV (D) I & III Only

2. The salt of which of the following four weak acids will be most hydrolysed :

(A) HA ; $K_a = 1 \times 10^{-6}$ (B) HB ; $K_a = 2 \times 10^{-6}$ (C) HC ; $K_a = 3 \times 10^{-8}$ (D) HD ; $K_a = 4 \times 10^{-10}$

3. $[H^+] = \sqrt{\frac{K_w}{K_a C}}$ is suitable for : (Considering h to be negligible)

(A) $(HCOO)_2$ Ba, NH_4Cl (B) CH_3COONa, NaCN

(C) $C_5H_5NH^+ Cl^-$, $(NH_4)_2SO_4$ (D) None of these

4. Determine pH of a 0.01 M aqueous solution of $ClC_6H_4NH_3Cl$:

Given : $K_b(ClC_6H_4NH_2) = 4 \times 10^{-13}$, log 19 = 1.3, log 2 = 0.3, $\sqrt{16.25} = 4.02$.

(A) 2.1 (B) 2.5 (C) 3.5 (D) 3.1

5. The pH of a solution containing 0.1 M CH_3COONa and 0.1 M $(C_2H_5COO)_2Ba$ will be :

$K_a(CH_3COOH) = 2 \times 10^{-5}$, $K_a(C_2H_5COOH) = 8 \times 10^{-6}$:

(A) 8.13 (B) 9.24 (C) 10.18 (D) 11.18

6. Which of the following salts does not undergo hydrolysis :

(A) CH_3COONH_4 (B) $FeCl_3.6H_2O$ (C) KCl (D) KCN

7. The pH value will be highest for the aqueous solution of :

(A) NaCl (B) KCN (C) $C_6H_5NH_3^+Cl^-$ (D) $NaHCO_3$

8. NH_4Cl (aq) is :

(A) acidic due to NH_4^+ (B) acidic due to Cl^- (C) basic due to NH_4^+ (D) basic due to Cl^-

9. Calculate $[H^+]$, $[HCOO^-]$ and $[OCN^-]$ in a solution that contains 0.1M HCOOH ($K_a = 2.4 \times 10^{-4}$) and 0.1 M HOCN ($K_a = 4 \times 10^{-4}$).

(A) $[H^+] = 8 \times 10^{-3}$ M (B) $[HCOO^-] = 3 \times 10^{-3}$ M

(C) $[OCN^-] = 5 \times 10^{-3}$ M (D) All of these

***10.** Nicotine, $C_{10}H_{14}N_2$, has two basic nitrogen atoms and both can react with water to give a basic solution.

$$Nic(aq) + H_2O(l) \rightleftharpoons NicH^+(aq) + OH^-(aq)$$

$$NicH^+(aq) + H_2O(l) \rightleftharpoons NicH_2^{2+}(aq) + OH^-(aq)$$

K_{b1} is 8.0×10^{-7} and K_{b2} is 1.1×10^{-10}. Calculate the approximate pH of a 0.2 M solution.

(A) 7.2 (B) 10.6 (C) 9.4 (D) 3.4

11. Find the concentration of H^+, HCO_3^- & CO_3^{2-} in a solution, which is 0.01 M wrt H_2CO_3 and may contain some other non-reactive solute also. pH of this solution is 2.74.

Given : $K_a(H_2CO_3) = 4 \times 10^{-7}$; $K_a(HCO_3^-) = 4.8 \times 10^{-11}$

(A) $[H^+] = 1.8 \times 10^{-3}$ M (B) $[HCO_3^-] = 2.22 \times 10^{-6}$ M

(C) $[CO_3^{2-}] = 6 \times 10^{-14}$ M (D) All of these

12. A solution containing 0.2 mole of HA acid ($K_a = 5 \times 10^{-2}$) and 0.1 mole NaA in one litre solution has $[H^+]$ equal to :

(A) 0.05 M (B) 0.025 M (C) 0.1 M (D) 0.005 M

*13. Hydrazine, N_2H_4, can interact with water in two stages.

$N_2H_4(aq) + H_2O(l) \rightleftharpoons N_2H_5^+ (aq) + OH^-(aq)$ $\qquad K_{b1} = 8.1 \times 10^{-7}$

$N_2H_5^+ (aq) + H_2O (l) \rightleftharpoons N_2H_6^{2+} (aq) + OH^-(aq)$ $\qquad K_{b2} = 9 \times 10^{-16}$

What are the concentrations of OH^-, $N_2H_5^+$ & $N_2H_6^{2+}$ and also pH value in a 0.01 M aqueous solution of hydrazine?

(A) $[OH^-] = 9 \times 10^{-5}$ M (B) $[N_2H_5^+] = 9 \times 10^{-5}$ M

(C) $[N_2H_6^{2+}] = 9 \times 10^{-16}$ M (D) pH = 4.04

14. How many of the following solutions will turn blue litmus red ?

(i) $Al(NO_3)_3$ (ii) NaH_2PO_4 (iii) KCN (iv) H_3BO_3

(v) H_3PO_3 (vi) $C_6H_5NH_3^+Cl^-$ (vii) K_2SO_4 (viii) NH_4HSO_4

(ix) CH_3CH_2OH

15. Calculate the % hydrolysis of a salt containing 10^{-3} M, A^- ions. Assume anionic hydrolysis only. K_a for HA is 2×10^{-11}.

16. Match the column :

Column-I	**Column-II**
(A) $NaHCO_3$ (aq.)	(p) Significant cationic hydrolysis
(B) CH_3COONH_4 (aq.)	(q) Significant anionic hydrolysis
(C) K_2SO_4. $Al_2(SO_4)_3.24H_2O$ (aq.)	(r) Acidic (pH < 7)
(D) NaCN (aq)	(s) Basic (pH > 7)
	(t) pH is independent of concentration

Given : $K_{a1} = 5 \times 10^{-7}$, $K_{a2} = 5 \times 10^{-11}$ for H_2CO_3

$K_a(CH_3COOH) = 1.8 \times 10^{-5}$; $K_b(NH_4OH) = 1.8 \times 10^{-5}$

17. (i) The smaller the value of K_a of a weak acid, lesser is the hydrolysis constant of its conjugate base.

(ii) A salt of strong acid with a strong base does not undergo hydrolysis .

(A) T T (B) T F (C) F T (D) F F

18. The ionisation constant of ammonium hydroxide is 1.77×10^{-5} at 298 K. Hydrolysis constant of ammonium chloride is :

(A) 5.65×10^{-10} (B) 6.50×10^{-12} (C) 5.65×10^{-13} (D) 5.65×10^{-12}

19. The dissociation constant of CH_3COOH is 1.8×10^{-5}. The hydrolysis constant of CH_3COO^- ions at 25°C would be

(A) 5.55×10^{-10}

(B) 1.8×10^{9}

(C) 5.55×10^{4}

(D) They would not undergo hydrolysis, since they come from a weak acid.

Home Work ***NCERT EXERCISE 7.54 To 7.63***

Buffer Solutions, Titration and Indicator:

1. pOH = 7 – 0.5 pK_a + 0.5 pK_b is true for aqueous solution containing which pair of cation and anion

(A) $C_6H_5NH_3^+$, CH_3COO^- (B) NH_4^+, F^-

(C) Both (A) and (B) (D) None of these

2. If pK_b of weak monoacidic base > pK_a of weak monobasic acid, then the solution of the salt of weak acid and weak base having equal concentrations of cation and anion will be

(A) Neutral (B) Acidic (C) Basic (D) Cannot be predicted

3. The pH of an basic buffer solution is

(A) > 7 (B) < 7 (C) = 7 (D) Depends upon K_b of base

4. Choose the correct statement(s) :

(A) pH of acidic buffer solution decrease if more salt is added

(B) pH of basic buffer solution increases if more base is added

(C) Both (A) and (B)

(D) None of these

5. The pH of buffer of $NH_4OH + NH_4Cl$ type at 25°C is given by

(A) pOH = pK_b (B) pH = pK_b + log [Salt]/[base]

(C) pH = 14 – pK_b – log [Salt]/[base] (D) pH = 7 – pK_b – log [Salt]/[base]

6. Which may be added to one litre of water to act a buffer

(A) One mole of CH_3COOH and one mole of NaOH

(B) One mole of NH_4Cl and two mole of NaOH

(C) One mole of $NaNO_3$ and one mole of HNO_3

(D) One mole of CH_3COOH and 0.5 mole of NaOH

7. Which of the following pairs constitutes a buffer ?

(A) HNO_3 and NH_4NO_3 (B) NaOH and NaCl (C) HNO_2 and $NaNO_2$ (D) HCl and KCl

8. To prepare a buffer solution of pH = 4.04, amount of Barium acetate to be added to 100 mL of 0.1 M acetic acid solution [$pK_b(CH_3COO^-)$ = 9.26] is :

(A) 0.05 mole (B) 0.025 mole (C) 0.1 mole (D) 0.005 mole

9. A 18 mL solution containing acetic acid and sodium acetate required 6 mL of 0.1 M NaOH for eutalization of the acid and 12 mL of 0.1 M HCl for reaction with salt, separately. If pK_a of the acid is 4.75, what is the pH of the initial mixture :

(A) 5.05 (B) 4.75 (C) 4.55 (D) 5.23

10. The pH of blood is 7.4. What is the ratio of $\frac{[HPO_4^{2-}]}{[H_2PO_4^-]}$ in the blood ? Given : K_{a1}, K_{a2} and K_{a3} for H_3PO_4 are 10^{-3}, 8×10^{-8} and 10^{-12} respectively.

(A) 2 : 1 (B) 1 : 2 (C) 3 : 1 (D) 1 : 3

11. When CO_2 dissolves in water, the following equilibrium is established :

$$CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$$

pH of solution is 5.9. The ratio of HCO_3^- to CO_2 would be :

(Given that K_{a1} & K_{a2} for H_2CO_3 are 4×10^{-7} and 4.8×10^{-11} respectively).

(A) 3.2 (B) 0.32 (C) 3.84×10^{-5} (D) 3.84×10^{-4}

12. Select the correct statement :

(A) Na_2HPO_3 is an acid salt.

(B) Water acts as a base, when ammonia is dissolved in it .

(C) All aqueous solutions, whether neutral, acidic or basic, contain both H^+ & OH^- ions.

(D) All of these

13. Does the pH of solution increases, decreases or remain same when you :

(a) add $NH_4Cl(s)$ to 100 ml of 0.1 M NH_3 ? (b) add sodium acetate(s) to 50 ml of 0.015 M acetic acid ?

(c) add NaCl(s) to 25 ml of 0.1 M NaOH ?

(A) decreases, increases, decreases (B) increases, decreases, no change

(C) increases, decreases, increases (D) decreases, increases, no change

14. Which of following solutions can act as buffer ?

(A) $NaHS + Na_2S$ (B) $NaNO_3 + HNO_3$ (C) $H_3PO_4 + NaH_2PO_4$ (D) KCl + KOH

15. When 0.1 mole arsenic acid (H_3AsO_4) is dissolved in a 1L buffer solution of pH = 8. Which of the following options is/are correct ? For arsenic acid : $K_{a1} = 2.5 \times 10^{-4}$, $K_{a2} = 5 \times 10^{-8}$, $K_{a3} = 2 \times 10^{-13}$. ['<<' sign denotes that the high concentration is at least more than 100 times the lower one]

(A) $[H_3AsO_4] << [H_2AsO_4^-]$ (B) $[H_2AsO_4^-] << [HAsO_4^{2-}]$

(C) $[HAsO_4^{2-}] << [H_2AsO_4^-]$ (D) $[AsO_4^{3-}] << [HAsO_4^{2-}]$

16. Solutions of two weak acids HA and HB with K_{a_1} and K_{a_2} as their dissociation constants are mixed to have equal concentration of both in final solutions. Which of the following is correct, if $K_{a_1} > K_{a_2}$?

(A) $[H^+]$ from HA > $[H^+]$ from HB (B) α of HA > α of HB

(C) $[A^-] > [B^-]$ (D) [HA] > [HB]

17. Which of the following is an acid salt :

(A) Na_2SO_4 (B) Ca(OH)Cl (C) $Ca(H_2PO_2)_2$ (D) Na_2HPO_4

18. What is $[H^+]$ in mol/L of a solution that is 0.20 M in CH_3COONa and 0.10 M in CH_3COOH ? (K_a for $CH_3COOH = 1.8 \times 10^{-5}$)

(A) 3.5×10^{-4} (B) 1.1×10^{-5} (C) 1.8×10^{-4} (D) 9×10^{-6}

19. A certain buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 1×10^{-10}. The pH of the buffer is :

(A) 4 (B) 7 (C) 10 (D) 14

20. Solution containing 0.1 N NH_4OH and 0.1 N NH_4Cl has pH = 9.25.Then find out pK_b of NH_4OH.

(A) 9.25 (B) 4.75 (C) 3.75 (D) 8.25

21. 1 L of a buffer solution contains one mole each of CH_3COOH & $(CH_3COO)_2Ba$. Which of the following options is correct ? (pK_a of CH_3COOH = 4.7)

(A) pH = 9 (B) pOH = 8.6 (C) pH = 4.7 (D) pOH = 9

22. What is the pH of the solution at half neutralization in the titration of 0.1 N CH_3COOH solution by 0.5 M KOH solution : (K_a of $CH_3COOH = 1.8 \times 10^{-5}$)

(A) 4.74 (B) 9.26 (C) 4.26 (D) None of these

23. The total number of different kind of buffers obtained during the titration of H_3PO_4 with NaOH are :

(A) 3 (B) 1 (C) 2 (D) No buffer is obtained

24. The rapid change of pH near the stoichiometric point of an acid-base titration is the basis of indicator detection. pH of the solution is related to ratio of concentration of conjugate acid (HIn) and the base (In^-) forms of the indicator by the expression :

(A) $\log \frac{[In^-]}{[HIn]} = pK_{In} - pH$

(B) $\log \frac{[HIn]}{[In^-]} = pK_{In} - pOH$

(C) $\log \frac{[HIn]}{[In^-]} = pH - pK_{In}$

(D) $\log \frac{[In^-]}{[HIn]} = pH - pK_{In}$

25. Which of the following indicators is best suited in the titration of a weak acid versus a strong base ?

(A) phenolpthalein (8.3 – 10.0)

(B) methyl orange (3.1 – 4.4)

(C) methyl red (4.2 – 6.3)

(D) litmus (4.5 – 8.3)

26. Which of the following indicators is best suited in the titration of a weak base versus a strong acid ?

(A) phenolphthalein (8.3 – 10.0)

(B) phenol red (6.8 – 8.4)

(C) methyl organe (3.1 – 4.4)

(D) litmus (4.5 – 8.3)

27. For the acid H_2X, $pK_{a_1} = 4$ and $pK_{a_2} = 10$. Which of the following indicators (with their ranges provided) is most suitable for the titration $H_2X + OH^- \longrightarrow HX^- + H_2O$?

(A) Methyl orange (3.1 to 4.4)

(B) Bromocresol green (3.8 to 5.4)

(C) p-nitrophenol (5.6 to 7.6)

(D) Phenolphthalein (8 to 9.6)

Home Work ***NCERT EXERCISE 7.64 To 7.66***

1. The indicator constant for an acidic indicator, HIn is 5×10^{-6} M. This indicator appears only in the colour of acidic form when $\frac{[In^-]}{[HIn]} \le \frac{1}{20}$ and it appears only in the colour of basic form when $\frac{[HIn]}{[In^-]} \le \frac{1}{40}$. The pH range of indicator is

(A) 3.7 – 6.9 (B) 4.0 – 6.6 (C) 4.0 – 6.9 (D) 3.7 – 6.6

2. What will be the pOH at the equivalence point during the titration of a 100 mL, 0.2 M solution of NH_4Cl with 0.2 M solution of NaOH ? K_b for ammonium hydroxide = 2×10^{-5}.

(A) $3 - \log\sqrt{2}$ (B) $3 + \log\sqrt{2}$ (C) $3 - \log 2$ (D) $3 + \log 2$

3. To a 100 mL of 0.1 M weak acid HA solution, 22.5 mL of 0.2 M solution of NaOH are added. Now, what volume of 0.1 M NaOH solution be added into above solution, so that pH of resulting solution be 4.7

[Given : $(K_b(A^-) = 5 \times 10^{-10}$]

(A) 5 mL (B) 20 mL (C) 10 mL (D) 15 mL

4. 60 mL of 0.2 M $Ba(OH)_2$ solution is added to 50 mL of 0.6 N H_3PO_4 solution. The pH of the mixture would be about : (K_{a1}, K_{a2} and K_{a3} for H_3PO_4 are 10^{-3}, 10^{-8} and 10^{-12} respectively).

(A) 11.82 (B) 3.6 (C) 12.18 (D)7.82

Solubility of sparingly soluble salt:

5. The solubility of A_2X salt (producing A^+ & X^{2-} ions in aqueous solution) in pure water is y mol dm^{-3}. Its solubility product is :

(A) 27 y^4 (B) 8 y^3 (C) 2 y^2 (D) 4 y^3

6. The solubility of sparingly soluble electrolyte M_mA_a (producing M^{a+} & A^{m-} ions in aqueous solution) in water is given by the expression :

(A) $s = \left(\frac{K_{sp}}{m^m a^a}\right)^{m+a}$ (B) $s = \left(\frac{K_{sp}}{m^m a^a}\right)^{1/m+a}$ (C) $s = \left(\frac{K_{sp}}{m^a a^m}\right)^{m+a}$ (D) $s = \left(\frac{K_{sp}}{m^a a^m}\right)^{1/m+a}$

7. Calculate the solubility of A_2X_3 (producing A^{3+} & X^{2-} ions in aqueous solution) in pure water, assuming that neither kind of ion reacts with water. For A_2X_3, K_{sp} = 1.08 x 10^{-23}.

(A) 8×10^{-7} mol L^{-1} (B) 10^{-5} mol L^{-1} (C) 6.4×10^{-5} mol L^{-1} (D) 2×10^{-5} mol L^{-1}

8. A particular saturated solution of silver chromate, Ag_2CrO_4, has $[Ag^+] = 5 \times 10^{-5}$ M and $[CrO_4^{2-}] = 4.4 \times 10^{-4}$ M. What is value of K_{sp} for Ag_2CrO_4 ?

(A) 1.1×10^{-12} (B) 2.2×10^{-8} (C) 9.68×10^{-12} (D) 4.4×10^{-12}

9. If the solubility of Ag_2SO_4 in 10^{-2} M Na_2SO_4 solution be 2×10^{-8} M, then K_{sp} of Ag_2SO_4 will be :

(A) 32×10^{-24} (B) 16×10^{-18} (C) 32×10^{-18} (D) 16×10^{-24}

10. A solution is saturated with respect to $SrCO_3$ & SrF_2. The $[CO_3^{2-}]$ was found to be 1.2 x 10^{-3} M. The concentration of F^- in the solution would be : $K_{sp}(SrCO_3) = 10^{-9}$, $K_{sp}(SrF_2) = 3 \times 10^{-11}$.

(A) 3×10^{-3} M (B) 3.6×10^{-5} M (C) 6×10^{-3} M (D) 6×10^{-2} M

11. Buffer solutions have constant acidity and alkalinity because :

(A) these give unionised acid or base on reaction with added acid or alkali.

(B) acids and alkalies in these solution are shielded from attack by other ions.

(C) they have large excess of H^+ or OH^- ions

(D) they have fixed value of pH.

12. Three sparingly soluble salts M_2X, MX and MX_3 have the solubility product are in the ratio of 4: 1 : 27. Their solubilities will be in the order :

(A) $MX_3 > MX > M_2X$
(B) $MX_3 > M_2X > MX$
(C) $MX > MX_3 > M_2X$
(D) $MX > M_2X > MX_3$

13. A student wants to prepare a saturated solution containing Ag^+ ion. He has got three salts : AgCl ($K_{sp} = 10^{-10}$), AgBr ($K_{sp} = 1.6 \times 10^{-13}$) and Ag_2CrO_4 ($K_{sp} = 3.2 \times 10^{-11}$). Which of the above compounds will be used by him in minimum weight to prepare 1 L of saturated solution ?

(A) AgCl
(B) AgBr
(C) Ag_2CrO_4
(D) any of the above

14. The solubility of CaF_2 in water at 1518°C is 2×10^{-4} mole/litre. Calculate K_{sp} of CaF_2 and its solubility in 0.1 M NaF solution. Assume no hydrolysis of cation or anion.

(A) $K_{sp} = 3.2 \times 10^{-11}$
(B) Solubility = 3.2×10^{-9} mole/litre
(C) Both (A) & (B)
(D) None of these

15. The precipitate of CaF_2 ($K_{sp} = 1.7 \times 10^{-10}$) is obtained when equal volumes of the following solutions are mixed:

(A) 10^{-3} M Ca^{2+} + 10^{-2} M F^-
(B) 10^{-2} M Ca^{2+} + 10^{-3} M F^-
(C) 10^{-4} M Ca^{2+} + 10^{-1} M F^-
(D) 10^{-1} M Ca^{2+} + 10^{-4} M F^-

16. If the amount given below is added in pure water, will all of the salt dissolve before equilibrium can be established, or will some salt remain undissolved ?

(a) 4.96 mg of MgF_2 in 125 mL of pure water, K_{sp} of $MgF_2 = 3.2 \times 10^{-8}$
(b) 3.9 mg of CaF_2 in 100 mL of pure water, K_{sp} of $CaF_2 = 4 \times 10^{-12}$

Also find the percentage saturation in each case. Assume no hydrolysis of cation or anion.

(A) MgF_2 will completely dissolve
(B) MgF_2 solution will have 32% saturation
(C) CaF_2 will not completely dissolve
(D) CaF_2 solution will have 100% saturation

17. The solubility product for silver iodate $AgIO_3$ is 1×10^{-8} . If 0.1 g of solid $AgIO_3$ is added to 100 mL of 0.02 M KIO_3 solution, what are the concentrations of K^+, IO_3^- & Ag^+ at equilibrium?

(A) $[Ag^+] = 5 \times 10^{-7}$ M
(B) $[IO_3^-] = 2 \times 10^{-2}$ M
(C) $[K^+] = 2 \times 10^{-2}$ M
(D) $[IO_3^-] = 0.023$ M

18. Calculate $[F^-]$ in a solution saturated with respect of both MgF_2 and SrF_2. Assume no hydrolysis of cation or anion.

$K_{sp}(MgF_2) = 9.5 \times 10^{-9}$, $K_{sp}(SrF_2) = 4 \times 10^{-9}$.

Report your answer after multiplying by 10^4.

19. pH of a saturated solution of $Ba(OH)_2$ is 12. The value of solubility product (K_{SP}) of $Ba(OH)_2$ is

(A) 3.3×10^{-7}
(B) 5.0×10^{-7}
(C) 4.0×10^{-6}
(D) 5.0×10^{-6}

20. The solubility product of As_2O_3 is 10.8×10^{-9}. It is 50% dissociated in saturated solution. The solubility of salt is :

(A) 10^{-2}
(B) 2×10^{-2}
(C) 5×10^{-3}
(D) 5.4×10^{-9}

21. On adding 0.1 M solution each of $[Ag^+]$, $[Ba^{2+}]$, $[Ca^{2+}]$ in a Na_2SO_4 solution, species first precipitated is

$[K_{sp}\ BaSO_4 = 10^{-11}$, $K_{sp}\ CaSO_4 = 10^{-6}$, $K_{sp}\ Ag_2SO_4 = 10^{-5}]$

(A) Ag_2SO_4
(B) $BaSO_4$
(C) $CaSO_4$
(D) All of these

Home Work ***NCERT EXERCISE 7.67, 7.68***

1. K_{sp} of AgCl is 1.96×10^{-10}. 100 mL of saturated AgCl solution is titrated with 1×10^{-5} M NH_4SCN solution. Minimum volume of NH_4SCN solution required to precipitate all Ag^+ from saturated AgCl solution as AgSCN is

(A) 140 mL (B) 14 mL (C) 1.4 L (D) None of these

2. 0.2 millimoles of Zn^{2+} ion is mixed with $(NH_4)_2S$ solution of molarity 0.02 M. The amount of Zn^{2+} that remains unprecipitated in 20 mL of this solution would be : (Given : K_{SP} ZnS = 4×10^{-24})

(A) 5.2×10^{-22} g (B) 2.6×10^{-22} g (C) 1.04×10^{-21} g (D) 5.2×10^{-23} g

3. To 0.1 mole of red precipitate Ag_2CrO_4 ($K_{sp} = 4 \times 10^{-12}\ M^3$) in 10 L solution, solid NaCl is added till practically all the red precipitate turns white. Find the concentrations of Ag^+, Cl^-, CrO_4^{2-} and Na^+ ions in the resulting solution. $K_{sp}(AgCl) = 10^{-10}$.

(A) $[Ag^+] = 2 \times 10^{-5}$ M (B) $[Cl^-] = 5 \times 10^{-6}$ M

(C) $[CrO_4^{2-}] = 0.01$ M (D) $[Na^+] = 0.02$ M

4.

Column I	**Column II**
(A) AgBr	(p) Solubility in water is more than expectation.
(B) AgCN	(q) Solubility in acidic solution is more than that in pure water.
(C) $Fe(OH)_3$	(r) Solubility in strongly basic solution is more than that in pure water.
(D) $Zn(OH)_2$	(s) Solubility decreases in presence of common anion.

Basics and FLOT:

5. A well stoppered thermoflask contains some ice cubes. This is an example of a-

(A) Closed system (B) Open system

(C) Isolated system (D) Non-thermodynamic system

6. Which of the following is an intensive property ?

(A) Temperature (B) Viscosity (C) Surface tension (D) All of these

7. Thermodynamic equilibrium involves

(A) Chemical equilibrium (B) Thermal equilibrium

(C) Mechanical (D) All the above

8. As per the First Law of thermodynamics, which of the following statement would be appropriate:

(A) Energy of the system remains constant (B) Energy of the surroundings remains constant

(C) Entropy of the universe remains constant (D) Energy of the universe remains constant

9. Which of the following expressions represents the first law of thermodynamics-

(A) $\Delta E = -q + W$ (B) $\Delta E = q - W$ (C) $\Delta E = q + W$ (D) $\Delta E = -q - W$

10. When ammonium chloride is dissolved in water, the solution becomes cold. The change is -

(A) Endothermic (B) Exothermic (C) Supercooling (D) None of the above

11. Choose the incorrect statement :

(A) system and surrounding are always separated by a real or imaginary boundary.

(B) perfectly isolated system can never be created.

(C) in reversible process, energy change in each step can be reversed.

(D) irreversible process is also called quasi-equilibrium state.

12. Select the incorrect statement :

(A) A continuous curve can be plotted between thermodynamic parameters for a reversible process.

(B) An irreversible process takes infinite time for completion.

(C) There is an infinitesimal difference between driving force and opposing force in a reversible process.

(D) None of these

13. Among the following, the respective number of state and path functions is :

(a) Enthalpy (b) Pressure (c) Reversible compression work

(d) Gibb's energy (e) Entropy (f) Specific heat capacity

(g) Height of a hill (h) Distance travelled in climbing the hill

(A) 7 & 1 (B) 6 & 2 (C) 5 & 3 (D) 4 & 4

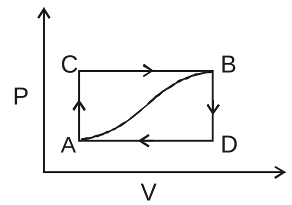
14. Which of the following is extensive property ?

(A) Enthalpy (B) Entropy (C) Specific heat (D) Volume

15. Which of the following is/are thermodynamic function ?

(A) Internal energy (B) Work done (C) Enthalpy (D) Entropy

Paragraph for Question Nos. 16 to 18



When a system is taken from state A to state B along path ACB as shown in figure below, 80 J of heat flows into the system and the system does 30 J of work.

16. How much heat flows into the system along path ADB if the work done by the system is 10 J :

(A) 40 J (B) 60 J (C) 80 J (D) 100 J

17. When the system is returned from state B to A along the curved path, the work done on the system is 20 J. Does the system absorb or liberate heat and by how much ?

(A) –70 J ; heat is liberated. (B) –60 J ; heat is liberated.

(C) +70 J ; heat is absorbed. (D) +60 J ; heat is absorbed.

18. If $E_D - E_A = -40$J, the heat absorbed in the processes AD and DB are respectively :

(A) $q_{AD} = 30$ J and $q_{DB} = -90$ J (B) $q_{AD} = -60$ J and $q_{DB} = 30$ J

(C) $q_{AD} = 30$ J and $q_{DB} = 90$ J (D) $q_{AD} = -30$ J and $q_{DB} = 90$ J

20. Categorise these properties into extensive and intensive property :

(a) Pressure of gas (b) Volume (c) Density (d) Temperature

(e) Heat capacity (f) Specific heat capacity (g) Molar heat capacity

(h) Molarity (i) Dielectric constant (j) Internal energy (k) Specific internal energy (l) Mass

19. Calculate work done when 1 mole of an ideal gas is expanded from 10 L to 20 L against a constant 1 atm pressure at constant temperature of 300 K :

(A) –2.5 kJ (B) –1.013 kJ (C) –0.01 kJ (D) –5.75 kJ

20. In a system, a piston caused an expansion against an external pressure of 1.25 bar giving a change in volume of 32 L for which $\Delta E = -51$ kJ. What was the value of heat involved :

(A) – 55 kJ (B) – 11 kJ (C) – 47 kJ (D) – 91 kJ

21. In which process, net work done by an ideal gas is zero :

(A) Cyclic (B) Isobaric (C) Free expansion (D) Adiabatic

22. 1 mole of ideal gas is expanded isothermally and reversibly from 10 L to 100 L. Which of the following options is/are incorrect for the process :

(A) $\Delta E = 0$ (B) $W = 0$ (C) heat supplied (q) = 0 (D) Both (B) & (C)

Home Work ***NCERT EXERCISE 7.58, 7.69 To 7.73***

FLOT

1. What is the difference between change in enthalpy and change in internal energy at constant volume :

(A) 0 (B) VdP (C) –VdP (D) PdV

2. The work done in an adiabatic process on an ideal gas by a constant external pressure would be :

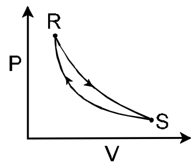
(A) Zero (B) ΔE (C) ΔH (D) q

3. Which of the following is correct option for free expansion of an ideal gas under adiabatic condition ?

(A) $q = 0, \Delta T \neq 0, w = 0$ (B) $q \neq 0, \Delta T = 0, w = 0$

(C) $q = 0, \Delta T = 0, w = 0$ (D) $q = 0, \Delta T < 0, w \neq 0$

4. Consider the cyclic process $R \rightarrow S \rightarrow R$ as shown in the fig. You are told that one of the path is adiabatic and the other one isothermal. Which one of the following is(are) true?



(A) Process $R \rightarrow S$ is isothermal

(B) Process $S \rightarrow R$ is adiabatic

(C) Process $R \rightarrow S$ is adiabatic

(D) Such a graph is not possible

5. Calculate the heat needed to raise the temperature of 20 g iron from 25ºC to 500ºC, if specific heat capacity of iron is 0.45 $JK^{-1}g^{-1}$.

(A) 4275 J (B) 225 J (C) 15.66 J (D) 2250 J

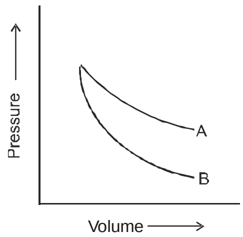
6. The ratio of slopes of P–V plots for reversible adiabatic process and reversible isothermal process of an ideal gas is equal to :

(A) γ (B) $1 - \gamma$ (C) $(\gamma - 1)$ (D) $1/\gamma$

7. For an ideal gas having molar mass M, specific heat at constant pressure can be given as :

(A) $\frac{\gamma RM}{\gamma - 1}$ (B) $\frac{R}{M(\gamma - 1)}$ (C) $\frac{RM}{\gamma - 1}$ (D) $\frac{\gamma R}{M(\gamma - 1)}$

8. P–V plots for two gases during an adiabatic process are given in the figure :



Plot A and plot B should correspond to : (Assume ideal behaviour)

(A) He and O_2 (B) SO_2 and Ar (C) O_2 and He (D) Both (B) and (C)

9. An ideal gas with $C_v = 3R$ expands adiabatically into a vacuum thus doubling its volume. The final temperatue is given by :

(A) $T_2 = T_1[2^{-1/3}]$ (B) $T_2 = T_1$ (C) $T_2 = 2T_1$ (D) $T_2 = \frac{T_1}{2}$

10. A monoatomic ideal gas ($C_V = \frac{3}{2}R$) is allowed to expand adiabatically and reversibly from initial volume of 8 L at 300 K to a volume of V_2 at 250 K. V_2 is : (Given $(4.8)^{1/2} = 2.2$)

(A) 10.56 L (B) 17.6 L

(C) 11.52 L (D) Expansion not possible since temperature has decreased

***11.** An ideal gas is expanded from volume V_0 to $4V_0$ by following two ways : (from same initial state)

(a) I[st] using reversible isothermal expansion from V_0 to $2V_0$, then using reversible adiabatic expansion from $2V_0$ to $4V_0$.

(b) I[st] using reversible adiabatic expansion from V_0 to $2V_0$, then from $2V_0$ to $4V_0$ using reversible isothermal expansion.

Then which of the following options is correct :

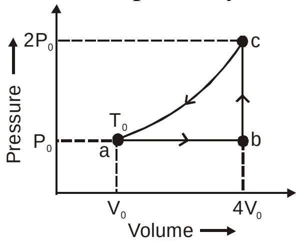
(A) Work done in (a) process > work done in (b) process

(B) Work done in (b) process > work done in (a) process

(C) Work done in (b) process = work done in (a) process

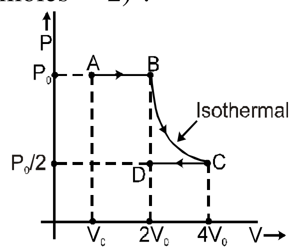
(D) Work done in both processes cannot be compared

12. One mole of an ideal monoatomic gas is caused to go through the cycle shown in figure. Then the change in the internal energy of gas from a to b and b to c is respectively :



(A) $\frac{9P_0V_0}{2}$, 6 RT_0 (B) $\frac{9P_0V_0}{2}$, 10 RT_0 (C) $\frac{15P_0V_0}{2}$, 6 RT_0 (D) $\frac{15P_0V_0}{2}$, 10 RT_0

***13.** q, W, ΔE and ΔH for the following process ABCD on an ideal monoatomic gas are (moles = 2) :



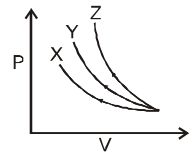
(A) $W = -2P_0V_0 \ln 2$, $q = 2P_0V_0 \ln 2$, $\Delta E = 0$, $\Delta H = 0$

(B) $W = -P_0V_0 \ln 2$, $q = P_0V_0 \ln 2$, $\Delta E = 0$, $\Delta H = 0$

(C) $W = -2P_0V_0 \ln 2$, $q = 0$, $\Delta E = -2P_0V_0 \ln 2$, $\Delta H = \frac{-10}{3} P_0V_0 \ln 2$

(D) $W = -2P_0V_0\left(\frac{1}{4} + \ln 2\right)$, $q = 2P_0V_0 \ln 2$, $\Delta E = \frac{-P_0V_0}{2}$, $\Delta H = \frac{-5P_0V_0}{6}$

***14.** P–V plots for three gases (assuming ideal behaviour and similar condition) for reversible adiabatic compression are given in the figure below :



Plots X, Y and Z should correspond to respectively :

(A) CO_2, Cl_2 and Ne (B) SO_2, N_2O and He

(C) He, N_2 and O_3 (D) NH_3, H_2S and Ar

15. **Assertion :** The increase in internal energy (ΔE) for the vaporisation of one mole of water at 1 atm and 373 K is zero.

Reason : For all isothermal processes on perfect gases, $\Delta E = 0$.

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

16. When the gas is an ideal one and expansion process is isothermal, then the correct relation is/are :

(A) $P_1V_1 = P_2V_2$ (B) $\Delta E = 0$ (C) $W = 0$ (D) $\Delta H = 0$

17. An ideal gas undergoes adiabatic expansion against constant external pressure. Which of the following are correct :

(A) Internal energy of the gas remains unchanged

(B) Temperature of the system decreases.

(C) $\Delta E + P\Delta V = 0$

(D) Internal energy of the gas decreases.

NCERT : 6.1, 6.2, 6.7

1. A sample of ideal gas is compressed from an initial volume of $2v_0$ to v_0 using three different processes (from same initial state).

First : Reversible isothermal process

Second : Reversible adiabatic process

Third : Irreversible adiabatic process by a constant external pressure

Then :

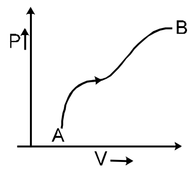
(A) Work done in first process will be greater than in second process.

(B) Final pressure of gas will be minimum in first process.

(C) Enthalpy change of sample will be highest in isothermal process.

(D) Final temperature of gas will be highest in third process.

2. The graph given below shows the P-V plot for a process on an ideal gas. Select the incorrect statement :



(A) Enthalpy content of the gas is constantly increasing and the process is carried out slowly.

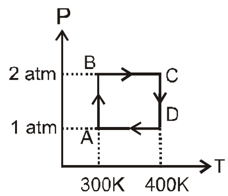
(B) Enthalpy content of the gas first increases, then decreases and the process is carried out quasistaticly.

(C) Enthalpy content of the gas is constant and the process takes infinite time for completion.

(D) Enthalpy content first decreases, then increases and the process is reversible.

Comprehension :

One mole of Helium gas undergoes a reversible cyclic process ABCDA as shown in the figure. Assuming gas to be ideal, answer the following questions :



3. What is the value of ΔH for the overall cyclic process :

(A) $-100\ R\ell n2$ (B) $+100R\ell n2$ (C) $+200R\ell n2$ (D) Zero

4. What is the value of 'q' for the overall cyclic process :

(A) $-100\ R\ell n2$ (B) $+100R\ell n2$ (C) $+200R\ell n2$ (D) $-200R\ell n2$

5. What is the net work involved in the process A to C :

(A) $-100\ R(1-\ell n8)$ (B) $300R\ell n2$ (C) $-100\ R(1+\ell n8)$ (D) 200 J

6. Which of the following are not state functions ?

(I) q + W (II) q (III) W (IV) H – TS

(A) (I) and (IV) (B) (II), (III) and (IV) (C) (I), (II) and (III) (D) (II) and (III)

SLOT:

7. Match the processes given in column-I performed on an ideal gas sample with changes in column-II :

Column-I	Column-II
(A) Isothermal expansion against constant external pressure	(p) $\Delta S_{univ} > 0$
(B) Free expansion against vacuum under isothermal conditions.	(q) $\Delta S_{sys} = 0$
(C) Reversible adiabatic compression	(r) $\Delta S_{surr} = 0$
(D) Free expansion against vacuum under adiabatic conditions.	(s) $\Delta S_{sys} > 0$

8. In which of the following cases, entropy decreases :

(A) Solid changing to liquid (B) Expansion of a gas

(C) Crystal dissolves (D) Polymerisation

9. The positive value of ΔS indicates that-

(A) The system becomes less disordered (B) The system becomes more disordered

(C) The system is in equilibrium position (D) The system tends to reach at equilibrium position

***10.** For a perfectly crystalline solid $C_{p.m.} = aT^3$, where a is constant. If $C_{p.m.}$ is 0.42 J/K–mol at 10 K, molar entropy at 10 K is

(A) 0.42 J/K-mol (B) 0.14 J/K-mol (C) 4.2 J/K-mol (D) zero

11. If a chemical reaction is non-spontaneous at high temperatures and spontaneous at low temperatures, then:

(A) $\Delta H > 0$ and $\Delta S > 0$ (B) $\Delta H > 0$ and$\Delta S < 0$

(C) $\Delta H < 0$ and $\Delta S > 0$ (D) $\Delta H < 0$ and $\Delta S < 0$

12. For a reaction to occur spontaneously

(A) $(\Delta H - T\Delta S)$ must be negative (B) $(\Delta H + T\Delta S)$ must be negative

(C) ΔH must be negative (D) ΔS must be negative

13. The entropy change in the fusion of one mole of a solid melting at 27°C (latent heat of fusion is 2930 J mol^{-1}) is

(A) 9.77 JK^{-1} mol^{-1} (B) 10.73 JK^{-1} mol^{-1} (C) 2930 JK^{-1} mol^{-1} (D) 108.5 JK^{-1} mol^{-1}

14. For the two reactions given below :

$$H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O\,(\ell) + x_1 \text{ kJ mol}^{-1}$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O\,(g) + x_2 \text{ kJ mol}^{-1}$$

Compare the magnitude of x_1 and x_2 : (x_1 and x_2 are the heat released in above two processes)

(A) $x_1 > x_2$ (B) $x_1 < x_2$ (C) $x_1 = x_2$ (D) cannot be compared

15. 2 moles of an ideal gas at 27ºC temperature is expanded reversibly from 2 L to 20 L. Find entropy change (R = 2 cal/mol K).

(A) 92.1 cal/K (B) 0 cal/K (C) 4 cal/K (D) 9.2 cal/K

16. A child bought a balloon which became very small in size the next day. Which is correct statement about ballon?

(A) It is isolated system (B) It is an open system.

(C) It is a closed system (D) It exchanges only energy with the surrounding.

17. If there are two compartments in a container of equal volume. 1L each, if one compartment is containing 4 gm of He and in other there is vaccum, then the ΔS when the separator between the compartments has been removed, will be

18. One mole of an ideal diatomic gas (C_v = 5 cal) was transformed from initial 25°C and 1 L to the state when temperature is 100°C and volume 10 L. The entropy change of the process can be expressed as :

(R = 2 calories/mol/K)

(A) $3 \ln \frac{298}{373} + 2 \ln 10$ (B) $5 \ln \frac{373}{298} + 2 \ln 10$

(C) $7 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$ (D) $5 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$

19. Two moles of an ideal gas ($\gamma = 3/2$) is made to expand reversibly and adiabatically to 4 times its initial volume. The change in entropy of the system during expansion is : (Given : R = 2 cal/K/mole, $\log_{10}2 = 0.3$, $\log_{10}3 = 0.48$)

(A) 5.6 cal/k (B) 11.2 cal/k (C) 2.8 cal/k (D) None of these

20. Predict which of the following reaction(s) have a positive entropy change ?

I. Ba^{2+} (aq) + SO_4^{2-} (aq) $\longrightarrow$ $BaSO_4$ (s)

II. $CaCO_3$ (s) $\longrightarrow$ CaO (s) + CO_2 (g)

III. $2SO_3$ (g) $\longrightarrow$ $2SO_2$ (g) + O_2 (g)

(A) I and II (B) III (C) II and III (D) II

21. Which statement regarding entropy is/are incorrect ?

(A) A completely ordered deck of cards has more entropy than a shuffled deck in which cards are arranged randomly.

(B) A perfect ordered crystal of solid nitrous oxide has more entropy than a disordered crystal in which the molecules are oriented randomly.

(C) 1 mole N_2 gas at STP has more entropy than 1 mole N_2 gas at 273 K in a volume of 11.2 litre.

(D) 1 mole N_2 gas at STP has more entropy than 1 mole N_2 gas at 273 K and 0.25 atm.

22. Which of the following occurs with increase in entropy.

(A) Dissolution of salt (B) Egg boiling

(C) Adsorption of gas on solid surface (D) Diffusion of gas

NCERT : 6.17, 6.18, 6.19

Thermochemistry:

1. In an endothermic reaction, the value of ΔH is :

(A) zero (B) positive (C) negative (D) constant

2. For the reaction, $C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(\ell)$

at constant temperature, $\Delta H - \Delta E$ is :

(A) + 3RT (B) – RT (C) + RT (D) – 3RT

3. Consider the following changes :

(i) $\frac{1}{2}H_2(g) + aq. \longrightarrow H^+(aq.)$ (ii) $2O(g) \longrightarrow O_2(g)$

(iii) $NH_4^+(g) + Cl^-(g) \longrightarrow NH_4Cl(s)$ (iv) P_4 (black) $+ 5O_2(g) \longrightarrow P_4O_{10}(s)$

Which of the above does not represent $\Delta H^\circ_{formation}$ of the product :

(A) II,III (B) I, IV (C) I, II, III (D) II , III, IV

4. Given that $S_{(s)} + \frac{3}{2}O_{2(g)} \longrightarrow SO_{3(g)} + 2x$ KCal; $SO_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow SO_{3(g)} + y$ KCal

What would be the enthalpy of formation of SO_2 :

(A) – 2x + y (B) 2x – y (C) – x + y (D) x – y

5. The combustion of 0.2 mole of liquid carbon disulphide CS_2 at constant pressure to give $CO_2(g)$ and $SO_2(g)$ releases 215 kJ of heat. What is ΔH_f° for $CS_{2(l)}$ in kJ mol^{-1} :

ΔH_f°	kJ mol^{-1}
$CO_{2(g)}$	–393.5
$SO_{2(g)}$	–296.8

(A) 772.1 (B) 87.9 (C) 384.7 (D) – 2062.1

6. Calculate the heat of combustion of carbon monoxide at 17ºC at constant volume from the given enthalpy of reactions at 17ºC :

(i) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H_1 = -97000$ Cal

(ii) $CO_2(g) + C(s) \longrightarrow 2CO(g)$; $\Delta H_2 = 39000$ Cal

(A) – 136000 Cal (B) – 135420 Cal (C) – 68000 Cal (D) – 67710 Cal

7. The enthalpy of neutralization of a strong base and strong acid is 57.0 kJ eq^{-1}. The heat evolved when 0.5 mole of HNO_3 are added to 1 L of 0.2 M NaOH solution is :

(A) ≈20 kJ (B) 28.5 kJ (C) 11.4 kJ (D) 34.9 kJ

8. The heat evolved in neutralizing a solution containing 1 mole of HCN with a strong alkali is 3 KCal. The enthalpy of dissociation of HCN is :

(A) 54.1 KCal/mol (B) 60.1 KCal/mol (C) 10.7 KCal/mol (D) 16.7 KCal/mol

9. Consider the following reactions :

(i) $H^+(aq) + OH^-(aq) = H_2O(\ell) - x_1$ kJ mol^{-1} (ii) $H_2(g) + \frac{1}{2}O_2(g) = H_2O(\ell) - x_2$ kJ mol^{-1}

(iii) $CO_2(g) + H_2(g) = CO(g) + H_2O(\ell) - x_3$kJ mol^{-1}

(iv) $C_2H_2(g) + \frac{5}{2}O_2(g) = 2CO_2(g) + H_2O(\ell) + x_4$ kJ mol^{-1}

Enthalpy of formation of $H_2O(\ell)$ is :

(A) $+ x_2$ kJ mol^{-1} (B) $+ x_3$ kJ mol^{-1} (C) $- x_4$ kJ mol^{-1} (D) $+ x_1$ kJ mol^{-1}

10. The value of heat of formation of SO_2 and SO_3 are –298.2 kJ and –98.2 kJ. The heat of reaction of the following reaction will be $SO_2 + \frac{1}{2} O_2 \rightarrow SO_3$

(A) – 200 kJ (B) –356.2 kJ (C) +200 kJ (D) –396.2 kJ

11. In the combustion of 2.0 gm of methane 25 kcal heat is liberated, heat of combustion of methane would be-

(A) 100 kcal (B) 200 kcal (C) 300 kcal (D) 400 kcal

12. For the equations : $C(diamond) + 2H_2(g) \rightarrow CH_4(g)$; ΔH_1 and $C(g) + 4H(g) \rightarrow CH_4(g)$; ΔH_2

Select the correct option (Magnitude) :

(A) $\Delta H_1 = \Delta H_2$ (B) $\Delta H_1 > \Delta H_2$ (C) $\Delta H_1 < \Delta H_2$ (D) Nothing can be said with certainty.

13. Consider the following changes :

$Na^+(g) \longrightarrow Na^+(aq)\ \Delta H_1$ .... (1)

$Na(s) \longrightarrow Na^+(g)\Delta H_2$ .... (2)

$Cl^-(g) \longrightarrow Cl^-(aq)\ \Delta H_3$ .... (3)

$Cl_2(g) \longrightarrow 2Cl^-(g)\ \Delta H_4$ .... (4)

$Na^+(g) + Cl^-(g) \longrightarrow NaCl(s)\ \Delta H_5$ .... (5)

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)\ \Delta H_6$ .... (6)

Hydration enthalpy of NaCl can be defined by sum of the following :

(A) $\Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4/2$ (B) ΔH_6 only

(C) $\Delta H_1 + \Delta H_3$ (D) $\Delta H_2 + \Delta H_4/2 + \Delta H_5 + \Delta H_6$

14. Enthalpy of atomisation of NH_3 and N_2H_4 are x KCal mol^{-1} and y KCal mol^{-1} respectively. Calculate average bond energy of N—N bond :

(A) $\frac{4y-3x}{3}$ KCal mol^{-1} (B) $\frac{2y-3x}{3}$ KCal mol^{-1} (C) $\frac{4y-3x}{4}$ KCal mol^{-1} (D) $\frac{3y-4x}{3}$ KCal mol^{-1}

***15.** If for reaction :

$N_2\ (g) + 3H_2\ (g) \longrightarrow 2NH_3\ (g)$, $\Delta H_1° = -30$ kJ/mole at temperature 300 K and if specific heat capacities of different species are $S_{P,\ N_2} = 1$ J/g°C, $S_{P,\ H_2} = 10$ J/g°C and $S_{P,\ NH_3} = 2$ J/g °C, then $\Delta H_2°$ at 400 K for the same reaction will be : (assume heat capacities to be constant in given temperature range)

(A) – 32 kJ/mole (B) – 28 kJ/mole (C) – 32.7 kJ/mole (D) – 27.3 kJ/mole

16. For which one of the following equation $\Delta H°_r$ not equal to $\Delta H°_f$ for the product ?

(A) $Xe(g) + 2F_2(g) \longrightarrow XeF_4(g)$ (B) $2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$

(C) $N_2(g) + O_3(g) \longrightarrow N_2O_3(g)$ (D) $CH_4(g) + 2Cl_2(g) \longrightarrow CH_2Cl_2(\ell) + 2HCl(g)$

***17.** For the process $H_2O(\ell)$ (1 bar, 373 K) $\rightleftharpoons$ $H_2O(g)$ (1 bar, 373 K), the correct set of thermodynamic parameters is :

(A) $\Delta G = 0$ (B) $\Delta S > 0$ (C) $\Delta H > 0$ (D) $\Delta G = -ve$

18. At 300 K, the reaction which have the following values of thermodynamic parameter, occur spontaneously.

(A) $\Delta G° = -400$ kJ mol^{-1} (B) $\Delta H° = 200$ kJ mol^{-1}, $\Delta S° = -4$ JK^{-1} mol^{-1}

(C) $\Delta H° = -200$ kJ mol^{-1}, $\Delta S° = 4$ JK^{-1} mol^{-1} (D) $\Delta H° = 200$ kJ mol^{-1}, $\Delta S° = -40$ JK^{-1} mol^{-1}

1. Standard enthalpies of formation of O_3, CO_2, NH_3 and HI are 142.2, –393.2, –46.2 and +25.9 kJ mol^{-1} respectively. The order of their increasing stabilities will be

(A) O_3, CO_2, NH_3, HI (B) CO_2, NH_3, HI, O_3 (C) O_3, HI, NH_3, CO_2 (D) NH_3, HI, CO_2, O_3

2. $Fe_2O_3(s) + \frac{3}{2}C(s) \longrightarrow \frac{3}{2}CO_2(g) + 2Fe(s)$

$\Delta H° = +234.1$ kJ

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H° = -393.5$ kJ

Use these equations and ΔH° values to calculate ΔH° for this reaction :

$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$

(A) –1648.7 kJ (B) –1255.3 kJ (C) –1021.2 kJ (D) –129.4 kJ

3. The enthalpy of neutralisation of a strong acid by a strong base is –57.32 kJ mol^{-1}. The enthalpy of formation of water is –285.84 kJ mol^{-1}. The enthalpy of formation of hydroxyl ion is $[\Delta_f H°[H^+(aq)] = 0]$

(A) + 228.52 kJ mol^{-1} (B) –114.26 kJ mol^{-1} (C) –228.52 kJ mol^{-1} (D) +114.2 kJ mol^{-1}

4. 4.48 L of an ideal gas at S.T.P. requires 12 calories to raise its temperature by 15°C at constant volume. The C_p of gas is

(A) 3 cal. (B) 6 cal. (C) 4 cal. (D) 9 cal.

5. The amount of heat released when 20 ml of 0.5 M NaOH is mixed with 100 mL of 0.1 M HCl is x kJ. The heat of neutralisation (in kJ mol^{-1}) is

(A) –100 x (B) –50 x (C) +100 x (D) +50x

6. What will be the heat of formation of methane, if the heat of combusition of carbon is '–x' kJ, heat of formation of water is '–y' kJ and heat of combustion of methane is '–z' kJ ?

(A) (– x – y + z) kJ (B) (– z + x + 2y) kJ (C) (– x – 2y – z) kJ (D) (– x – 2y + z) kJ

7. How much energy is absorbed by 10 mol of an ideal gas if it expands reversibly from an initial pressure of 8 atm. to 4 atm at a constant temperature of 27°C ?

(A) 1.728×10^4 J (B) 2×10^5 J (C) 8×10^6 J (D) 1.728×10^5 J

8. Which reaction occurs with the greatest increase in entropy ?

(A) $2H_2O(\ell) \rightarrow 2H_2(g) + O_2(g)$ (B) $2NO(g) \rightarrow N_2(g) + O_2(g)$

(C) $C(s) + O_2(g) \rightarrow CO_2(g)$ (D) $Br_2(g) + Cl_2(g) \rightarrow 2BrCl(g)$

9. In which of the reactions at contant pressure sign of work done is –ve i.e. work done by the system.

(A) $H_2O(s) \longrightarrow H_2O(g)$ (B) $H_2O(l) \longrightarrow H_2O(s)$

(C) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ (D) All of these

10. The enthalpy change for the reaction of 50 ml of ethylene with 50.0 mL of H_2 at 1.5 atm. pressure is ΔH = –0.31 KJ. What is the ΔU?

(A) –0.3024 KJ (B) –0.3214 KJ (C) –0.2852 KJ (D) –20.84 KJ

***11.** The entropies of $H_2(g)$ and H(g) are 130.6 and 114.6 Jmol^{-1} K^{-1} respectively at 298K. Bond energy of H_2 (in KJ mol^{-1}) J is

Given : $H_2(g) \longrightarrow 2H(g)$; $\Delta G° = 406.62$ KJ mol^{-1}

(A) 364 KJ (B) 643 KJ (C) 436 KJ (D) 346 KJ

12. C (diamond) $\rightleftharpoons$ C (graphite) is in equilibrium at 300 K.

$\Delta S_{300K} = 10$ cal K^{-1}

C (diamond) + $O_2 \longrightarrow CO_2$

$\Delta H = -91$K cal mol^{-1} at 300 K

C (graphite) + $O_2 \longrightarrow CO_2$

$\Delta H = X$ at 300 K

X is

(A) –81 K cal (B) 101 K cal (C) –94 K cal (D) 88 K cal

13. The latent heat of vaporization of a liquid at 500 K and 1 atm pressure is 10 Kcal/mol. What will be the change in internal energy (ΔU) of 3 moles of the liquid at the same temperature and pressure?

(A) 13 Kcal (B) – 13 Kcal (C) 27 Kcal (D) –27 Kcal

14. Following reaction occurs in an automobile

$2C_8H_{18}(g) + 25O_2(g) \longrightarrow 16CO_2(g) + 18H_2O(g)$ The sign of ΔH , ΔS and ΔG will be

(A) + , – , + (B) – , + , – (C) – , +, + (D) + , +, –

15. In a closed insulated container a liquid is stirred with a paddle to increase the temperature which of the following is true

(A) $\Delta E = W \neq 0, q = 0$ (B) $\Delta E = W = q \neq 0$ (C) $\Delta E = 0, W = q \neq 0$ (D) $W = 0$ $\Delta E = q \neq 0$

***16.** The densities of graphite and diamond at 298 K are 2.25 and 3.31 g cm^{-3}, respectively. If the standard free energy difference ($\Delta G°$) is equal to 1895 J mol^{-1}, the pressure at which graphite will be transformed into diamond at 298 K is

(A) 11.1×10^8 Pa (B) 11.1×10^7 Pa (C) 11.1×10^6 Pa (D) 11.1×10^5 Pa

17. The molar heat capacity of water at constant pressure, is 75 JK^{-1} mol^{-1}. When 1.0 KJ of heat is supplied to 100 g of water which is free to expand, the increase in temperature of water is

(A) 1.2 K (B) 2.4 K (C) 4.8 K (D) 6.6 K

18. One gram sample of NH_4NO_3 is decomposed in a bomb calorimeter. The temperature of the calorimeter increase by 6.12 K. The heat capacity of the system is 1.23 kJ/g/K. What is the molar heat of decomposition for NH_4NO_3

(A) 7.53 kJ/mol (B) 398.1 kJ/mol (C) 16.1 kJ/mol (D) 602 kJ/mol

***19.** In an irreversible process taking place at constant T and P and in which only pressure–volume work is being done, the change in Gibbs free energy (dG) and change in entropy (dS), satisfy the criteria

(A) $(dS)_{V, E} = 0, (dG)_{T, P} = 0$ (B) $(dS)_{V, E} = 0, (dG)_{T, P} > 0$

(C) $(dS)_{V, E} < 0, (dG)_{T, P} < 0$ (D) $(dS)_{V, E} > 0, (dG)_{T, P} < 0$

***20.** One mole of a non-ideal gas undergoes a change of state (2.0 atm, 3.0L,73.08 K) $\rightarrow$ (4.0 atm, 5.0L, 243.6 K) with a change in internal energy, ΔU = 30.0L atm. The change in enthalpy (ΔH) of the process in litre atm is

(A) 40.0 (B) 42.3

(C) 44.0 (D) not define,because pressure is not constant

NCERT: 6.3, 6.4, 6.5, 6.6, 6.8, 6.11, 6.14, 6.15

1. Hydrogen resembles halogens in many respects for which several factors are responsible. Of the following factors which one is most important in this respect?

(i) Its tendency to lose an electron to form a cation.

(ii) Its tendency to gain a single electron in its valence shell to attain stable electronic configuration.

(iii) Its low negative electron gain enthalpy value.

(iv) Its small size.

2. Why does H^+ ion always get associated with other atoms or molecules?

(i) Ionisation enthalpy of hydrogen resembles that of alkali metals.

(ii) Its reactivity is similar to halogens.

(iii) It resembles both alkali metals and halogens.

(iv) Loss of an electron from hydrogen atom results in a nucleus of very small size as compared to other atoms or ions. Due to smal size it cannot exist free.

3. Metal hydrides are ionic, covalent or molecular in nature. Among LiH, NaH, KH, RbH, CsH, the correct order of increasing ionic character is

(i) LiH > NaH > CsH > KH>RbH (ii) LiH < NaH < KH < RbH < CsH

(iii) RbH > CsH > NaH > KH > LiH (iv) NaH > CsH > RbH > LiH > KH

4. Which of the following hydrides is electron-precise hydride?

(i) B_2H_6 (ii) NH_3 (iii) H_2O (iv) CH_4

5. Radioactive elements emit α, β and γ rays and are characterised by their halflives. The radioactive isotope of hydrogen is

(i) Protium (ii) Deuterium (iii) Tritium (iv) Hydronium

6. Consider the reactions

(A) $H_2O_2 + 2HI \longrightarrow I_2 + 2H_2O$ (B) $HOCl + H_2O_2 \longrightarrow H_3O^+ + Cl^- + O_2$

Which of the following statements is correct about H_2O_2 with reference to these reactions? Hydrogen perioxide is ________.

(i) an oxidising agent in both (A) and (B)

(ii) an oxidising agent in (A) and reducing agent in (B)

(iii) a reducing agent in (A) and oxidising agent in (B)

(iv) a reducing agent in both (A) and (B)

7. The oxide that gives H_2O_2 on treatment with dilute H_2SO_4 is

(i) PbO_2 (ii) $BaO_2 .8H_2O + O_2$ (iii) MnO_2 (iv) TiO_2

8. Which of the following equations depict the oxidising nature of H_2O_2?

(i) $2MnO_4^- + 6H^+ + 5H_2O_2 \longrightarrow 2Mn^{2+} + 8H_2O + 5O_2$

(ii) $2Fe^{3+} + 2H^+ + H_2O_2 \longrightarrow 2Fe^{2+} + 2H_2O + O_2$

(iii) $2I^- + 2H^+ + H_2O_2 \longrightarrow I_2 + 2H_2O$

(iv) $KIO_4 + H_2O_2 \longrightarrow KIO_3 + H_2O + O_2$

9. Which of the following equation depicts reducing nature of H_2O_2?

(i) $2[Fe(CN)_6]^{4-} + 2H^+ + H_2O_2 \longrightarrow 2[Fe(CN)_6]^{3-} + 2H_2O$

(ii) $I_2 + H_2O_2 + 2OH^- \longrightarrow 2I^- + 2H_2O + O_2$

(iii) $Mn^{2+} + H_2O_2 \longrightarrow Mn^{4+} + 2OH^-$

(iv) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$

10. Hydrogen peroxide is _________.

(i) an oxidising agent (ii) a reducing agent

(iii) both an oxidising and a reducing agent (iv) neither oxidising nor reducing agent

11. Which of the following reactions increases production of dihydrogen from synthesis gas?

(i) $CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270\,K} CO(g) + 3H_2(g)$

(ii) $C(s) + H_2O(g) \xrightarrow{1270\,K} CO(g) + H_2(g)$

(iii) $CO(g) + H_2O(g) \xrightarrow[Catalyst]{673\,K} CO_2(g) + H_2(g)$

(iv) $C_2H_6 + 2H_2O \xrightarrow[Ni]{1270\,K} 2CO + 5H_2$

12. When sodium peroxide is treated with dilute sulphuric acid, we get ______.

(i) sodium sulphate and water (ii) sodium sulphate and oxygen

(iii) sodium sulphate, hydrogen and oxygen (iv) sodium sulphate and hydrogen peroxide

13. Hydrogen peroxide is obtained by the electrolysis of ______.

(i) water (ii) sulphuric acid (iii) hydrochloric acid (iv) fused sodium peroxide

14. Which of the following reactions is an example of use of water gas in the synthesis of other compounds?

(i) $CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270\,K} CO(g) + H_2(g)$

(ii) $CO(g) + H_2O(g) \xrightarrow[Catalyst]{673\,K} CO_2(g) + H_2(g)$

(iii) $C_nH_{2n+2} + nH_2O(g) \xrightarrow[Ni]{1270\,K} nCO + (2n+1)H_2$

(iv) $CO(g) + 2H_2(g) \xrightarrow[Catalyst]{Cobalt} CH_3OH(l)$

15. Which of the following ions will cause hardness in water sample ?

(i) Ca^{2+} (ii) Na^+ (iii) Cl^- (iv) K^+

16. Which of the following compounds is used for water softening?

(i) $Ca_3(PO_4)_2$ (ii) Na_3PO_4 (iii) $Na_6P_6O_{18}$ (iv) Na_2HPO_4

17. Elements of which of the following group(s) of periodic table do not form hydrides.

(i) Groups 7, 8, 9 (ii) Group 13 (iii) Groups 15, 16, 17 (iv) Group 14

18. Only one element of ________ forms hydride.

(i) group 6 (ii) group 7 (iii) group 8 (iv) group 9

19. Which of the following statements are not true for hydrogen?

(i) It exists as diatomic molecule.

(ii) It has one electron in the outermost shell.

(iii) It can lose an electron to form a cation which can freely exist

(iv) It forms a large number of ionic compounds by losing an electron.

20. Dihydrogen can be prepared on commercial scale by different methods. In its preparation by the action of steam on hydrocarbons, a mixture of CO and H_2 gas is formed. It is known as ____________.

(i) Water gas (ii) Syngas (iii) Producer gas (d) Industrial gas

21. Which of the following statement(s) is/are correct in the case of heavy water ?

(i) Heavy water is used as a moderator in nuclear reactor.

(ii) Heavy water is more effective as solvent than ordinary water.

(iii) Heavy water is more associated than ordinary water.

(iv) Heavy water has lower boiling point than ordinary water.

22. Which of the following statements about hydrogen are correct ?

(i) Hydrogen has three isotopes of which protium is the most common

(ii) Hydrogen never acts as cation in ionic salts

(iii) Hydrogen ion, H+, exists freely in solution.

(iv) Dihydrogen does not act as a reducing agent.

23. Some of the properties of water are described below. Which of them is/are not correct?

(i) Water is known to be a universal solvent.

(ii) Hydrogen bonding is present to a large extent in liquid water.

(iii) There is no hydrogen bonding in the frozen state of water.

(iv) Frozen water is heavier than liquid water.

24. Hardness of water may be temporary or permanent. Permanent hardness is due to the presence of

(i) Chlorides of Ca and Mg in water (ii) Sulphates of Ca and Mg in water

(iii) Hydrogen carbonates of Ca and Mg in water (iv) Carbonates of alkali metals in water

25. Which of the following statements is correct?

(i) Elements of group 15 form electron deficient hydrides.

(ii) All elements of group 14 form electron precise hydrides.

(iii) Electron precise hydrides have tetrahedral geometries.

(iv) Electron rich hydrides can act as Lewis acids.

26. Which of the following statements is correct ?

(i) Hydrides of group 13 act as Lewis acids. (ii) Hydrides of group 14 are electron deficient hydrides.

(iii) Hydrides of group 14 act as Lewis acids. (iv) Hydrides of group 15 act as Lewis bases.

27. Which of the following statements is correct?

(i) Metallic hydrides are deficient of hydrogen.

(ii) Metallic hydrides conduct heat and electricity.

(iii) Ionic hydrides do not conduct electricity in solid state.

(iv) Ionic hydrides are very good conductors of electricity in solid state.

1. The alkali metals are low melting. Which of the following alkali metal is expected to melt if the room temperature rises to 30°C?

(i) Na (ii) K (iii) Rb (iv) Cs

2. Alkali metals react with water vigorously to form hydroxides and dihydrogen. Which of the following alkali metals reacts with water least vigorously ?

(i) Li (ii) Na (iii) K (iv) Cs

3. The reducing power of a metal depends on various factors. Suggest the factor which makes Li, the strongest reducing agent in aqueous solution.

(i) Sublimation enthalpy (ii) Ionisation enthalpy (iii) Hydration enthalpy (iv) Electron-gain enthalpy

4. Metal carbonates decompose on heating to give metal oxide and carbon dioxide. Which of the metal carbonates is most stable thermally?

(i) $MgCO_3$ (ii) $CaCO_3$ (iii) $SrCO_3$ (iv) $BaCO_3$

5. Which of the carbonates given below is unstable in air and is kept in CO_2 atmosphere to avoid decomposition.

(i) $BeCO_3$ (ii) $MgCO_3$ (iii) $CaCO_3$ (iv) $BaCO_3$

6. Metals form basic hydroxides. Which of the following metal hydroxide is the least basic?

(i) $Mg(OH)_2$ (ii) $Ca(OH)_2$ (iii) $Sr(OH)_2$ (iv) $Ba(OH)_2$

7. Some of the Group 2 metal halides are covalent and soluble in organic solvents. Among the following metal halides, the one which is soluble in ethanol is

(i) $BeCl_2$ (ii) $MgCl_2$ (iii) $CaCl_2$ (iv) $SrCl_2$

8. The order of decreasing ionisation enthalpy in alkali metals is

(i) Na > Li > K > Rb (ii) Rb < Na < K < Li (iii) Li > Na > K > Rb (iv) K < Li < Na < Rb

9. The solubility of metal halides depends on their nature, lattice enthalpy and hydration enthalpy of the individual ions. Amongst fluorides of alkali metals, the lowest solubility of LiF in water is due to

(i) Ionic nature of lithium fluoride (ii) High lattice enthalpy

(iii) High hydration enthalpy for lithium ion. (iv) Low ionisation enthalpy of lithium atom

10. Amphoteric hydroxides react with both alkalies and acids. Which of the following Group 2 metal hydroxides is soluble in sodium hydroxide?

(i) $Be(OH)_2$ (ii) $Mg(OH)_2$ (iii) $Ca(OH)_2$ (iv) $Ba(OH)_2$

11. In the synthesis of sodium carbonate, the recovery of ammonia is done by treating NH_4Cl with $Ca(OH)_2$. The by-product obtained in this process is

(i) $CaCl_2$ (ii) NaCl (iii) NaOH (iv) $NaHCO_3$

12. When sodium is dissolved in liquid ammonia, a solution of deep blue colour is obtained. The colour of the solution is due to

(i) ammoniated electron (ii) sodium ion (iii) sodium amide (iv) ammoniated sodium ion

13. By adding gypsum to cement

(i) setting time of cement becomes less. (ii) setting time of cement increases.

(iii) colour of cement becomes light. (iv) shining surface is obtained.

14. Dead burnt plaster is

(i) $CaSO_4$ (ii) $CaSO_4 \cdot \frac{1}{2} H_2O$ (iii) $CaSO_4.H_2O$ (iv) $CaSO_4.2H_2O$

15. Suspension of slaked lime in water is known as

(i) lime water (ii) quick lime

(iii) milk of lime (iv) aqueous solution of slaked lime

16. Which of the following elements does not form hydride by direct heating with dihydrogen?

(i) Be (ii) Mg (iii) Sr (iv) Ba

17. The formula of soda ash is

(i) $Na_2CO_3.10H_2O$ (ii) $Na_2CO_3.2H_2O$ (iii) $Na_2CO_3.H_2O$ (iv) Na_2CO_3

18. A substance which gives brick red flame and breaks down on heating to give oxygen and a brown gas is

(i) Magnesium nitrate (ii) Calcium nitrate (iii) Barium nitrate (iv) Strontium nitrate

19. Which of the following statements is true about $Ca(OH)_2$?

(i) It is used in the preparation of bleaching powder

(ii) It is a light blue solid

(iii) It does not possess disinfectant property.

(iv) It is used in the manufacture of cement.

20. A chemical A is used for the preparation of washing soda to recover ammonia. When CO_2 is bubbled through an aqueous solution of A, the solution turns milky. It is used in white washing due to disinfectant nature. What is the chemical formula of A?

(i) $Ca\ (HCO_3)_2$ (ii) CaO (iii) $Ca(OH)_2$ (iv) $CaCO_3$

21. Dehydration of hydrates of halides of calcium, barium and strontium i.e.,

$CaCl_26H_2O$, $BaCl_2.2H_2O$, $SrCl_2.2H_2O$, can be achieved by heating. These become wet on keeping in air. Which of the following statements is correct about these halides?

(i) act as dehydrating agent

(ii) can absorb moisture from air

(iii) Tendency to form hydrate decreases from calcium to barium

(iv) All of the above

II. Multiple Choice Questions (Type-II)

In the following questions two or more options may be correct.

22. Metallic elements are described by their standard electrode potential, fusion enthalpy, atomic size, etc. The alkali metals are characterised by which of the following properties?

(i) High boiling point (ii) High negative standard electrode potential

(iii) High density (iv) Large atomic size

23. Several sodium compounds find use in industries. Which of the following compounds are used for textile industry?

(i) Na_2CO_3 (ii) $NaHCO_3$ (iii) $NaOH$ (iv) $NaCl$

24. Which of the following compounds are readily soluble in water ?

(i) $BeSO_4$ (ii) $MgSO_4$ (iii) $BaSO_4$ (iv) $SrSO_4$

25. When Zeolite, which is hydrated sodium aluminium silicate is treated with hard water, the sodium ions are exchanged with which of the following ion(s) ?

(i) H^+ ions (ii) Mg^{2+} ions (iii) Ca^{2+} ions (iv) SO_4^{2-} ions

1. Which of the following statements is not correct?
(A) Alkaline earth metals are harder and denser than alkali metals
(B) The melting and boiling points of alkaline earth metals are higher than those of alkali metals
(C) Alkaline earth metals are more reactive than alkali metals
(D) The reactivities of alkaline earth metals increase down the group

2. On combustion in air Li forms
(A) Li_2O and Li_2O_2 (B) Li_3N (C) Li_2O and Li_3N (D) Li_2O_2 and LiN_3

3. What is the characteristic of LiCl
(A) LiCl is the deliquescent (B) LiCl is soluble in pyridine
(C) LiCl is available as $LiCl.2H_2O$ (D) all

4. Na and Li are placed in dry air. We get
(A) $NaOH, Na_2O, Li_2O$ (B) Na_2O, Li_2O
(C) $Na_2O, Li_2O, Li_3N, NH_3$ (D) $Na_2O, Li_3N, Li_2O, Na_2O_2$

5. Solvay process can not be used for the preparation of K_2CO_3 because
(A) K_2CO_3 soluble (B) $KHCO_3$ can be precipitated.
(C) $KHCO_3$ can not be precipitated (D) KCl does not react with NH_4HCO_3

6. What product is formed when CO is passed through sodium hydroxide under high pressure?
(A) Sodium carbonate (B) sodium bicarbonate (C) sodium formate (D) (A) and (B)

7. Which of the following is not found in solid state?
(A) $LiHCO_3$ (B) $NaHCO_3$ (C) $KHCO_3$ (D) NH_4HCO_3

8. When dry ammonia gas is passed over heated sodium (out of contact of air) the product formed is
(A) sodium hydride (B) sodium nitride (C) sodamide (D) sodium cynamide

9. Which of the following is best CO_2 absorber as well as source of O_2 in submarines?
(A) KO_2 (B) NaOH (C) KOH (D) LiOH

10. On dissolving moderate amount of sodium metal in liquid NH3 at low temperature which one of the following does not occur
(A) Blue coloured solution is obtained (B) Na^+ ions are formed in the solution
(C) Liquid NH_3 becomes good conductor of electricity (D) Liquid ammonia remains diamagnetic

11. A metal M reacts with N_2 give a compound 'A(M_3N)
'A' on heating at high temperature gives back 'M' and 'A' on reacting with H_2O gives a gas 'B'. 'B' turns $CuSO_4$ solution blue on passing through it. A and B can be
(A) Al and NH_3 (B) Li and NH_3 (C) Na and NH_3 (D) Mg and NH_3

12. Which sequence of reactions shows correct chemical relation between sodium and its compounds
(A) $Na+O_2 \longrightarrow Na_2O \xrightarrow{HCl_{(aq)}} NaCl \xrightarrow{CO_2} Na$
(B) $Na \xrightarrow{O_2} Na_2O \xrightarrow{H_2O} NaOH \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$
(C) $Na + H_2O \longrightarrow NaOH \xrightarrow{HCl} NaCl \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$
(D) $Na+H_2O \rightarrow NaOH \xrightarrow{CO_2} NaCO_3 \xrightarrow{HCl} \underset{(Molten)}{NaCl} \xrightarrow{Electrolysis} Na$

13. Select the correct statement with respect to alkali metals.
(A) Melting points decrease with increasing atomic number.
(B) Potassium is lighter than sodium.
(C) Salts of Li to Cs impart characteristic colour to an oxidising flame. (of Bunsen burner).
(D) All of these.

14. The alkali metal hydroxides dissolve freely in water with evolution of much heat on account of :
(A) intense hydration (B) strongest base (C) (A) and (B) both (D) none of these.

15. Alkali metals are characterized by
(A) Good conductor of heat and electricity (B) High oxidation potentials
(C) High melting points (D) Solubility in liquid ammonia.

16. Highly pure dilute solution of sodium in liquid ammonia
(A) Shows blue colour (B) Exhibits electrical conductivity
(C) Produces sodium amide instantly (D) Produces hydrogen gas instantly

17. Sodium nitrate decomposes above 800°C to give
(A) N_2 (B) O_2 (C) NO_2 (D) Na_2O

18. Property of the alkaline earth metals that increases with their atomic number is -
(A) Ionisation energy (B) solubility of their hydroxides
(C) solubility of their sulphates (D) Electronegativity

19. **match appropriate according to reaction with air**

List I	**List II**
(A) Na	(P) Super oxide
(B) Cs	(Q) Coloured oxide
(C) K	(R) Stable peroxide
(D) Li	(S) Normal oxide

20. Which of the following gives H_2O_2 on hydrolysis–
(A) Na_2O (B) Na_2O_2 (C) KO_2 (D) B & C both

21. Which of the following is coloured–
(A) Li_2O (B) Na_2O_2 (C) KO_2 (D) Na_2O

22. Correct order of melting points & boiling point is– (M = Alkali metal)
(A) MF > MBr > MI > MCl (B) MF > MCl > MBr > MI
(C) MI > MBr > MCI > MF (D) MI > MBr > MF > MCl

23. Which of the following releases reddish brown gas on heating–
(A) $NaNO_3$ (B) $LiNO_3$ (C) KNO_3 (D) $RbNO_3$

24. In Downs cell KCl is added in NaCl to–
(A) Lower its melting point (B) Dissolve more of NaCl(C) Increase Conductivity (D) Increase the dissociation

25. Identify the product [X] & [Y]

$NaNO_3 \xrightarrow{500°C} [X] + O_2$; $NaNO_3 \xrightarrow{800°C} [Y]+O_2+N_2$
(A) [X] = $NaNO_2$ [Y] = Na_2O_2 (B) [X] = $NaNO_2$ [Y] = NaOH
(C) [X] = $NaNO_2$, [Y] = Na_2O (D) [X] Na_2O [Y] = $NaNO_2$

26. Select incorrect statement :-

(A) Stability of peroxides and super oxides of alkali metals increases with increase in size of metal ion.

(B) Increase in stability in (A) is due to stablisation of large anions by larger cations through lattice energy effects.

(C) The low solubility of LiF is due to its high lattice energy whereas low solubility of CsI is due to smaller hydration energy.

(D) NaOH does not form hydrated salt.

27. Which of following on hydrolysis gives the cooresponding hydroxide, H_2O_2 & O_2 :-

(A) Li_2O (B) Na_2O_2 (C) NaO_2 (D) Na_2O

28. The alkali halide that is soluble in Pyridine is

(A) NaCl (B) LiCl (C) KCl (D) CsI

29. In Solvay process of manufacture of Na2CO3 , the by products obtained from recovery tower are :

(A) NH_4Cl, CaO, CO_2 (B) CaO, Na_2CO_3, $CaCl_2$

(C) $CaCl_2$, CO_2 , NH_3 (D) Na_2CO_3 ,$CaCl_2$, CO_2

30. Which of the following is incorrect for sodium peroxide ?

(A) It reacts with water and acid giving hydrogen peroxide.

(B) It is pale yellow in colour and is used to purify the air in submarine and confined space.

(C) It is used in many industries as bleaching

(D) It can be prepared by burning the metal in dioxygen at atmos pheric pressure.

31. Which of the following does not exist in solid form–

(A) Na HCO_3 (B) $Ba(HCO_3)_2$ (C) KH CO_3 (D) Rb HCO_3

32. If NaOH is added to an aqueous solution of Zn^{+2} ions, a white precipitate appears and on adding excess NaOH, the precipitate dissolves. In this solution zinc exists in the

(A) Cationic part (B) Anionic part

(C) Both in cationic and anionic part (D) There is no zinc left in the solution

33. When a standard solution of NaOH is left in air for a few hours

(A) A precipitate is formed (B) Strength of NaOH decreases

(C) Strength of NaOH increases (D) Concentration of Na^+ ions will remain constant

34. On allowing ammonia solution of s-block metals to stand for a long time, blue colour becomes fade. The reason is:-

(A) Formation of NH_3 gas (B) Formation of metal amide

(C) Cluster formation of metal ions (D) Formation of metal nitrate

35. KO_2 finds use in oxygen cylinders used for space and submarines. The fact(s) related to such use of KO_2 is/are

(A) it produces O_2 (B) it produces O_3

(C) it absorbs CO_2 (D) it absorbs both CO and CO_2

36. Sodium metal is highly reactive and cannot be stored under

(A) toluene (B) kerosene oil (C) alcohol (D) benzene

37. Nitrogen dioxide cannot be prepared by heating

(A) KNO_3 (B) $AgNO_3$ (C) $Pb(NO_3)_2$ (D) $Cu(NO_3)_2$

38. $X + C + Cl_2 \xrightarrow[\text{of about 1000K}]{\text{High temperature}} Y + CO$; $Y + 2H_2O \rightarrow Z + 2HCl$

Compound Y is found in polymeric chain structure and is an electron deficient molecule. Y must be

(A) BeO (B) $BeCl_2$ (C) BeH_2 (D) $AlCl_3$

39. The order of thermal stability of carbonates of IIA group is

(A) $BaCO_3 > SrCO_3 > CaCO_3 > MgCO_3$
(B) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
(C) $CaCO_3 > SrCO_3 > BaCO_3 > MgCO_3$
(D) $MgCO_3 = CaCO_3 > SrCO_3 = BaCO_3$

40. $Be_2C + H_2O \longrightarrow BeO + X$

$CaC_2 + H_2O \longrightarrow Ca(OH)_2 + Y$; then X and Y are respectively

(A) CH_4, CH_4 (B) CH_4, C_2H_6 (C) CH_4, C_2H_2 (D) C_2H_2, CH_4

41. Which of the following statements are false?

(A) $BeCl_2$ is a linear molecule in the vapour state but it is polymeric form in the solid state

(B) Calcium hydride is called hydrolith.

(C) Carbides of both Be and Ca react with water to form acetylene

(D) Oxides of both Be and Ca are amphoteric.

42. Which of the following are ionic carbides?

(A) CaC_2 (B) Al_4C_3 (C) SiC (D) Be_2C

43. Which of the following is/are the characteristic of barium?

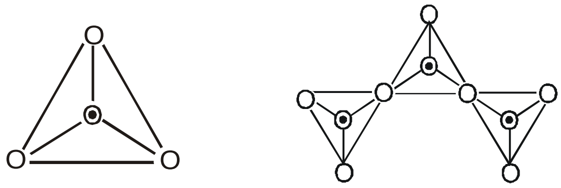
(A) It emits electrons on exposure to light

(B) It is a silvery white metal

(C) It forms $Ba(NO_3)_2$ which is used in preparation of green fire

(D) All of these

1. The following structures represent various silicate anions. Their formulas are respectively



O = Oxygen
• = Silicon

(A) SiO_4^{4-} & $Si_3O_8^{8-}$ (B) SiO_4^{4-} & $Si_3O_{10}^{8-}$ (C) SiO_4^{2-} & $Si_3O_9^{2-}$ (D) SiO_3^{4-} & $Si_3O_{10}^{8-}$

2. Select the correct statements
(A) Oxides of boron (B_2O_3) and silicon (SiO_2) are acidic in nature.
(B) Oxides of aluminium (Al_2O_3) and gallium (Ga_2O_3) are amphoteric in nature.
(C) Oxides of germanium (GeO_2) and tin (SnO_2) are acidic in nature.
(D) both (1) and (2)

3. $BC\ell_3$ is more stable than $T\ell C\ell_3$ because :
(A) The difference between electronegativity of B and $C\ell$ is larger than that of $T\ell$ and $C\ell$.
(B) The higher oxidation state of $T\ell$ (+ 3) is less stable than that of B (+ 3).
(C) Thallium being larger in size is able to accomodate more $C\ell$ atoms around it.
(D) B^{3+} is more easily formed than $T\ell^{3+}$.

4. The straight chain polymer is formed by :
(A) hydrolysis of $(CH_3)_3$ SiCl followed by condensation polymerization.
(B) hydrolysis of $(CH_3)_2$ $SiCl_2$ followed by condensation polymerization.
(C) hydrolysis of (CH_3) $SiCl_3$ followed by condensation polymerization.
(D) hydrolysis of $(CH_3)_4$ Si followed by condensation polymerization.

5. Which gas is formed when CaC_2 is allowed to react with dilute HCl ?
(A) Ethane (B) Ethene (C) Ethyne (D) Methane

6. The function of fluorspar in the electrolytic reduction of alumina dissolved is fused cryolite (Na_3AlF_6) is
(A) as a catalyst
(B) to lower the temperature of the melt and to make the fused mixture very conducting
(C) to decrease the rate of oxidation of carbon at the anode
(D) none of these above

7. Which of the following is bauxite?
(A) $Al(NO_3)_3$ (B) $AlCl_3$ (C) $Al_2(SO_4)_3 . xH_2O$ (D) $Al_2O_3 . xH_2O$

8. Which of the following oxidation states are the most characterstics for lead and tin, respectively ?
(A) + 2 , + 4 (B) + 4 , + 4 (C) + 2 , + 2 (D) + 4 , + 2

9. Which of the following anions is present in the chain structure of silicates ?
(A) $(Si_2O_5^{2-})_n$ (B) $(SiO_3^{2-})_n$ (C) SiO_4^{4-} (D) $Si_2O_7^{6-}$

10. Glass reacts with HF to produce
(A) H_2SIO_3 (B) SiF_4 (C) Na_3AlF_6 (D) H_2SiF_6

11. The oxide which is not a reducing agent is :
(A) CO_2 (B) CO (C) SO_2 (D) Both (A) & (C)

12. Aqueous solution of potash alum is :
(A) alkaline (B) acidic (C) neutral (D) soapy

13. From B_2H_6, all the following can be prepared except :
(A) H_3BO_3 (B) $[BH_2(NH_3)_2]^+ [BH_4]^-$ (C) $B_2(CH_3)_6$ (D) $NaBH_4$

1. Moderate electrical conductivity is shown by :

(A) silica (B) graphite (C) diamond (D) $(BN)_X$

2. Which of the following halides is least stable and has doubtful existence?

(A) CI_4 (B) GeI_4 (C) SnI_4 (D) PbI_4

3. Which one of the following oxides is neutral ?

(A) CO (B) SnO_2 (C) ZnO (D) SiO_2

4. Which of the following is bauxite?

(A) $Al(NO_3)_3$ (B) $AlCl_3$ (C) $Al_2(SO_4)_3.\ xH_2O$ (D) $Al_2O_3.\ xH_2O$

5. Name the type of the structure of silicate in which one oxygen atom of $[SiO_4]^{4-}$ is shared ?

(A) Three – dimensional (B) Linear chain silicate

(C) Sheet silicate (D) Pyrosilicate

6. Which of the following is correct

(A) B_2H_6 reacts with excess of ammonia at low temperature to form an ionic compound.

(B) Borax, when heated with oxides of certain transition metals, forms coloured beads.

(C) One mole borax in aqueous solution will require one mole HCl for titration

(D) B_2H_6 can be methylated completely to give $B_2(CH_3)_6$

7. Which of the following can produce B_2O_3 ?

(A) Heating borax with conc. H_2SO_4 (B) Passing CO_2 through aq. $NaBO_2$

(C) Combustion of diborane, B_2H_6. (D) Warming H_3BO_3 crystals till red hot.

8. Which of the following is correct about B_2H_6

(A) each 'B' atom is sp^3 hybridised

(B) It consists of two-"3 centre 2 electron bonds"

(C) The two bridging H-atoms are in the plane of the molecules

(D) All of these

9. Which of the following hydrides react with water ?

(A) B_2H_6 (B) NH_3 (C) NaH (D) C_2H_6

10. Which of the following oxides are acidic :

(A) CaO (B) Na_2O (C) B_2O_3 (D) CO_2

11. In borax ion how many B–O–B bonds present :

12. In diborane maximum how many 2c – 2e bonds present :

13. How many oxides in the following are basic in water:

B_2O_3, Al_2O_3, CO_2, SO_2, NO_2 CaO, Na_2O, K_2O, Cr_2O, CaO

14. Basicity of H_3BO_3 is :

1. Stable valency of Carbon is

(A) Bivalency (B) Trivalency (C) Tetravalency (D) Monovalency

2. Which of the following is saturated hydrocarbon?

(A)  (B)  (C)  (D) 

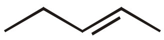
3. Carbon has strong tendency to show catenation due to :

(A) Its tetravalency (B) small size

(C) Its high C – C bond energy (D) All of these

4. Which of the following is the empirical formula of C_6H_6 ?

(A) CH (B) C_2H_2 (C) C_6H_6 (D) none of these

5.  and  are :

(A) Identical (B) Homologous (C) Alkane (D) Saturated hydrocarbon

6. Which hybridisation of carbon involve in ethene ?

(A) sp (B) sp^2 (C) sp^3 (D) None of these

7. What is the general formula of alkene ?

(A) C_nH_{2n+2} (B) C_nH_{2n} (C) C_nH_{2n-2} (D) C_nH_{2n+4}

8. How many π and σ bonds are in the given compound ?

$HC \equiv C — CH = CH_2$

(A) 3π and 6σ bonds (B) 3π and 7σ bonds (C) 2π and 7σ bonds (D) 4π and 8σ bonds

9. Which of the following has 28 molecular mass ?

(A) C_2H_2 (B) C_2H_6 (C) C_2H_4 (D) C_3H_4

10. Which of the following represents a homologous series ?

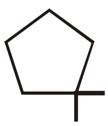
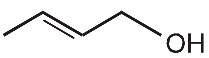
(A) ethane, ethene and ethyne (B) methane and methanol

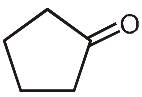
(C) methane, ethane and propane (D) none of these

11. The state of hybridization of the asterisked carbon in $CH_3–CH = C^* = CH_2$ is :

(A) sp (B) sp^2 (C) sp^3 (D) none of these

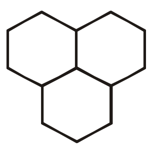
12. Which of the following is unsaturated hydrocarbon ?

(A)  (B) 

(C) 

(D) 

13. How many 3° carbon atoms are present in the given compound ?



(A) 4 (B) 3 (C) 2 (D) 1

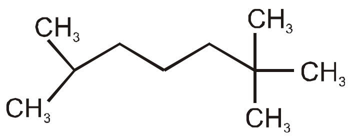
14. Ketene $CH_2 = C = O$ has

(A) Only sp^2 carbon atom
(B) Only sp carbon atom
(C) sp^2 and sp carbon atoms
(D) sp^3, sp^2 and sp carbon atoms

15. Which of the following is third member of C_nH_{2n-2} homologous series ?

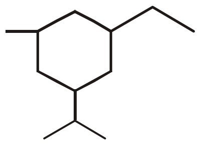
(A) C_2H_2 (B) C_3H_4 (C) C_4H_6 (D) C_4H_8

16. Number of secondary H in the following compound



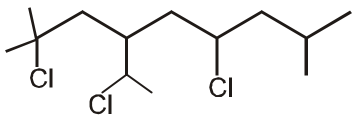
(A) 4 (B) 6 (C) 8 (D) 10

17. Number of tertiary H in the following compound



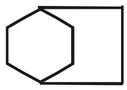
(A) 4 (B) 6 (C) 7 (D) 8

18. Number of secondary halogen in the given compound are



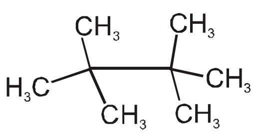
(A) 2 (B 3 (C) 3 (D) 4

19. Number of tertiary carbon is the given compound are



(A) 1 (B) 2 (C) 3 (D) 4

20. Number of primary H is the given compound are



(A) 10 (B) 18 (C) 14 (D) 18

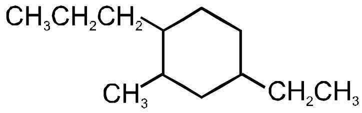
21. In the organic compound $\overset{1}{C}H_2 = \overset{2}{C}H - \overset{3}{C}H_2 - \overset{4}{C}H_2 - \overset{5}{C}H = \overset{6}{C}H_2$, the pair of hybridised orbitals involved in the formation of C_2–C_3 bond is :

(A) sp–sp^2 (B) sp–sp^3 (C) sp^2–sp^3 (D) sp^3–sp^3

22. Which of the following has 3º carbon.

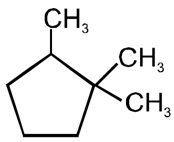
(A) $CH_3-\underset{NH_3}{\underset{|}{CH}}-CH_2-CH_3$ (B) $CH_3-\overset{CH_3}{\overset{|}{\underset{\underset{CH_3}{|}}{C}}}-OH$ (C)  (D) $CH_3-\overset{CH_3}{\overset{|}{\underset{\underset{CH_3}{|}}{C}}}-CH_3$

1. The IUPAC name of the above compound ?



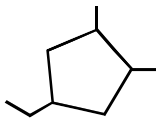
(A) 5-ethyl-1-methyl-2-propylcyclohexane (B) 1-ethyl-3-methyl-4-propylcyclohexane
(C) 4-ethyl-2-methyl-1-propylcyclohexane (D) 1-propyl-3-ethyl-2-methylcyclohexane

2. What will be the IUPAC name of the above compound ?



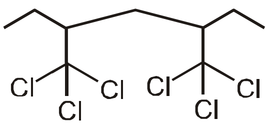
(A) 1,1,2-trimethylcyclopentane (B) 1,2,2-trimethylcyclopentane
(B) 1,1,5-trimethylcyclopentane (D) None of the above

3. Select the correct IUPAC name of the following compound



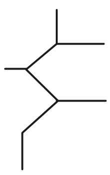
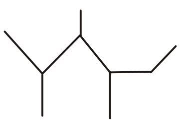
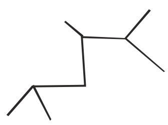
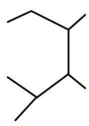
(A) 3-ethyl-1,5-dimethylcyclopentane (B) 1-ethyl-3,4-dimethylcyclopentane
(C) 4-ethyl-1,2-dimethylcyclopentane (D) 2-ethyl-4,5-dimethylcyclopentane

***4.** The correct IUPAC name of the following compound is :

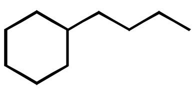
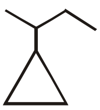
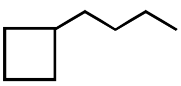
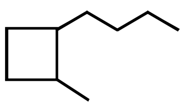


(A) 1,1,1,5,5,5-Hexachloro-2,4-diethylpentane. (B) 3,5-Hexachlorodimethylheptane.
(C) 3,3,3,5,5,5-Hexachloromethylheptane. (D) 3,5-Bis(trichloromethyl)heptane.

5. The correct structure of 2,3,4-Trimethylhexane is

(A)  (B)  (C)  (D) 

6. In which of the following cyclic ring is a parent chain

(A)  (B)  (C)  (D) 

7. Which of the following are used as prefix :
(A) Bromo (B) Nitro (C) Methoxy (D) Sulphonic acid

***8.** The name of $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-\overset{\overset{CH_3}{|}}{CH}-CH_3$ is/are

(A) Isooctane (B) 2,2,4- Trimethylpentane

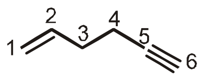
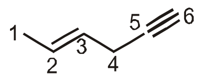
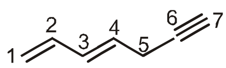
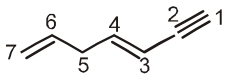
(C) Neoocatne (D) Isoheptane

Comprehensions

Paragraph for Question Nos. 9 to 10

While naming unsaturated hydrocarbons, lowest locant is alloted for multiple bond than any substituent, since the priorty of multiple bond is more than any substituent present. When the locant for unsaturated is same from either side then double bond is given priority over triple bond. For main chain selection if the size of main chain and number of unsaturation are equal, then priority is given to the lowest set of locants for unsaturation.

***9.** Which of the following represents incorrect numbering ?

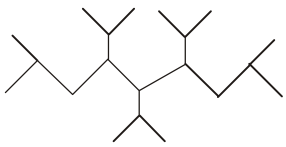
(A)  (B)  (C)  (D) 

10. The structure of 4-Methylpent-2-ene is :

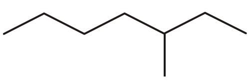
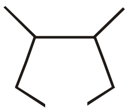
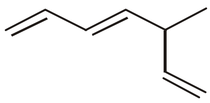
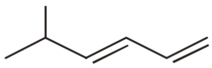
(A) $(CH_3)_2CH-CH_2CH=CH_2$ (B) $(CH_3)_2CH-CH=CH-CH_3$

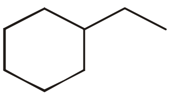
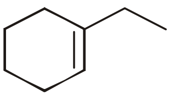
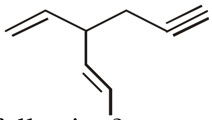
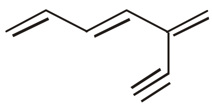
(C) $(CH_3)_2CH-CH_2CH=CH-CH_3$ (D) $(CH_3)_2C=CHCH_2CH_3$

11. How many isopropyl groups are present in the parent chain ?

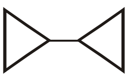
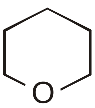
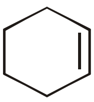
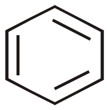


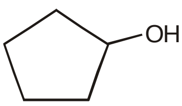
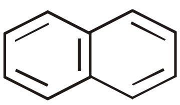
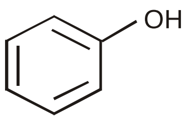
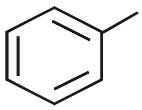
12. How many parent chains have root word hept or hepta in the following structures ?

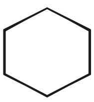
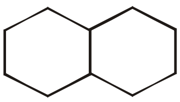
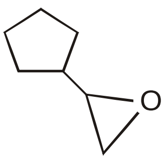
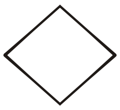
   

13. How many compounds are hydrocarbon in the following?

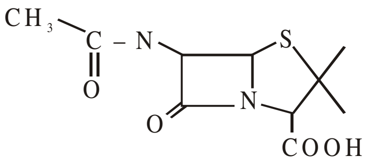
   

NCERT Questions: Pg 361, (12.4, 12.5)

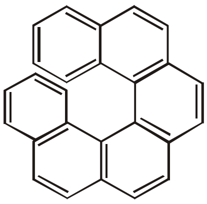
***1.** Match the column

Column-I (Compound) **Column-II (Double bond equivalent)**

(A) 

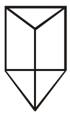
(penicillin)

(P) 4

(B) 

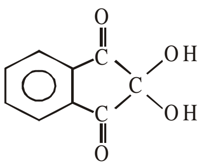
(Hexahelicene)

(Q) 5

(C) 

(Prismane)

(R) 7

(D) 

(Ninhydrin)

(S) 19

2. The correct IUPAC name of the compound is :

$$CH_3-CH_2-CH_2-\underset{\substack{|\\ CH(CH_3)_2}}{CH}-\overset{\substack{CH_3\\ |}}{CH}-C(C_2H_5)_3$$

(A) 3,3-Diethyl-4-methyl 5-(1-methyl ethyl) octane

(B) 6,6-Diethyl-4-methyl-5-isopropyloctane

(C) 6,6-Diethyl-3-methyl 5-(1-methylethyl) octane

(D) 6,6-Diethyl-4-isopropyl-5-methyloctane

***3.** IUPAC name of the compound

$$CH_3-CH_2-\underset{\substack{|\\ CH_3}}{CH}-CH_2-\underset{\substack{|\\ CH-CH_3\\ |\\ CH_3}}{CH}-CH_2CH_2CH_3$$

(A) 4-Isopropyl-6-methyloctane

(B) 3-Methyl-5-(1-methylethyl) octane

(C) 3-Methyl-5-isopropyloctane

(D) 6-Methyl-4-(1-methylethyl) octane

4. The IUPAC name of the crotonaldehyde is: (CH_3–CH = CH–CHO) is -

(A) 2-Propenal (B) Propenal (C) 2-Butenal (D) Butenals

5. Which of the following compound has wrong IUPAC name?

(A) $CH_3-CH_2-CH_2-COOCH_2CH_3$

Ethyl butanoate

(B) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CHO$

3-Methylbutanol

(C) $CH_3-\underset{\underset{OH}{|}}{CH}-\underset{\underset{CH_3}{|}}{CH}-CH_3$

2-Methyl-3-butanol

(D) $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{O}{||}}{C}-CH_2-CH_3$

2-Methyl-3-pentanone

6. C_5H_{12} has a symmetrical structure with one quaternary carbon. Its IUPAC name is :

(A) n-pentane (B) 2-methylbutane

(C) 2,2-dimethylpropane (D) 3-methylbutane

7. Which of the following compound has all four type (1°,2°,3°, 4°) of carbon atoms :

(A) 2,3,4-Trimethylpentane (B) 2,2,3-Trimethylpentane

(C) neo-pentane (D) none of the three

8. In following compound

$$CH_3-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-\underset{\underset{CH_3}{|}}{CH}-\overset{\overset{CH_3}{|}}{CH}-CH_2-CH_3$$

The correct lowest set of locant is :

(A) 3,3,4,5 (B) 3,4,5,5 (C) 4,5,3,3 (D) 5,5,4,3

9. The IUPAC name of

$$CH_3-\underset{\underset{Cl}{|}}{C}=\underset{\underset{CH_3}{|}}{C}-\underset{\underset{C_2H_5}{|}}{CH}-CH_2-C\equiv CH$$

(A) 2-Chloro-4-ethyl-3-methyl-6-heptyn-2 ene (B) 2-Chloro-4-ethyl-3-methyl-2-heptyn-6 yne

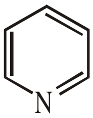
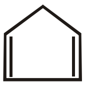
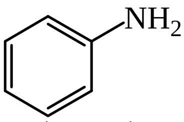
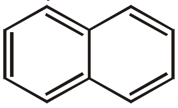
(C) 6-Chloro-4-ethyl-5-methyl-1-heptyn-5 ene (D) 6-Chloro-4-ethyl-5-methyl-5-hepten-1 yne

10. IUPAC name of $CH_3-\overset{\overset{H}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{C_4H_9}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3$ is:

(A) 3,4,4- Trimethylheptane (B) 4-Ethyl-3,4-dimethyl octane

(C) 2-Butyl-2-methyl-3-ethylbutane (D) None of these

1. Match the column:

Column I (Compounds)		**Column II (Property)**	
(A)		(P)	Homocyclic compound
(B)		(Q)	Homocyclic hydrocarbon
(C)		(R)	Heterocyclic
(D)		(S)	Even number of π-bond
		(T)	Odd number of σ-bond

***2.** **Statement 1 :** 3-Bromo-4-ethyl pentanoytl bromide is not the correct IUPAC name

Statement 2 : Selection of carbon chain in 3-Bromo-4-ethyl pentanoyl bromide is not according to IUPAC rule.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

***3.**

$$\begin{array}{c} \qquad\qquad\qquad\qquad\qquad\qquad\quad CH_2-CH_3 \\ \qquad\qquad\qquad\qquad\qquad\qquad\quad | \\ CH_3-CH_2-CH_2-CH_2-\underset{\displaystyle CH_3-\overset{|}{\underset{\underset{\displaystyle CH_3}{|}}{C}}-CH_3}{\overset{}{\underset{|}{CH}}}-CH-CH_2-CH_2-CH_2-CH_3 \end{array}$$

(A) 5-(1, 1-Dimethyl Ethyl)-6-Ethyl decane (B) 5-Ethyl-6-(1, 1-Dimethyl Ethyl) decane
(C) 6-Ethyl-5-(1, 1-Dimethyl Ethyl) decane (D) 6-(1, 1-Dimethyl Ethyl)-5 Ethyl decane

4. IUPAC name of $CH_3-CH=CH-C\equiv CH$ is:
(A) 2-Penten-4-yne (B) 1-Pentyn-3-ene (C) 3-Penten-1-yne (D) None of these

5. The incorrect IUPAC name is:

(A) $CH_3-\underset{\underset{\displaystyle O}{||}}{C}-CH(CH_3)_2$
2-Methyl-3-butanone

(B) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle CH_2-CH_3}{\underset{|}{CH}}-CH_3$
2,3-Dimethylpentane

(C) $CH_3-C\equiv C-CH(CH_3)_2$
4-Methyl-2-pentyne

(D) $CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$
3-Chloro-2-bromobutane

***6.** Find out the correct statement(s) about given molecule.

$CH_3 - C \equiv C - CH = CH - CH = CH_2$

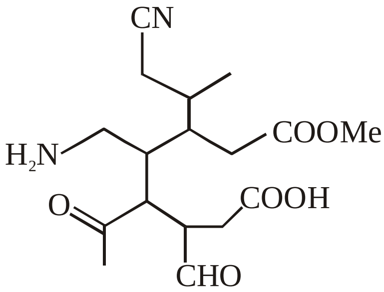
(A) All the carbons in given molecule are not in same plane
(B) IUPAC name is hepta-1,3-dien-5-yne
(C) It has 4 sp^2 hybrid carbons, 2 sp hybrid carbons and 1 sp^3 hybrid carbon
(D) All the carbons in given molecule are in same plane

7. IUPAC name of given organic compound:

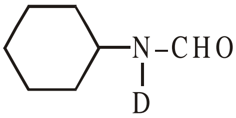
$(CH_3)_2C(CH_2CH_3)CH_2CH(Cl)CH_3$ is:

(A) 5-Chloro-3,3-dimethylhexane (B) 5-Chloro-2-ethyl-2-methylpentane
(C) 2-Chloro-4-ethyl-4-methylpentane (D) 2-Chloro-4,4-dimethylhexane

8. Number of functional groups and double bond equivalent are respectivily.



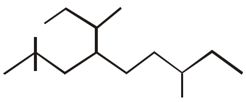
(A) 6, 1 (B) 6, 5 (C) 6, 6 (D) 6, 7

9. IUPAC name of 

is

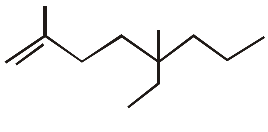
(A) N-deutero-N-formylcyclohexanamine (B) N-cyclohexylamino-N-deuteromethanal
(C) N-cyclohexyl-N-deuteromethanamide (D) N-deutero cyclohexanaminal

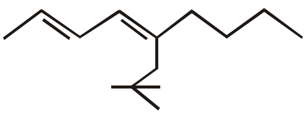
***10.** Provide the IUPAC name of following compounds :

(1) 

(2)
CH_3
H_3C-CH
$CH_3-(CH_2)_2-CH-CH-CH_2-CH_2-CH_3$
$CH_2-CH_2-CH_3$

(3)
CH_3
H_3C-CH C_2H_5
$CH_3-CH_2-CH_2-CH-CH-C-CH_2-CH_3$
CH_3 C_2H_5

(4) 

(5) 

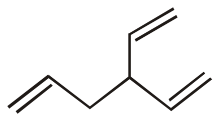
***1.** Match the column

Column I (Compound)	**Column II (Number of carbons in selected carbon chain for IUPAC nomenclature)**
(A) $CH_2(OH)-CH(CH=CH_2)-CH_2-COOH$	(P) 4
(B) $CH_3-CH(CH_2-CH_3)-CH_2-C(CH_2CH_3)=CH_2$	(Q) 5
(C) $CH_3-CH_2-CH(CHO)-CH_2-CH_2-CH_2-COOH$	(R) 6
(D) $CH_3-CH_2-CH_2-CH(CH_2COOH)-COOH$	(S) 7

2. The DBE in bezoic acid is :

(A) 3 (B) 4 (C) 5 (D) 6

3. IUPAC name for compound

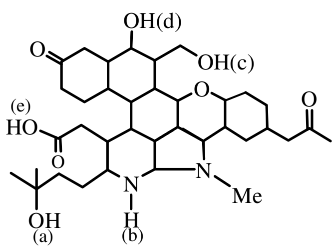
 is :

(A) 3-Ethenyl hexa-1,5-diene (B) 3-Propenyl penta-1,4-diene

(C) 4-Ethenyl hexa-1,5-diene (D) None of these

Paragraph for Question 04 and 05

Observe following compound and answer questions given below :



4. Total number of different types of functional gorups in this compound are

(A) 5 (B) 6 (C) 7 (D) 8

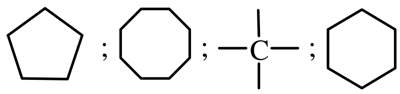
5. Degree of unsaturation of this compound is

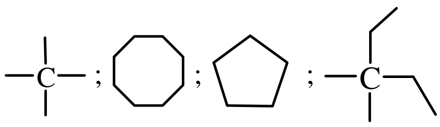
(A) 8 (B) 9 (C) 10 (D) 11

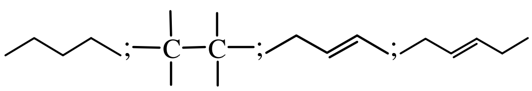
6.

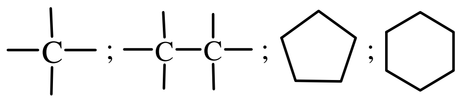
Compounds	**Property**
C_5H_{12} (A)	All hydrogens are identical & 1°.
C_8H_{18} (B)	All hydrogens are identical & 1°.
C_5H_{10} (C)	All hydrogens are identical & 2°.
C_6H_{12} (D)	All hydrogens are identical & 2°.

A,B,C,D are respectively :

(A) 

(B) 

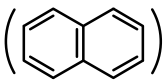
(C) 

(D) 

7. Write correct IUPAC names of following compounds

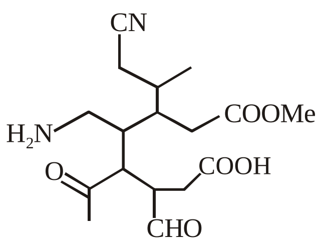
(i) 2-Ethyl-5-hexyne (ii) 1-amino-3-isopropylpentan-1-one

(iii) 1-methyl-2-vinyl cyclohex-1(6)-ene

8. Find the total number of sigma bond(s) in naphthalene 

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

9. How many functional groups are presents.



10. $$\begin{array}{c} \quad\quad\quad\quad\quad CH_3\;\; CH_3 \\ CH_3-CH_2-\underset{\displaystyle CH_3}{CH}-CH-\underset{\displaystyle CH_3}{C}-CH_3 \end{array}$$ compound (M)

(a) No. of 1° carbon atoms in compound (M) is X

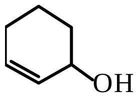
(b) No. of 2° carbon atoms in compound (M) is Y

(c) No. of 3° carbon atoms in compound (M) is Z

Write answer of part (a), (b) & (c) in the same order and present the three digit number XYZ as your answer in OMR sheet. for example : If all these answers are 9 (X = 9, Y = 9, Z =9) then fill 999 in OMR sheet.

NCERT Questions: Pg 362, (12.8)

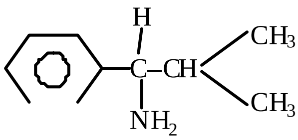
RACE # 59 — IUPAC NOMENCLATURE & ISOMERISM — CHEMISTRY

1. The IUPAC name of  is -

(A) 3-hydroxycyclohexene (B) cyclohexen-3-ol

(C) 2-cyclohexenol (D) 2-ene-cyclohexan-1-ol

2. The IUPAC name of the compound is :



(A) 1-amino-1-phenyl-2-methyl propane (B) 1-amino-2-methyl-1-phenyl propane

(C) 2-methyl-1-amino-1-phenyl propane (D) 1-isopropyl-1-phenyl methyl amine

3. The IUPAC name of $BrCH_2-\underset{\underset{CONH_2}{|}}{CH}-CO-CH_2-CH_2CH_3$ is :

(A) 2-bromo methyl-3-oxo hexanamide (B) 1-bromo-2-amido-3-oxo hexane

(C) 1-bromo-2-amido-*n*-propyl ketone (D) 3-bromo-2-proponyl propanamide

4. The IUPAC name of $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-COOC_2H_5$ is

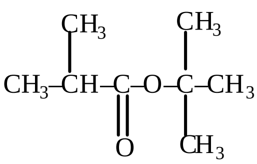
(A) 2-ethyl-ethyl acetate (B) ethyl 3-methyl butanoate

(C) ethyl 2-methyl butanoate (D) 2-methyl butanoic acid ethylester

5. The IUPAC name of $N \equiv C-CH_2-CH_2-OH$ is;

(A) 1-hydroxy ethanenitrile (B) 3-hydroxy propanenitrile

(C) 2-hydroxy ethyl cyanide (D) 1-hydroxy-2-cyanoethane

6. 

The common name of given ester is -

(A) neo butyl iso butyrate (B) t-butyl n- butyrate

(C) t- butyl iso butyrate (D) iso butyl iso butyrate

7. Ethyl methyl vinyl amine has the structure -

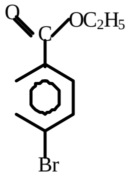
(A) $CH_3CH_2-\underset{\underset{CH_3}{|}}{N}-CH_2CH=CH_2$ (B) $CH_3CH_2-\underset{\underset{CH_3}{|}}{N}-CH=CH_2$

(C) $CH_2=CH-\underset{\underset{CH_3}{|}}{N}-CH=CH_2$ (D) $CH_3-\underset{\underset{CH_3}{|}}{N}-CH=CH_2$

8. Which of the following compound's prefix 'iso' is not correct –

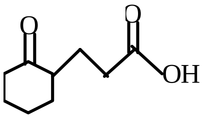
(A) Iso pentane (B) Iso Hexane (C) Iso butane (D) Iso octane

9. Give IUPAC name of the following :



(A) Ethyl-1-bromo 4-benzoate (B) Ethyl-4-bromobenzoate

(C) Bromo benzoate (D) Bromo benzene ester

10. The IUPAC name of  is -

(A) 2-carboxyethylcyclohexanone (B) 3-(2-oxocyclohexyl) propanoic acid

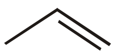
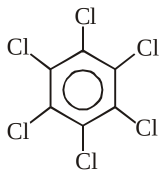
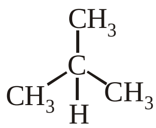
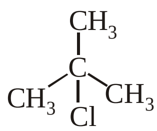
(C) cyclohexanone-2-propanoic acid (D) 2-propanoic cyclohexanone

11. The IUPAC name of $CH_3CH_2-\underset{\displaystyle CH_3}{\underset{|}{N}}-CH_2CH_3$ is:

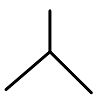
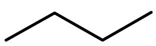
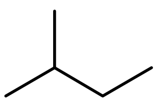
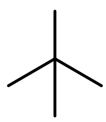
(A) N-methyl-N-ethylethyl amine (B) diethyl methyl amine

(C) N-ethyl-N-methyl ethan amine (D) methyl diethyl amine

12. Compound which can show position isomerism ?

(A)  (B)  (C)  (D) 

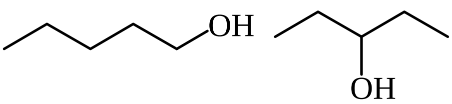
13. The structural isomers of isopentane is :-

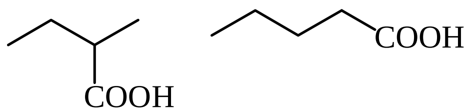
(A)  (B)  (C)  (D) 

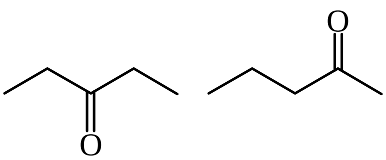
14. How many cyclic structural (excluding bicyclo) isomers are possible from the molecular formula C_4H_6 ?

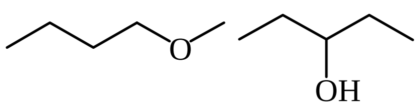
(A) 2 (B) 3 (C) 4 (D) 5

15. Which of the following represent functional isomers ?

(A) 

(B) 

(C) 

(D) 

16. How many structural isomers are possible for molecular formula $C_4H_{10}O$?

(A) 3 (B) 4 (C) 7 (D) 10

17. Which one of the following is a functional isomer of $CH_3–CH_2–CH_2–CH_2–NH_2$?

(A) $CH_3\text{-}CH_2\text{-}\underset{\displaystyle H}{\underset{|}{N}}\text{-}CH_2\text{-}CH_3$

(B) $CH_3\text{-}\overset{\displaystyle CH_3}{\overset{|}{C}}H\text{-}CH_2\text{-}NH_2$

(C) $CH_3\text{-}\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}\text{-}NH_2$

(D) $CH_3\text{-}CH_2\text{-}\underset{\displaystyle CH_3}{\underset{|}{N}}\text{-}CH_2\text{-}CH_3$

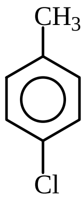
18. Which of the following represent metamer of $CH_3–CH_2–O–CH_2–CH_3$?

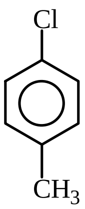
(A) $CH_3–CH_2–CH_2–CH_2OH$

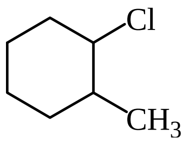
(B) $CH_3\text{-}\underset{\displaystyle CH_3}{\underset{|}{C}}H\text{-}CH_2\text{-}OH$

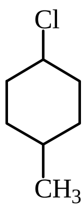
(C) $CH_3–O–CH_2–CH_2–CH_3$

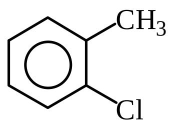
(D) $CH_3–CH_2–O–CH_2–CH_2–CH_3$

19. Which of the following is positional isomer of  ?

(A) 

(B) 

(C) 

(D) 

20. $CH_3–CH_2–CH_2–CHO$ & $CH_3\text{-}\underset{\displaystyle CHO}{\underset{|}{C}}H\text{-}CH_3$ are

(A) Position isomer (B) Chain isomer (C) Metamer (D) Functional isomer

21. Number of structure isomer of molecular formula C_5H_{10} having one π-bond.

(A) 2 (B) 3 (C) 5 (D) 4

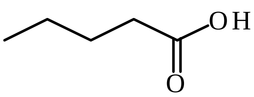
22. In which of the following, functional group isomerism is not possible ?

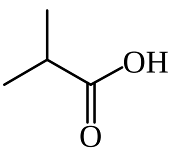
(A) Aldehyde (B) Alcohols (C) Alkyl halides (D) Cyanides

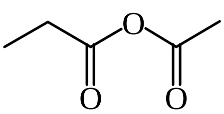
23. Which compound never exist ?

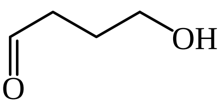
(A) Iso-octane (B) Neo-pentane (C) Iso-propane (D) n-butane

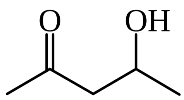
24. Compound which is functional isomer of :



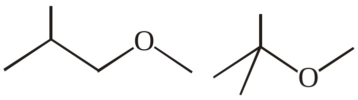
(A) 

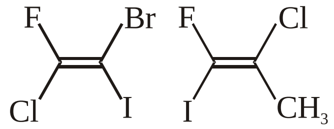
(B) 

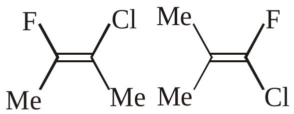
(C) 

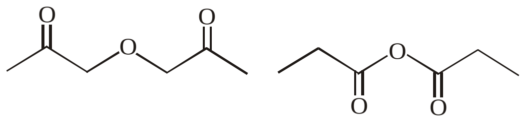
(D) 

25. Select pair of position isomers.

(A) 

(B) 

(C) 

(D) 

26. How many amines are possible with the molecular formula $C_4H_{11}N$ (structural isomers only) ?

(A) 6 (B) 7 (C) 8 (D) 9

27. How many benzenoid isomers are possible for cresol ?

(A) 3 (B) 4 (C) 5 (D) 6

28. How many benzenoid isomers are possible for $C_6H_3(Cl)(Br)(I)$?

(A) 6 (B) 8 (C) 9 (D) 10

29. How may total structural isomers of 3° amines are possible with the molecular formula $C_6H_{15}N$?

(A) 4 (B) 7 (C) 6 (D) 5

30. The total no. of carboxylic acids and esters having molecular formula $C_4H_8O_2$

(A) 3 (B) 4 (C) 5 (D) 6

31. An isomer of ethanol is –

(A) Methanol (B) Diethyl ether (C) Acetone (D) Dimethyl ether

32. What is the total no. of structural isomers with the formula C_3H_6O, that are cyclic

(A) 3 (B) 4 (C) 6 (D) 7

1. A position isomer of butyne-1 is -

(A) Butene-1 (B) Butene-2 (C) Butyne-2 (D) Buta-1, 3-diene

2. Which of the following is an isomer of propanal ?

(A) Ethanol (B) Acetone (C) Propanol (D) Popanoic acid

3. Compounds $CH_3NHC_3H_7$ and $C_2H_5NHC_2H_5$ exhibit –

(A) Geometrical isomerism (B) Optical isomerism

(C) Position isomerism (D) Metamerism

4. Which type of isomerism is shown by diethyl ether and methyl propyl ether ?

(A) chain (B) function (C) metamerism (D) position

5. How many primary amines are possible for the formula $C_4H_{11}N$?

(A) 1 (B) 2 (C) 3 (D) 4

6. C_7H_7Cl shows how many isomers ?

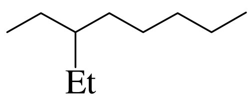
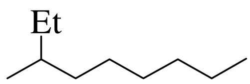
(A) 4 (B) 5 (C) 3 (D) 2

7. Which of the following are chain isomers :

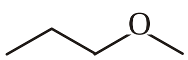
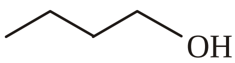
(A) $CH_3CH_2CH_2-Cl$ & $CH_3-\overset{\overset{\displaystyle CH_3}{|}}{C}H-Cl$

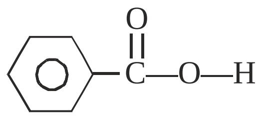
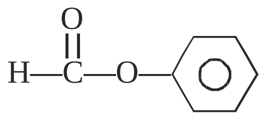
(B) $CH_3CH_2CH_2-CN$ & $CH_3-\overset{\overset{\displaystyle CN}{|}}{C}H-CH_3$

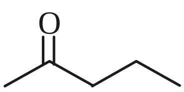
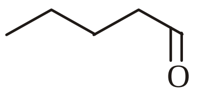
(C) $CH_3CH_2CH_2CH_3$ & $CH_3-\underset{\underset{\displaystyle CH_3}{|}}{C}H-CH_3$

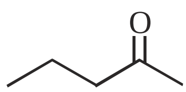
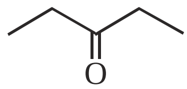
(D)  & 

8. One of the following is not the pair of functional isomers.

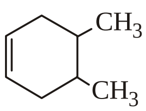
(A)  and 

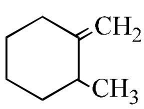
(B)  and 

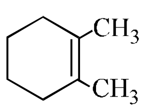
(C)  and 

(D)  and 

9. Write increasing order of heat of hydrogenation :

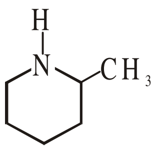
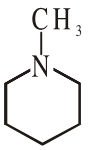

(i)


(ii)


(iii)

10. How many total number of structural isomer are possible for C_4H_7Cl having parent chain of four carbon.

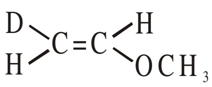
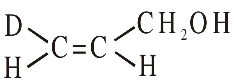
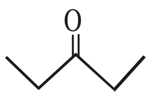
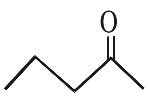
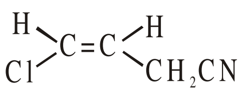
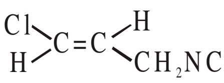
11. Identify the relationship between the given compound :

 and 

12. Which of the following is correct matchings?

Column –I	**Column -II**
(A) $CH_3-\overset{\overset{O}{\parallel}}{C}-OH$ & $H-\overset{\overset{O}{\parallel}}{C}-OCH_3$	metamers
(B) $CH_3-CH_2-C\equiv CH$ & $CH_3-C\equiv C-CH_3$	position isomers
(C) $CH_3CH_2CH_2NH_2$ & $CH_3-\underset{\underset{NH_2}{\vert}}{CH}-CH_3$	tautomers
(D) CH_3CH_2OH & $(CH_3)_2O$	tautomers

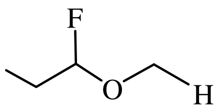
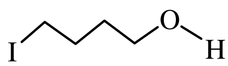
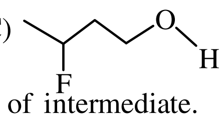
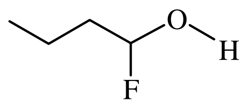
13. Match the column :

Column-I (Compound)	**Column-II (Isomerism)**
(A)  and 	(P) Functional isomers
(B)  and 	(Q) Geometrical isomers
(C)  and 	(R) Position isomers
(D)  and 	(S) Chain isomers
	(T) Metamer

1. Match the column :

Column-I	**Column-II**
(A) Carbocation $\overset{\oplus}{C}H_3$	(P) Lewis acid
(B) Carbanion $\overset{\ominus}{C}H_3$	(Q) Lewis base
(C) Carbon free radical $\overset{\bullet}{C}H_3$	(R) Electron deficient central atom
(D) Ammonia $\overset{\bullet\bullet}{N}H_3$	(S) sp^2 hybridisation
	(T) sp^3 hybridisation

2. Which is most acidic ?

(A)  (B)  (C)  (D) 

3. Which of the following is correct order of stability of intermediate.

(A) $Me_3C^\ominus < Me_2CH^\ominus < Me-CH_2^\ominus$ (B) $Me-CH_2^\oplus < Me_2CH^\oplus < Me_3C^\oplus$

(C) $Me-\dot{C}H_2 < Me_2\dot{C}H < Me_3C^\bullet$ (D) $Me-O^\ominus < MeCH_2-O^\ominus < Me_2CHO^\ominus < Me_3C-O^\ominus$

Passage for Q.04 to Q.06

In acid mixture stronger acid acts as acid, weaker acid acts as base, so in a mixture of HNO_3 & H_2SO_4 ; H_2SO_4 acts as acid & HNO_3 acts as base where as when HNO_3 reacts with H_2O, HNO_3 acts as acid. So same compound can act as acid or base depending on which compound it react with. If an acid mixture HA_1, HA_2, HA_3 are present & Acidity order $HA_1 > HA_2 > HA_3$, then

***4.** Reaction when HA_1 & HA_2 mixed with each other -

(A) $HA_1 + HA_2 \longrightarrow H^+ + A_1^- + H^+ + A_2^-$ (B) $HA_1 + HA_2 \longrightarrow H_2A_2^+ + A_1^-$

(C) $HA_1 + HA_2 \longrightarrow H_2A_1^+ + A_2^-$ (D) $HA_1 + HA_2 \longrightarrow H^+ + A_1^+ + H^- + A_2^-$

***5.** Reaction when HA_2 & HA_3 mixed with each other -

(A) $HA_2 + HA_3 \longrightarrow H^+ + A_2^- + H^+ + A_3^-$ (B) $HA_2 + HA_3 \longrightarrow H^- + A_2^+ + H^- + A_3^+$

(C) $HA_2 + HA_3 \longrightarrow H^+ + A_2^+ + H^- + A_3^-$ (D) $HA_2 + HA_3 \longrightarrow A_2^- + H_2A_3^+$

6. HA_2 during reaction with HA_1 & HA_3 acts respectively as -

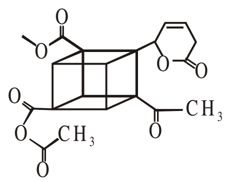
(A) base & acid (B) acid & base (C) acid & acid (D) base & base

7. Which of the following is/are correct IUPAC name -

(A) 1,4-dichloro cyclopentane (B) 1,4-dichloro-4-ethyl pentane

(C) 5-bromo-2-chloro hexane (D) 5-ethyl-2-methyl heptane

8. The degree of unsaturation in compound is:



(A) 10 (B) 12 (C) 13 (D) 15

9. How many of the following are –I effect (negative inductive effect) showing groups.

(a) $-NO_2$ (b) $-CO_2H$ (c) $-CH_3$ (d) $-\overset{\oplus}{N}H_3$

(e) $-O^-$ (f) –CN (h) $-CMe_3$ (i) –Cl

(j) –H

10. The number of groups showing +I effect = X.

The number of groups showing –I effect = Y.

Find value XY e.g. if X = 3, Y = 4. Answer is 034

groups are given as follows :

(a) —Cl (b) $-\overline{N}H$ (c) $-\underset{\underset{O}{\|}}{C}-OH$ (d) $-N\!<$

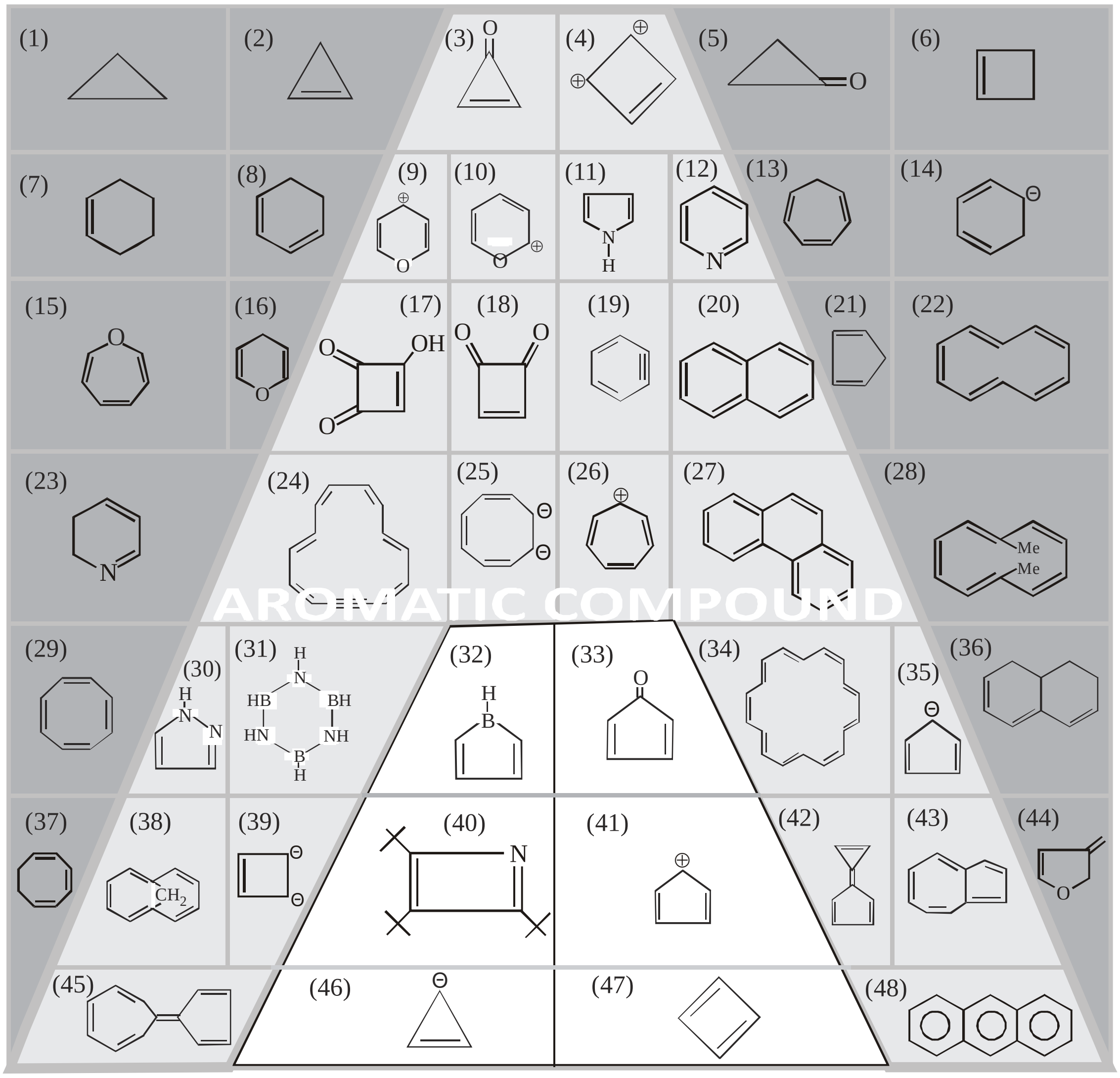
(e) $-CH\!<$ (f) $-\underset{\underset{O}{\|}}{C}-O^-$ (g) $-CCl_3$ (h) $-\overset{\oplus}{P}H_3$

(i) –OH

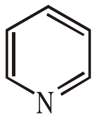
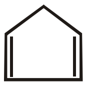
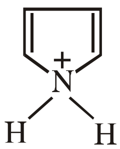
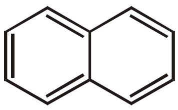
Paragraph Type :

	Aromatic		Anti-aromatic	Non-aromatic compounds
(1)	Cyclic	(1)	Cyclic	
(2)	Planar	(2)	Planar	
(3)	Complete conjugation over the entire cycle	(3)	Complete conjugation over the entire cycle	If any one of the four or more violates then non-aromatic
(4)	Huckel's rule $(4n+2)\pi$ electrons should be in cyclic conjugation $(4n+2)\ \pi$ electrons $= 2,6,10,14$ etc	(4)	$(4n)\pi$ electrons should be in cyclic conjugation $(4n)\pi$ electrons $= 4,8,12$,etc.	

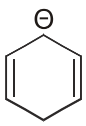
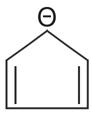
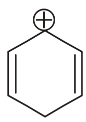
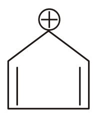
Q. Darken the complete area by pencil, having aromatic compounds in it.



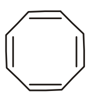
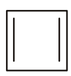
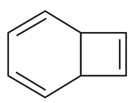
1. Match the column:

Column I (Compounds)	**Column II (Properties)**
(A) 	(P) Aromatic
(B) 	(Q) Nonaromatic
(C) 	(R) Heterocyclic
(D) 	(S) Even number of π-bond
	(T) Odd number of σ-bond

2. Which of the following is aromatic compound?

(A)  (B)  (C)  (D) 

3. Which of the following is anti-aromatic compound.

(A)  (B)  (C)  (D) None of these

4. HO_2C–CH=CH–OH, the direction of electron movement would be :

(A) toward –OH group

(B) neither –OH nor towards –CO_2H

(C) toward –CO_2H group

(D) at low temperature –OH group and at high temperature toward –CO_2H group

5. Polarisation of electron in acrolein (CH_2=CH–CH=O) or R.H. can be written as :

(A) $\overset{\partial^+}{CH_2}$ ==== CH ==== CH ==== $\overset{\partial^-}{O}$

(B) $\overset{\partial^-}{CH_2}$ ==== CH ==== CH ==== $\overset{\partial^+}{O}$

(C) $\overset{\partial^-}{CH_2}$ ==== CH — CH ==== $\overset{\partial^-}{O}$

(D) $\overset{\partial^+}{CH_2}$ ==== CH ==== CH ==== $\overset{\partial^+}{O}$

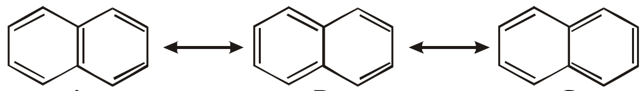
Paragraph for Q.06 to Q.08

Determine the stability of resonating structure, the following points are considered

(i) Resonating structure with complete octet is more stable than resonating structure similar with incomplete octet.

(ii) Resonating structure with negative charge on more electronegative element is more stable with respect to resonating structure with negative charge on less electronegative elements.

(iii) Resonating structure with benzonoid structure is more stable with respect to resonating structure with non-benzonoid structure

6. 

A B C

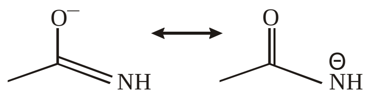
The correct order of stability -

(A) A > B > C (B) A < B < C (C) A = B = C (D) A > B = C

7. $\underset{\text{I}}{H_3C{-}C{\equiv}\overset{+}{O}} \longleftrightarrow \underset{\text{I'}}{H_3C{-}\overset{+}{C}{=}O}$; $\underset{\text{II}}{H_2\overset{+}{C}{-}O{-}CH_3} \longleftrightarrow \underset{\text{II'}}{CH_2{=}\overset{+}{O}{-}CH_3}$

The correct order of stability -

(A) I = I' , II = II' (B) I > I' , II > II' (C) I > I' , II' > II (D) I' > I , II' > II

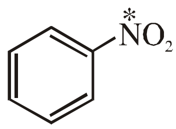
8.  I II ; $\underset{\text{I'}}{CH_2{=}CH{-}O^-} \longleftrightarrow \underset{\text{II'}}{\overset{\ominus}{C}H_2{-}CH{=}O}$ **[3]**

The correct order of stability -

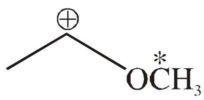
(A) I > II , I' > II' (B) I < II , I' < II' (C) I = II , I' = II' (D) I > II , II' > I'

9. How many compounds having marked group show both –I & + R / + M effects.

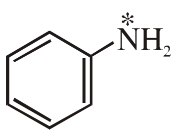
(a) $CH_3{-}CH{=}CH{-}\overset{*}{C}HO$

(b) 

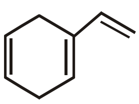
(c) $H_2C{=}CH{-}\overset{*}{F}$

(d) 

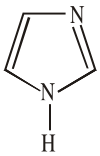
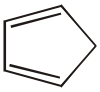
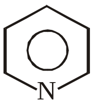
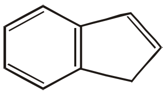
(e) $H_3C{-}CH{=}CH{-}\overset{*}{C}H_3$

(f) 

10. Number of π electrons in conjugation.



1. Match the column-

Column-I (Compounds)	**Column-II (Properties)**
(A) 	(P) Aromatic
(B) 	(Q) Conjugate base is aromatic
(C) 	(R) Most acidic hydrogen in column-I
(D) 	(S) Most basic compound in column-I
	(T) Non-aromatic

2. **Statement 1 :** Correct order of acidic strength is $CH_3CH_3 > CH_2{=}CH_2 > HC{\equiv}CH$

Statement 2 : C–H bond energy in ethene is more than ethane but less than ethyne.

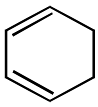
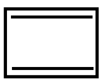
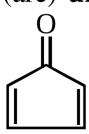
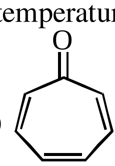
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

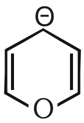
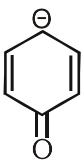
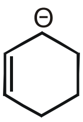
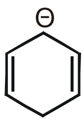
3. Which of the following molecules, in pure from , is (are) **unstable** at room temperature ?

(A)  (B)  (C)  (D) 

4. Which of the following group will increase the basicity of aniline when attached to its para position ?

(A) –OH (B) $-OCH_3$ (C) $-CH_3$ (D) $-NO_2$

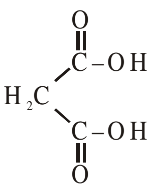
5. Identify most stable anion :

(A)  (B)  (C)  (D) 

6. Select true statement(s) :

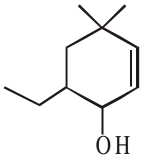
(A) Resonance affects bond length

(B) Inductive affect is a permanent effect.

(C) In  most acidic H is connected directly to oxygen not on carbon.

(D) In urea both C–N bond length are identical

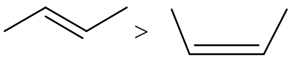
7. IUPAC name of compound is :

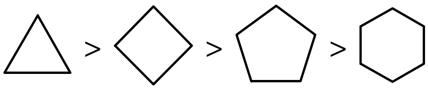


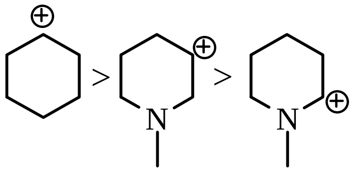
(A) 5-ethyl-3,3-dimethyl cyclohex-en-6-ol (B) 6-ethyl-4,4-dimethyl cyclohex-2-enol

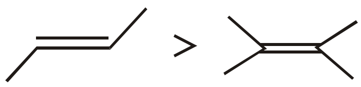
(C) 2-ethyl-4,4-dimethyl cyclohex-5-enol (D) 6-ethyl-4,4-dimethyl cyclohexenol

8. Which of the following are incorrect :

(A) Order of heat of hydrogenation 

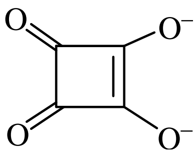
(B) Order of heat of combustion 

(C) Order of stability of carbocation 

(D) Order of heat of combustion is 

9. Which of the following options are correct :

(A) p chloro phenol is more acidic than p fluoro phenol

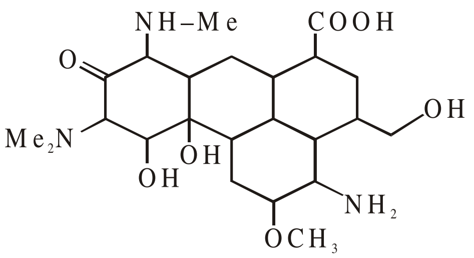
(B) Squaric acid dianion 

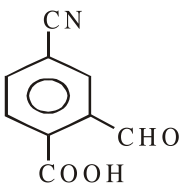
have identical R.S.

(C) Squaric acid dianion have all C–O bonds identical

(D) Squaric acid dianion have all C–C bonds identical

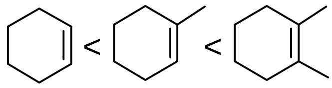
10. Number of different types of functional groups in the following compound is/are

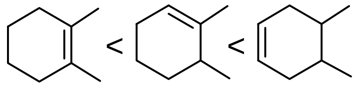
(A) 

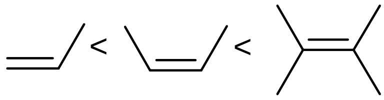
(B) Double bond equivalent for the given compound is 

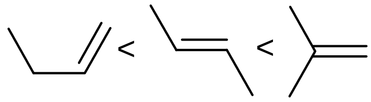
1. Match the column :

Column - I

(A) 

(B) 

(C) 

(D) 

Column - II

(P) Correct order of stability of alkene

(Q) Correct order of heat of combustion

(R) Correct order of heat of hydrogenation

(S) Correct order of C=C bond length

(T) Correct order of C=C bond energy

2. **Statement-1 :** $Me-\overset{\oplus}{C}H_2$ is more stable than $MeO-\overset{\oplus}{C}H_2$

Statement-2 : Me is a +I group where as MeO is a –I group.

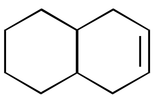
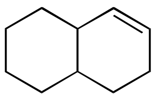
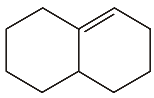
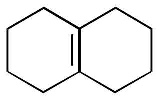
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false.

3. Arrange the given alkenes in order of stability.

(A) 
(B) 
(C) 
(D) 

4. Correct order of basic strength :

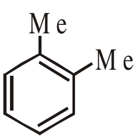
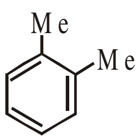
(A) $CH_3NH_2 > CH_3-\overset{\oplus}{N}H_3 > CH_3-\overset{\ominus}{N}H$

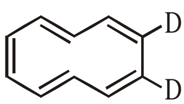
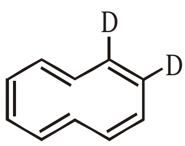
(B) $\overset{\ominus}{O}H > CH_3CO_2^{\ominus} > Cl^{\ominus}$

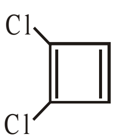
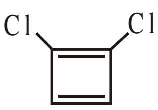
(C) $CH_3\overset{\ominus}{O} > CH_3\overset{\ominus}{N}H > CH_3-\overset{\ominus}{C}H_2$

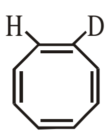
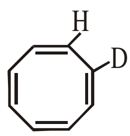
(D) $CH_3CH_2NH_2 > CH_2=CH-NH_2 > HC\equiv C-NH_2$

5. Which one of following represents different molecules ?

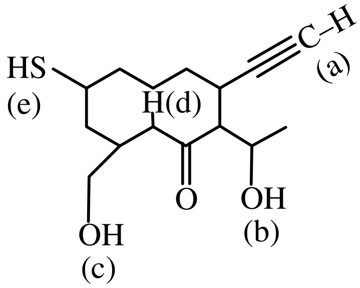
(A)  and 

(B)  and 

(C)  and 

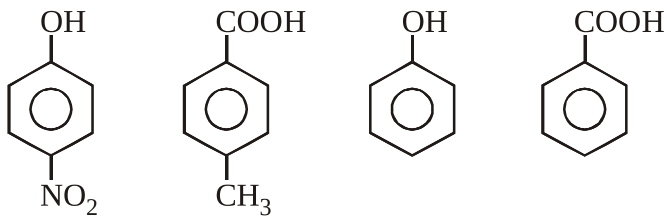
(D)  and 

6. Correct desending order of deprotonation in the following compound :



(A) e > d > c > b > a (B) e > d > c > a > b (C) e > c > b > d > a (D) e > c > b > a > d

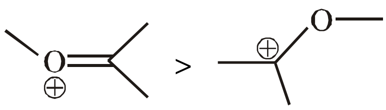
7. Acidic strength order of given compound

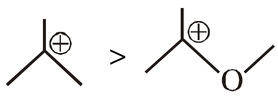


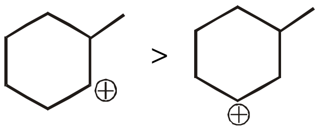
(I) (II) (III) (IV)

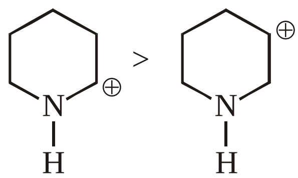
(A) I > IV > III > II (B) I > IV > II > III (C) IV > II > I > III (D) IV > I > II > III

8. Incorrect stability order is/are

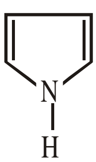
(A) 

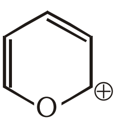
(B) 

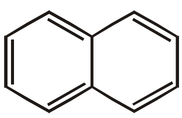
(C) 

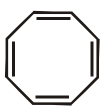
(D) 

9. Which of the following is / are aromatic species.

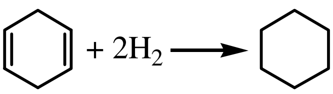
(A) 

(B) 

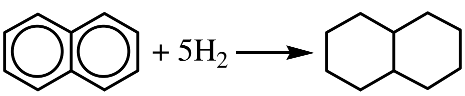
(C) 

(D) 

10. Analyse the following hydrogenation reactions and calculate resonance energy of 'X' in kcal/mole:



$\Delta H = -58$ kcal/mole



'X'

$\Delta H = -94$ kcal/mole

ANSWER KEY

RACE # 01

1. (D) **2.** (D) **3.** (C) **4.** (A) **5.** (C) **6.** (C)

7. (a) C (b) E (c) M (d) C (e) M (f) C (g) E (h) M (i) M (j) C

8. (D) **9.** (C) **10.** (A) **11.** (C) **12.** (B) **13.** (B)

14. (A)→P,T; (B)→QRT; (C)→QRT; (D)→P,R

15. (C) **16.** (C) **17.** (B) **18.** (D) **19.** (D) **20.** (C) **21.** (B) **22.** (B)

RACE # 02

1. (C) **2.** (C) **3.** (C) **4.** (A) **5.** (D) **6.** (D) **7.** (C) **9.** (D) **10.** (B) **11.** (C)

12. (B) **13.** (A) **14.** **15.** (C) **16.** (D) **17.** (B) **18.** (C) **19.** (D) **20.** (ABD)

21. (A)→P,Q; (B)→P; (C)→S; (D)→R

RACE # 03

1. (B) **2.** (A) **3.** (A) **4.** (C) **5.** (C) **6.** (A) **7.** (B) **8.** (A) **9.** (D) **10.** (B)

11. (B) **12.** (A) **13.** (A) **14.** (C) **15.** (A) **16.** (C) **17.** (C) **18.** (C) **19.** (C) **20.** (A)

21. (B) **22.** (C)

RACE # 04

1. (A) **2.** (C) **3.** (A) **4.** (D) **5.** (D) **6.** (D) **7.** (D) **8.** (C) **9.** (D) **10.** (C)

11. (D) **12.** (D) **13.** (C) **14.** (C) **15.** (B) **16.** (A) **17.** (B) **18.** (A) **19.** (C) **20.** (C)

21. (C) **22.** (A) **23.** (C)

RACE # 05

1. (A–P, B–R, C– Q) **2.** (18,32) **3.** (D) **4.** (B) **5.** (A) **6.** (ABC) **7.** 4

8. 3 **9.** (A – R,S), (B –R,S, (C – P,Q), (D- P,Q) **10.** (A–PS) ; (B–R); (C– R); (D – Q)

11. (A – P,R), (B – P, Q, R), (C – P, R, S), (D – P, R, S) **12.** (A – P,S), (B – P), (C – Q), (D – R)**13.** (C)

RACE # 06

1. (A) **2.** (A) **3.** (C) **4.** (C) **5.** (D) **6.** (B) **7.** (D) **8.** (B) **9.** (B) **10.** (B)

11. (C) **12.** (D) **13.** (A) **14.** (C) **15.** (A) **16.** (D) **17.** (B) **18.** (C) **19.** (A) **20.** (B)

21. (C) **22.** (D) **23.** (A) **24.** (C) **25.** (A) **26.** (D) **27.** (C) **28.** (C)

RACE # 07

1. (D) **2.** (D) **3.** (D) **4.** (B) **5.** (B) **6.** (C) **7.** (D) **8.** (D) **9.** (D) **10.** (B)

11. (C) **12.** (C) **13.** (B) **14.** (B) **15.** (B) **16.** (A) **17.** (B) **18.** (C) **19.** (B) **20.** (C)

21. $N^{3-} > O^{2-} > F^{-} > Na^{+} > Mg^{2+}$ **23.** Cl < Si < Mg < Na < Cs**24.** r_{HCl} =5.919 A^0

RACE # 08

1. (C) **2.** (C) **3.** (A) **4.** (A) **5.** (AC) **6.** (D) **7.** (B) **8.** (B) **9.** (D) **10.** (B)

11. (C) **12.** (A) **13.** (B) **14.** (D) **15.** (D) **16.** (C) **17.** (A)

19. IE_2 = 1825 kJ/mol , IE_3 = 2737.5 kJ/mol

RACE # 09

1. (B) **2.** (D) **3.** (C) **4.** (D) **5.** (C) **6.** (A) **7.** (D) **8.** (D) **9.** (C) **10.** (B)
11. (B) **12.** (C) **13.** (C) **14.** (D) **15.** (C) **16.** (B) **17.** (D) **18.** (C) **19.** (B) **20.** (B)
21. (C) **22.** (D) **23.** (A) **24.** (D) **25.** (A)

RACE # 10

1. (B) **2.** (B) **3.** (A) **4.** (A) **5.** (B) **6.** (A) **7.** (A) **8.** (A) **9.** (D)

10* **1.** (+5) **2.** (+8) **3.** (-3) **4.** (+6) **5.** (+6) **6.** (+6) **7.** (+5) **8.** (+2) **9.** (0)
10. (+6) **11.** (+3) **12.** (+3) **13.** (0) **14.** (0) **15.** (+5) **16.** (8/3) **17.** (-1/3)
18. (-1/3) **19.** (+1) **20.** (+4) **21.** (+3) **22.** (+5) **23.** (+4) **24.** (-4/3) **25.** (-1) **26.** (-4)
27. (+3) **28.** (2.08)

11*. **1.** (8/3, -2) **2.** (+2, -2) **3.** (+1, 2.5, -2) **4.** (-2, +1, -2, +1)
5. (+2, +6, -2, -3, +1, +6, -2, +1, -2) **6.** (+4, -2) **7.** (+2, -1)
8. (+2, -2) **9.** (+4, -2) **10.** (+6, –1, –2) **11.** (-2, +1, +6, -2)
12. (+5, -2) **13.** (+1, +2, -3) **14.** (+1, -3, +2) **15.** (+2, +1, +1, -2)
16. (+8, -2) **17.** (+1, –2, +6, –2) **18.** (-2, +1, +4, -2, +1)

12. (A) **13.** (D) **14.** (ACD) **15.** (A–P, Q,T) ; (B R, T); (C– T); (D – S)

16. (i) Cl (ii) Cs (iii) Al (iv) F (v) Xe **17.** BaO

18. (a) Basic (b) Acidic (c) Basic (d) Acidic (e) Basic (f) Acidic

19. $CaO < CO < CO_2 < N_2O_5 < SO_3$

RACE # 11

1. (i) $Na_2S \rightarrow$ ionic, (ii) $SnCl_4 \rightarrow$ ionic, (iii) Diamond $\rightarrow$ covalent, (iv) $CaC_2 \rightarrow$ ionic
(v) $NaH \rightarrow$ ionic, (vi) $C_2H_4 \rightarrow$ covalent, (vii) $CaCl_2 \rightarrow$ ionic, (viii) HCl(g) $\rightarrow$ covalent
(ix) $NH_4^+ \rightarrow$ covalent, (x) KBr $\rightarrow$ ionic

2. (i) ionic $\rightarrow$ NaCl, (ii) Covalent $\rightarrow$ CS_2, (iii) Covalent $\rightarrow$ SO_2, (iv) ionic $\rightarrow$ CaH_2

3. (i) ionic, (ii) covalent, (iii) covalent, (iv) covalent, (v) covalent, (vi) covalent

4. (A) **5.** (C) **6.** (B) **7.** (C) **8.** (A) **9.** (A) **10.** (B) **11.** (D) **12.** (D) **13.** (A)
14. (D) **15.** (B) **16.** (B) **17.** (B) **18.** (C) **19.** (D) **20.** (D) **21.** (C) **22.** (B) **23.** (A)
24. (A) **25.** (B) **26.** (D)

RACE # 12

1. (D) **2.** (B) **3.** (B) **4.** (C) **5.** (D) **6.** (D) **8.** (B) **9.** (D) **10.** (C) **11.** (C)
12. (C) **13.** (A) **14.** (4) **15.** (3) **16.** (C) **17.** (D) **18.** (C) **19.** (A)

RACE # 13

1. (B) **2.** (B) **3.** (D) **4.** (BC) **5.** (B) **6.** (A) **7.** (B) **8.** (D) **9.** (D) **10.** (C)
11. (A) **12.** (B) **13.** (B) **14.** (i) 4 (ii) Two types (i) Covalent three (ii) Co-ordinate one
15. (B) **16.** (C) **17.** (A) **18** (i) $\sigma = 7, \pi = 0$ (ii) $\sigma = 5, \pi = 1$ (iii) $\sigma = 3, \pi = 2$
(iv) $\sigma = 8, \pi = 2$ (v) $\sigma = 5, \pi = 6$ (vi) $\sigma = 11, \pi = 4$ (vii) $\sigma = 11, \pi = 4$ **19.** (D)
20. (D) **21.** (D) **22.** (C) **23.** (A) **24.** (B) **25.** (B) **26.** (C) **27.** (B) **28.** (C) **29.** (C)
30. (C) **31.** (A) **32.** (D) **33.** (C) **34.** (A) **35.** (B) **36.** (3) **37.** (B)

38. (1) sp^3d^2 (2) sp (3) sp (4) sp^2 (36) sp^3d (7) sp^2 (9) sp^2 (10) sp^2 (11) sp^2
(12) sp^2 (13) sp^2 (14) sp^2 (15) sp^2 (16) sp^2 (17) sp^2 (18) sp^3 (19) sp^3 (20) sp^3
(22) sp^3 (23) sp^3 (24) sp^3 (25) sp^3 (27) sp^3 (28) sp^3 (30) sp^3d (31) sp^3d (32) sp^3d^2
(33) sp^3d^2 (34) sp^3d^2 (37) sp^3d^2 (38) sp^3d^2

RACE # 14

1. (C) **2.** (D) **3.** (A) **4.** (A) **5.** (C) **6.** (C) **7.** (B) **10.** (C) **11.** (D) **12.** (D)
13. (C) **14.** $H_2O > H_2S > BF_3$ **15.** (A) **16.** (B) **17.** a–qr, b–qs, c–q, d–prs **18.** (C) **19.** (C)
20. (D) **21.** (D) **22.** (C) **23.** (C) **24.** (D) **25.** (A) **26.** a–S, b–p, c–Q, d–TR **27.** (B)
28. (C) **29.** (D) **31.** (B) **32.** (A) **33.** (A)

34. (i) diple-dipole (ii) H-bonding (iii) H-bonding (iv) dipole-dipole
(v) H-bonding (vi) dipole-dipole (vii) Metallic (viii) H-bonding
(ix) dipole-dipole (x) ion-dipole (xi) London-forces (xii) co-valent bond
(xiii) London forces (xiv) Ionic (xv) dipole-induced dipole (xvi) London forces

RACE # 15

1. (B) **2.** A–p, B–r, C–q, D–s **3.** (A) **4.** (C) **5.** (B) **8.** (C) **9.** NO^+ **10.** (B)
11. (A) **12.** (B) **13.** (B)

RACE # 16

1. (D) **2.** (B) **3.** (C) **4.** (D) **5.** (ABCD) **6.** (D) **7.** (C) **8.** (C) **9.** (D)
10. (C) **11.** (C) **12.** (B) **13.** (A) **14.** (B) **15.** (D) **16.** (C) **17.** (B) **18.** (C) **19.** (C)
20. (A) **21.** (C)

RACE # 17

1. (A) **2.** (B) **3.** (A) **4.** (C) **5.** (B) **6.** (D) **7.** (B) **8.** (C) **9.** (B) **10.** (C)
11. (B) **12.** (A) **13.** (B)
14. (i) P_4O_6= 0.5 mole, P_4O_{10}= 0.5 mole, (ii) P_4O_6= 2 mole, P_4O_{10}= 1 mole, (iii) P_4O_6= 1 mole, P_4O_{10}= 2 mole
15. 640 gram

RACE # 18

1. (C) **2.** (D) **3.** (B) **4.** (C) **5.** (D) **6.** (A) **7.** (B) **8.** (B) **9.** (B) **10.** (B)
11. (A) **12.** (B) **13.** (A) **14.** (D) **15.** (a) 0.1M (b) 2 M **15.** (B) **17.** (A)
19. (A)→P,Q; (B)→P,Q; (C)→S (D)→R **20.** (A)→Q (B)→P (C)→S (D)→R

RACE # 19

1. 2.79 **2.** (D) **3.** (D) **4.** (B) **5.** (A) **6.** 0.3275 **7.** 1.5 M **8.** 0.34 **9.** (B) **10.** 60 ml **11.** (A) **12.** (A) **13.** (B) **14.** (C) **15.** (B) **16.** (C) **17.** (D) **18.** (C) **19.** (D) **20.** (B) **21.** (C) **22.** (A)

RACE # 20

1. (+4/2, +2) **2.** (10/4) **3.** (+6) **4.** (+6) **5.** (+6) **6.** (0) **7.** (0) **8.** (+5)
9. (+4/3) **10.** (+8) **11.** (-3) **12.** (+6) **13.** (+6) **14.** (+6) **15.** (+6) **16.** (+5) **17.** (+2)
18. (0) **19.** (-3, +5)
20. **1.** (+5) **2.** (–1,1,0) **3.** (+8) **4.** (+3) **5.** (+3) **6.** (0) **7.** (0)
8. (+3, +5, 4.5) **9.** (8/3) **10.** (-1/3) **11.** (-1/3) **12.** (+1) **13.** (+4)
14. (+3) **15.** (+5) **16.** (+4) **17.** (-4/3) **18.** (-1) **19.** (-4)
3. (C) **4.** (C) **5.** (C) **6.** (D) **7.** (D) **8.** (A) **9.** (A) **10.** (D)

11. (A) (B) (C) (D) (E)

Oxidant - $KMnO_4$ H_2O_2 HNO_3 $KBrO_3$ I_2

Reductant- KCl $FeCl_2$ Cu Na_2HAsO_3 $Na_2S_2O_3$

12. (a) (i) 2 (ii) 2 (b) (i) 1 (ii) 2 **13.** (i) 2, 6, 2, 3 (ii) $3,2,\frac{6}{5}$ (iii) 3,1,3 **14.** (B)

15. (D) **16.** (A) **17.** (B) **18.** (A) **19.** (B) **20.** (A)

21. (a) $FeSO_4 = \frac{152}{1} = 152$ $KClO_3 = \frac{122.5}{6} = 20.4$ (b) $Na_2SO_3 = \frac{126}{2} = 63$ $Na_2CrO_4 = \frac{162}{4} = 40.5$

(c) $Fe_3O_4 = \frac{232}{1} = 232$ $KMnO_4 = \frac{158}{3} = 52.67$ (d) $KI = \frac{166}{1}$ $K_2Cr_2O_7 = \frac{294}{6} = 49$

RACE # 21

1. (i) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$; (ii) $3Cl_2 + 6OH^- \rightarrow 5Cl^- + ClO_3^- + 3H_2O$

(iii) $3AsO_3^{3-} + 2MnO_4^- + 2H^+ \rightarrow 3AsO_4^{3-} + 2MnO_2 + H_2O$; (iv) $FeC_2O_4 \rightarrow Fe^{3+} + 2CO_2 + 3e^-$;

(v) $5P_2H_4 \rightarrow P_4H_2 + 6PH_3$

2. (i) $K_2\underset{6}{Cr_2}O_7 + 3\underset{2}{H_2O_2} + 4H_2SO_4 \rightarrow K_2SO_4 + \underset{3}{Cr_2}(SO_4)_3 + 7H_2O + 3\underset{2}{O_2}$

(ii) $4\underset{2}{Zn} + Na\underset{2}{N}O_3 + 7NaOH \rightarrow 4Na_2\underset{2}{Zn}O_2 + 2H_2O + \underset{8}{N}H_3$

(iii) $\underset{(6)}{ClO_3^-} + 6\underset{(1)}{Fe^{2+}} + 6H^+ \rightarrow \underset{(6)}{Cl^-} + 6\underset{(1)}{Fe^{3+}} + 3H_2O$

(iv) $3\underset{2}{N_2O_4} + \underset{6}{BrO_3^-} + 6OH^- \rightarrow 6\underset{1}{NO_3^-} + \underset{6}{Br^-} + 3H_2O$

(v) $3\underset{4}{S_2O_3^{-2}} + 2\underset{8}{Sb_2O_5} + 6H^+ + 3H_2O \rightarrow 4\underset{8}{Sb}O + 6H_2\underset{2}{S}O_3$

(vi) $\underset{6}{Cr_2O_7^{2-}} + 6\underset{1}{I^-} + 14H^+ \rightarrow 2\underset{3}{Cr^{3+}} + 3\underset{2}{I_2} + 7H_2O$

(vii) $\underset{5}{IO_3^-} + 5\underset{1}{I^-} + 6H^+ \rightarrow 3\underset{10}{I_2} + 3\underset{6}{H_2O}$

3. (B) **4.** (B) **5.** (C) **6.** (C) **7.** (D) **8.** (B) **9.** (A) **10.** (B) **11.** (C) **12.** (B)
13. (A) **14.** (B)

RACE # 22

1. (C) **2.** (C) **3.** (A) **4.** (A) **5.** (C) **6.** (C) **7.** (B) **8.** (B) **9.** (B) **10.** (B)
11. (A) **12.** (C) **13.** (B) **14.** (C) **15.** (D) **16.** (C) **17.** (C) **18.** (A) **19.** (B)

20. A–R, B–S, C–P, D–Q **21.** (B) **22.** (B) **23.** (D)

RACE # 23

1. (B) **2.** (C) **3.** (A) **4.** (C) **5.** (BC) **6.** (C) **7.** (C) **8.** (D) **9.** (A) **10.** (C)
11. (ABD) **12.** (i) 20 gm (ii) 35.4gm (iii) 39.6gm **13.** (a) 0.169 (b) 118%
14. (C) **15.** (B) **16.** (A) **17.** (A) **18.** (C) **19.** (D)

RACE # 24

1. (D) **2.** (D) **3.** (ABCD) **4.** (AD) **5.** (C) **6.** **7.** (A) **8.** (30) **9.** (40)
10. A–PR, B–QS, C–QS, D–PR **11.** (D) **12.** (D) **13.** (B) **14.** (D) **15.** (B) **16.** (C)
17. (ABD) **18.** (C) **19.** (A) **20.** (A)

RACE # 25

1. (C) **2.** (D) **3.** (D) **4.** (C) **5.** (C) **6.** (A) **7.** (D) **8.** (A) **9.** (B) **10.** (B)
11. (C) **12.** (C) **13.** (A) **14.** (D) **15.** (ABD) **16.** (ABC) **17.** (1000m**)**

RACE # 26

1. (B) **2.** (B) **3.** (A) **4.** (A) **5.** (D) **6.** (C) **7.** **(9)** **8.** (5) **9.** (D) **10.** (A)
11. (C) **12.** (D) **13.** (D) **14.** (C) **15.** (B) **16.** (B) **17.** (A) **18.** (A) **19.** (C) **20.** (A)
21. (A) **22.** (C) **23.** (C)

RACE # 27

1. (C) **2.** (D) **3.** (B) **4.** (B) **5.** (D) **6.** (B) **7.** (D) **8.** (C) **9.** (C) **10.** (C)
11. (A) **12.** (B) **13.** (D) **14.** (B) **15.** (C) **16.** (ABD) **17.** (C) **18.** (ABCD)
19. (BD) **20.** (A) **21.** (A) **22.** (D)

RACE # 28

1. (B) **2.** (A) **3.** (D) **4.** (B) **5.** (B) **6.** (C) **7.** (D) **8.** (A) **9.** (B) **10.** (AB)
11. A–RS, B–PS, C–QRS, D–PQS **12.** (A) **13.** (B) **14.** (B) **15.** (A) **16.** (A) **17.** (D)
18. (9) **19.** (8)

RACE # 29

1. (C) **2.** (C) **3.** (A) **4.** (A) **5.** (C) **6.** (A) **7.** (D) **8.** (C) **9.** (C) **10.** (B)
11. (D) **12.** (D) **13.** (A) **14.** (B) **15.** (A) **16.** (D) **17.** (C) **18.** (A) **19.** (D) **20.** (C)

RACE # 30

1. (A) **2.** (C) **3.** (D) **4.** (A) **5.** (C) **6.** (B) **7.** (A) **8.** (ABC)
9. (ABC) **10.** (C) **11.** (C) **12.** (D) **13.** (D) **14.** (D)

RACE # 31

1. (C) **2.** (A) **3.** (A) **4.** (A) **5.** (B) **6.** (B) **7.** (A) ***8.** (C) **9.** (C) **10.** (A)
***11.** (B) **12.** (A) **13.** (D) **14.** (B) **15.** (C) **16.** (A) ***17.** (C) **18.** (B) **19.** (A) **20.** (B)

RACE # 32

1. (C) **2.** (A) ***3.** (B) **4.** (D) ***5.** (D) ***6.** (A) **7.** (D) **8.** (A) **9.** (D)
*__10.__ (ABCD) **11.** (A) **12.** (D) ***13.** (D) **14.** (A) **15.** (A) **16.** (A) **17.** (A) **18.** (C)
19. (A)

RACE # 33

1. (D) **2.** (B) **3.** (C) **4.** (B) **5.** (B) **6.** (A) **7.** (D) **8.** (C) **9.** (A) **10.** (B)
11. (C) **12.** (A) **13.** (B) **14.** (B) **15.** (C) ***16.** (C) **17.** (B) **18.** (B) ***19.** (B) **20.** (C)
21. (B)

RACE # 34

1. (B) **2.** (B) **3.** (B) ***4.** (B) ***5.** (B) **6.** (C) **7.** (C) **8.** (C) **9.** (B) **10.** (A)
11. (A) **12.** (D) ***13.** (B) **14.** (C) **15.** (C) **16.** (A) **17.** (C)

RACE # 35

1. (D) **2.** (A) **3.** (ABCD) **4.** (A) **5.** (B) **6.** (C) **7.** (D) **8.** (B) **9.** (B)
10. (B) **11.** (C) **12.** (C) **13.** (D) **14.** (D) **15.** (C) **16.** (A) **17.** (D) **18.** (C)
19. (C) **20.** (A) **21.** (B) **22.** (A) **23.** (A)

RACE # 36

1. (D) **2.** (B) **3.** (C) **4.** (B) **5.** (B) **6.** (C) **7.** (D) **8.** (D) **9.** (BC) **10.** (CD)
11. (C) **12.** (C) **13.** (C) **14.** (D) **15.** (D) **16.** (A) **17.** (A) **18.** (C) **19.** (A) **20.** (B)
21. (C) **22.** (D)

RACE # 37

2. (B) **3.** (A) **4.** (A) **5.** (C) **6.** (A) **7.** (B) **8.** (D) **9.** (A) **10.** (A) **11.** (C)
12. (C) **13.** (D) **14.** (D) **15.** (D) **16.** (A) **17.** 0.320 **18.** 0.0833,4M,1.5M
19. (BCD) **20.** (ABD)

RACE # 38

1. (ABCD) **2.** (AC) **3.** (ABCD) **4.** (B) **5.** (BC) **6.** (ABD) **7.** (ABCD)
8. (AC) **9.** (AC) **10.** (CD) **11.** (BC) **12.** (BD) **13.** (ACD) **14.** (C) **15.** (C)
16. (A) **17.** (A) **18.** (A) **19.** (A)

RACE # 39

1. (AB) **2.** (C) **3.** (B) **4.** (B) **5.** (BC) **6.** (B) **7.** (B) **9.** (D) **10.** (B) **11.** (D)
12. (A) **13.** (C) **14.** (B) **15.** (D) **16.** (B) **17.** (B) **18.** 50 **19.** (C) **20.** (D) **21.** (C)
22. (B) **23.** (C) **24.** (C) **25.** (A) **26.** (D)

RACE # 40

1. (B) **2.** (C) **3.** (B) **4.** (C) **5.** (D) **6.** (C) **7.** (B) **8.** (B) **9.** (A) **10.** (B)
11. (C) **12.** (C) **13.** (ACD) **14.** (ABCD) **15.** (C) **16.** (B) **17.** (A) **18.** (C)
19. (C) **20.** (D) **21.** (B)

RACE # 41

1. (C) **2.** (D) **3.** (D) **4.** (A) **5.** (B) **6.** (C) **7.** (B) **8.** (A) **9.** (D) **10.** (B)
11. (D) **12.** (A) **13.** (ABC) **14.** (4) **15.** (50) **16.** (A) q,s,t ; (B) p,q,t ; (C) p,r ; (D) q,s
17. (C) **18.** (A) **19.** (A)

RACE # 42

1. (C) **2.** (B) **3.** (D) **4.** (B) **5.** (C) **6.** (D) **7.** (C) **8.** (B) **9.** (A) **10.** (A)
11. (B) **12.** (C) **13.** (D) **14.** (AC) **15.** (AD) **16.** (ABC) **17.** (D) **18.** (D) **19.** (A)
20. (B) **21.** (D) **22.** (A) **23.** (A) **24.** (D) **25.** (A) **26.** (C) **27.** (C)

RACE # 43

1. (C) **2.** (A) **3.** (A) **4.** (A) **5.** (D) **6.** (B) **7.** (B) **8.** (A) **9.** (B) **10.** (C)
11. (A) **12.** (B) **13.** (B) **14.** (C) **15.** (ABC) **16.** (ABCD) **17.** (ABC)
18. (30) **19.** (B) **20.** (B) **21.** (B)

RACE # 44

1. (A) **2.** (A) **3.** (ABCD) **4.** (A – s); (B – p, q); (C – q, s) ; (D – q, r) **5.** (C) **6.** (D)
7. (D) **8.** (D) **9.** (C) **10.** (A) **11.** (D) **12.** (B) **13.** (C) **14.** (ABD) **15.** (ACD)
16. (B) **17.** (A) **18.** (D) **19.** (B) **20.** (C) **21.** (C) **22.** (D)

RACE # 45

1. (B) **2.** (B) **3.** (C) **4.** (D) **5.** (A) **6.** (A) **7.** (D) **8.** (D) **9.** (B) **10.** (A)
11. (A) **12.** (A) **13.** (A) **14.** (B) **15.** (D) **16.** (ABD) **17.** (BCD)

RACE # 46

1. (BD) **2.** (BCD) **3.** (D) **4.** (B) **5.** (A) **6.** (D)
7. (A) $\rightarrow$ p,s (B) $\rightarrow$ p,r,s (C) $\rightarrow$ q,r (D) $\rightarrow$ p,r,s **8.** (D) **9.** (B) **10.** (B) **11.** (D) **12.** (A)
13. (A) **14.** (A) **15.** (D) **16.** (B) **17** 5.76 J/K **18.** (B) **19.** (D) **20.** (C)
21. (ABD) **22.** (ABD)

RACE # 47

1. (B) **2.** (D) **3.** (D) **4.** (A) **5.** (B) **6.** (D) **7.** (C) **8.** (C) **9.** (A) **10.** (C)
11. (B) **12.** (B) **13.** (C) **14.** (D) **15.** (A) **16.** (BCD) **17.** (ABC) **18.** (AC)

RACE # 48

1. (C) **2.** (A) **3.** (C) **4.** (B) **5.** (A) **6.** (D) **7.** (A) **8.** (A) **9.** (D) **10.** (A)
11. (C) **12.** (C) **13.** (C) **14.** (B) **15.** (A) **16.** (A) **17.** (B) **18.** (D) **19.** (D) **20.** (C)

RACE # 49

1. (ii) **2.** (iv) **3.** (ii) **4.** (iv) **5.** (iii) **6.** (ii) **7.** (ii) **8.** (iii) **9.** (ii)
10. (iii) **11.** (iii) **12.** (iv) **13.** (ii) **14.** (iv) **15.** (i) **16.** (iii) **17.** (i) **18.** (i)

II. Multiple Choice Questions (Type-II)

19. (iii), (iv) **20.** (i), (ii) **21.** (i), (iii) **22.** (i), (ii) **23.** (iii), (iv)
24. (i), (ii) **25.** (ii), (iii) **26.** (i), (iv) **27.** (i), (ii), (iii)

RACE # 50

I. Multiple Choice Questions (Type-I)

1. (iv) **2.** (i) **3.** (iii) **4.** (iv) **5.** (i) **6.** (i) **7.** (i) **8.** (iii) **9.** (ii)
10. (i) **11.** (i) **12.** (i) **13.** (ii) **14.** (i) **15.** (iii) **16.** (i) **17.** (iv) **18.** (ii)
19. (i) **20.** (iii) **21.** (iv)

II. Multiple Choice Questions (Type-II)

22. (ii), (iv) **23.** (i), (iii) **24.** (i), (ii) **25.** (ii), (iii)

RACE # 51

1. (C) **2.** (C) **3.** (D) **4.** (D) **5.** (C) **6.** (C) **7.** (A) **8.** (C) **9.** (A) **10.** (D)
11. (B) **12.** (D) **13.** (D) **14.** (A) **15.** (ABD) **16.** (AB) **17.** (A) **18.** (B)
19. A- R, B- P,Q. C-P,Q. D- S **20.** (D) **21.** (C) **22.** (B) **23.** (B) **24.** (A) **25.** (C) **26.** (D)
27. (C) **28.** (B) **29.** (C) **30.** (C) **31.** (B) **32.** (B) **33.** (B) **34.** (B) **35.** (D) **36.** (C)
37. (A) **38.** (B) **39.** (A) **40.** (C) **41.** (C) **42.** (ABD) **43.** (D)

RACE # 52

1. (B) **2.** (D) **3.** (B) **4.** (B) **5.** (C) **6.** (B) **7.** (D) **8.** (A) **9.** (B) **10.** (D)
11. (A) **12.** (B) **13.** (C)

RACE # 53

1. (B) **2.** (D) **3.** (A) **4.** (D) **5.** (D) **6.** (AB) **7.** (ACD) **8.** (AB) **9.** (ABC)
10. (D) **11.** (5) **12.** (4) **13.** (5) **14.** (1)

RACE # 54

1. (C) **2.** (D) **3.** (D) **4.** (A) **5.** (B) **6.** (B) **7.** (B) **8.** (B) **9.** (C) **10.** (C)
11. (A) **12.** (D) **13.** (A) **14.** (C) **15.** (C) **16.** (B) **17.** (A) **18.** (A) **19.** (B) **20.** (B)
21. (C) **22.** (B)

RACE # 55

1. (C) **2.** (A) **3.** (C) **4.** (D) **5.** (ABD) **6.** (ACD) **7.** (ABC) **8.** (AB)
9. (B) **10.** (B) **11.** **(3)** **12.** (4) **13.** (8)

RACE # 56

1. (A) Q; (B) S; (C) P; (D) R 2. (A) **3.** (B) **4.** (C) **5.** (BC) **6.** (C) 7. (B) **8.** (A)
9. (D) **10.** (D)

RACE # 57

1. (A)-R,T ; (B)-P,Q,S,T ; (C)-P, (D) - P,Q,T **2.** (A) **3.** (A) **4.** (C) **5.** (AD) **6.** (BCD)
7. (D) **8.** (C) **9.** (C)
10. (1) 2,2,7 Trimethyl-4-(methyl)propyl nonane (2) 4-(methylethyl)-5-n-propyl octane
(3) 3,3-Diethyl-5-isopropyl-4-methyl octane (4) 5-ethyl-2,5-dimethyl oct-1-ene
(5) 5-(2,2-dimethylpropyl)2,4-nonadiene

RACE # 58

1. (A)→Q ; (B)→R ; (C)→S ; (D)→P **2.** (C) **3.** (A) **7.** (i) 5-Methyl hept-1-yne
(ii) 3-Ethyl-4-methylpentanamide (iii) 1-Methyl-6-vinylcyclohexene **8.** (19) **9.** (6) **10.** (612)

RACE # 59

1. (C) **2.** (B) **3.** (A) **4.** (C) **5.** (B) **6.** (C) **7.** (B) **8.** (D) **9.** (B) **10.** (B)
11. (C) **12.** (D) **13.** (D) **14.** (C) **15.** (D) **16.** (C) **17.** (A) **18.** (C) **19.** (D) **20.** (B)
21. (C) **22.** (C) **23.** (C) **24.** (D) **25.** (A) **26.** (C) **27.** (C) **28.** (D) **29.** (B) **30.** (D)
31. (D) **32.** (A)

RACE # 60

1. (C) **2.** (B) **3.** (D) **4.** (C) **5.** (D) **6.** (A) **7.** (BCD) **8.** (D)
9. (ii) > (i) > (iii) **10.** (7) **11.** (FGI) **12.** (B) **13.** (A) → P ; (B) → T ; (C) → P ; (D) → S

RACE # 61

1. (A)→P, R, S ; (B)→Q, T ; (C)→R, S ; (D)→Q, T **2.** (D) **3.** (ABC) **4.** (B) **5.** (D)
6. (A) **7.** (D) **8.** (B) **9.** (005) **10.** (036)

RACE # 62

(1) Grey-non Aromatic (2) Light Grey-Aromatic (3) White-Anti Aromatic

RACE # 63

1. (A)-P,R,T ; (B)-Q,S,T ; (C)-Q,R,S,T , (D)-P,T **2.** (B) **3.** (B) **4.** (C) **5.** (A) **6.** (D)
7. (C) **8.** (A) **9.** (003) **10.** (4)

RACE # 64

1. (A) – P, Q,R,S ; (B) – T, Q ; (C) – P ; (D) - P, Q **2.** (D) **3.** (BC) **4.** (ABC) **5.** (B)
6. (ABCD) **7.** (B) **8.** (ABCD) **9.** (ABCD) **10.** (A) (7) , (B) Ans. (8)

RACE # 65

1. (A)$\rightarrow$P, Q, S (B)$\rightarrow$Q, R, T (C)$\rightarrow$P, Q, S (D)$\rightarrow$P, S **2.** (C) **3.** (D > C > A > B) **4.** (BD)
5. (B) **6.** (C) **7.** (C) **8.** (BC) **9.** (ABC) **10.** (51)

Important Notes